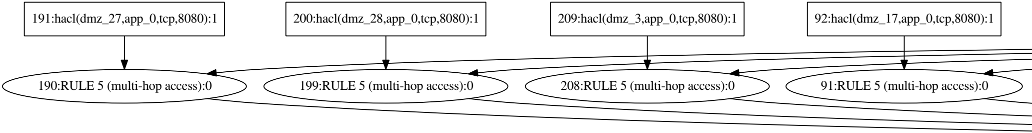
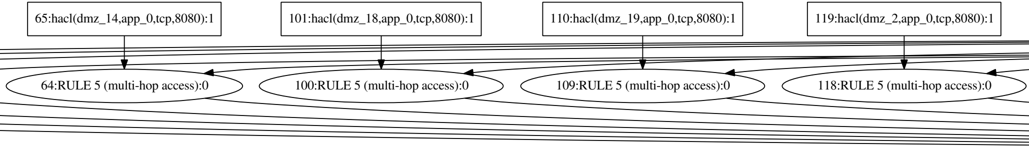
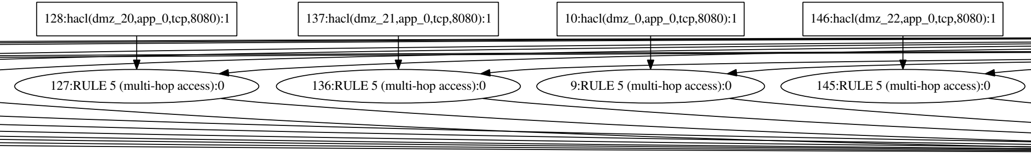
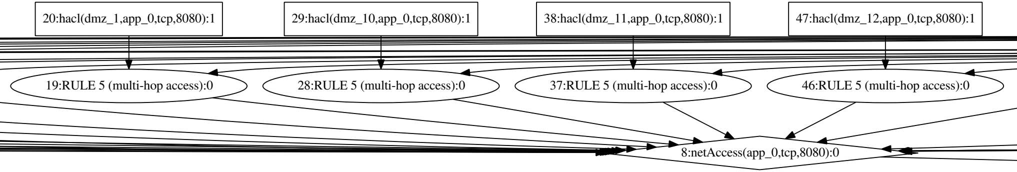
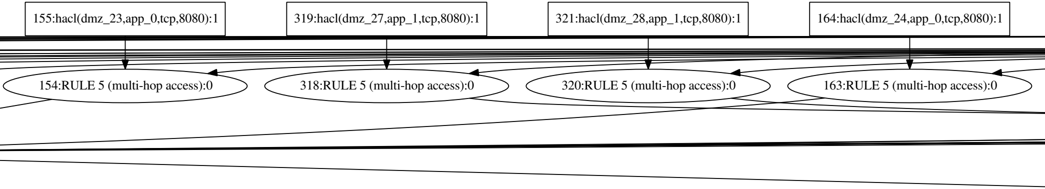
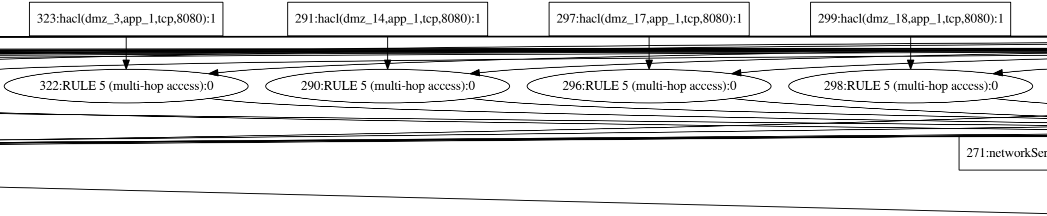

191:hacl(dmz_27,app_0,tcp,8080):1
200:hacl(dmz_28,app_0,tcp,8080):1
209:hacl(dmz_3,app_0,tcp,8080):1
92:hacl(dmz_17,app_0,tcp,8080):1
190:RULE 5 (multi-hop access):0
199:RULE 5 (multi-hop access):0
208:RULE 5 (multi-hop access):0
91:RULE 5 (multi-hop access):0

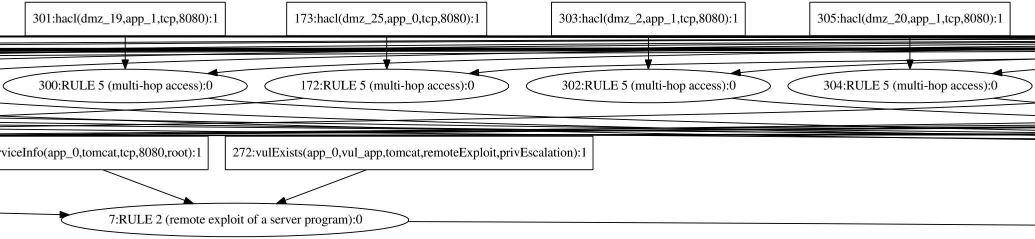

65:hacl(dmz_14,app_0,tcp,8080):1
101:hacl(dmz_18,app_0,tcp,8080):1
110:hacl(dmz_19,app_0,tcp,8080):1
119:hacl(dmz_2,app_0,tcp,8080):1
64:RULE 5 (multi-hop access):0
100:RULE 5 (multi-hop access):0
109:RULE 5 (multi-hop access):0
118:RULE 5 (multi-hop access):0

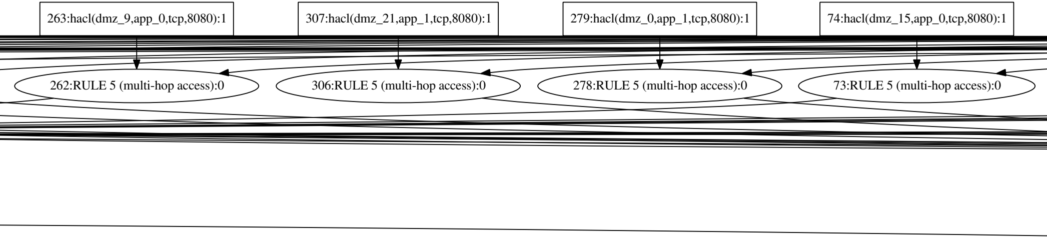

128:hacl(dmz_20,app_0,tcp,8080):1
137:hacl(dmz_21,app_0,tcp,8080):1
10:hacl(dmz_0,app_0,tcp,8080):1
146:hacl(dmz_22,app_0,tcp,8080):1
127:RULE 5 (multi-hop access):0
136:RULE 5 (multi-hop access):0
9:RULE 5 (multi-hop access):0
145:RULE 5 (multi-hop access):0

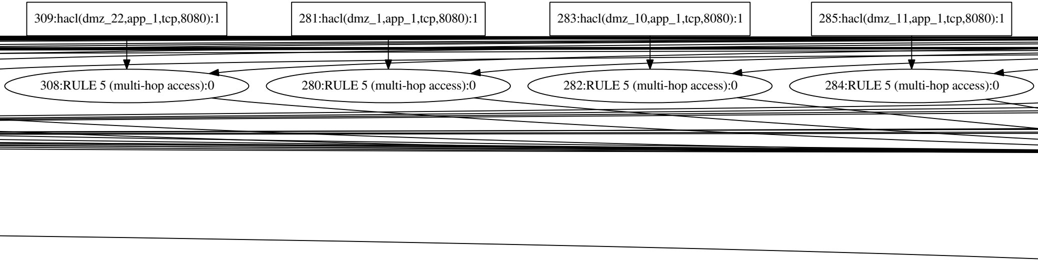

20:hacl(dmz_1,app_0,tcp,8080):1
29:hacl(dmz_10,app_0,tcp,8080):1
38:hacl(dmz_11,app_0,tcp,8080):1
47:hacl(dmz_12,app_0,tcp,8080):1
19:RULE 5 (multi-hop access):0
28:RULE 5 (multi-hop access):0
37:RULE 5 (multi-hop access):0
46:RULE 5 (multi-hop access):0
8:netAccess(app_0,tcp,8080):0

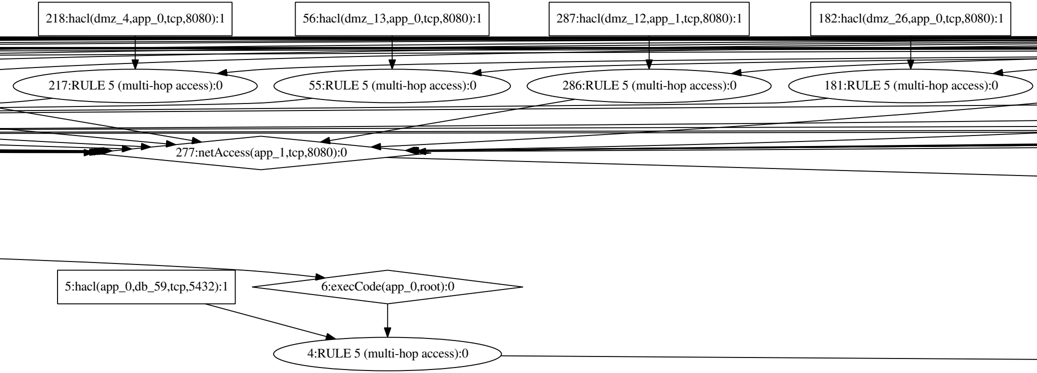

155:hacl(dmz_23,app_0,tcp,8080):1
319:hacl(dmz_27,app_1,tcp,8080):1
321:hacl(dmz_28,app_1,tcp,8080):1
164:hacl(dmz_24,app_0,tcp,8080):1
154:RULE 5 (multi-hop access):0
318:RULE 5 (multi-hop access):0
320:RULE 5 (multi-hop access):0
163:RULE 5 (multi-hop access):0

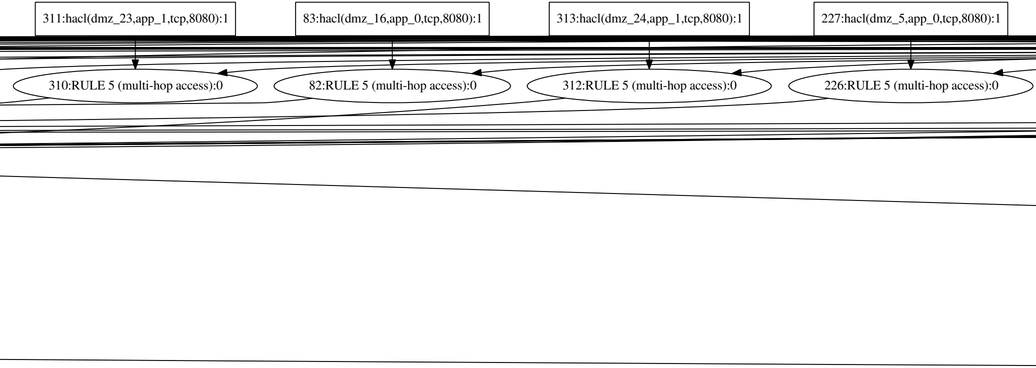

323:hacl(dmz_3,app_1,tcp,8080):1
291:hacl(dmz_14,app_1,tcp,8080):1
297:hacl(dmz_17,app_1,tcp,8080):1
299:hacl(dmz_18,app_1,tcp,8080):1
322:RULE 5 (multi-hop access):0
290:RULE 5 (multi-hop access):0
296:RULE 5 (multi-hop access):0
298:RULE 5 (multi-hop access):0
271:networkSer

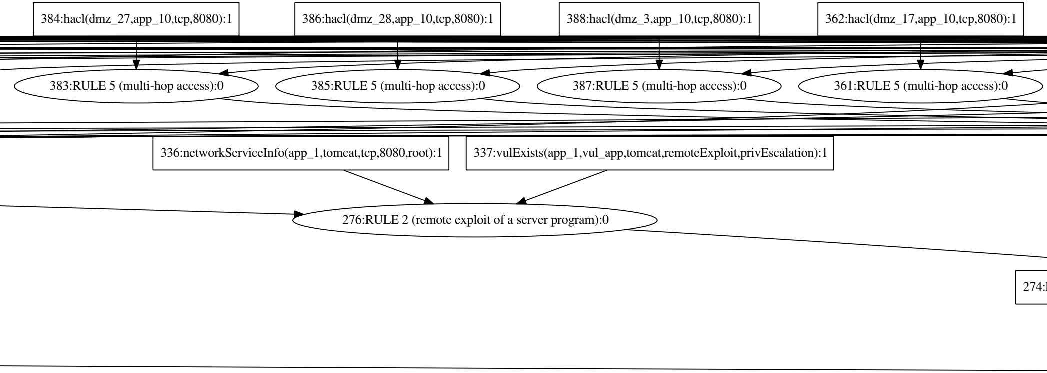

301:hacl(dmz_19,app_1,tcp,8080):1
173:hacl(dmz_25,app_0,tcp,8080):1
303:hacl(dmz_2,app_1,tcp,8080):1
305:hacl(dmz_20,app_1,tcp,8080):1
300:RULE 5 (multi-hop access):0
172:RULE 5 (multi-hop access):0
302:RULE 5 (multi-hop access):0
304:RULE 5 (multi-hop access):0
viceInfo(app_0,tomcat,tcp,8080,root):1
272:vulExists(app_0,vul_app,tomcat,remoteExploit,privEscalation):1
7:RULE 2 (remote exploit of a server program):0

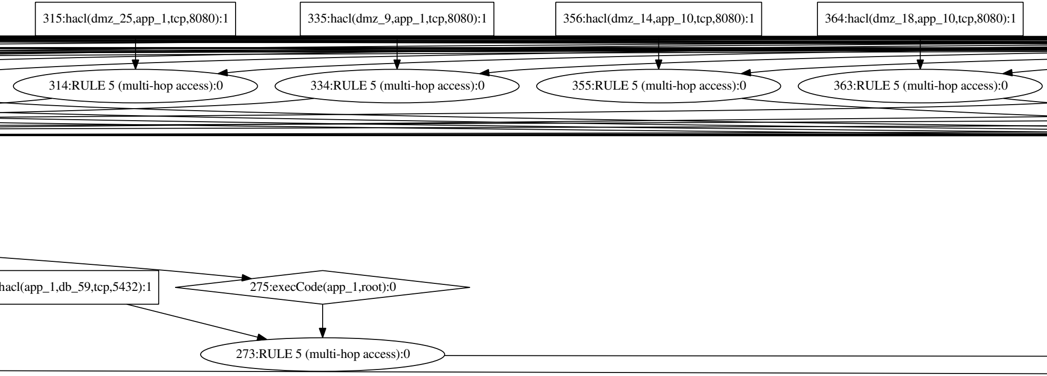

263:hacl(dmz_9,app_0,tcp,8080):1
307:hacl(dmz_21,app_1,tcp,8080):1
279:hacl(dmz_0,app_1,tcp,8080):1
74:hacl(dmz_15,app_0,tcp,8080):1
262:RULE 5 (multi-hop access):0
306:RULE 5 (multi-hop access):0
278:RULE 5 (multi-hop access):0
73:RULE 5 (multi-hop access):0

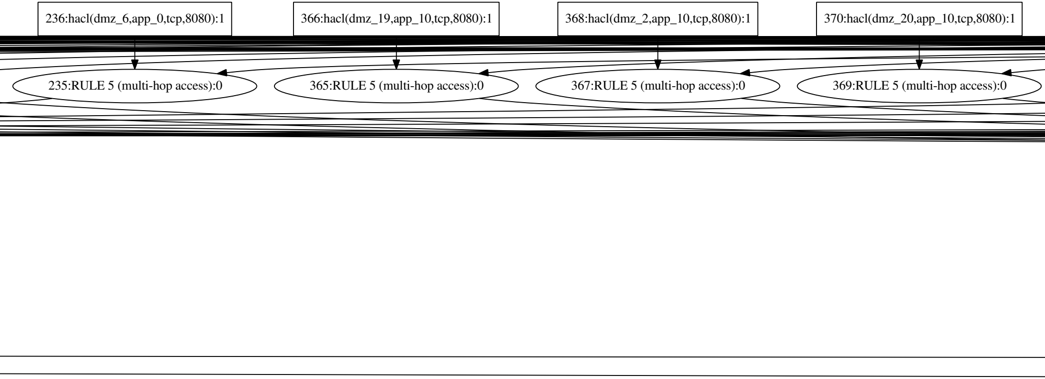

309:hacl(dmz_22,app_1,tcp,8080):1
281:hacl(dmz_1,app_1,tcp,8080):1
283:hacl(dmz_10,app_1,tcp,8080):1
285:hacl(dmz_11,app_1,tcp,8080):1
308:RULE 5 (multi-hop access):0
280:RULE 5 (multi-hop access):0
282:RULE 5 (multi-hop access):0
284:RULE 5 (multi-hop access):0

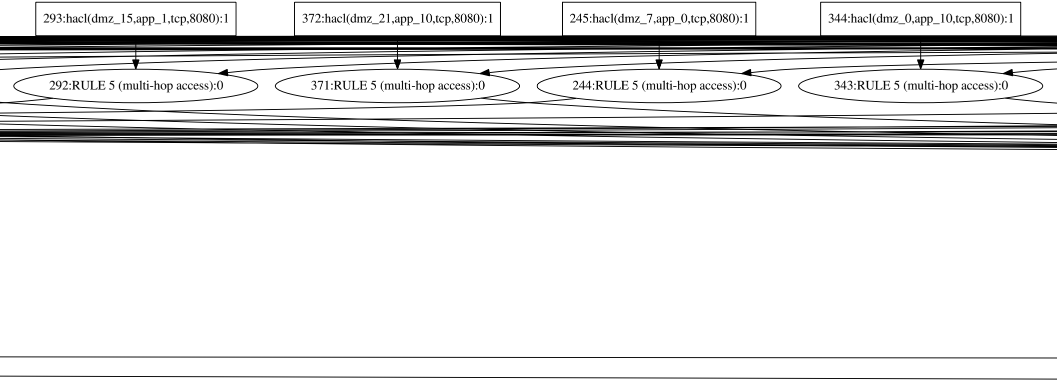

218:hacl(dmz_4,app_0,tcp,8080):1
56:hacl(dmz_13,app_0,tcp,8080):1
287:hacl(dmz_12,app_1,tcp,8080):1
182:hacl(dmz_26,app_0,tcp,8080):1
217:RULE 5 (multi-hop access):0
55:RULE 5 (multi-hop access):0
286:RULE 5 (multi-hop access):0
181:RULE 5 (multi-hop access):0
277:netAccess(app_1,tcp,8080):0
5:hacl(app_0,db_59,tcp,5432):1
6:execCode(app_0,root):0
4:RULE 5 (multi-hop access):0

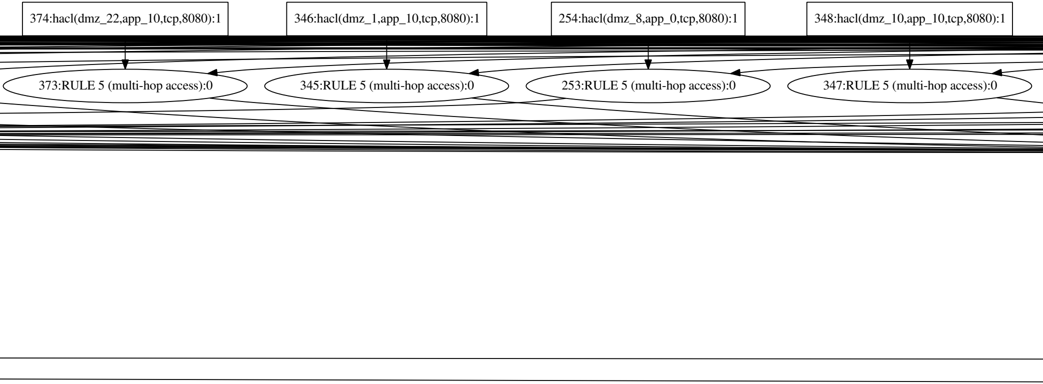

311:hacl(dmz_23,app_1,tcp,8080):1
83:hacl(dmz_16,app_0,tcp,8080):1
313:hacl(dmz_24,app_1,tcp,8080):1
227:hacl(dmz_5,app_0,tcp,8080):1
310:RULE 5 (multi-hop access):0
82:RULE 5 (multi-hop access):0
312:RULE 5 (multi-hop access):0
226:RULE 5 (multi-hop access):0

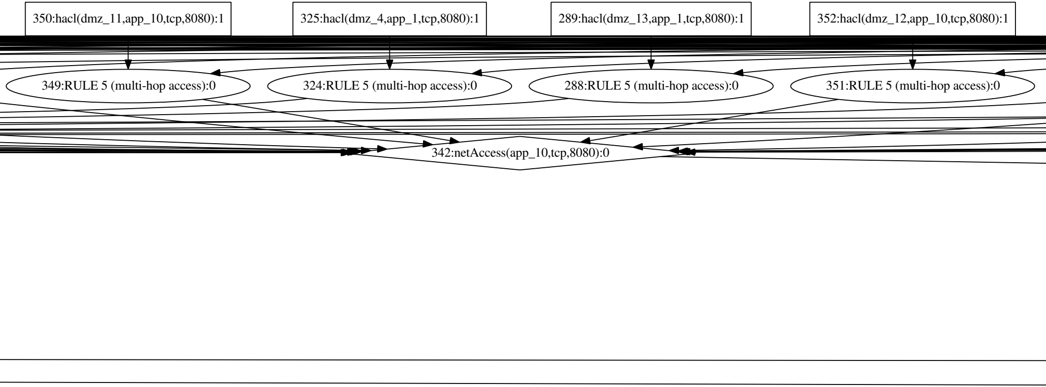

384:hacl(dmz_27,app_10,tcp,8080):1
386:hacl(dmz_28,app_10,tcp,8080):1
388:hacl(dmz_3,app_10,tcp,8080):1
362:hacl(dmz_17,app_10,tcp,8080):1
383:RULE 5 (multi-hop access):0
385:RULE 5 (multi-hop access):0
387:RULE 5 (multi-hop access):0
361:RULE 5 (multi-hop access):0
336:networkServiceInfo(app_1,tomcat,tcp,8080,root):1
337:vulExists(app_1,vul_app,tomcat,remoteExploit,privEscalation):1
276:RULE 2 (remote exploit of a server program):0
274:

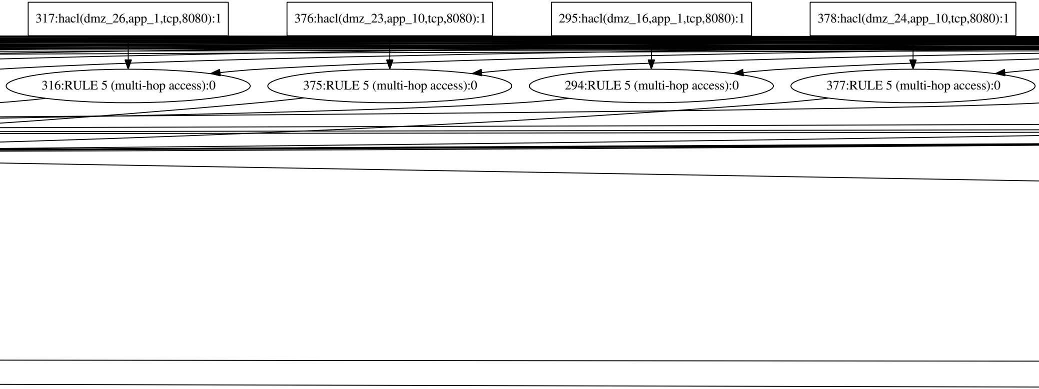

315:hacl(dmz_25,app_1,tcp,8080):1
335:hacl(dmz_9,app_1,tcp,8080):1
356:hacl(dmz_14,app_10,tcp,8080):1
364:hacl(dmz_18,app_10,tcp,8080):1
314:RULE 5 (multi-hop access):0
334:RULE 5 (multi-hop access):0
355:RULE 5 (multi-hop access):0
363:RULE 5 (multi-hop access):0
hacl(app_1,db_59,tcp,5432):1
275:execCode(app_1,root):0
273:RULE 5 (multi-hop access):0

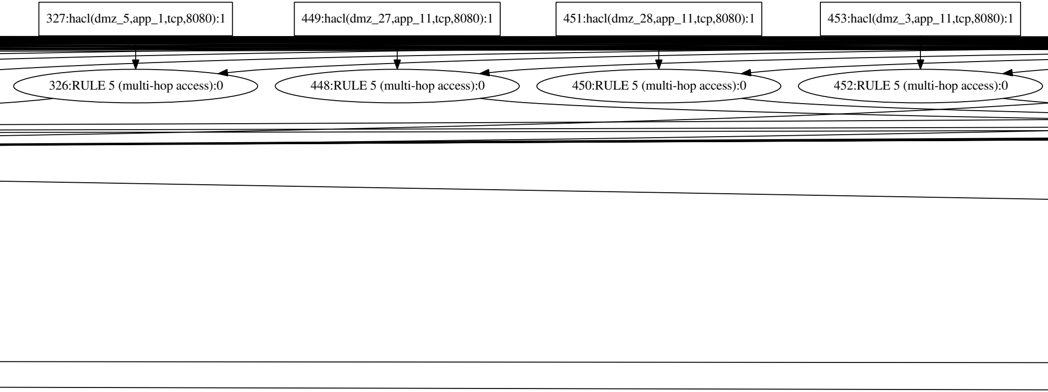

236:hacl(dmz_6,app_0,tcp,8080):1
366:hacl(dmz_19,app_10,tcp,8080):1
368:hacl(dmz_2,app_10,tcp,8080):1
370:hacl(dmz_20,app_10,tcp,8080):1
235:RULE 5 (multi-hop access):0
365:RULE 5 (multi-hop access):0
367:RULE 5 (multi-hop access):0
369:RULE 5 (multi-hop access):0

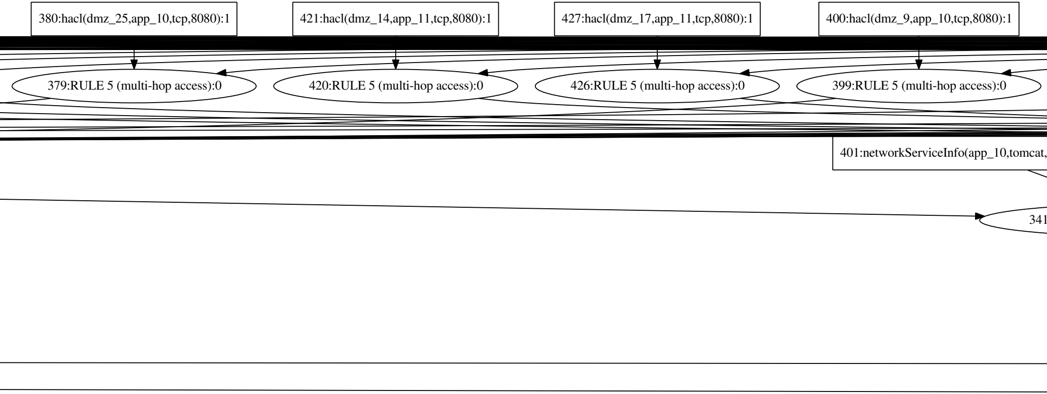

293:hacl(dmz_15,app_1,tcp,8080):1
372:hacl(dmz_21,app_10,tcp,8080):1
245:hacl(dmz_7,app_0,tcp,8080):1
344:hacl(dmz_0,app_10,tcp,8080):1
292:RULE 5 (multi-hop access):0
371:RULE 5 (multi-hop access):0
244:RULE 5 (multi-hop access):0
343:RULE 5 (multi-hop access):0

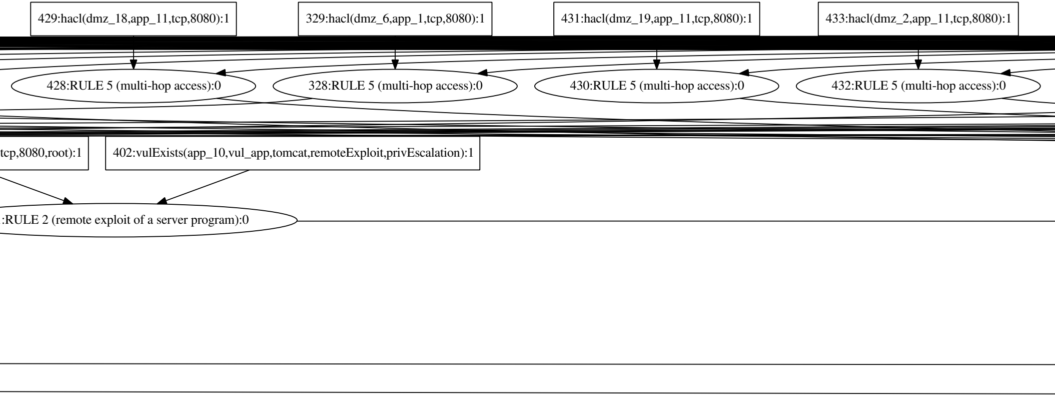

374:hacl(dmz_22,app_10,tcp,8080):1
346:hacl(dmz_1,app_10,tcp,8080):1
254:hacl(dmz_8,app_0,tcp,8080):1
348:hacl(dmz_10,app_10,tcp,8080):1
373:RULE 5 (multi-hop access):0
345:RULE 5 (multi-hop access):0
253:RULE 5 (multi-hop access):0
347:RULE 5 (multi-hop access):0

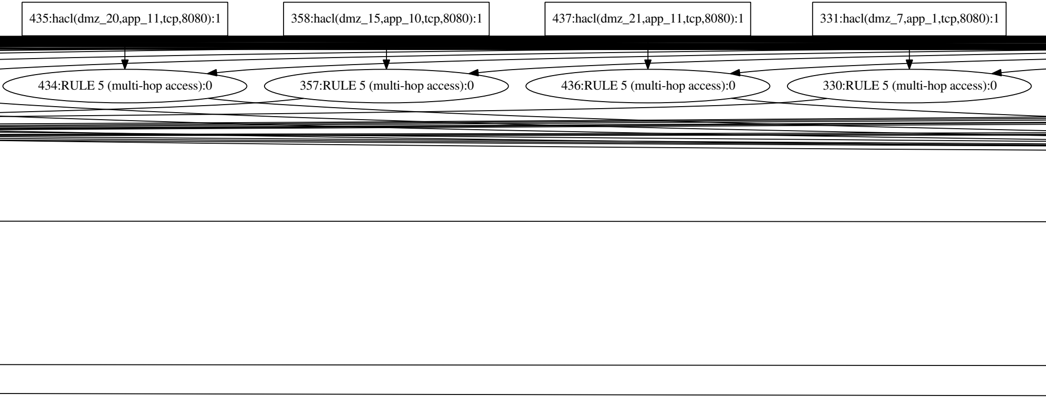

350:hacl(dmz_11,app_10,tcp,8080):1
325:hacl(dmz_4,app_1,tcp,8080):1
289:hacl(dmz_13,app_1,tcp,8080):1
352:hacl(dmz_12,app_10,tcp,8080):1
349:RULE 5 (multi-hop access):0
324:RULE 5 (multi-hop access):0
288:RULE 5 (multi-hop access):0
351:RULE 5 (multi-hop access):0
342:netAccess(app_10,tcp,8080):0

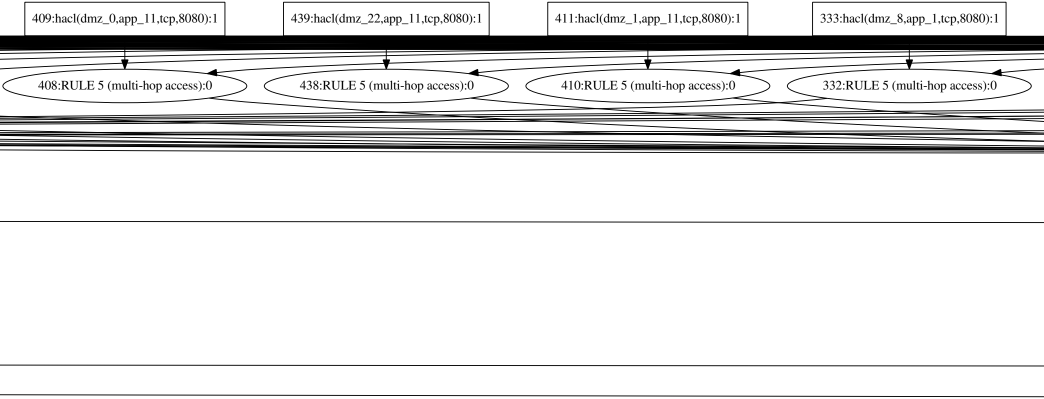

317:hacl(dmz_26,app_1,tcp,8080):1
376:hacl(dmz_23,app_10,tcp,8080):1
295:hacl(dmz_16,app_1,tcp,8080):1
378:hacl(dmz_24,app_10,tcp,8080):1
316:RULE 5 (multi-hop access):0
375:RULE 5 (multi-hop access):0
294:RULE 5 (multi-hop access):0
377:RULE 5 (multi-hop access):0

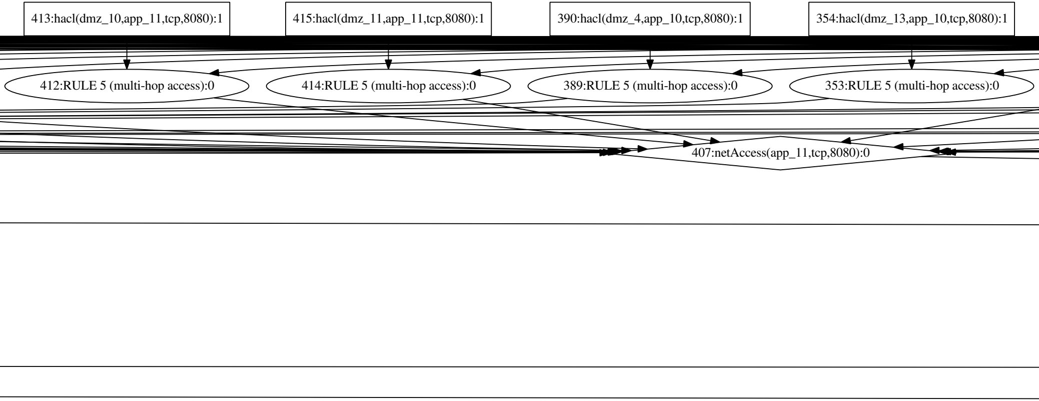

327:hacl(dmz_5,app_1,tcp,8080):1
449:hacl(dmz_27,app_11,tcp,8080):1
451:hacl(dmz_28,app_11,tcp,8080):1
453:hacl(dmz_3,app_11,tcp,8080):1
326:RULE 5 (multi-hop access):0
448:RULE 5 (multi-hop access):0
450:RULE 5 (multi-hop access):0
452:RULE 5 (multi-hop access):0

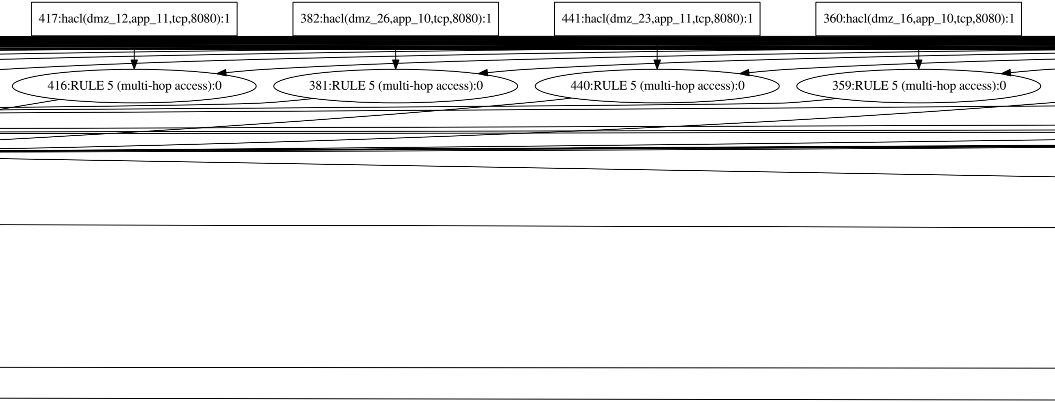

380:hacl(dmz_25,app_10,tcp,8080):1
421:hacl(dmz_14,app_11,tcp,8080):1
427:hacl(dmz_17,app_11,tcp,8080):1
400:hacl(dmz_9,app_10,tcp,8080):1
379:RULE 5 (multi-hop access):0
420:RULE 5 (multi-hop access):0
426:RULE 5 (multi-hop access):0
399:RULE 5 (multi-hop access):0
401:networkServiceInfo(app_10,tomcat,
341

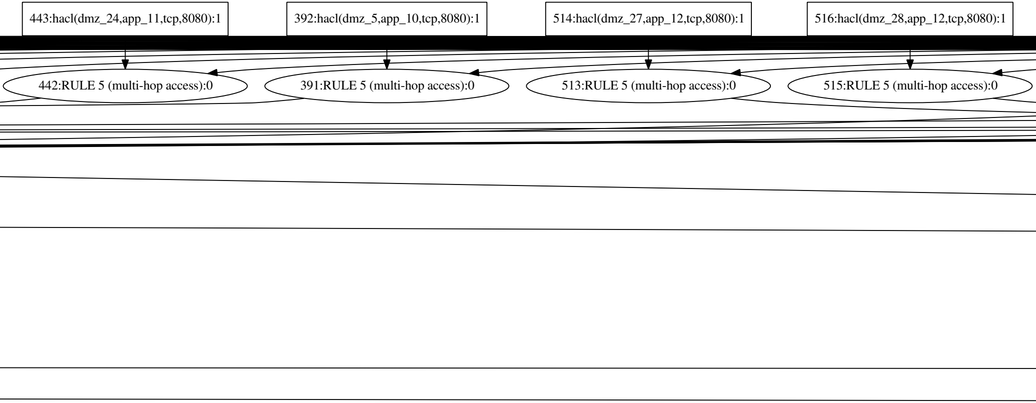

429:hacl(dmz_18,app_11,tcp,8080):1
329:hacl(dmz_6,app_1,tcp,8080):1
431:hacl(dmz_19,app_11,tcp,8080):1
433:hacl(dmz_2,app_11,tcp,8080):1
428:RULE 5 (multi-hop access):0
328:RULE 5 (multi-hop access):0
430:RULE 5 (multi-hop access):0
432:RULE 5 (multi-hop access):0
tcp,8080,root):1
402:vulExists(app_10,vul_app,tomcat,remoteExploit,privEscalation):1
:RULE 2 (remote exploit of a server program):0

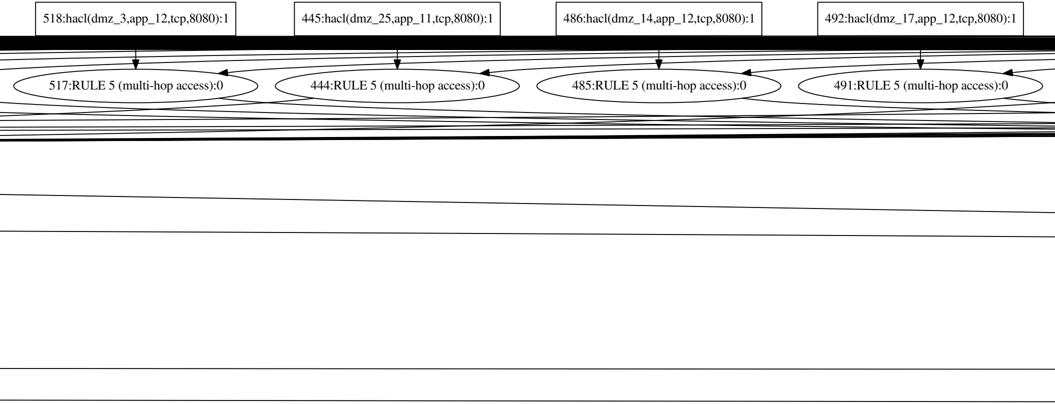

435:hacl(dmz_20,app_11,tcp,8080):1
358:hacl(dmz_15,app_10,tcp,8080):1
437:hacl(dmz_21,app_11,tcp,8080):1
331:hacl(dmz_7,app_1,tcp,8080):1
434:RULE 5 (multi-hop access):0
357:RULE 5 (multi-hop access):0
436:RULE 5 (multi-hop access):0
330:RULE 5 (multi-hop access):0

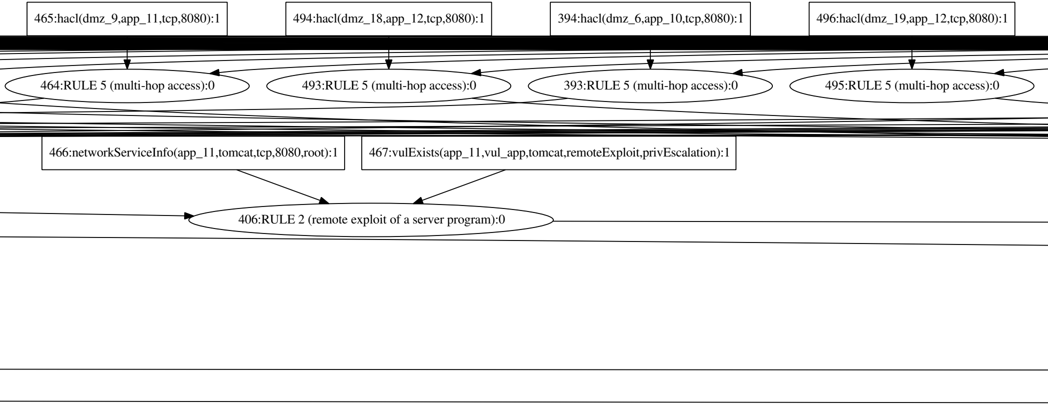

409:hacl(dmz_0,app_11,tcp,8080):1
439:hacl(dmz_22,app_11,tcp,8080):1
411:hacl(dmz_1,app_11,tcp,8080):1
333:hacl(dmz_8,app_1,tcp,8080):1
408:RULE 5 (multi-hop access):0
438:RULE 5 (multi-hop access):0
410:RULE 5 (multi-hop access):0
332:RULE 5 (multi-hop access):0

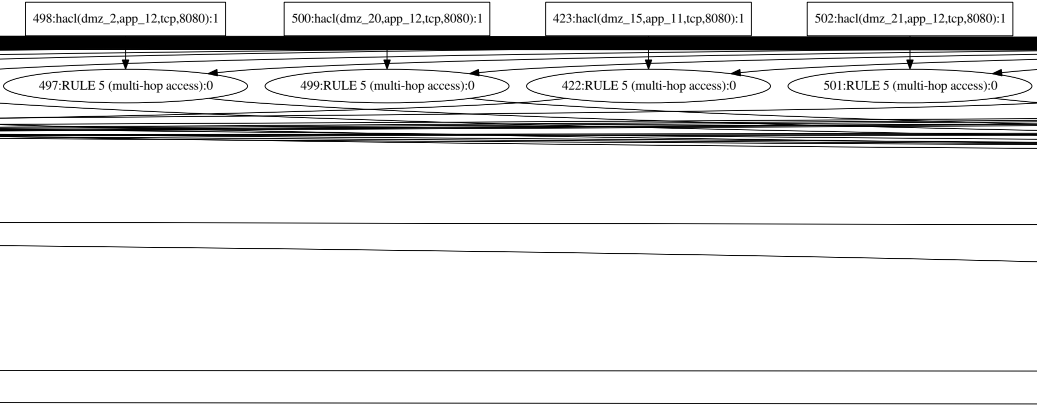

413:hacl(dmz_10,app_11,tcp,8080):1
415:hacl(dmz_11,app_11,tcp,8080):1
390:hacl(dmz_4,app_10,tcp,8080):1
354:hacl(dmz_13,app_10,tcp,8080):1
412:RULE 5 (multi-hop access):0
414:RULE 5 (multi-hop access):0
389:RULE 5 (multi-hop access):0
353:RULE 5 (multi-hop access):0
407:netAccess(app_11,tcp,8080):0

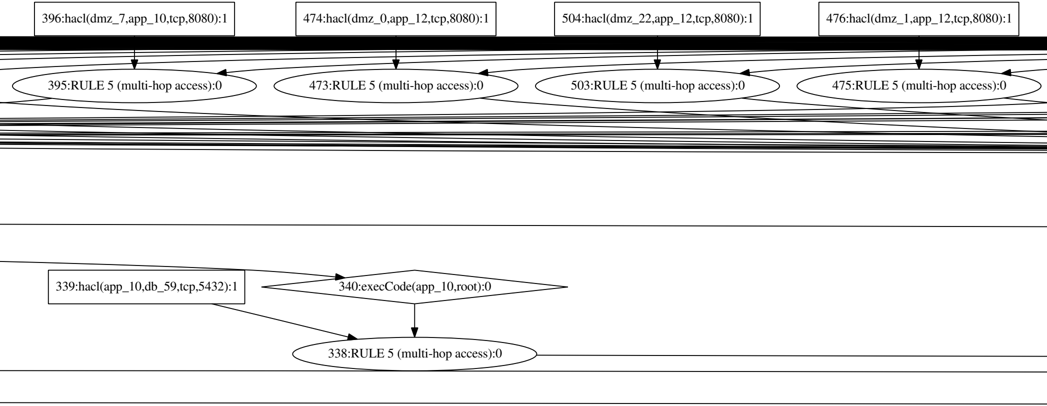

417:hacl(dmz_12,app_11,tcp,8080):1
382:hacl(dmz_26,app_10,tcp,8080):1
441:hacl(dmz_23,app_11,tcp,8080):1
360:hacl(dmz_16,app_10,tcp,8080):1
416:RULE 5 (multi-hop access):0
381:RULE 5 (multi-hop access):0
440:RULE 5 (multi-hop access):0
359:RULE 5 (multi-hop access):0

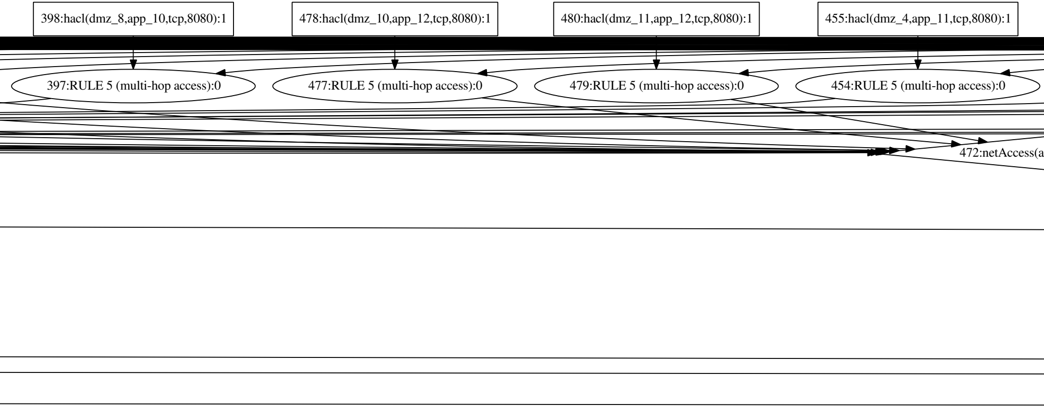

443:hacl(dmz_24,app_11,tcp,8080):1
392:hacl(dmz_5,app_10,tcp,8080):1
514:hacl(dmz_27,app_12,tcp,8080):1
516:hacl(dmz_28,app_12,tcp,8080):1
442:RULE 5 (multi-hop access):0
391:RULE 5 (multi-hop access):0
513:RULE 5 (multi-hop access):0
515:RULE 5 (multi-hop access):0

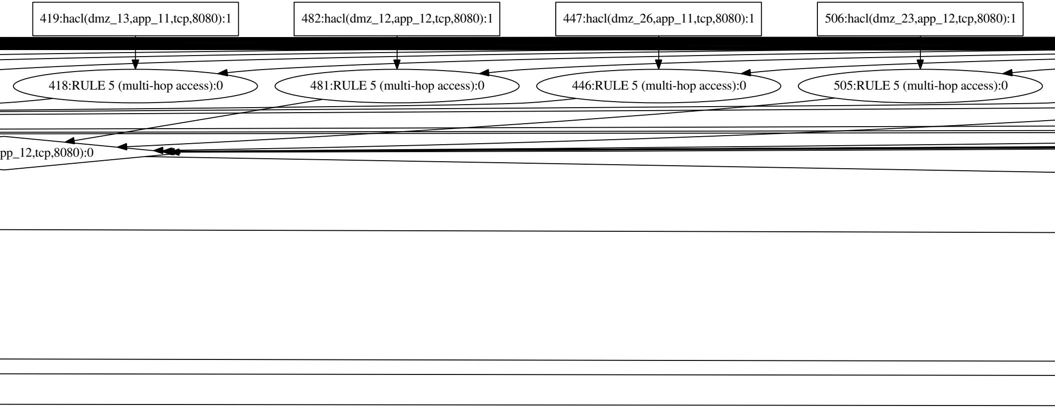

518:hacl(dmz_3,app_12,tcp,8080):1
445:hacl(dmz_25,app_11,tcp,8080):1
486:hacl(dmz_14,app_12,tcp,8080):1
492:hacl(dmz_17,app_12,tcp,8080):1
517:RULE 5 (multi-hop access):0
444:RULE 5 (multi-hop access):0
485:RULE 5 (multi-hop access):0
491:RULE 5 (multi-hop access):0

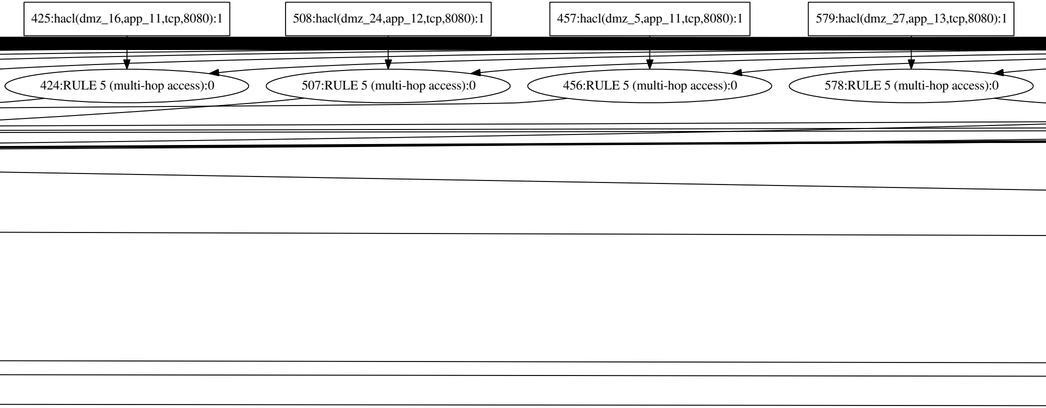

465:hacl(dmz_9,app_11,tcp,8080):1
494:hacl(dmz_18,app_12,tcp,8080):1
394:hacl(dmz_6,app_10,tcp,8080):1
496:hacl(dmz_19,app_12,tcp,8080):1
464:RULE 5 (multi-hop access):0
493:RULE 5 (multi-hop access):0
393:RULE 5 (multi-hop access):0
495:RULE 5 (multi-hop access):0
466:networkServiceInfo(app_11,tomcat,tcp,8080,root):1
467:vulExists(app_11,vul_app,tomcat,remoteExploit,privEscalation):1
406:RULE 2 (remote exploit of a server program):0

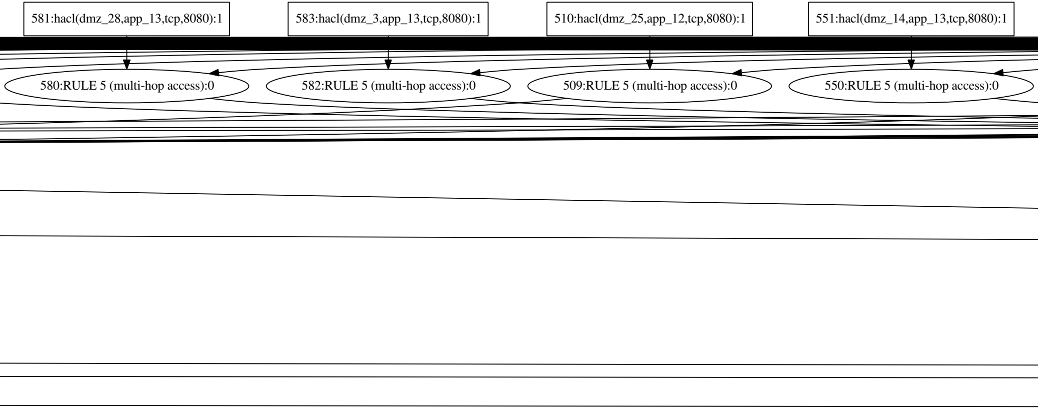

498:hacl(dmz_2,app_12,tcp,8080):1
500:hacl(dmz_20,app_12,tcp,8080):1
423:hacl(dmz_15,app_11,tcp,8080):1
502:hacl(dmz_21,app_12,tcp,8080):1
497:RULE 5 (multi-hop access):0
499:RULE 5 (multi-hop access):0
422:RULE 5 (multi-hop access):0
501:RULE 5 (multi-hop access):0

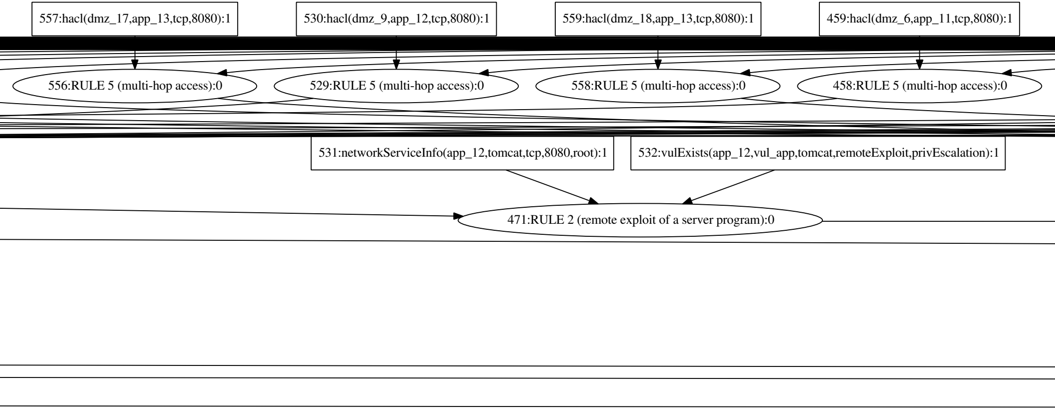

396:hacl(dmz_7,app_10,tcp,8080):1
474:hacl(dmz_0,app_12,tcp,8080):1
504:hacl(dmz_22,app_12,tcp,8080):1
476:hacl(dmz_1,app_12,tcp,8080):1
395:RULE 5 (multi-hop access):0
473:RULE 5 (multi-hop access):0
503:RULE 5 (multi-hop access):0
475:RULE 5 (multi-hop access):0
339:hacl(app_10,db_59,tcp,5432):1
340:execCode(app_10,root):0
338:RULE 5 (multi-hop access):0

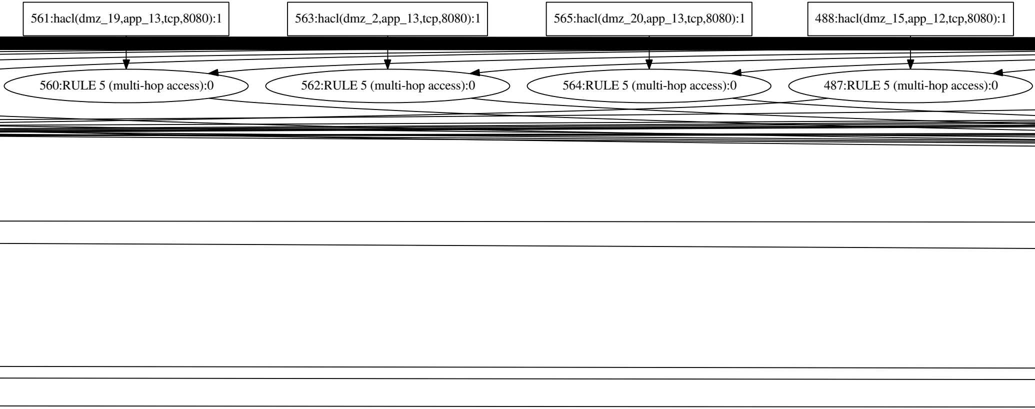

398:hacl(dmz_8,app_10,tcp,8080):1
478:hacl(dmz_10,app_12,tcp,8080):1
480:hacl(dmz_11,app_12,tcp,8080):1
455:hacl(dmz_4,app_11,tcp,8080):1
397:RULE 5 (multi-hop access):0
477:RULE 5 (multi-hop access):0
479:RULE 5 (multi-hop access):0
454:RULE 5 (multi-hop access):0
472:netAccess(a

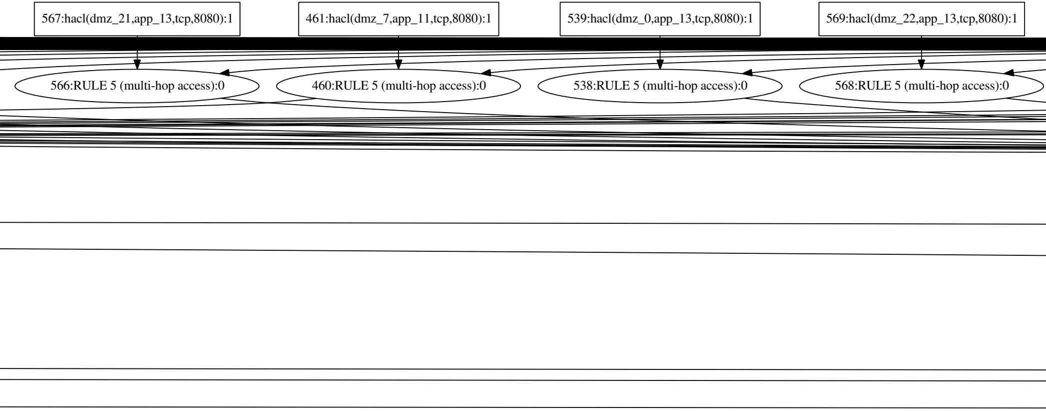

419:hacl(dmz_13,app_11,tcp,8080):1
482:hacl(dmz_12,app_12,tcp,8080):1
447:hacl(dmz_26,app_11,tcp,8080):1
506:hacl(dmz_23,app_12,tcp,8080):1
418:RULE 5 (multi-hop access):0
481:RULE 5 (multi-hop access):0
446:RULE 5 (multi-hop access):0
505:RULE 5 (multi-hop access):0
pp_12,tcp,8080):0

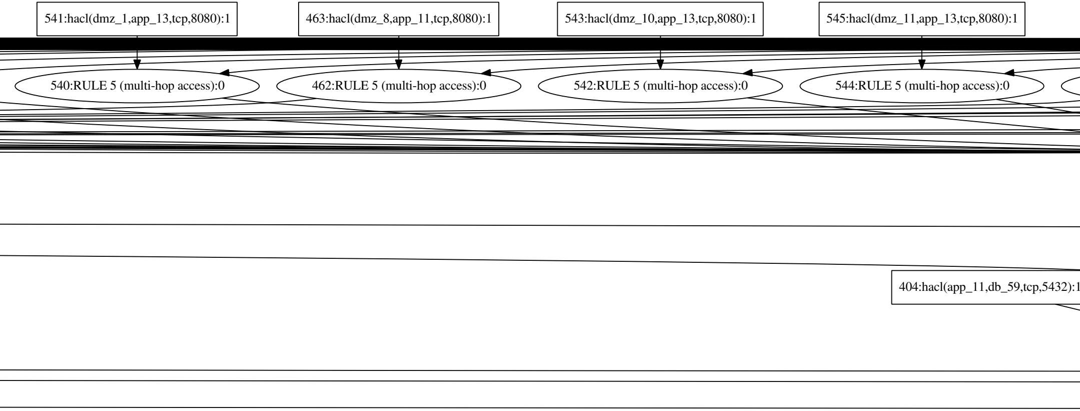

425:hacl(dmz_16,app_11,tcp,8080):1
508:hacl(dmz_24,app_12,tcp,8080):1
457:hacl(dmz_5,app_11,tcp,8080):1
579:hacl(dmz_27,app_13,tcp,8080):1
424:RULE 5 (multi-hop access):0
507:RULE 5 (multi-hop access):0
456:RULE 5 (multi-hop access):0
578:RULE 5 (multi-hop access):0

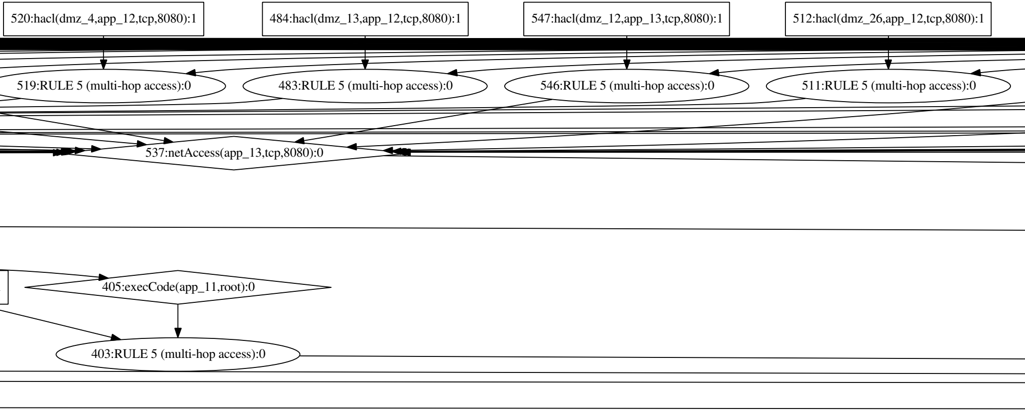

581:hacl(dmz_28,app_13,tcp,8080):1
583:hacl(dmz_3,app_13,tcp,8080):1
510:hacl(dmz_25,app_12,tcp,8080):1
551:hacl(dmz_14,app_13,tcp,8080):1
580:RULE 5 (multi-hop access):0
582:RULE 5 (multi-hop access):0
509:RULE 5 (multi-hop access):0
550:RULE 5 (multi-hop access):0

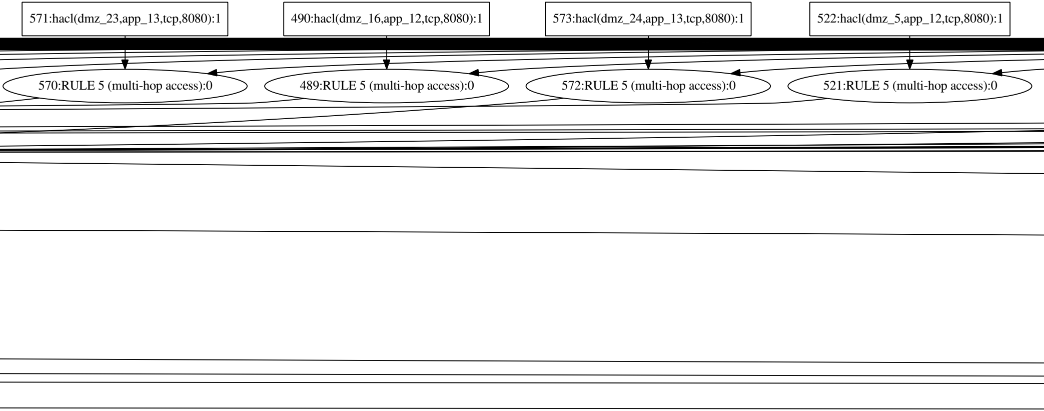

557:hacl(dmz_17,app_13,tcp,8080):1
530:hacl(dmz_9,app_12,tcp,8080):1
559:hacl(dmz_18,app_13,tcp,8080):1
459:hacl(dmz_6,app_11,tcp,8080):1
556:RULE 5 (multi-hop access):0
529:RULE 5 (multi-hop access):0
558:RULE 5 (multi-hop access):0
458:RULE 5 (multi-hop access):0
531:networkServiceInfo(app_12,tomcat,tcp,8080,root):1
532:vulExists(app_12,vul_app,tomcat,remoteExploit,privEscalation):1
471:RULE 2 (remote exploit of a server program):0

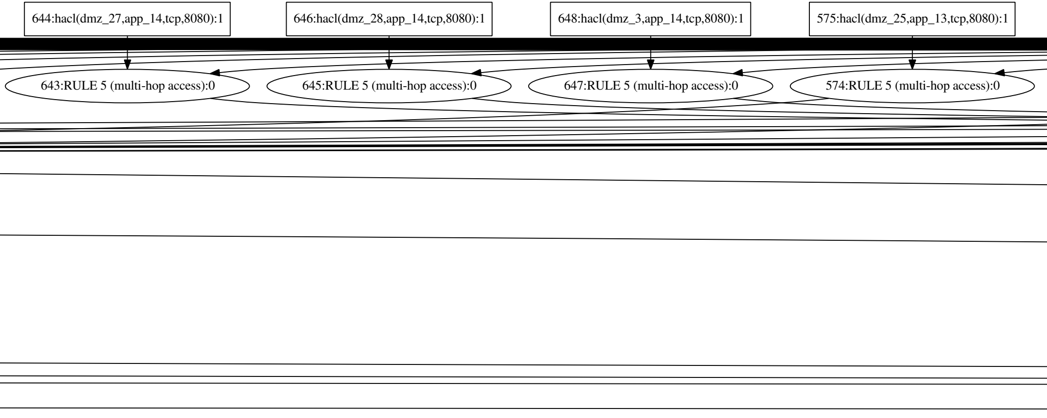

561:hacl(dmz_19,app_13,tcp,8080):1
563:hacl(dmz_2,app_13,tcp,8080):1
565:hacl(dmz_20,app_13,tcp,8080):1
488:hacl(dmz_15,app_12,tcp,8080):1
560:RULE 5 (multi-hop access):0
562:RULE 5 (multi-hop access):0
564:RULE 5 (multi-hop access):0
487:RULE 5 (multi-hop access):0

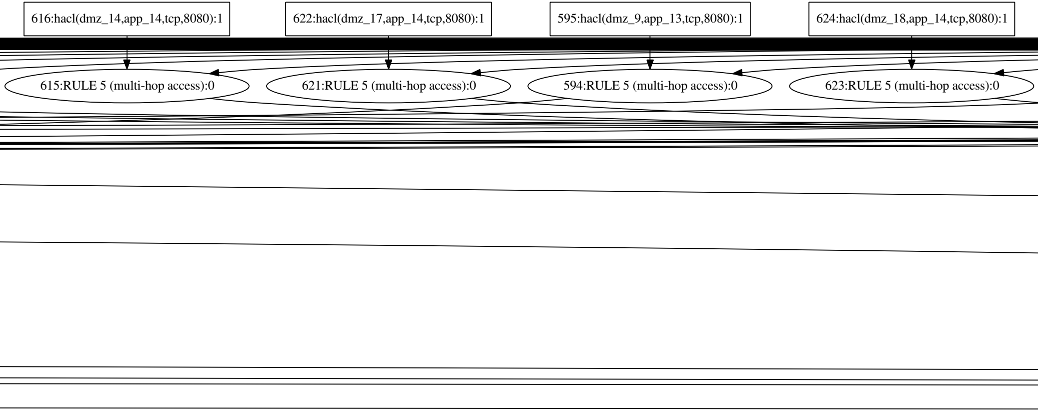

567:hacl(dmz_21,app_13,tcp,8080):1
461:hacl(dmz_7,app_11,tcp,8080):1
539:hacl(dmz_0,app_13,tcp,8080):1
569:hacl(dmz_22,app_13,tcp,8080):1
566:RULE 5 (multi-hop access):0
460:RULE 5 (multi-hop access):0
538:RULE 5 (multi-hop access):0
568:RULE 5 (multi-hop access):0

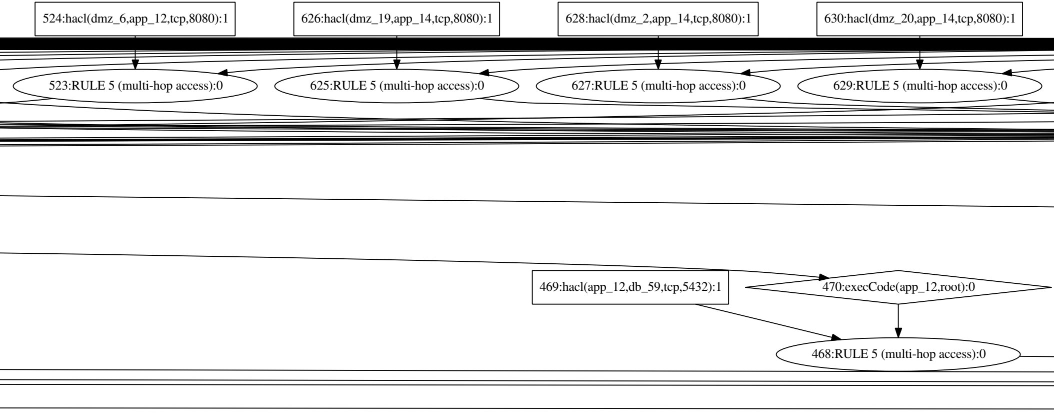

541:hacl(dmz_1,app_13,tcp,8080):1
463:hacl(dmz_8,app_11,tcp,8080):1
543:hacl(dmz_10,app_13,tcp,8080):1
545:hacl(dmz_11,app_13,tcp,8080):1
540:RULE 5 (multi-hop access):0
462:RULE 5 (multi-hop access):0
542:RULE 5 (multi-hop access):0
544:RULE 5 (multi-hop access):0
404:hacl(app_11,db_59,tcp,5432):1

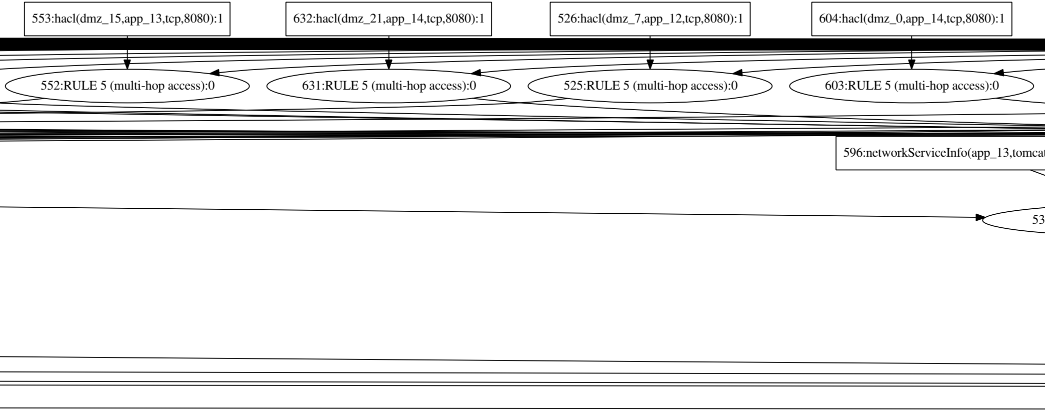

520:hacl(dmz_4,app_12,tcp,8080):1
484:hacl(dmz_13,app_12,tcp,8080):1
547:hacl(dmz_12,app_13,tcp,8080):1
512:hacl(dmz_26,app_12,tcp,8080):1
519:RULE 5 (multi-hop access):0
483:RULE 5 (multi-hop access):0
546:RULE 5 (multi-hop access):0
511:RULE 5 (multi-hop access):0
537:netAccess(app_13,tcp,8080):0
405:execCode(app_11,root):0
403:RULE 5 (multi-hop access):0

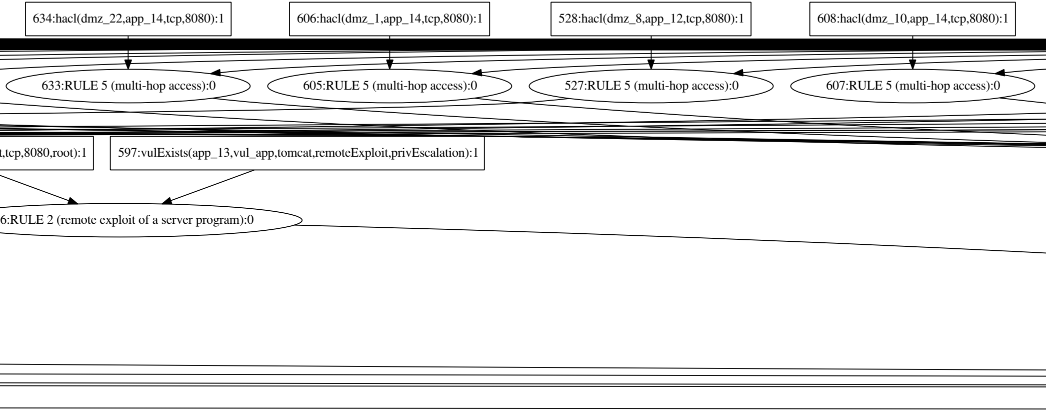

571:hacl(dmz_23,app_13,tcp,8080):1
490:hacl(dmz_16,app_12,tcp,8080):1
573:hacl(dmz_24,app_13,tcp,8080):1
522:hacl(dmz_5,app_12,tcp,8080):1
570:RULE 5 (multi-hop access):0
489:RULE 5 (multi-hop access):0
572:RULE 5 (multi-hop access):0
521:RULE 5 (multi-hop access):0

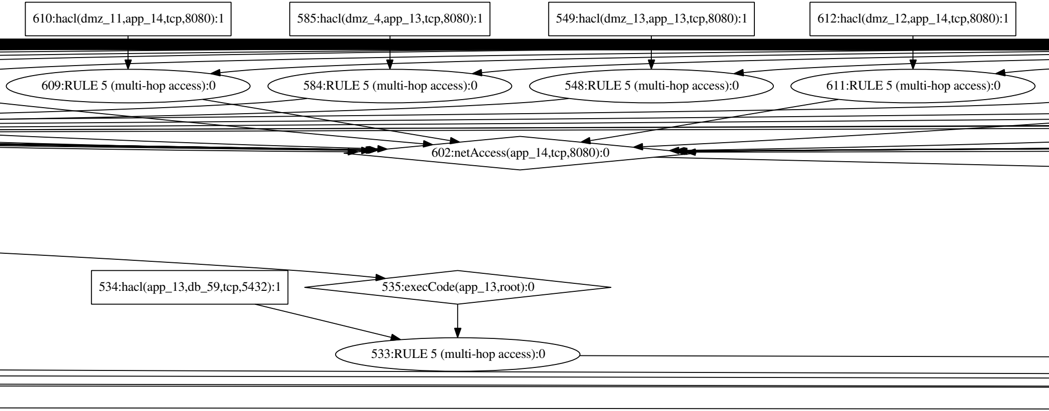

644:hacl(dmz_27,app_14,tcp,8080):1
646:hacl(dmz_28,app_14,tcp,8080):1
648:hacl(dmz_3,app_14,tcp,8080):1
575:hacl(dmz_25,app_13,tcp,8080):1
643:RULE 5 (multi-hop access):0
645:RULE 5 (multi-hop access):0
647:RULE 5 (multi-hop access):0
574:RULE 5 (multi-hop access):0

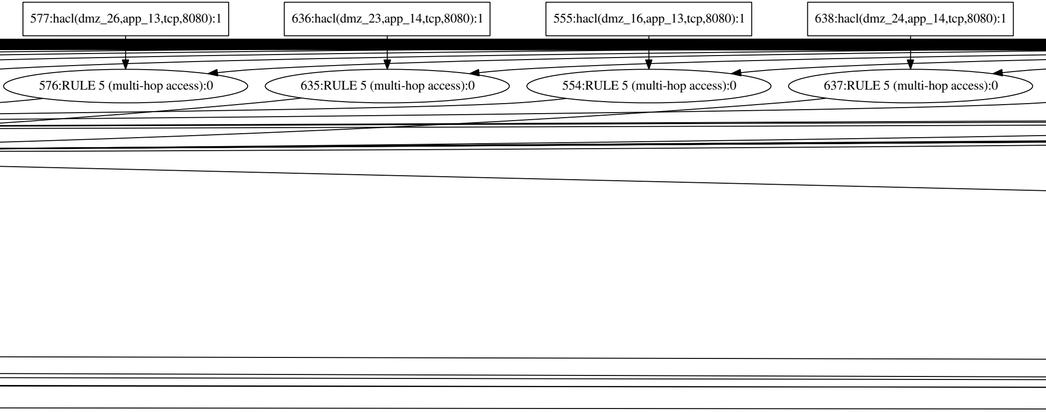

616:hacl(dmz_14,app_14,tcp,8080):1
622:hacl(dmz_17,app_14,tcp,8080):1
595:hacl(dmz_9,app_13,tcp,8080):1
624:hacl(dmz_18,app_14,tcp,8080):1
615:RULE 5 (multi-hop access):0
621:RULE 5 (multi-hop access):0
594:RULE 5 (multi-hop access):0
623:RULE 5 (multi-hop access):0

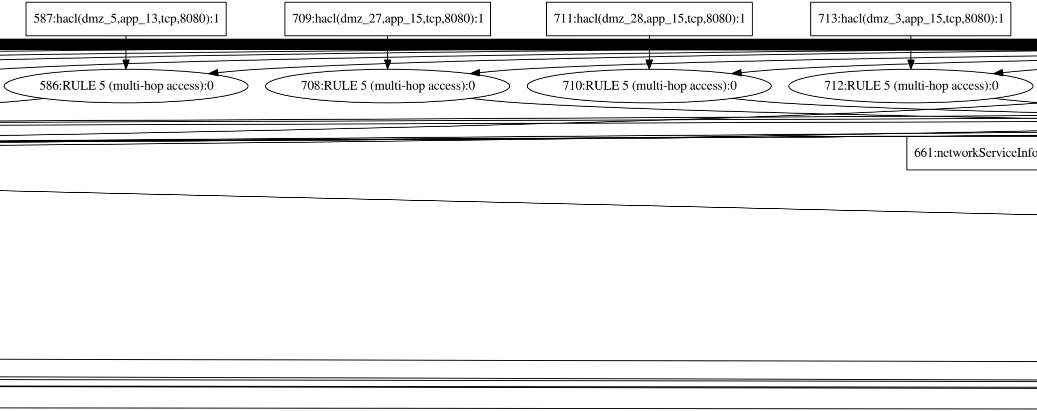

524:hacl(dmz_6,app_12,tcp,8080):1
626:hacl(dmz_19,app_14,tcp,8080):1
628:hacl(dmz_2,app_14,tcp,8080):1
630:hacl(dmz_20,app_14,tcp,8080):1
523:RULE 5 (multi-hop access):0
625:RULE 5 (multi-hop access):0
627:RULE 5 (multi-hop access):0
629:RULE 5 (multi-hop access):0
469:hacl(app_12,db_59,tcp,5432):1
470:execCode(app_12,root):0
468:RULE 5 (multi-hop access):0

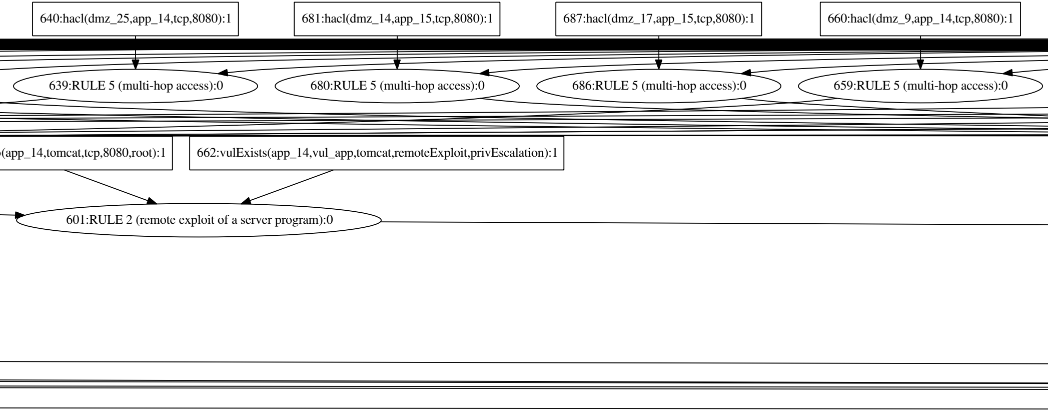

553:hacl(dmz_15,app_13,tcp,8080):1
632:hacl(dmz_21,app_14,tcp,8080):1
526:hacl(dmz_7,app_12,tcp,8080):1
604:hacl(dmz_0,app_14,tcp,8080):1
552:RULE 5 (multi-hop access):0
631:RULE 5 (multi-hop access):0
525:RULE 5 (multi-hop access):0
603:RULE 5 (multi-hop access):0
596:networkServiceInfo(app_13,tomca
53

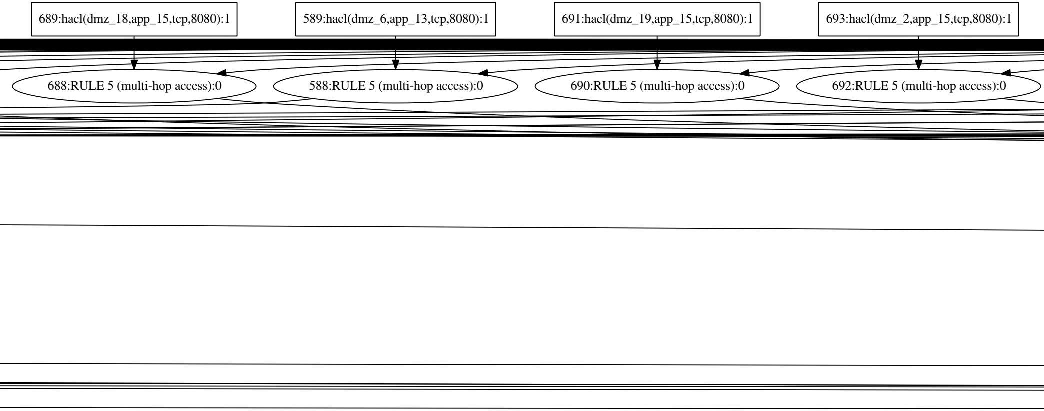

634:hacl(dmz_22,app_14,tcp,8080):1
606:hacl(dmz_1,app_14,tcp,8080):1
528:hacl(dmz_8,app_12,tcp,8080):1
608:hacl(dmz_10,app_14,tcp,8080):1
633:RULE 5 (multi-hop access):0
605:RULE 5 (multi-hop access):0
527:RULE 5 (multi-hop access):0
607:RULE 5 (multi-hop access):0
t,tcp,8080,root):1
597:vulExists(app_13,vul_app,tomcat,remoteExploit,privEscalation):1
6:RULE 2 (remote exploit of a server program):0

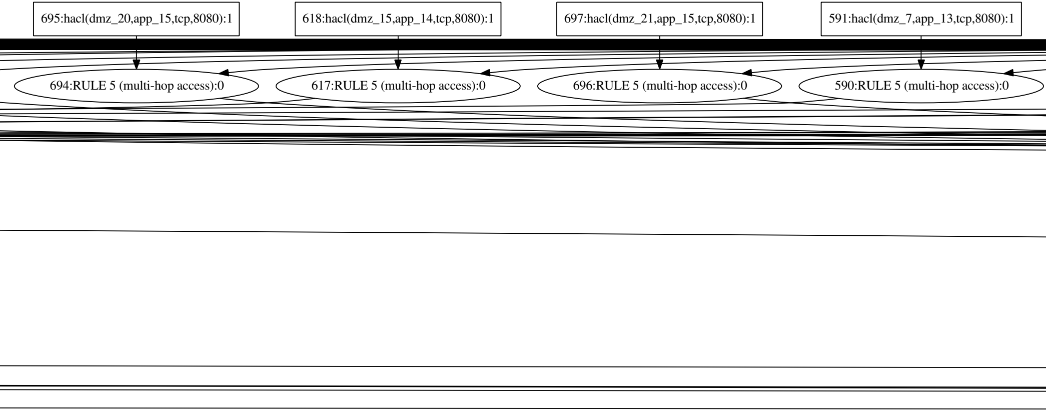

610:hacl(dmz_11,app_14,tcp,8080):1
585:hacl(dmz_4,app_13,tcp,8080):1
549:hacl(dmz_13,app_13,tcp,8080):1
612:hacl(dmz_12,app_14,tcp,8080):1
609:RULE 5 (multi-hop access):0
584:RULE 5 (multi-hop access):0
548:RULE 5 (multi-hop access):0
611:RULE 5 (multi-hop access):0
602:netAccess(app_14,tcp,8080):0
534:hacl(app_13,db_59,tcp,5432):1
535:execCode(app_13,root):0
533:RULE 5 (multi-hop access):0

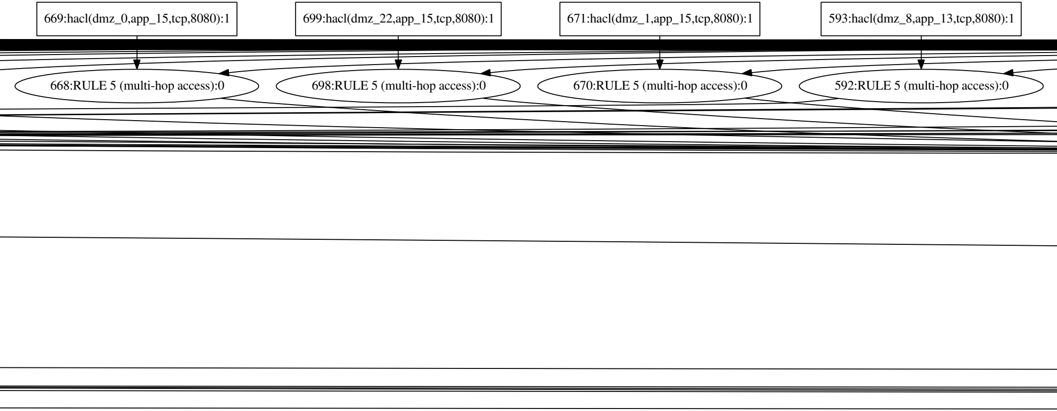

577:hacl(dmz_26,app_13,tcp,8080):1
636:hacl(dmz_23,app_14,tcp,8080):1
555:hacl(dmz_16,app_13,tcp,8080):1
638:hacl(dmz_24,app_14,tcp,8080):1
576:RULE 5 (multi-hop access):0
635:RULE 5 (multi-hop access):0
554:RULE 5 (multi-hop access):0
637:RULE 5 (multi-hop access):0

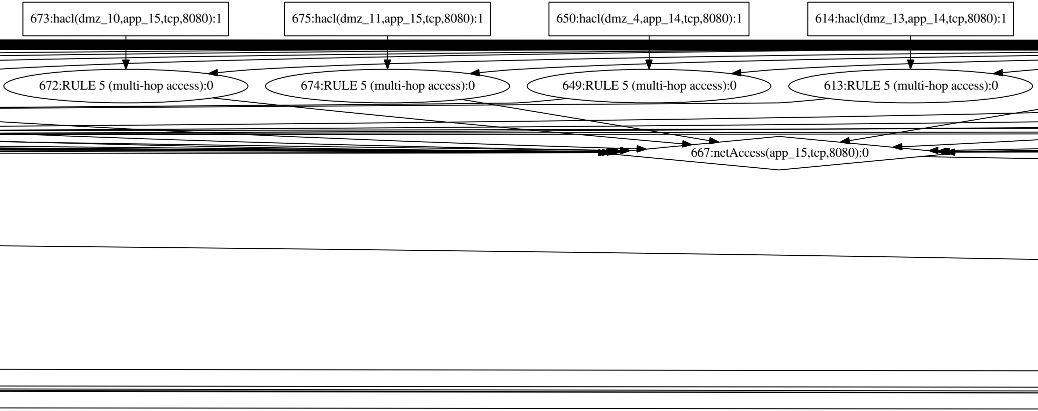

587:hacl(dmz_5,app_13,tcp,8080):1
709:hacl(dmz_27,app_15,tcp,8080):1
711:hacl(dmz_28,app_15,tcp,8080):1
713:hacl(dmz_3,app_15,tcp,8080):1
586:RULE 5 (multi-hop access):0
708:RULE 5 (multi-hop access):0
710:RULE 5 (multi-hop access):0
712:RULE 5 (multi-hop access):0
661:networkServiceInfo

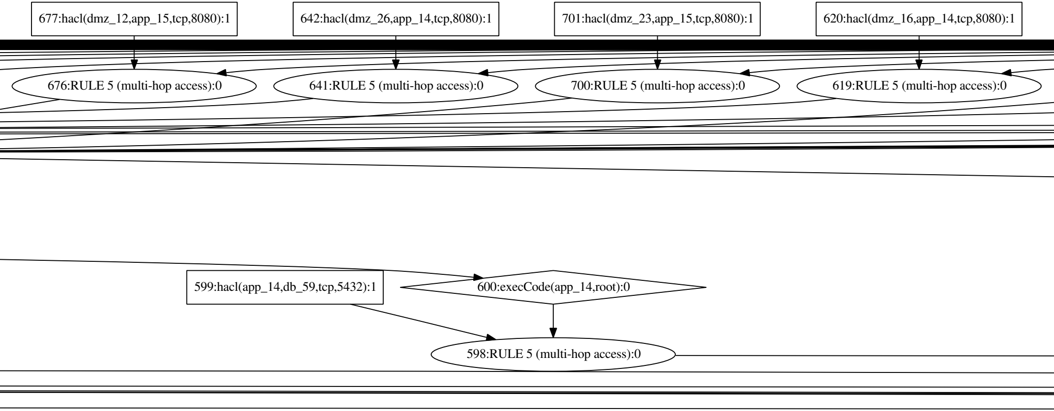

640:hacl(dmz_25,app_14,tcp,8080):1
681:hacl(dmz_14,app_15,tcp,8080):1
687:hacl(dmz_17,app_15,tcp,8080):1
660:hacl(dmz_9,app_14,tcp,8080):1
639:RULE 5 (multi-hop access):0
680:RULE 5 (multi-hop access):0
686:RULE 5 (multi-hop access):0
659:RULE 5 (multi-hop access):0
(app_14,tomcat,tcp,8080,root):1
662:vulExists(app_14,vul_app,tomcat,remoteExploit,privEscalation):1
601:RULE 2 (remote exploit of a server program):0

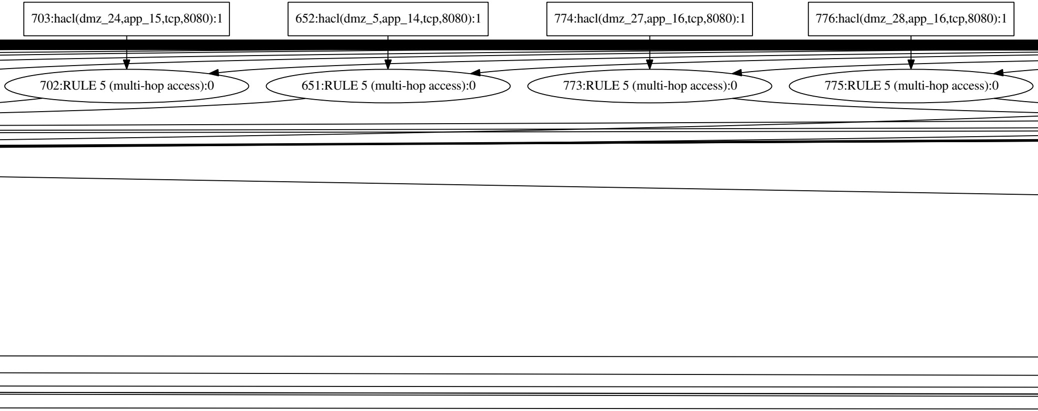

689:hacl(dmz_18,app_15,tcp,8080):1
589:hacl(dmz_6,app_13,tcp,8080):1
691:hacl(dmz_19,app_15,tcp,8080):1
693:hacl(dmz_2,app_15,tcp,8080):1
688:RULE 5 (multi-hop access):0
588:RULE 5 (multi-hop access):0
690:RULE 5 (multi-hop access):0
692:RULE 5 (multi-hop access):0

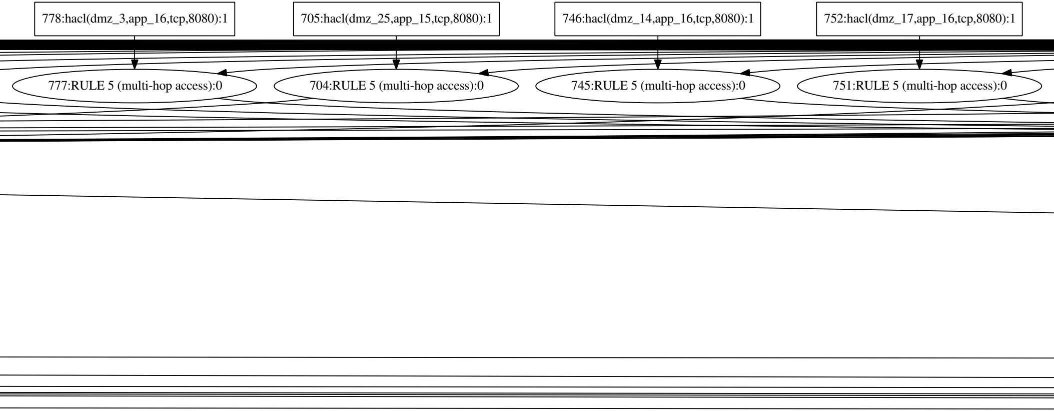

695:hacl(dmz_20,app_15,tcp,8080):1
618:hacl(dmz_15,app_14,tcp,8080):1
697:hacl(dmz_21,app_15,tcp,8080):1
591:hacl(dmz_7,app_13,tcp,8080):1
694:RULE 5 (multi-hop access):0
617:RULE 5 (multi-hop access):0
696:RULE 5 (multi-hop access):0
590:RULE 5 (multi-hop access):0

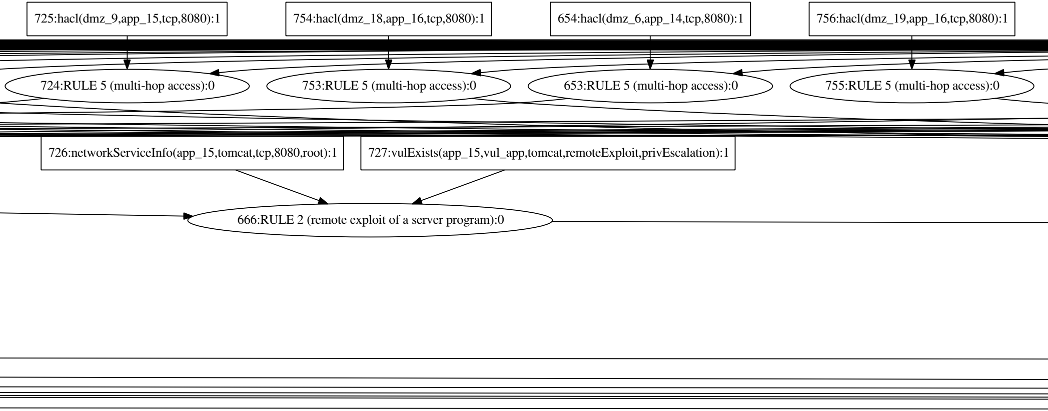

669:hacl(dmz_0,app_15,tcp,8080):1
699:hacl(dmz_22,app_15,tcp,8080):1
671:hacl(dmz_1,app_15,tcp,8080):1
593:hacl(dmz_8,app_13,tcp,8080):1
668:RULE 5 (multi-hop access):0
698:RULE 5 (multi-hop access):0
670:RULE 5 (multi-hop access):0
592:RULE 5 (multi-hop access):0

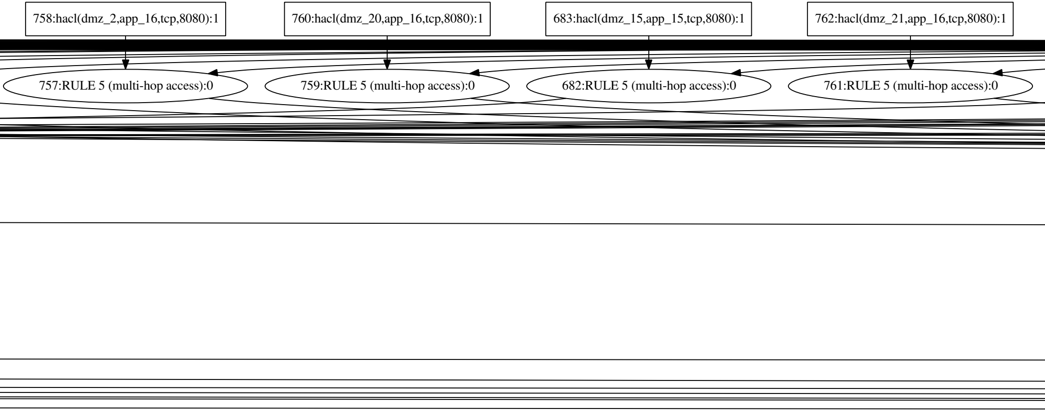

673:hacl(dmz_10,app_15,tcp,8080):1
675:hacl(dmz_11,app_15,tcp,8080):1
650:hacl(dmz_4,app_14,tcp,8080):1
614:hacl(dmz_13,app_14,tcp,8080):1
672:RULE 5 (multi-hop access):0
674:RULE 5 (multi-hop access):0
649:RULE 5 (multi-hop access):0
613:RULE 5 (multi-hop access):0
667:netAccess(app_15,tcp,8080):0

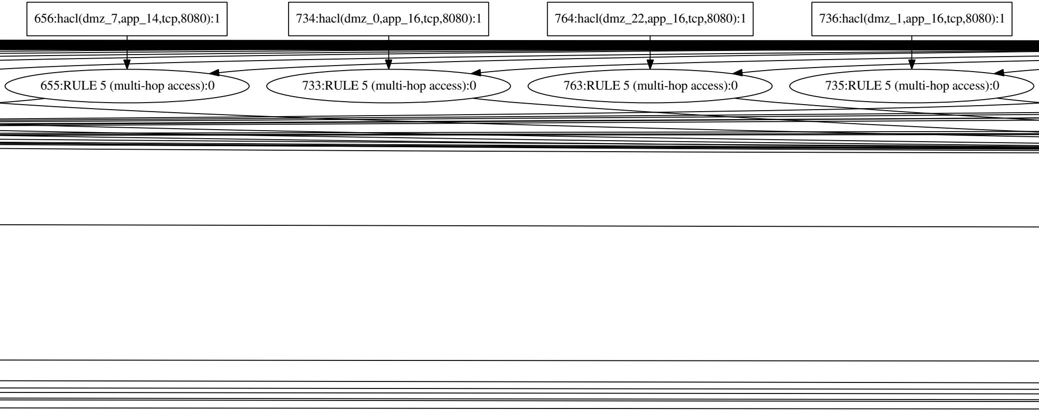

677:hacl(dmz_12,app_15,tcp,8080):1
642:hacl(dmz_26,app_14,tcp,8080):1
701:hacl(dmz_23,app_15,tcp,8080):1
620:hacl(dmz_16,app_14,tcp,8080):1
676:RULE 5 (multi-hop access):0
641:RULE 5 (multi-hop access):0
700:RULE 5 (multi-hop access):0
619:RULE 5 (multi-hop access):0
599:hacl(app_14,db_59,tcp,5432):1
600:execCode(app_14,root):0
598:RULE 5 (multi-hop access):0

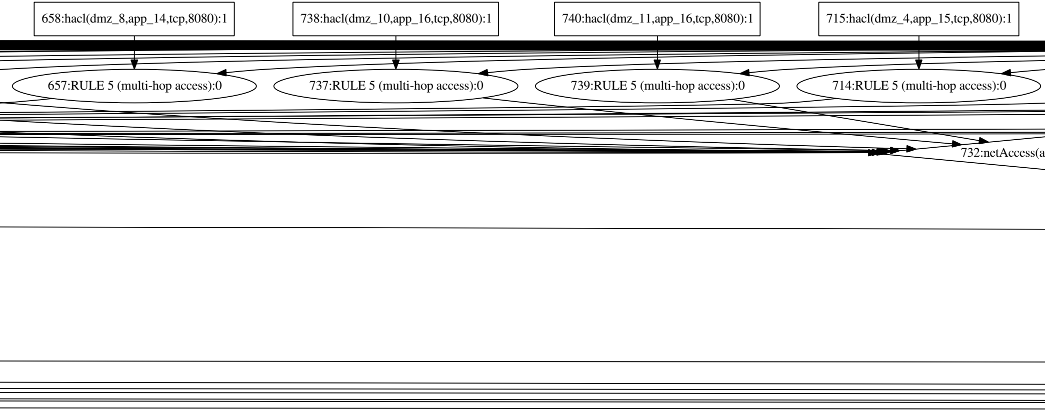

703:hacl(dmz_24,app_15,tcp,8080):1
652:hacl(dmz_5,app_14,tcp,8080):1
774:hacl(dmz_27,app_16,tcp,8080):1
776:hacl(dmz_28,app_16,tcp,8080):1
702:RULE 5 (multi-hop access):0
651:RULE 5 (multi-hop access):0
773:RULE 5 (multi-hop access):0
775:RULE 5 (multi-hop access):0

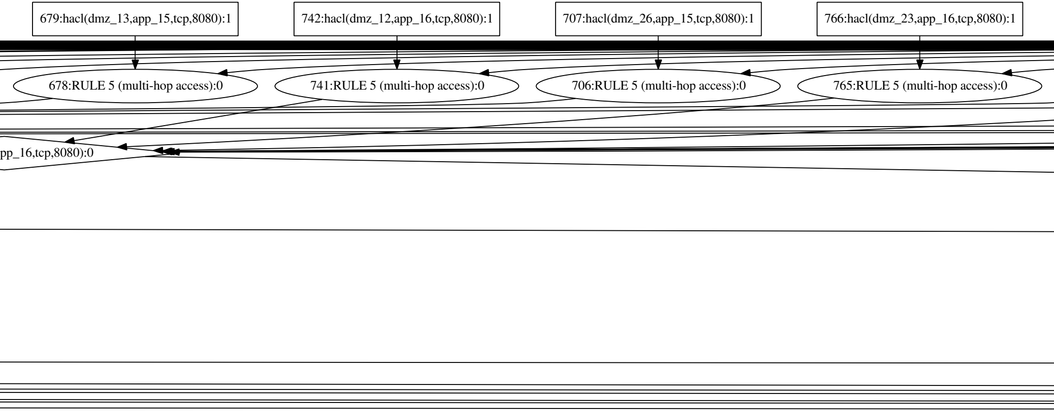

778:hacl(dmz_3,app_16,tcp,8080):1
705:hacl(dmz_25,app_15,tcp,8080):1
746:hacl(dmz_14,app_16,tcp,8080):1
752:hacl(dmz_17,app_16,tcp,8080):1
777:RULE 5 (multi-hop access):0
704:RULE 5 (multi-hop access):0
745:RULE 5 (multi-hop access):0
751:RULE 5 (multi-hop access):0

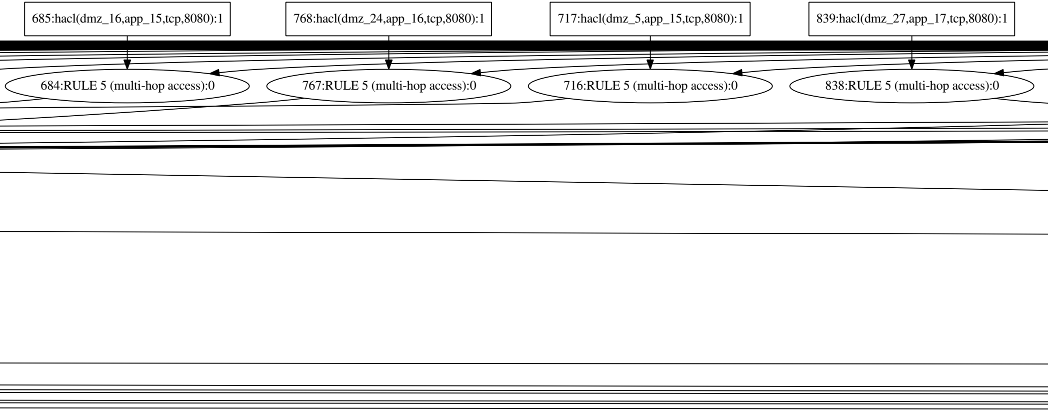

725:hacl(dmz_9,app_15,tcp,8080):1
754:hacl(dmz_18,app_16,tcp,8080):1
654:hacl(dmz_6,app_14,tcp,8080):1
756:hacl(dmz_19,app_16,tcp,8080):1
724:RULE 5 (multi-hop access):0
753:RULE 5 (multi-hop access):0
653:RULE 5 (multi-hop access):0
755:RULE 5 (multi-hop access):0
726:networkServiceInfo(app_15,tomcat,tcp,8080,root):1
727:vulExists(app_15,vul_app,tomcat,remoteExploit,privEscalation):1
666:RULE 2 (remote exploit of a server program):0

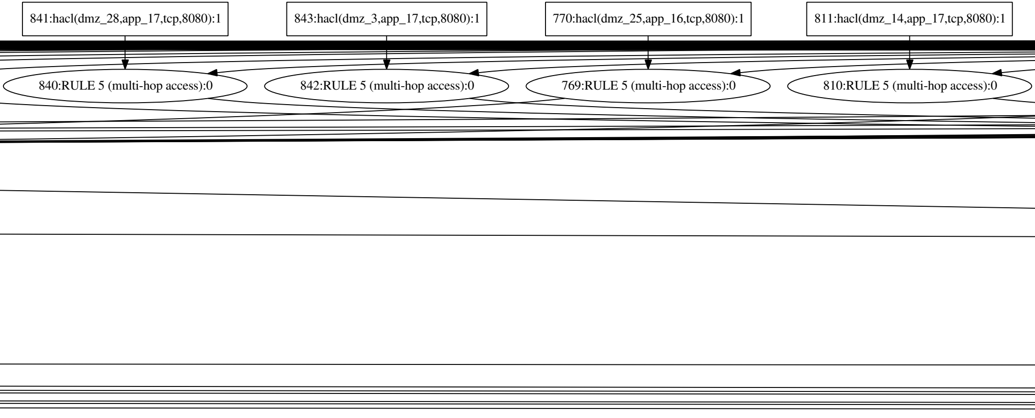

758:hacl(dmz_2,app_16,tcp,8080):1
760:hacl(dmz_20,app_16,tcp,8080):1
683:hacl(dmz_15,app_15,tcp,8080):1
762:hacl(dmz_21,app_16,tcp,8080):1
757:RULE 5 (multi-hop access):0
759:RULE 5 (multi-hop access):0
682:RULE 5 (multi-hop access):0
761:RULE 5 (multi-hop access):0

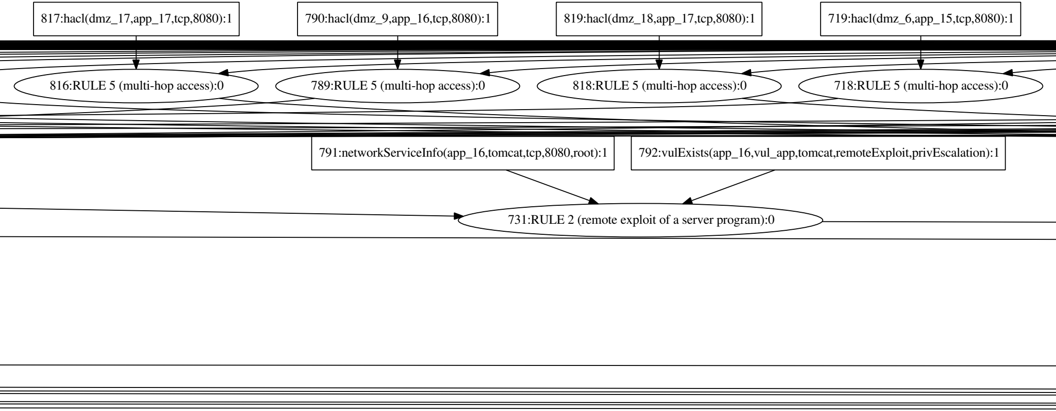

656:hacl(dmz_7,app_14,tcp,8080):1
734:hacl(dmz_0,app_16,tcp,8080):1
764:hacl(dmz_22,app_16,tcp,8080):1
736:hacl(dmz_1,app_16,tcp,8080):1
655:RULE 5 (multi-hop access):0
733:RULE 5 (multi-hop access):0
763:RULE 5 (multi-hop access):0
735:RULE 5 (multi-hop access):0

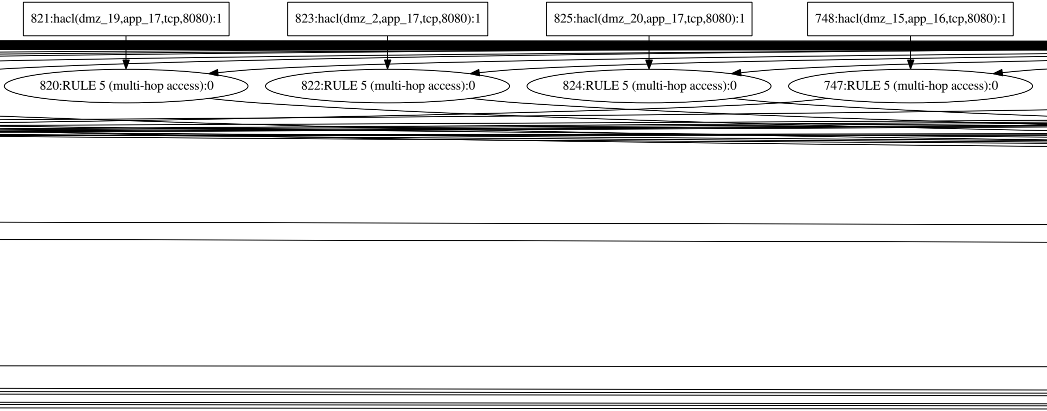

658:hacl(dmz_8,app_14,tcp,8080):1
738:hacl(dmz_10,app_16,tcp,8080):1
740:hacl(dmz_11,app_16,tcp,8080):1
715:hacl(dmz_4,app_15,tcp,8080):1
657:RULE 5 (multi-hop access):0
737:RULE 5 (multi-hop access):0
739:RULE 5 (multi-hop access):0
714:RULE 5 (multi-hop access):0
732:netAccess(a

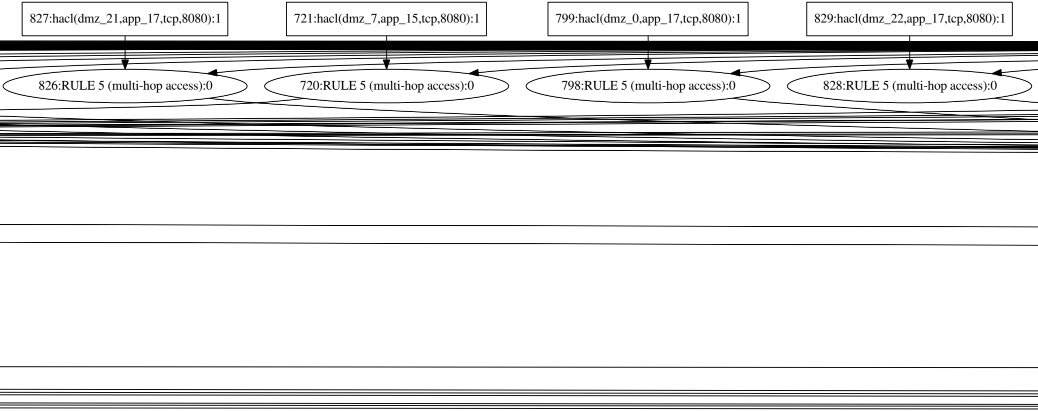

679:hacl(dmz_13,app_15,tcp,8080):1
742:hacl(dmz_12,app_16,tcp,8080):1
707:hacl(dmz_26,app_15,tcp,8080):1
766:hacl(dmz_23,app_16,tcp,8080):1
678:RULE 5 (multi-hop access):0
741:RULE 5 (multi-hop access):0
706:RULE 5 (multi-hop access):0
765:RULE 5 (multi-hop access):0
pp_16,tcp,8080):0

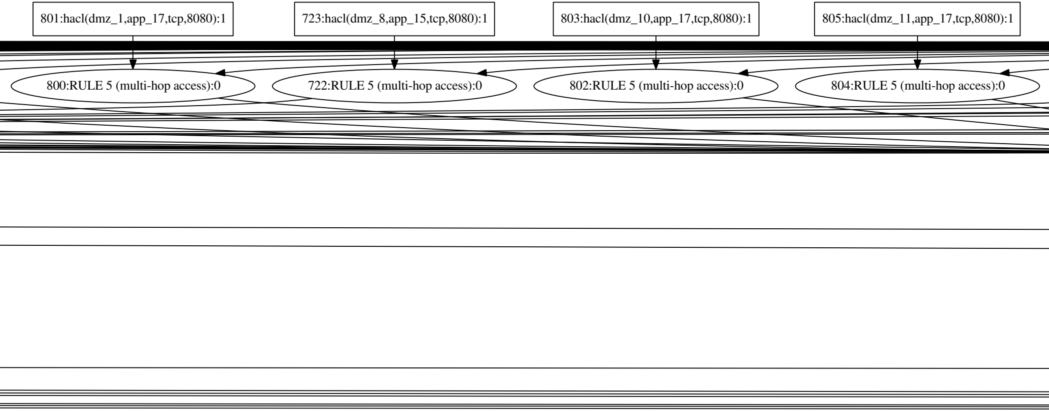

685:hacl(dmz_16,app_15,tcp,8080):1
768:hacl(dmz_24,app_16,tcp,8080):1
717:hacl(dmz_5,app_15,tcp,8080):1
839:hacl(dmz_27,app_17,tcp,8080):1
684:RULE 5 (multi-hop access):0
767:RULE 5 (multi-hop access):0
716:RULE 5 (multi-hop access):0
838:RULE 5 (multi-hop access):0

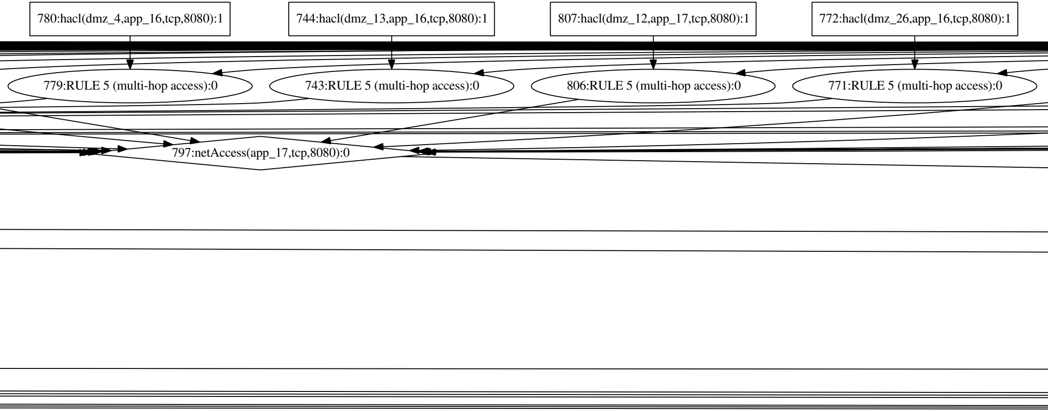

841:hacl(dmz_28,app_17,tcp,8080):1
843:hacl(dmz_3,app_17,tcp,8080):1
770:hacl(dmz_25,app_16,tcp,8080):1
811:hacl(dmz_14,app_17,tcp,8080):1
840:RULE 5 (multi-hop access):0
842:RULE 5 (multi-hop access):0
769:RULE 5 (multi-hop access):0
810:RULE 5 (multi-hop access):0

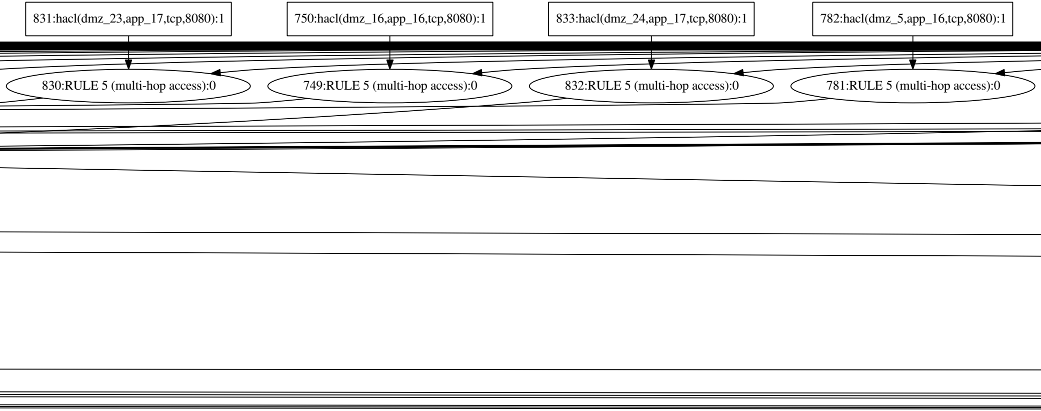

817:hacl(dmz_17,app_17,tcp,8080):1
790:hacl(dmz_9,app_16,tcp,8080):1
819:hacl(dmz_18,app_17,tcp,8080):1
719:hacl(dmz_6,app_15,tcp,8080):1
816:RULE 5 (multi-hop access):0
789:RULE 5 (multi-hop access):0
818:RULE 5 (multi-hop access):0
718:RULE 5 (multi-hop access):0
791:networkServiceInfo(app_16,tomcat,tcp,8080,root):1
792:vulExists(app_16,vul_app,tomcat,remoteExploit,privEscalation):1
731:RULE 2 (remote exploit of a server program):0

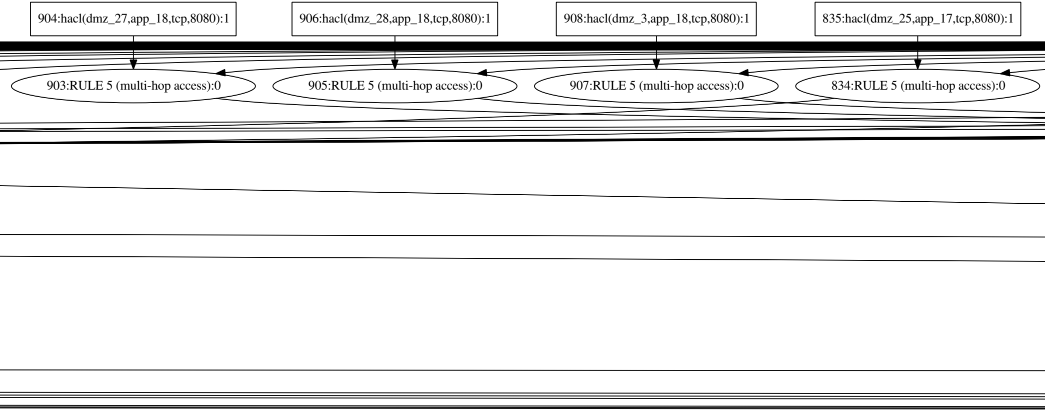

821:hacl(dmz_19,app_17,tcp,8080):1
823:hacl(dmz_2,app_17,tcp,8080):1
825:hacl(dmz_20,app_17,tcp,8080):1
748:hacl(dmz_15,app_16,tcp,8080):1
820:RULE 5 (multi-hop access):0
822:RULE 5 (multi-hop access):0
824:RULE 5 (multi-hop access):0
747:RULE 5 (multi-hop access):0

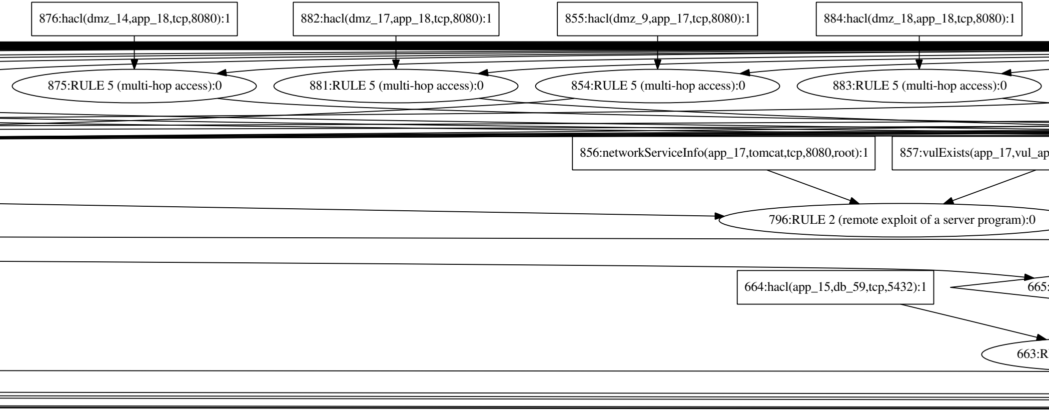

827:hacl(dmz_21,app_17,tcp,8080):1
721:hacl(dmz_7,app_15,tcp,8080):1
799:hacl(dmz_0,app_17,tcp,8080):1
829:hacl(dmz_22,app_17,tcp,8080):1
826:RULE 5 (multi-hop access):0
720:RULE 5 (multi-hop access):0
798:RULE 5 (multi-hop access):0
828:RULE 5 (multi-hop access):0

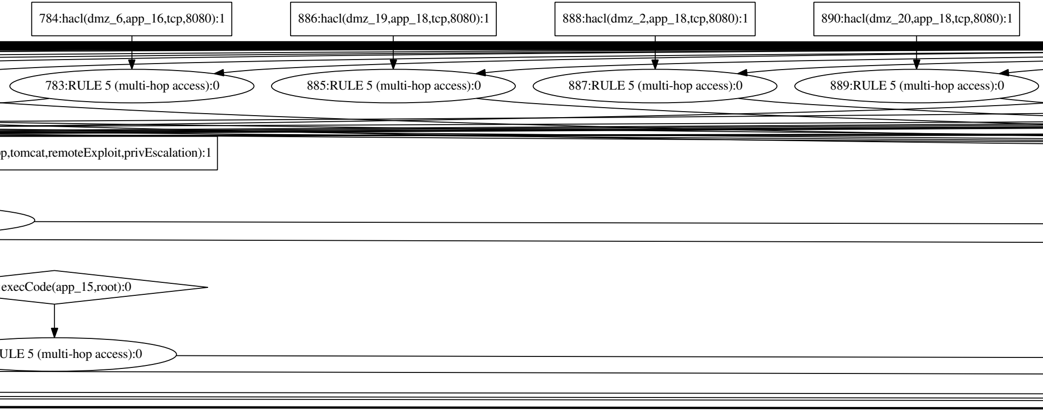

801:hacl(dmz_1,app_17,tcp,8080):1
723:hacl(dmz_8,app_15,tcp,8080):1
803:hacl(dmz_10,app_17,tcp,8080):1
805:hacl(dmz_11,app_17,tcp,8080):1
800:RULE 5 (multi-hop access):0
722:RULE 5 (multi-hop access):0
802:RULE 5 (multi-hop access):0
804:RULE 5 (multi-hop access):0

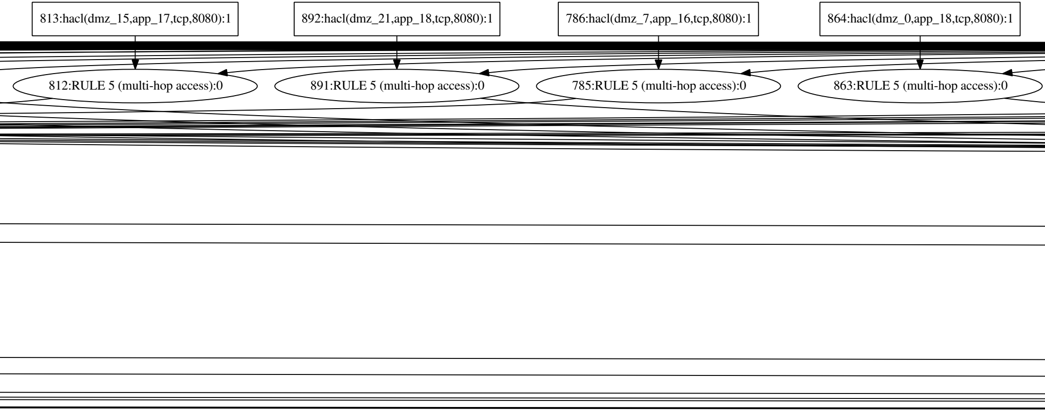

780:hacl(dmz_4,app_16,tcp,8080):1
744:hacl(dmz_13,app_16,tcp,8080):1
807:hacl(dmz_12,app_17,tcp,8080):1
772:hacl(dmz_26,app_16,tcp,8080):1
779:RULE 5 (multi-hop access):0
743:RULE 5 (multi-hop access):0
806:RULE 5 (multi-hop access):0
771:RULE 5 (multi-hop access):0
797:netAccess(app_17,tcp,8080):0

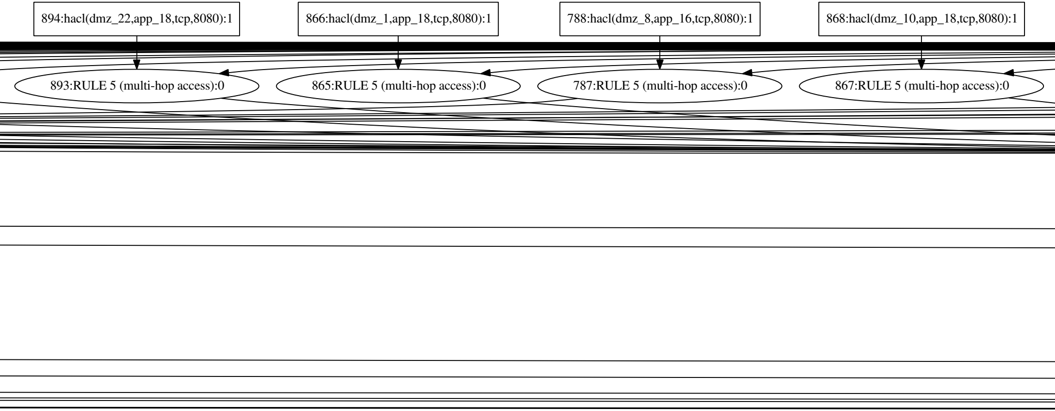

831:hacl(dmz_23,app_17,tcp,8080):1
750:hacl(dmz_16,app_16,tcp,8080):1
833:hacl(dmz_24,app_17,tcp,8080):1
782:hacl(dmz_5,app_16,tcp,8080):1
830:RULE 5 (multi-hop access):0
749:RULE 5 (multi-hop access):0
832:RULE 5 (multi-hop access):0
781:RULE 5 (multi-hop access):0

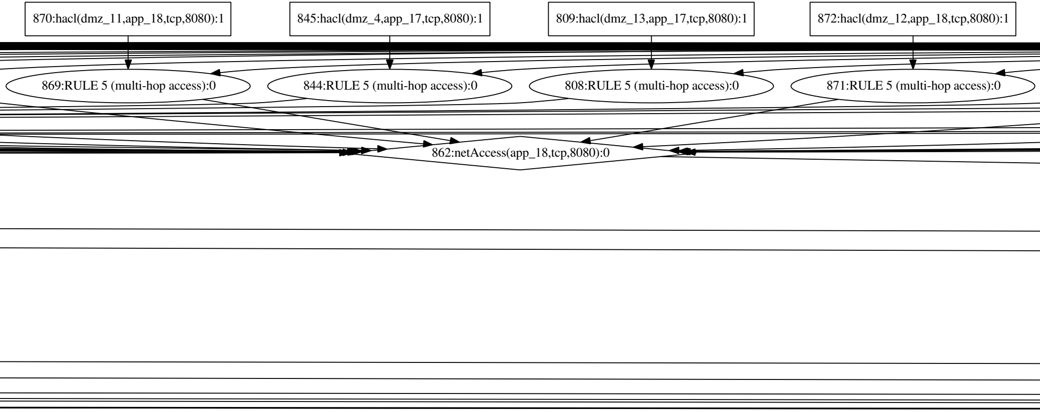

904:hacl(dmz_27,app_18,tcp,8080):1
906:hacl(dmz_28,app_18,tcp,8080):1
908:hacl(dmz_3,app_18,tcp,8080):1
835:hacl(dmz_25,app_17,tcp,8080):1
903:RULE 5 (multi-hop access):0
905:RULE 5 (multi-hop access):0
907:RULE 5 (multi-hop access):0
834:RULE 5 (multi-hop access):0

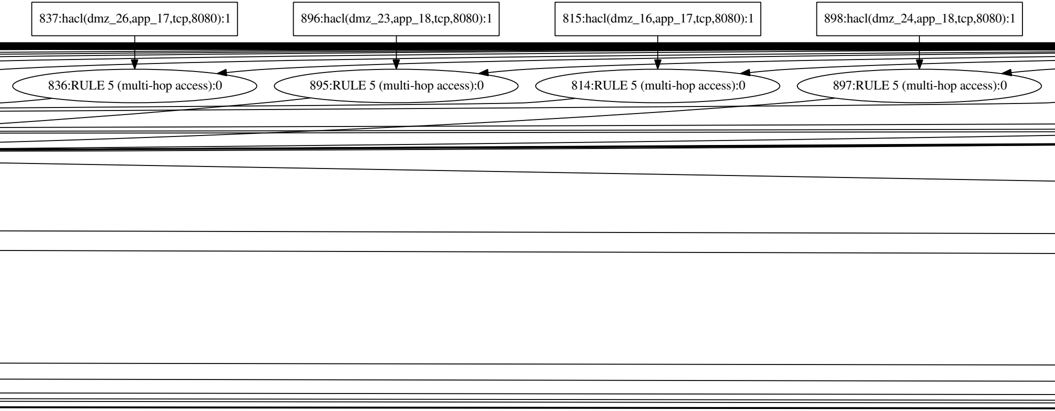

876:hacl(dmz_14,app_18,tcp,8080):1
882:hacl(dmz_17,app_18,tcp,8080):1
855:hacl(dmz_9,app_17,tcp,8080):1
884:hacl(dmz_18,app_18,tcp,8080):1
875:RULE 5 (multi-hop access):0
881:RULE 5 (multi-hop access):0
854:RULE 5 (multi-hop access):0
883:RULE 5 (multi-hop access):0
856:networkServiceInfo(app_17,tomcat,tcp,8080,root):1
857:vulExists(app_17,vul_ap
796:RULE 2 (remote exploit of a server program):0
664:hacl(app_15,db_59,tcp,5432):1
665:
663:R

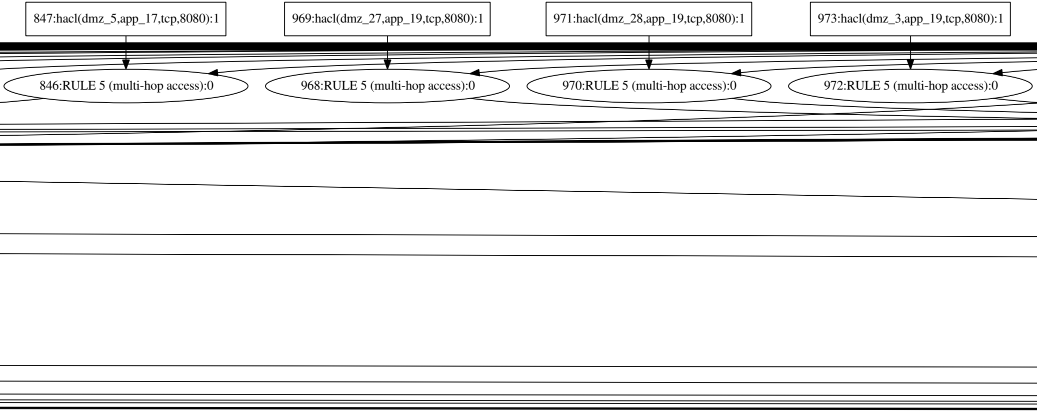

784:hacl(dmz_6,app_16,tcp,8080):1
886:hacl(dmz_19,app_18,tcp,8080):1
888:hacl(dmz_2,app_18,tcp,8080):1
890:hacl(dmz_20,app_18,tcp,8080):1
783:RULE 5 (multi-hop access):0
885:RULE 5 (multi-hop access):0
887:RULE 5 (multi-hop access):0
889:RULE 5 (multi-hop access):0
p,tomcat,remoteExploit,privEscalation):1
execCode(app_15,root):0
ULE 5 (multi-hop access):0

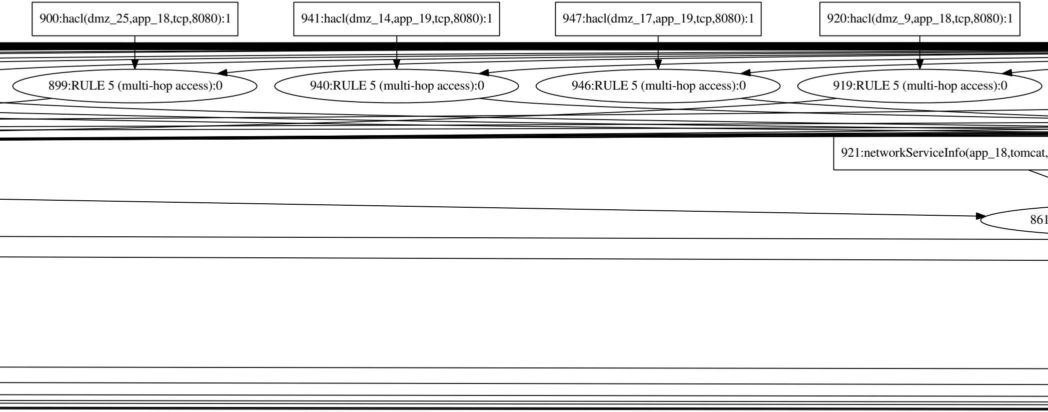

813:hacl(dmz_15,app_17,tcp,8080):1
892:hacl(dmz_21,app_18,tcp,8080):1
786:hacl(dmz_7,app_16,tcp,8080):1
864:hacl(dmz_0,app_18,tcp,8080):1
812:RULE 5 (multi-hop access):0
891:RULE 5 (multi-hop access):0
785:RULE 5 (multi-hop access):0
863:RULE 5 (multi-hop access):0

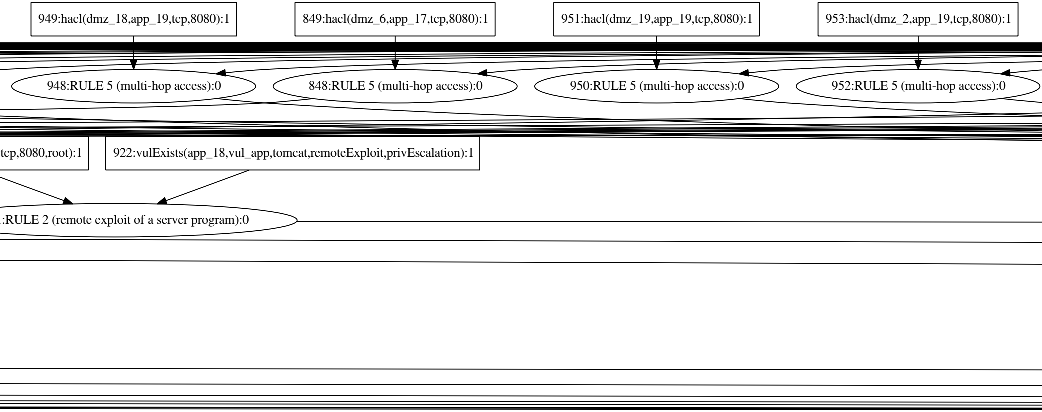

894:hacl(dmz_22,app_18,tcp,8080):1
866:hacl(dmz_1,app_18,tcp,8080):1
788:hacl(dmz_8,app_16,tcp,8080):1
868:hacl(dmz_10,app_18,tcp,8080):1
893:RULE 5 (multi-hop access):0
865:RULE 5 (multi-hop access):0
787:RULE 5 (multi-hop access):0
867:RULE 5 (multi-hop access):0

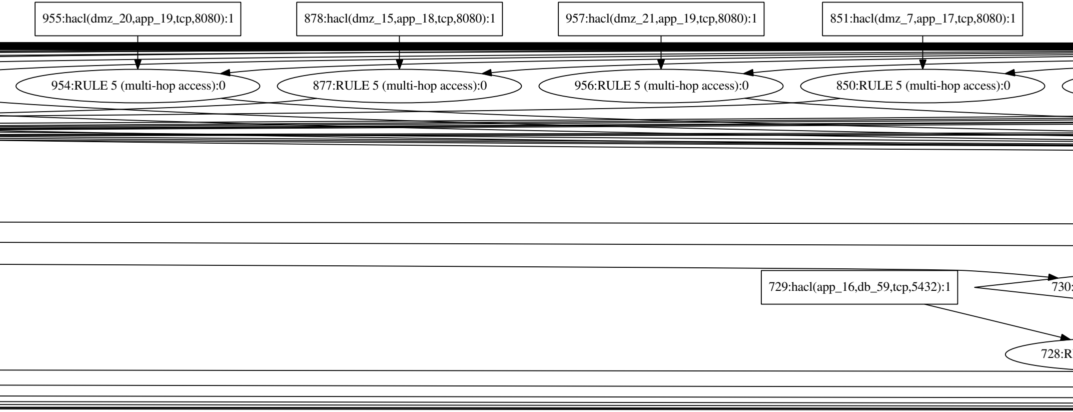

870:hacl(dmz_11,app_18,tcp,8080):1
845:hacl(dmz_4,app_17,tcp,8080):1
809:hacl(dmz_13,app_17,tcp,8080):1
872:hacl(dmz_12,app_18,tcp,8080):1
869:RULE 5 (multi-hop access):0
844:RULE 5 (multi-hop access):0
808:RULE 5 (multi-hop access):0
871:RULE 5 (multi-hop access):0
862:netAccess(app_18,tcp,8080):0

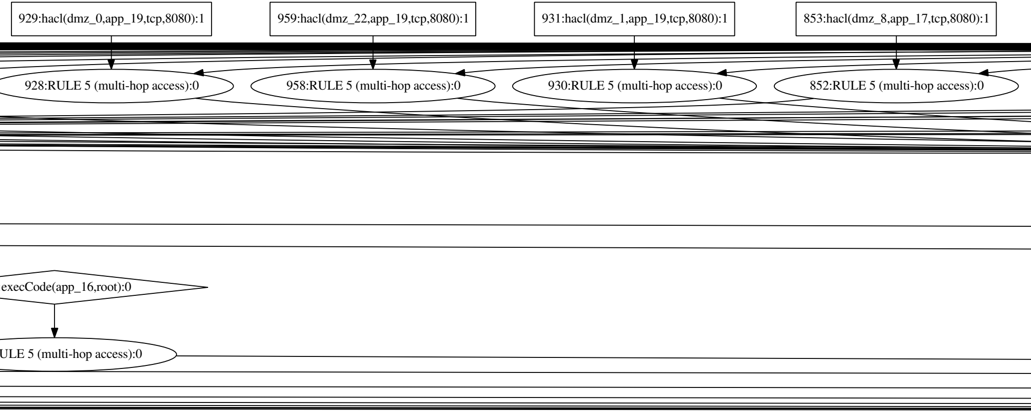

837:hacl(dmz_26,app_17,tcp,8080):1
896:hacl(dmz_23,app_18,tcp,8080):1
815:hacl(dmz_16,app_17,tcp,8080):1
898:hacl(dmz_24,app_18,tcp,8080):1
836:RULE 5 (multi-hop access):0
895:RULE 5 (multi-hop access):0
814:RULE 5 (multi-hop access):0
897:RULE 5 (multi-hop access):0

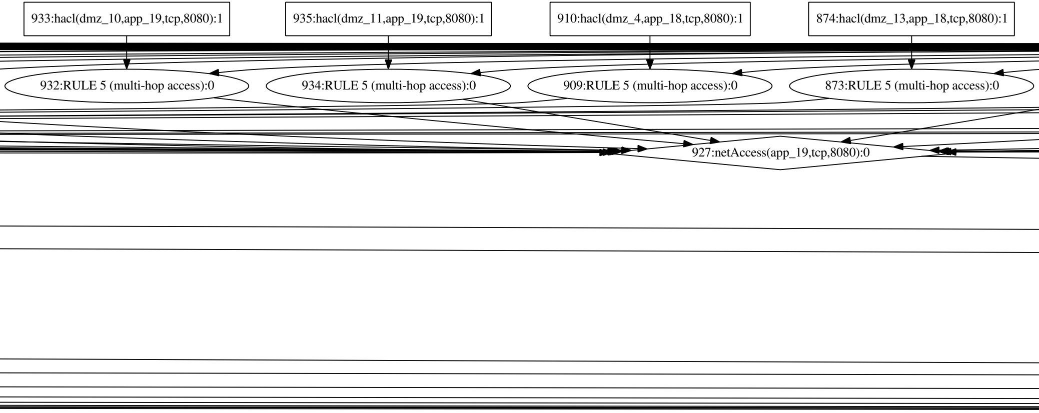

847:hacl(dmz_5,app_17,tcp,8080):1
969:hacl(dmz_27,app_19,tcp,8080):1
971:hacl(dmz_28,app_19,tcp,8080):1
973:hacl(dmz_3,app_19,tcp,8080):1
846:RULE 5 (multi-hop access):0
968:RULE 5 (multi-hop access):0
970:RULE 5 (multi-hop access):0
972:RULE 5 (multi-hop access):0

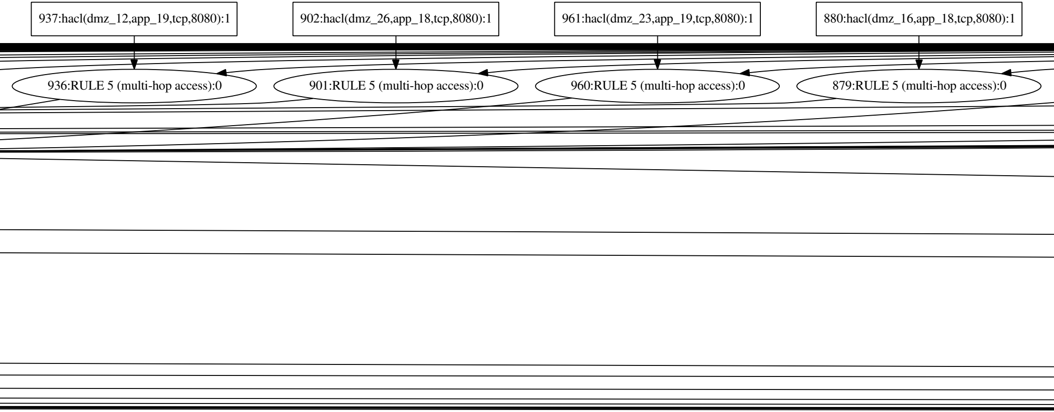

900:hacl(dmz_25,app_18,tcp,8080):1
941:hacl(dmz_14,app_19,tcp,8080):1
947:hacl(dmz_17,app_19,tcp,8080):1
920:hacl(dmz_9,app_18,tcp,8080):1
899:RULE 5 (multi-hop access):0
940:RULE 5 (multi-hop access):0
946:RULE 5 (multi-hop access):0
919:RULE 5 (multi-hop access):0
921:networkServiceInfo(app_18,tomcat,
861

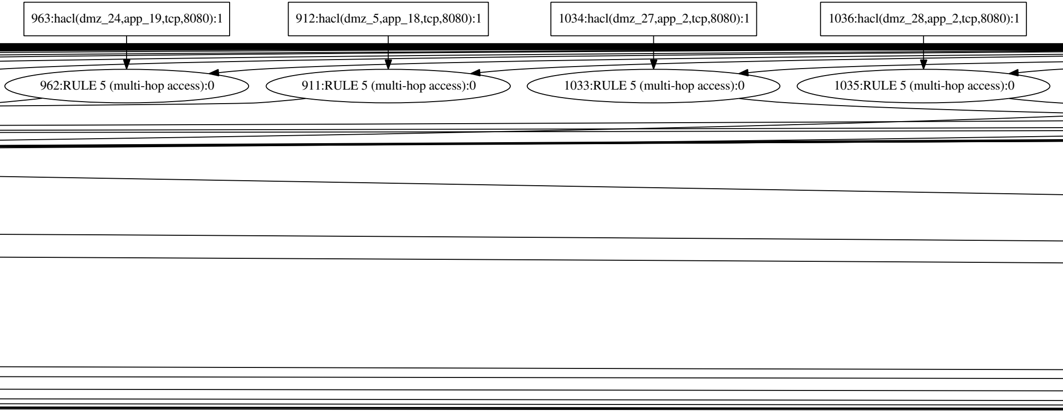

949:hacl(dmz_18,app_19,tcp,8080):1
849:hacl(dmz_6,app_17,tcp,8080):1
951:hacl(dmz_19,app_19,tcp,8080):1
953:hacl(dmz_2,app_19,tcp,8080):1
948:RULE 5 (multi-hop access):0
848:RULE 5 (multi-hop access):0
950:RULE 5 (multi-hop access):0
952:RULE 5 (multi-hop access):0
tcp,8080,root):1
922:vulExists(app_18,vul_app,tomcat,remoteExploit,privEscalation):1
:RULE 2 (remote exploit of a server program):0

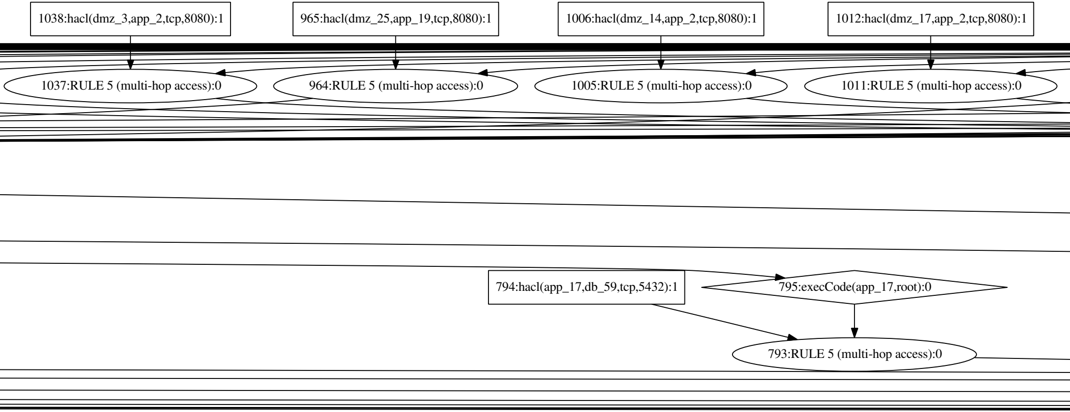

955:hacl(dmz_20,app_19,tcp,8080):1
878:hacl(dmz_15,app_18,tcp,8080):1
957:hacl(dmz_21,app_19,tcp,8080):1
851:hacl(dmz_7,app_17,tcp,8080):1
954:RULE 5 (multi-hop access):0
877:RULE 5 (multi-hop access):0
956:RULE 5 (multi-hop access):0
850:RULE 5 (multi-hop access):0
729:hacl(app_16,db_59,tcp,5432):1
730:
728:R

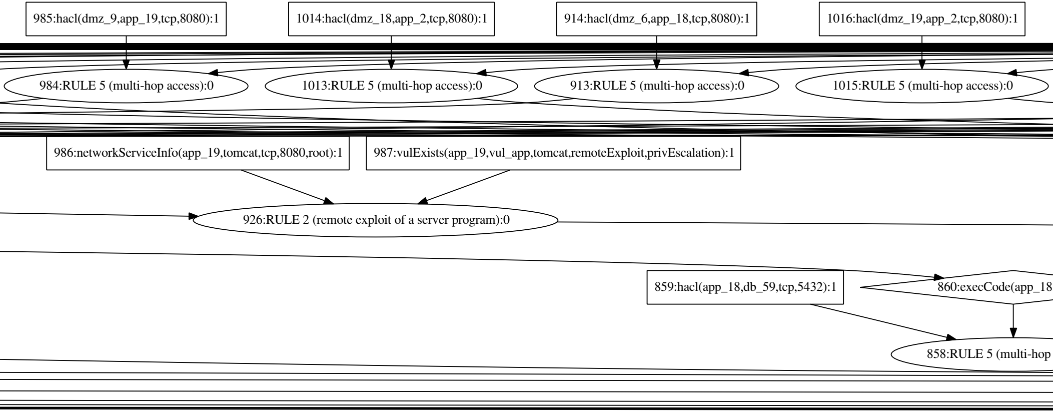

929:hacl(dmz_0,app_19,tcp,8080):1
959:hacl(dmz_22,app_19,tcp,8080):1
931:hacl(dmz_1,app_19,tcp,8080):1
853:hacl(dmz_8,app_17,tcp,8080):1
928:RULE 5 (multi-hop access):0
958:RULE 5 (multi-hop access):0
930:RULE 5 (multi-hop access):0
852:RULE 5 (multi-hop access):0
execCode(app_16,root):0
ULE 5 (multi-hop access):0

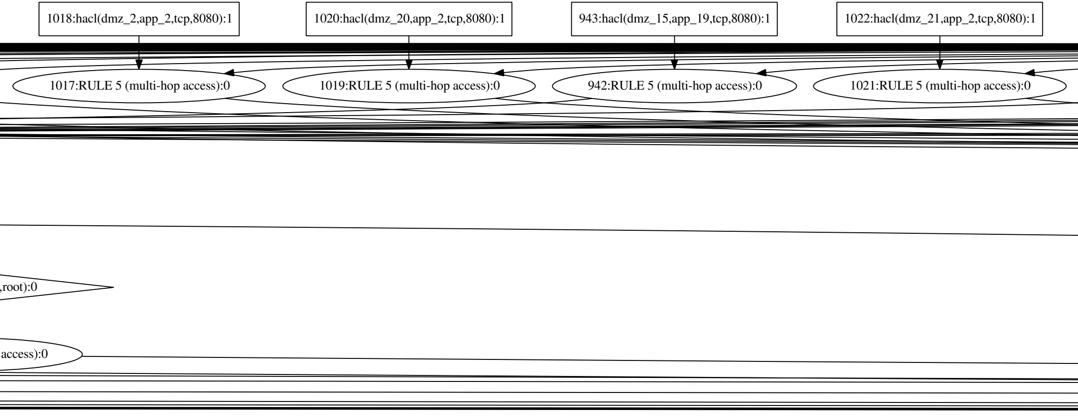

933:hacl(dmz_10,app_19,tcp,8080):1
935:hacl(dmz_11,app_19,tcp,8080):1
910:hacl(dmz_4,app_18,tcp,8080):1
874:hacl(dmz_13,app_18,tcp,8080):1
932:RULE 5 (multi-hop access):0
934:RULE 5 (multi-hop access):0
909:RULE 5 (multi-hop access):0
873:RULE 5 (multi-hop access):0
927:netAccess(app_19,tcp,8080):0

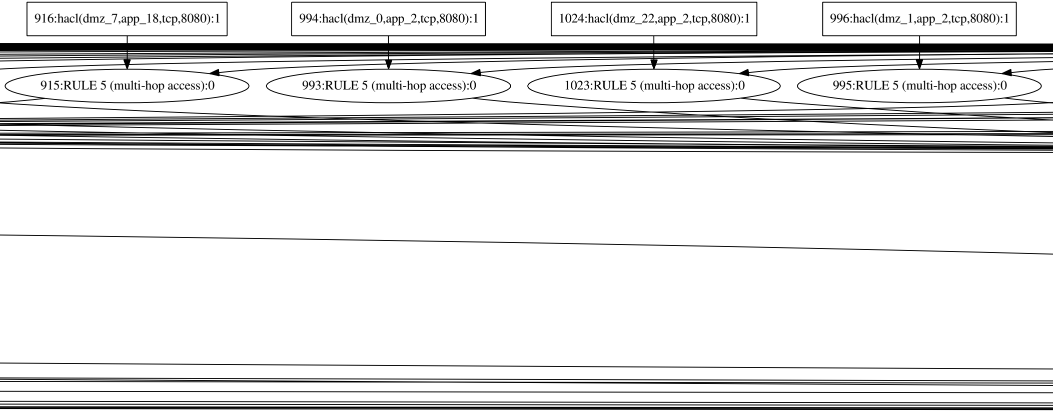

937:hacl(dmz_12,app_19,tcp,8080):1
902:hacl(dmz_26,app_18,tcp,8080):1
961:hacl(dmz_23,app_19,tcp,8080):1
880:hacl(dmz_16,app_18,tcp,8080):1
936:RULE 5 (multi-hop access):0
901:RULE 5 (multi-hop access):0
960:RULE 5 (multi-hop access):0
879:RULE 5 (multi-hop access):0

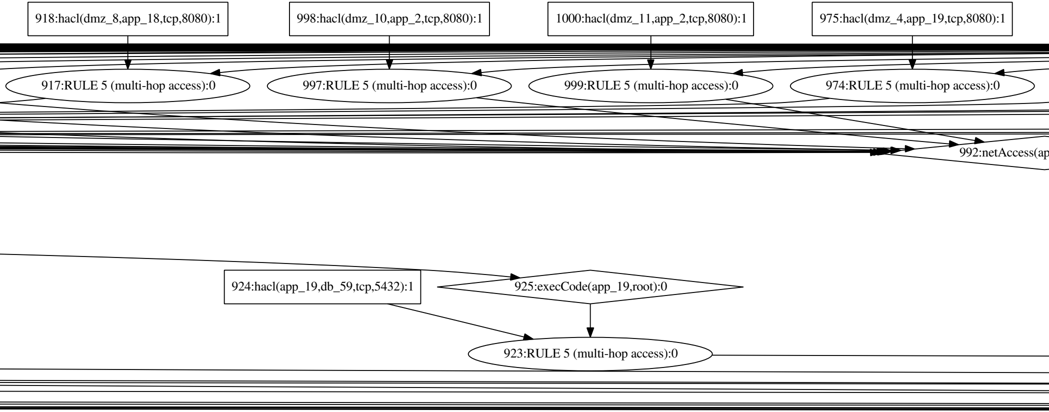

963:hacl(dmz_24,app_19,tcp,8080):1
912:hacl(dmz_5,app_18,tcp,8080):1
1034:hacl(dmz_27,app_2,tcp,8080):1
1036:hacl(dmz_28,app_2,tcp,8080):1
962:RULE 5 (multi-hop access):0
911:RULE 5 (multi-hop access):0
1033:RULE 5 (multi-hop access):0
1035:RULE 5 (multi-hop access):0

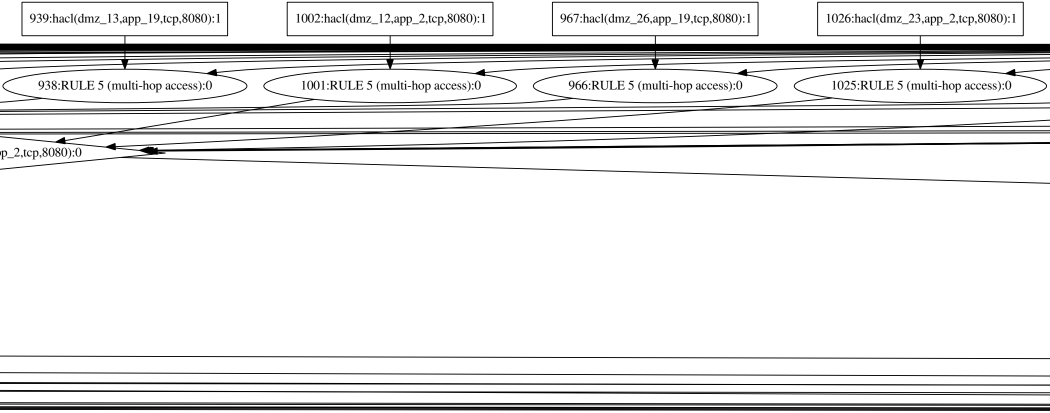

1038:hacl(dmz_3,app_2,tcp,8080):1
965:hacl(dmz_25,app_19,tcp,8080):1
1006:hacl(dmz_14,app_2,tcp,8080):1
1012:hacl(dmz_17,app_2,tcp,8080):1
1037:RULE 5 (multi-hop access):0
964:RULE 5 (multi-hop access):0
1005:RULE 5 (multi-hop access):0
1011:RULE 5 (multi-hop access):0
794:hacl(app_17,db_59,tcp,5432):1
795:execCode(app_17,root):0
793:RULE 5 (multi-hop access):0

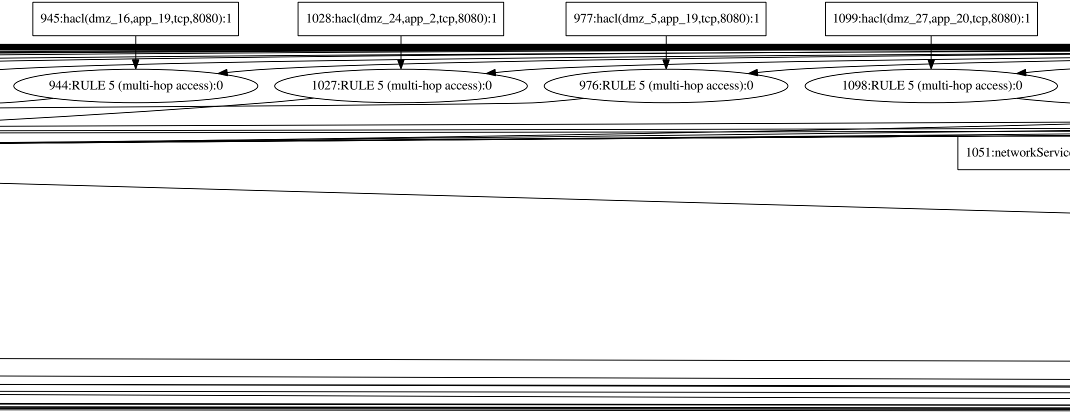

985:hacl(dmz_9,app_19,tcp,8080):1
1014:hacl(dmz_18,app_2,tcp,8080):1
914:hacl(dmz_6,app_18,tcp,8080):1
1016:hacl(dmz_19,app_2,tcp,8080):1
984:RULE 5 (multi-hop access):0
1013:RULE 5 (multi-hop access):0
913:RULE 5 (multi-hop access):0
1015:RULE 5 (multi-hop access):0
986:networkServiceInfo(app_19,tomcat,tcp,8080,root):1
987:vulExists(app_19,vul_app,tomcat,remoteExploit,privEscalation):1
926:RULE 2 (remote exploit of a server program):0
859:hacl(app_18,db_59,tcp,5432):1
860:execCode(app_18
858:RULE 5 (multi-hop

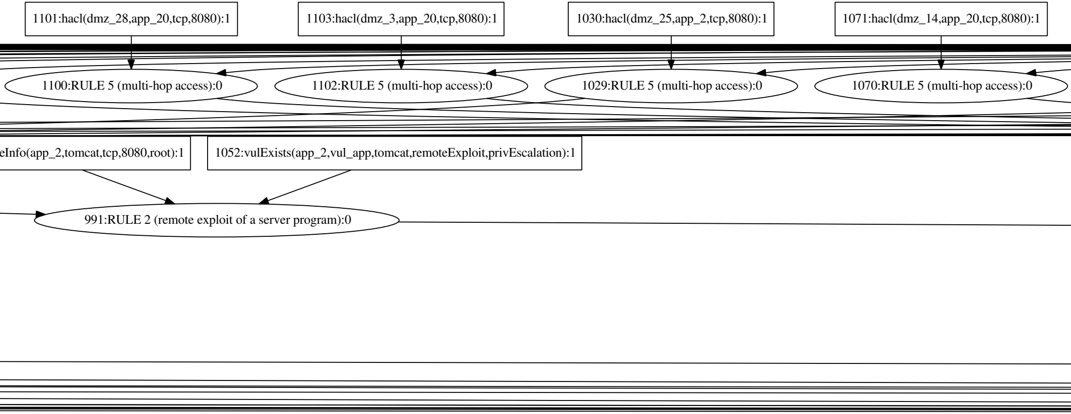

1018:hacl(dmz_2,app_2,tcp,8080):1
1020:hacl(dmz_20,app_2,tcp,8080):1
943:hacl(dmz_15,app_19,tcp,8080):1
1022:hacl(dmz_21,app_2,tcp,8080):1
1017:RULE 5 (multi-hop access):0
1019:RULE 5 (multi-hop access):0
942:RULE 5 (multi-hop access):0
1021:RULE 5 (multi-hop access):0
,root):0
access):0

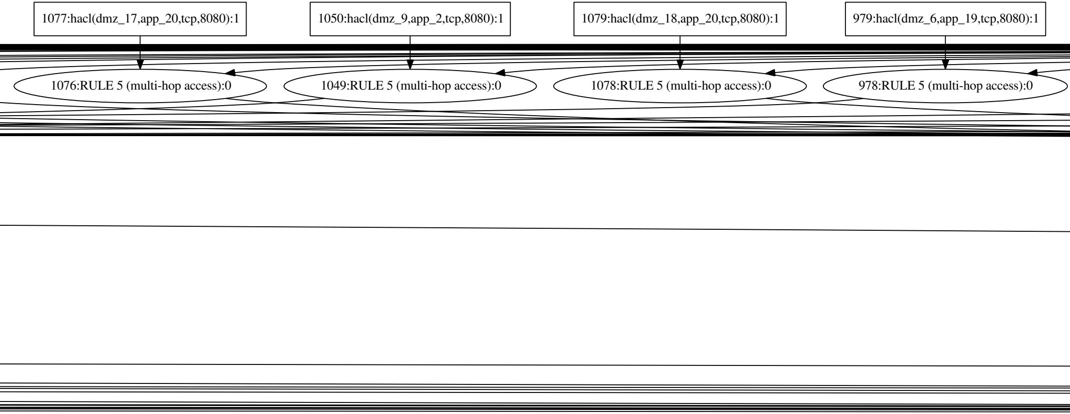

916:hacl(dmz_7,app_18,tcp,8080):1
994:hacl(dmz_0,app_2,tcp,8080):1
1024:hacl(dmz_22,app_2,tcp,8080):1
996:hacl(dmz_1,app_2,tcp,8080):1
915:RULE 5 (multi-hop access):0
993:RULE 5 (multi-hop access):0
1023:RULE 5 (multi-hop access):0
995:RULE 5 (multi-hop access):0

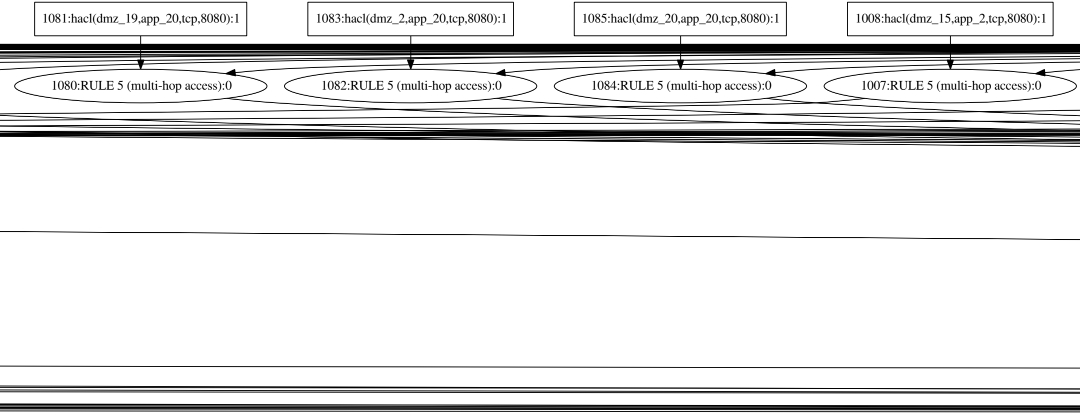

918:hacl(dmz_8,app_18,tcp,8080):1
998:hacl(dmz_10,app_2,tcp,8080):1
1000:hacl(dmz_11,app_2,tcp,8080):1
975:hacl(dmz_4,app_19,tcp,8080):1
917:RULE 5 (multi-hop access):0
997:RULE 5 (multi-hop access):0
999:RULE 5 (multi-hop access):0
974:RULE 5 (multi-hop access):0
992:netAccess(ap
924:hacl(app_19,db_59,tcp,5432):1
925:execCode(app_19,root):0
923:RULE 5 (multi-hop access):0

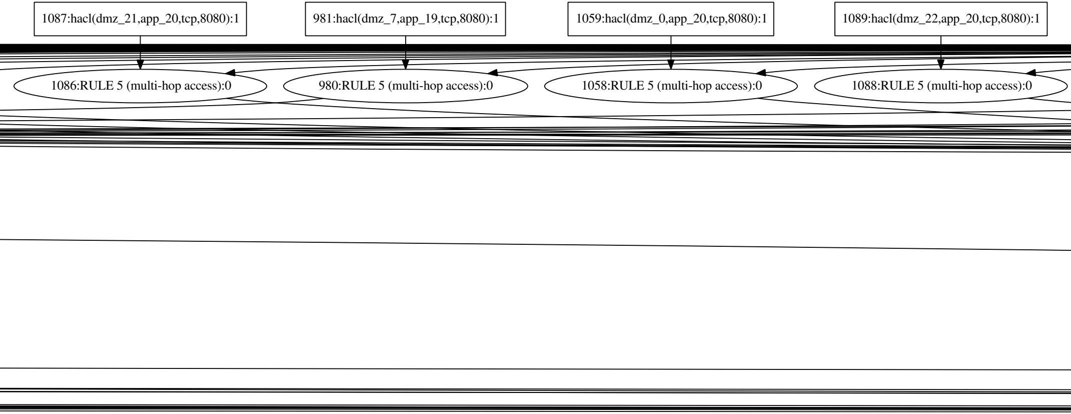

939:hacl(dmz_13,app_19,tcp,8080):1
1002:hacl(dmz_12,app_2,tcp,8080):1
967:hacl(dmz_26,app_19,tcp,8080):1
1026:hacl(dmz_23,app_2,tcp,8080):1
938:RULE 5 (multi-hop access):0
1001:RULE 5 (multi-hop access):0
966:RULE 5 (multi-hop access):0
1025:RULE 5 (multi-hop access):0
p_2,tcp,8080):0

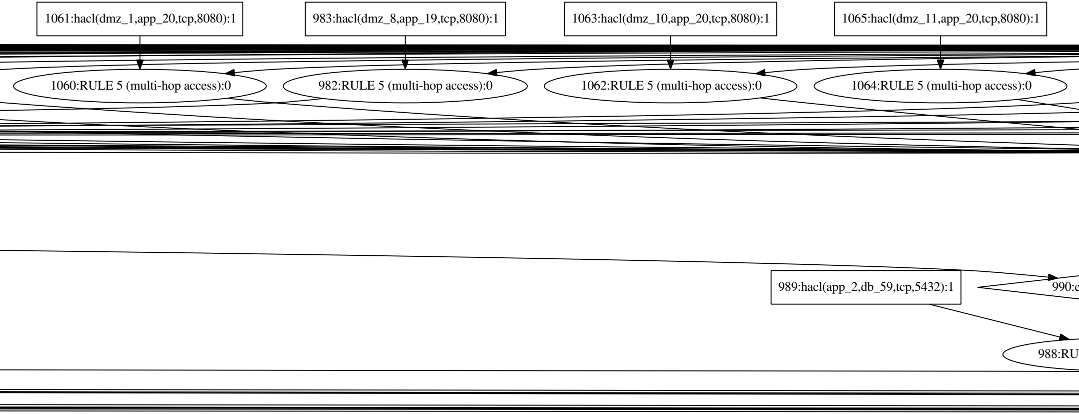

945:hacl(dmz_16,app_19,tcp,8080):1
1028:hacl(dmz_24,app_2,tcp,8080):1
977:hacl(dmz_5,app_19,tcp,8080):1
1099:hacl(dmz_27,app_20,tcp,8080):1
944:RULE 5 (multi-hop access):0
1027:RULE 5 (multi-hop access):0
976:RULE 5 (multi-hop access):0
1098:RULE 5 (multi-hop access):0
1051:networkServic

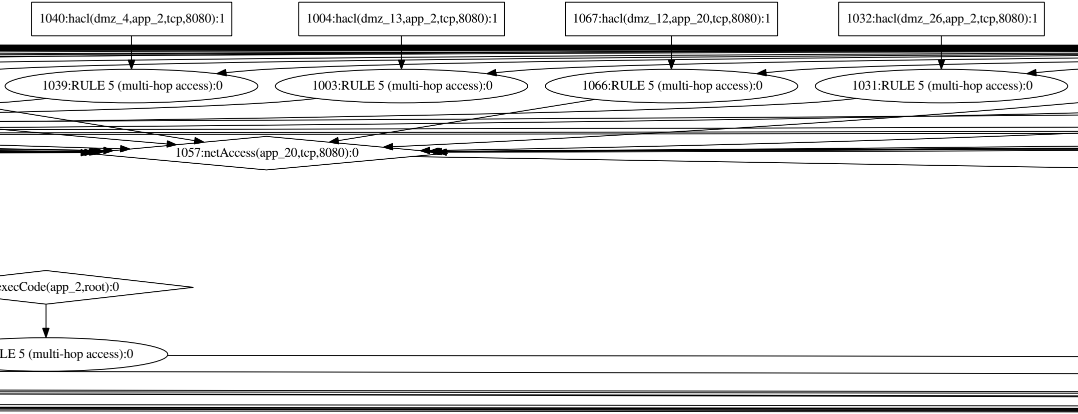

1101:hacl(dmz_28,app_20,tcp,8080):1
1103:hacl(dmz_3,app_20,tcp,8080):1
1030:hacl(dmz_25,app_2,tcp,8080):1
1071:hacl(dmz_14,app_20,tcp,8080):1
1100:RULE 5 (multi-hop access):0
1102:RULE 5 (multi-hop access):0
1029:RULE 5 (multi-hop access):0
1070:RULE 5 (multi-hop access):0
eInfo(app_2,tomcat,tcp,8080,root):1
1052:vulExists(app_2,vul_app,tomcat,remoteExploit,privEscalation):1
991:RULE 2 (remote exploit of a server program):0

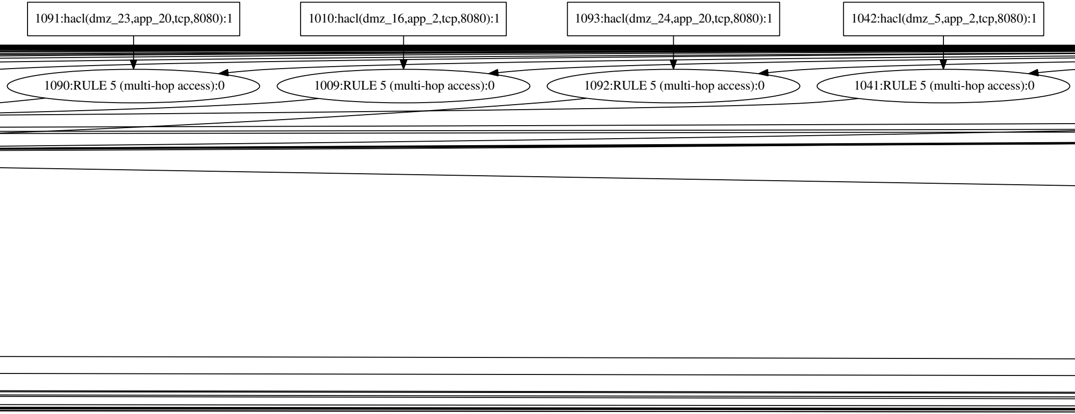

1077:hacl(dmz_17,app_20,tcp,8080):1
1050:hacl(dmz_9,app_2,tcp,8080):1
1079:hacl(dmz_18,app_20,tcp,8080):1
979:hacl(dmz_6,app_19,tcp,8080):1
1076:RULE 5 (multi-hop access):0
1049:RULE 5 (multi-hop access):0
1078:RULE 5 (multi-hop access):0
978:RULE 5 (multi-hop access):0

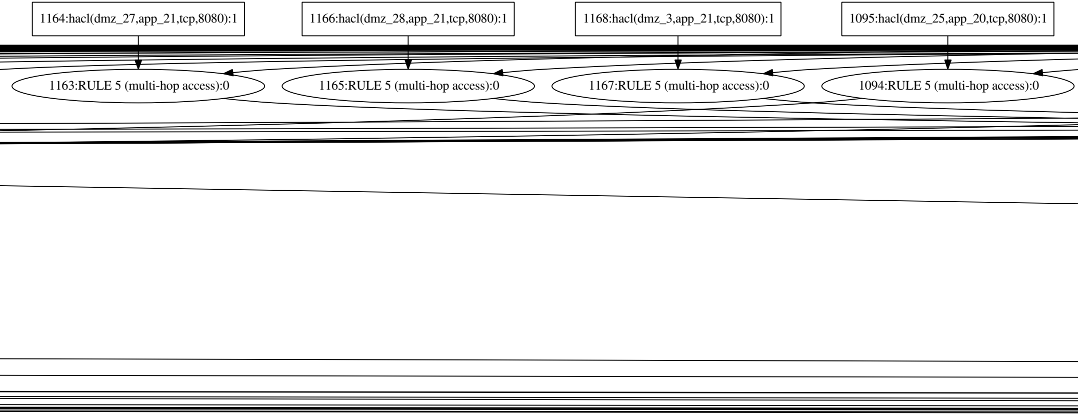

1081:hacl(dmz_19,app_20,tcp,8080):1
1083:hacl(dmz_2,app_20,tcp,8080):1
1085:hacl(dmz_20,app_20,tcp,8080):1
1008:hacl(dmz_15,app_2,tcp,8080):1
1080:RULE 5 (multi-hop access):0
1082:RULE 5 (multi-hop access):0
1084:RULE 5 (multi-hop access):0
1007:RULE 5 (multi-hop access):0

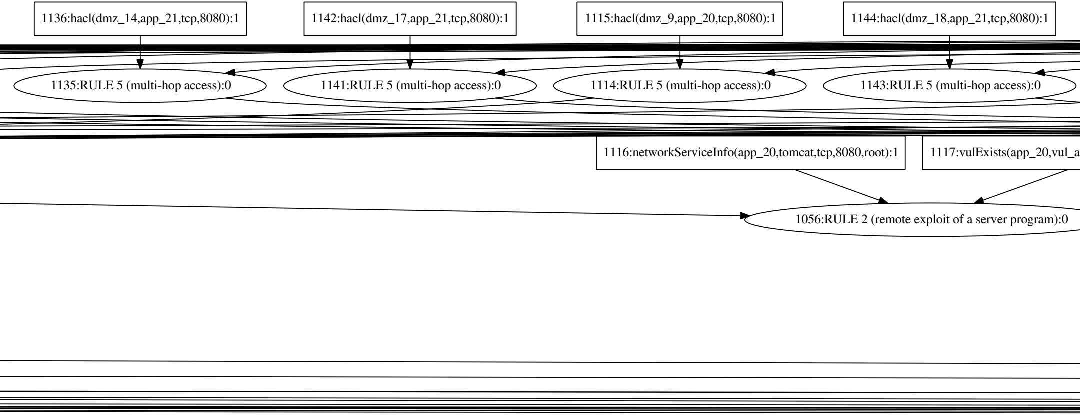

1087:hacl(dmz_21,app_20,tcp,8080):1
981:hacl(dmz_7,app_19,tcp,8080):1
1059:hacl(dmz_0,app_20,tcp,8080):1
1089:hacl(dmz_22,app_20,tcp,8080):1
1086:RULE 5 (multi-hop access):0
980:RULE 5 (multi-hop access):0
1058:RULE 5 (multi-hop access):0
1088:RULE 5 (multi-hop access):0

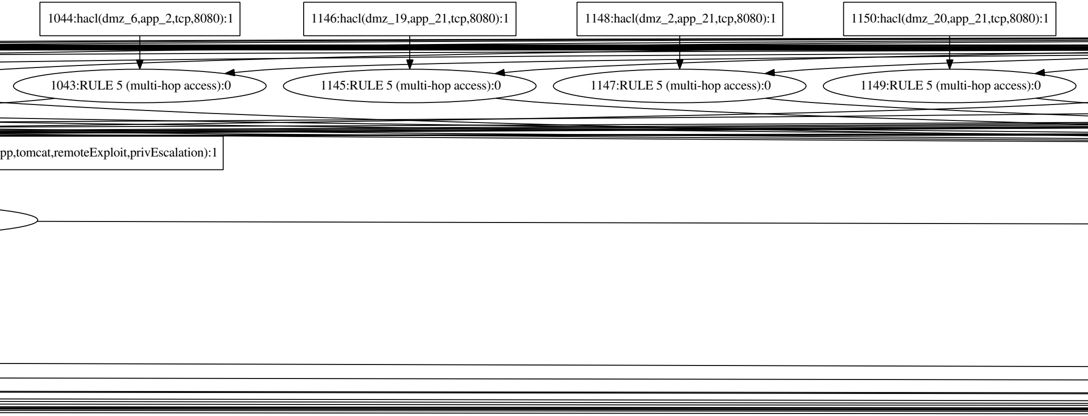

1061:hacl(dmz_1,app_20,tcp,8080):1
983:hacl(dmz_8,app_19,tcp,8080):1
1063:hacl(dmz_10,app_20,tcp,8080):1
1065:hacl(dmz_11,app_20,tcp,8080):1
1060:RULE 5 (multi-hop access):0
982:RULE 5 (multi-hop access):0
1062:RULE 5 (multi-hop access):0
1064:RULE 5 (multi-hop access):0
989:hacl(app_2,db_59,tcp,5432):1
990:e
988:RU

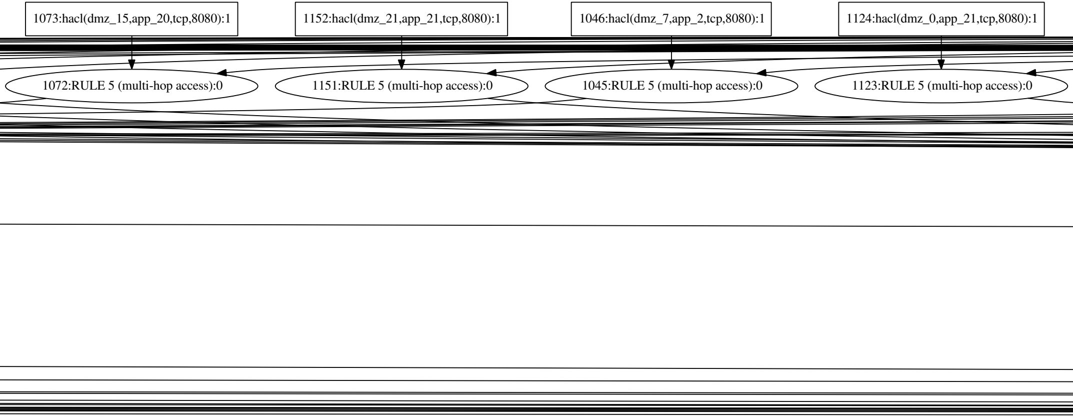

1040:hacl(dmz_4,app_2,tcp,8080):1
1004:hacl(dmz_13,app_2,tcp,8080):1
1067:hacl(dmz_12,app_20,tcp,8080):1
1032:hacl(dmz_26,app_2,tcp,8080):1
1039:RULE 5 (multi-hop access):0
1003:RULE 5 (multi-hop access):0
1066:RULE 5 (multi-hop access):0
1031:RULE 5 (multi-hop access):0
1057:netAccess(app_20,tcp,8080):0
execCode(app_2,root):0
LE 5 (multi-hop access):0


1091:hacl(dmz_23,app_20,tcp,8080):1
1010:hacl(dmz_16,app_2,tcp,8080):1
1093:hacl(dmz_24,app_20,tcp,8080):1
1042:hacl(dmz_5,app_2,tcp,8080):1
1090:RULE 5 (multi-hop access):0
1009:RULE 5 (multi-hop access):0
1092:RULE 5 (multi-hop access):0
1041:RULE 5 (multi-hop access):0

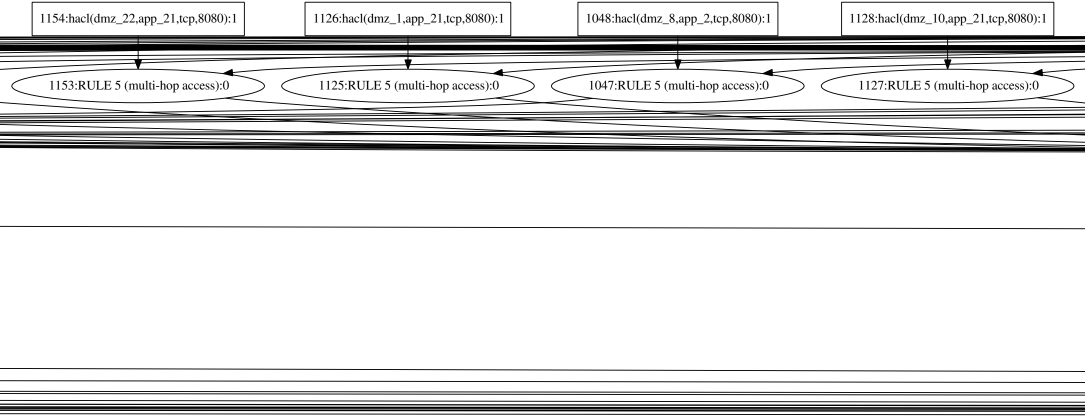

1164:hacl(dmz_27,app_21,tcp,8080):1
1166:hacl(dmz_28,app_21,tcp,8080):1
1168:hacl(dmz_3,app_21,tcp,8080):1
1095:hacl(dmz_25,app_20,tcp,8080):1
1163:RULE 5 (multi-hop access):0
1165:RULE 5 (multi-hop access):0
1167:RULE 5 (multi-hop access):0
1094:RULE 5 (multi-hop access):0

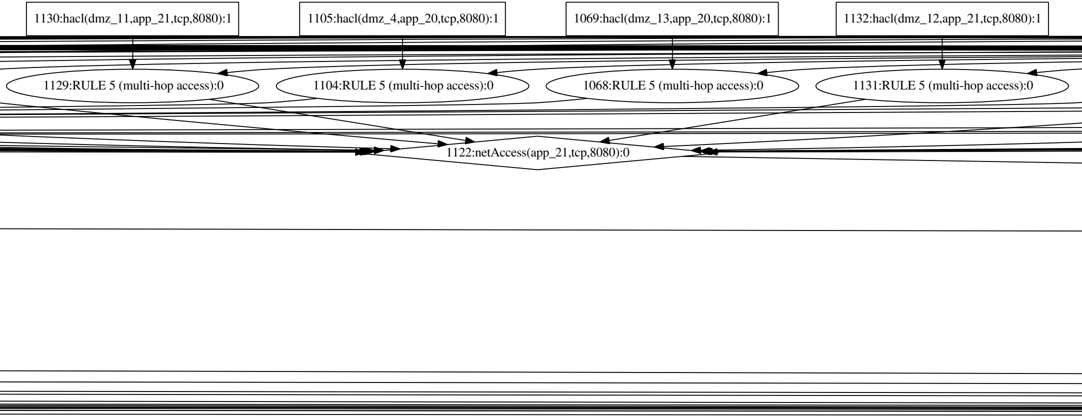

1136:hacl(dmz_14,app_21,tcp,8080):1
1142:hacl(dmz_17,app_21,tcp,8080):1
1115:hacl(dmz_9,app_20,tcp,8080):1
1144:hacl(dmz_18,app_21,tcp,8080):1
1135:RULE 5 (multi-hop access):0
1141:RULE 5 (multi-hop access):0
1114:RULE 5 (multi-hop access):0
1143:RULE 5 (multi-hop access):0
1116:networkServiceInfo(app_20,tomcat,tcp,8080,root):1
1117:vulExists(app_20,vul_a
1056:RULE 2 (remote exploit of a server program):0

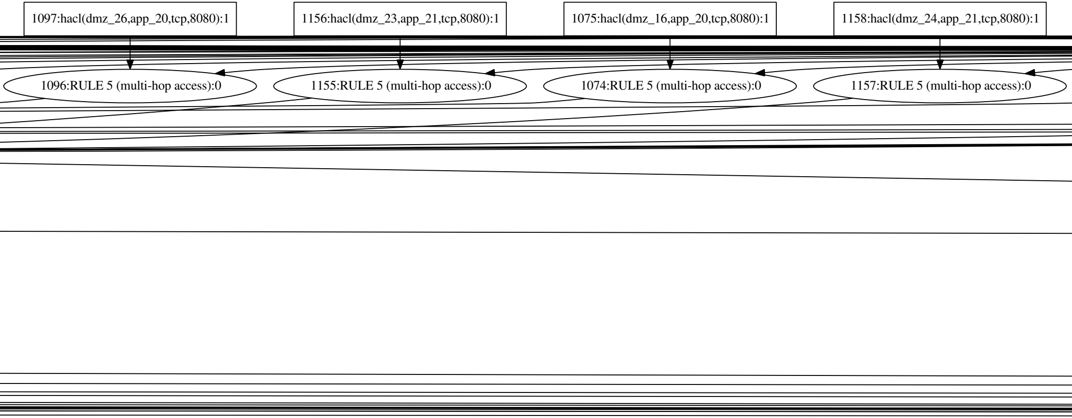

1044:hacl(dmz_6,app_2,tcp,8080):1
1146:hacl(dmz_19,app_21,tcp,8080):1
1148:hacl(dmz_2,app_21,tcp,8080):1
1150:hacl(dmz_20,app_21,tcp,8080):1
1043:RULE 5 (multi-hop access):0
1145:RULE 5 (multi-hop access):0
1147:RULE 5 (multi-hop access):0
1149:RULE 5 (multi-hop access):0
pp,tomcat,remoteExploit,privEscalation):1

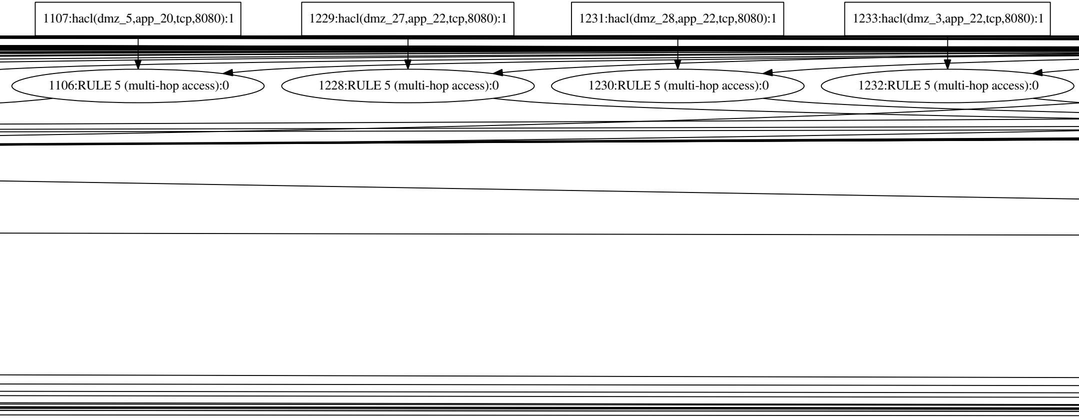


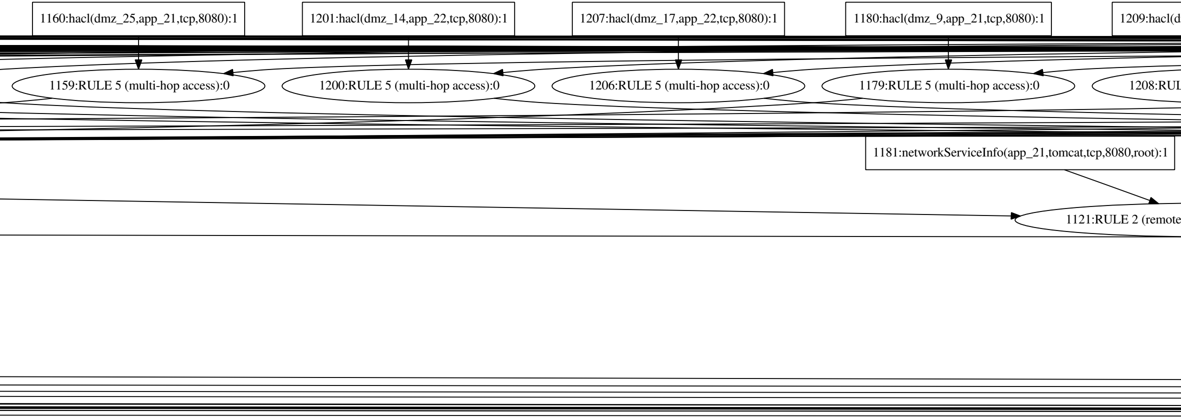
1073:hacl(dmz_15,app_20,tcp,8080):1
1152:hacl(dmz_21,app_21,tcp,8080):1
1046:hacl(dmz_7,app_2,tcp,8080):1
1124:hacl(dmz_0,app_21,tcp,8080):1
1072:RULE 5 (multi-hop access):0
1151:RULE 5 (multi-hop access):0
1045:RULE 5 (multi-hop access):0
1123:RULE 5 (multi-hop access):0

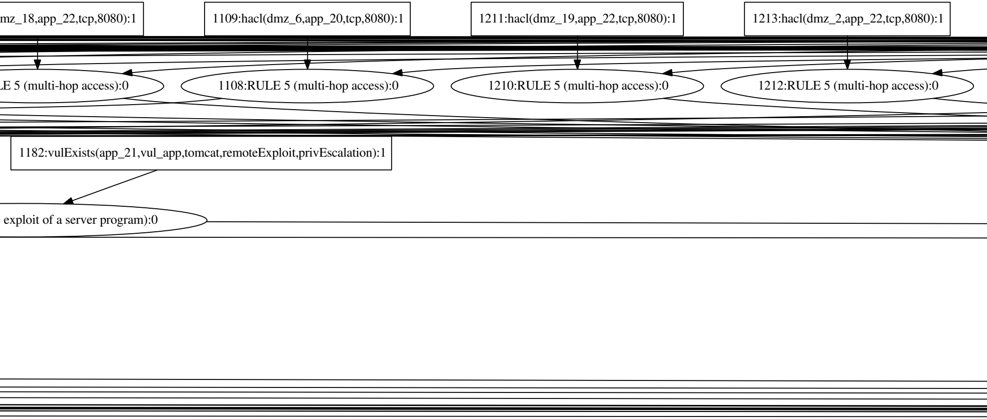

1154:hacl(dmz_22,app_21,tcp,8080):1
1126:hacl(dmz_1,app_21,tcp,8080):1
1048:hacl(dmz_8,app_2,tcp,8080):1
1128:hacl(dmz_10,app_21,tcp,8080):1
1153:RULE 5 (multi-hop access):0
1125:RULE 5 (multi-hop access):0
1047:RULE 5 (multi-hop access):0
1127:RULE 5 (multi-hop access):0

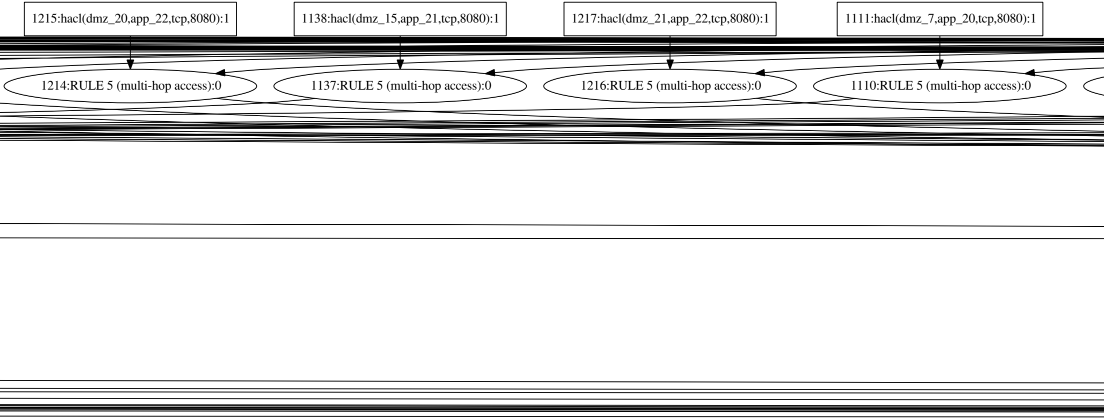

1130:hacl(dmz_11,app_21,tcp,8080):1
1105:hacl(dmz_4,app_20,tcp,8080):1
1069:hacl(dmz_13,app_20,tcp,8080):1
1132:hacl(dmz_12,app_21,tcp,8080):1
1129:RULE 5 (multi-hop access):0
1104:RULE 5 (multi-hop access):0
1068:RULE 5 (multi-hop access):0
1131:RULE 5 (multi-hop access):0
1122:netAccess(app_21,tcp,8080):0

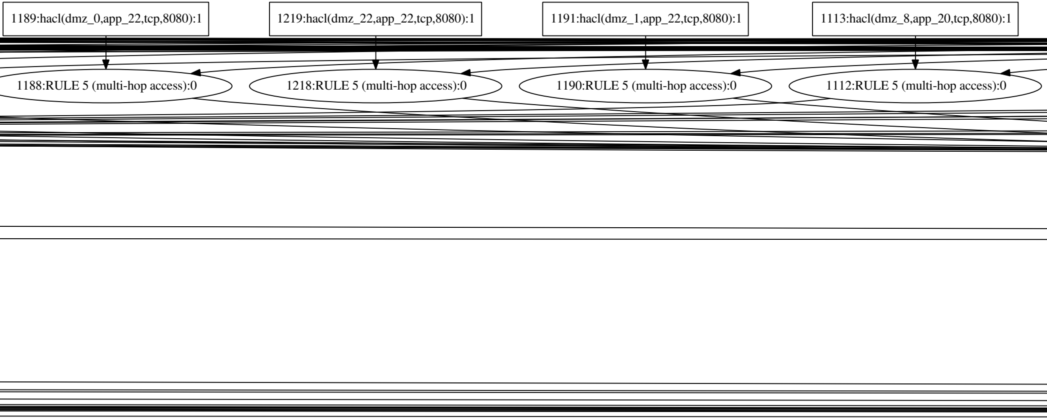

1097:hacl(dmz_26,app_20,tcp,8080):1
1156:hacl(dmz_23,app_21,tcp,8080):1
1075:hacl(dmz_16,app_20,tcp,8080):1
1158:hacl(dmz_24,app_21,tcp,8080):1
1096:RULE 5 (multi-hop access):0
1155:RULE 5 (multi-hop access):0
1074:RULE 5 (multi-hop access):0
1157:RULE 5 (multi-hop access):0

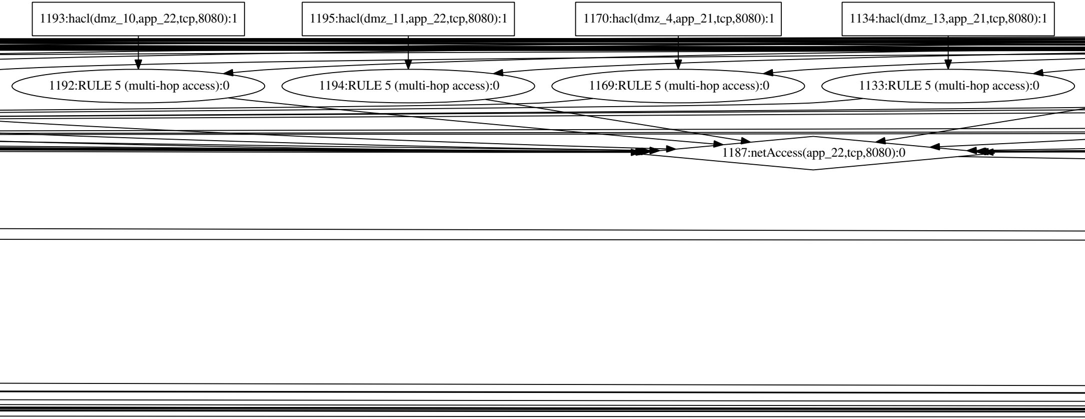

1107:hacl(dmz_5,app_20,tcp,8080):1
1229:hacl(dmz_27,app_22,tcp,8080):1
1231:hacl(dmz_28,app_22,tcp,8080):1
1233:hacl(dmz_3,app_22,tcp,8080):1
1106:RULE 5 (multi-hop access):0
1228:RULE 5 (multi-hop access):0
1230:RULE 5 (multi-hop access):0
1232:RULE 5 (multi-hop access):0

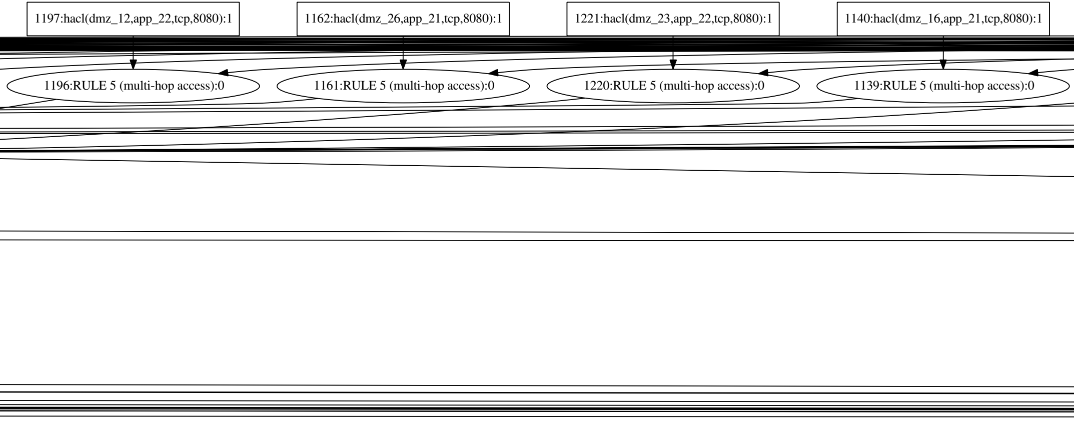

1160:hacl(dmz_25,app_21,tcp,8080):1
1201:hacl(dmz_14,app_22,tcp,8080):1
1207:hacl(dmz_17,app_22,tcp,8080):1
1180:hacl(dmz_9,app_21,tcp,8080):1
1209:hacl(d
1159:RULE 5 (multi-hop access):0
1200:RULE 5 (multi-hop access):0
1206:RULE 5 (multi-hop access):0
1179:RULE 5 (multi-hop access):0
1208:RUL
1181:networkServiceInfo(app_21,tomcat,tcp,8080,root):1
1121:RULE 2 (remote

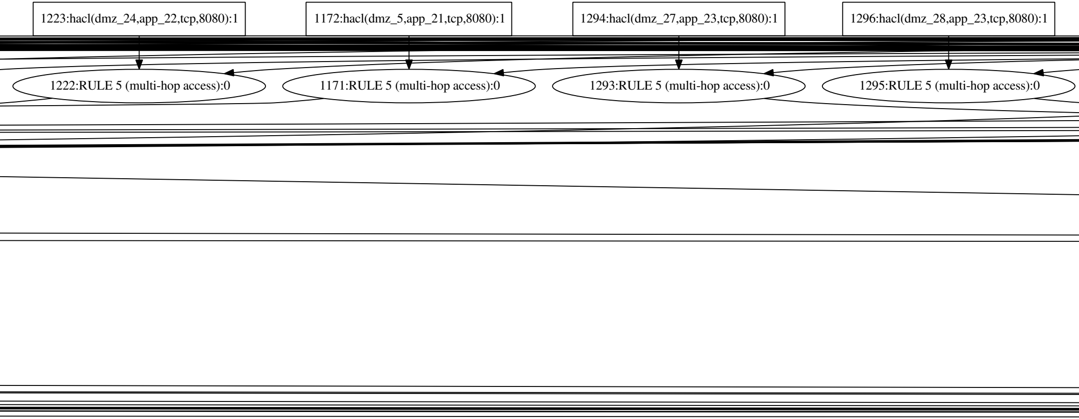

mz_18,app_22,tcp,8080):1
1109:hacl(dmz_6,app_20,tcp,8080):1
1211:hacl(dmz_19,app_22,tcp,8080):1
1213:hacl(dmz_2,app_22,tcp,8080):1
LE 5 (multi-hop access):0
1108:RULE 5 (multi-hop access):0
1210:RULE 5 (multi-hop access):0
1212:RULE 5 (multi-hop access):0
1182:vulExists(app_21,vul_app,tomcat,remoteExploit,privEscalation):1
exploit of a server program):0

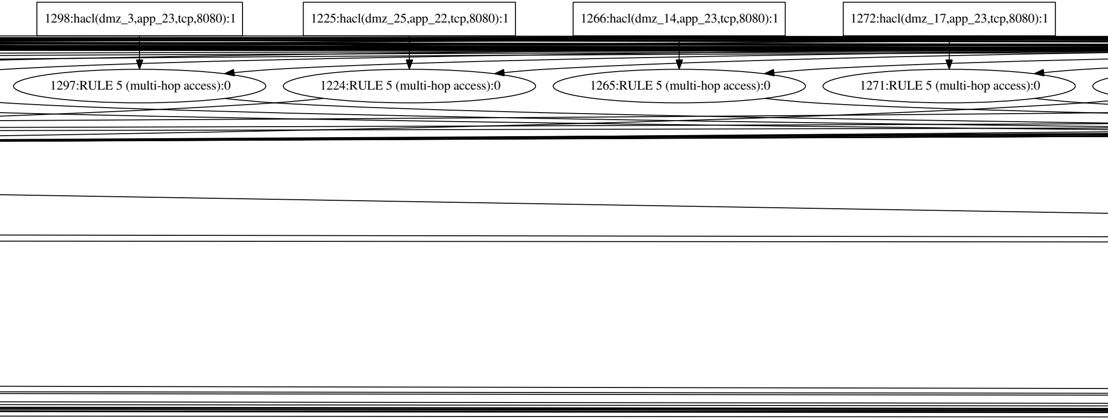

1215:hacl(dmz_20,app_22,tcp,8080):1
1138:hacl(dmz_15,app_21,tcp,8080):1
1217:hacl(dmz_21,app_22,tcp,8080):1
1111:hacl(dmz_7,app_20,tcp,8080):1
1214:RULE 5 (multi-hop access):0
1137:RULE 5 (multi-hop access):0
1216:RULE 5 (multi-hop access):0
1110:RULE 5 (multi-hop access):0

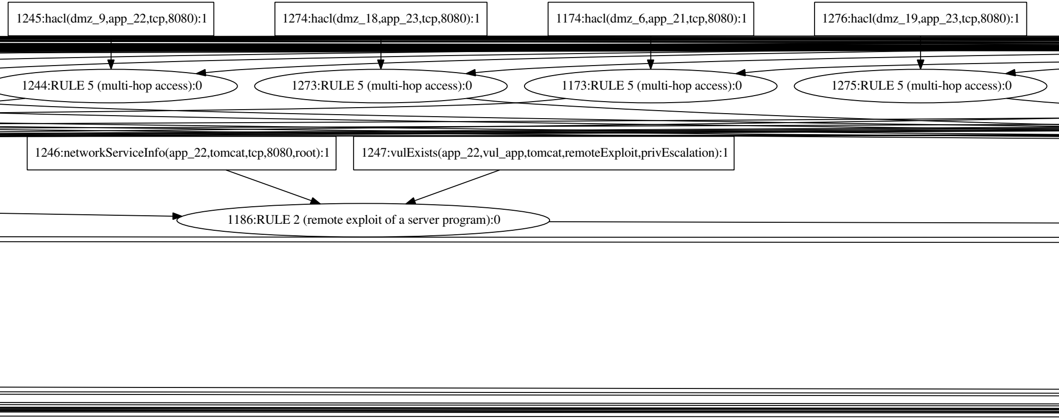

1189:hacl(dmz_0,app_22,tcp,8080):1
1219:hacl(dmz_22,app_22,tcp,8080):1
1191:hacl(dmz_1,app_22,tcp,8080):1
1113:hacl(dmz_8,app_20,tcp,8080):1
1188:RULE 5 (multi-hop access):0
1218:RULE 5 (multi-hop access):0
1190:RULE 5 (multi-hop access):0
1112:RULE 5 (multi-hop access):0

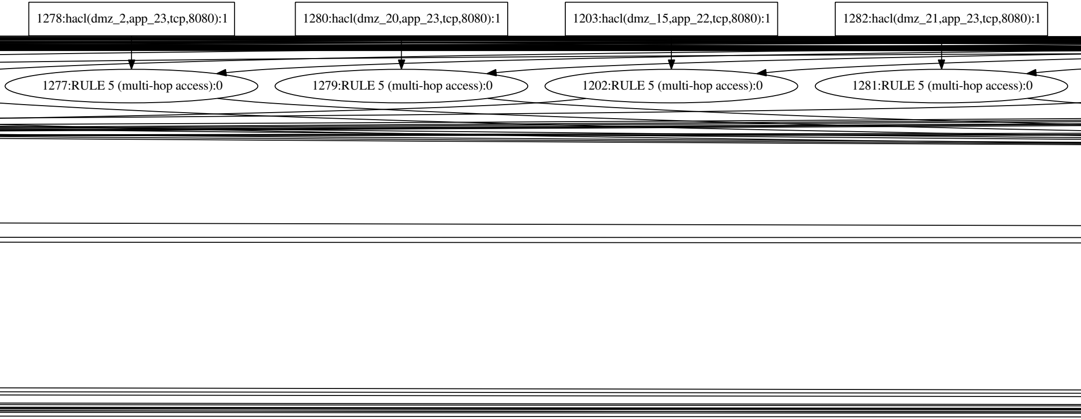

1193:hacl(dmz_10,app_22,tcp,8080):1
1195:hacl(dmz_11,app_22,tcp,8080):1
1170:hacl(dmz_4,app_21,tcp,8080):1
1134:hacl(dmz_13,app_21,tcp,8080):1
1192:RULE 5 (multi-hop access):0
1194:RULE 5 (multi-hop access):0
1169:RULE 5 (multi-hop access):0
1133:RULE 5 (multi-hop access):0
1187:netAccess(app_22,tcp,8080):0

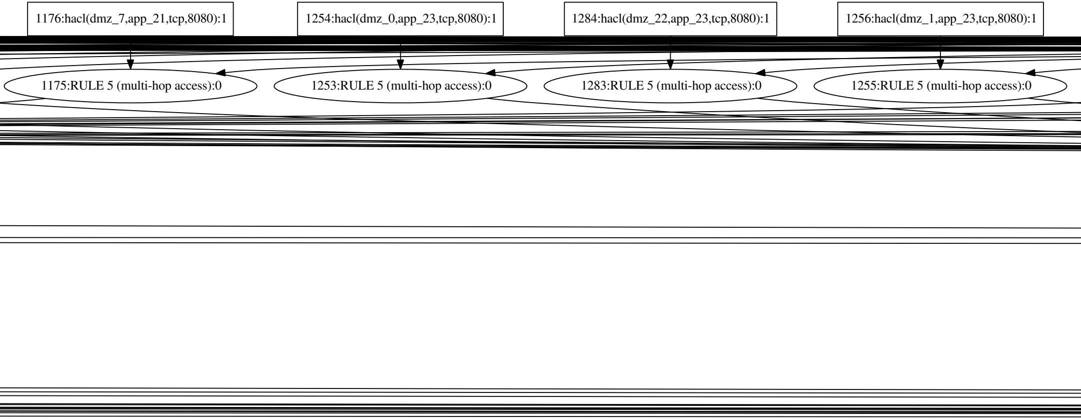

1197:hacl(dmz_12,app_22,tcp,8080):1
1162:hacl(dmz_26,app_21,tcp,8080):1
1221:hacl(dmz_23,app_22,tcp,8080):1
1140:hacl(dmz_16,app_21,tcp,8080):1
1196:RULE 5 (multi-hop access):0
1161:RULE 5 (multi-hop access):0
1220:RULE 5 (multi-hop access):0
1139:RULE 5 (multi-hop access):0

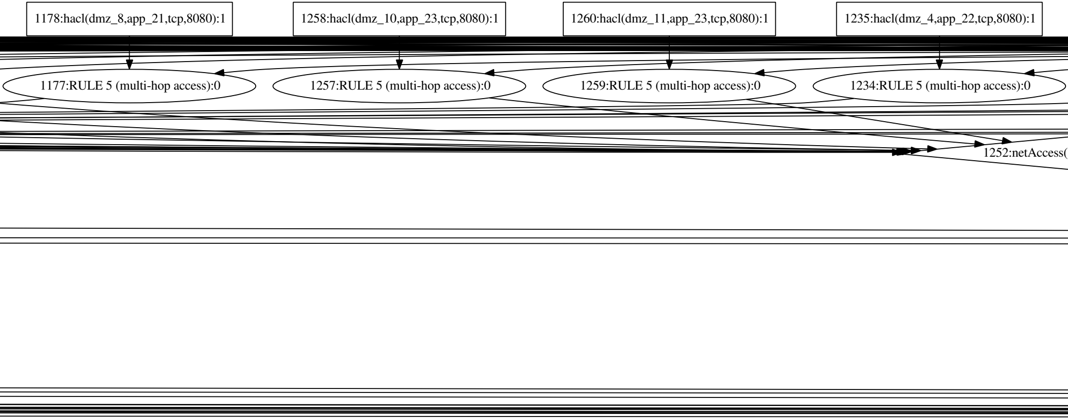

1223:hacl(dmz_24,app_22,tcp,8080):1
1172:hacl(dmz_5,app_21,tcp,8080):1
1294:hacl(dmz_27,app_23,tcp,8080):1
1296:hacl(dmz_28,app_23,tcp,8080):1
1222:RULE 5 (multi-hop access):0
1171:RULE 5 (multi-hop access):0
1293:RULE 5 (multi-hop access):0
1295:RULE 5 (multi-hop access):0

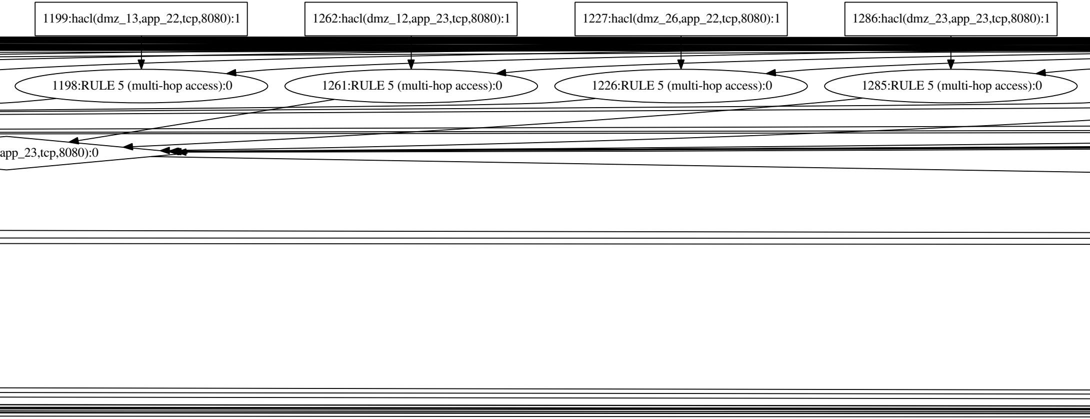

1298:hacl(dmz_3,app_23,tcp,8080):1
1225:hacl(dmz_25,app_22,tcp,8080):1
1266:hacl(dmz_14,app_23,tcp,8080):1
1272:hacl(dmz_17,app_23,tcp,8080):1
1297:RULE 5 (multi-hop access):0
1224:RULE 5 (multi-hop access):0
1265:RULE 5 (multi-hop access):0
1271:RULE 5 (multi-hop access):0

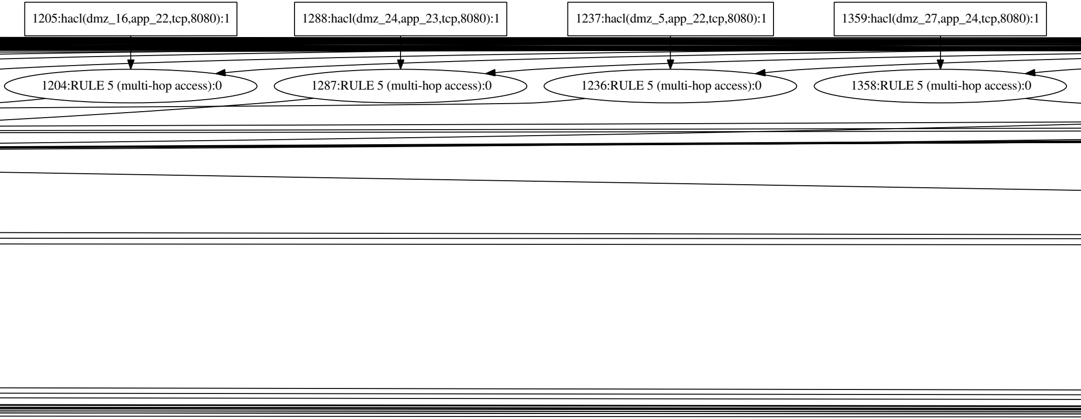

1245:hacl(dmz_9,app_22,tcp,8080):1
1274:hacl(dmz_18,app_23,tcp,8080):1
1174:hacl(dmz_6,app_21,tcp,8080):1
1276:hacl(dmz_19,app_23,tcp,8080):1
1244:RULE 5 (multi-hop access):0
1273:RULE 5 (multi-hop access):0
1173:RULE 5 (multi-hop access):0
1275:RULE 5 (multi-hop access):0
1246:networkServiceInfo(app_22,tomcat,tcp,8080,root):1
1247:vulExists(app_22,vul_app,tomcat,remoteExploit,privEscalation):1
1186:RULE 2 (remote exploit of a server program):0

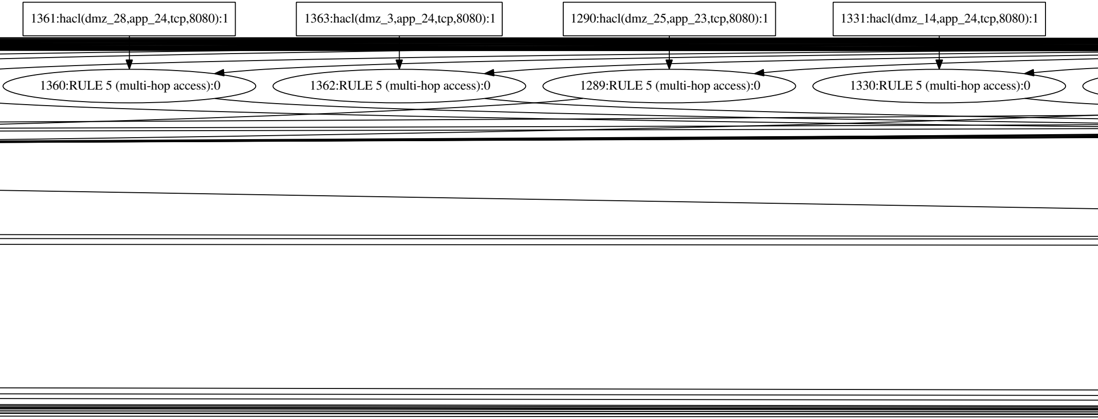

1278:hacl(dmz_2,app_23,tcp,8080):1
1280:hacl(dmz_20,app_23,tcp,8080):1
1203:hacl(dmz_15,app_22,tcp,8080):1
1282:hacl(dmz_21,app_23,tcp,8080):1
1277:RULE 5 (multi-hop access):0
1279:RULE 5 (multi-hop access):0
1202:RULE 5 (multi-hop access):0
1281:RULE 5 (multi-hop access):0

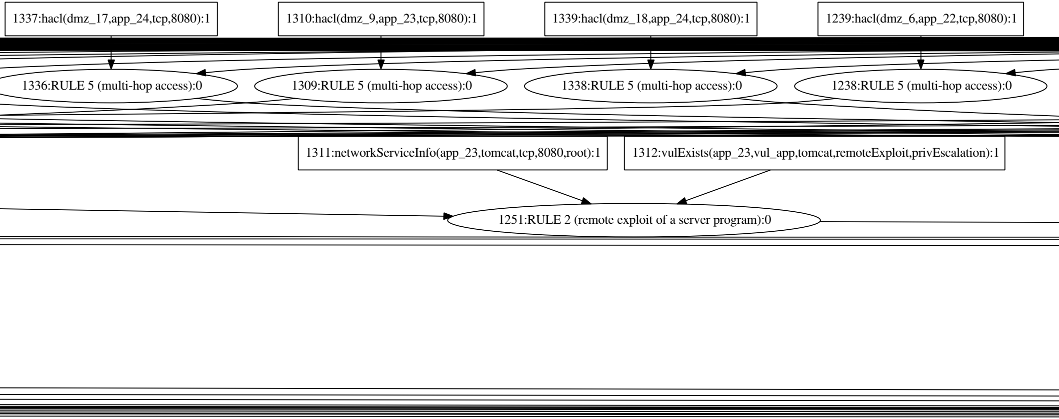

1176:hacl(dmz_7,app_21,tcp,8080):1
1254:hacl(dmz_0,app_23,tcp,8080):1
1284:hacl(dmz_22,app_23,tcp,8080):1
1256:hacl(dmz_1,app_23,tcp,8080):1
1175:RULE 5 (multi-hop access):0
1253:RULE 5 (multi-hop access):0
1283:RULE 5 (multi-hop access):0
1255:RULE 5 (multi-hop access):0

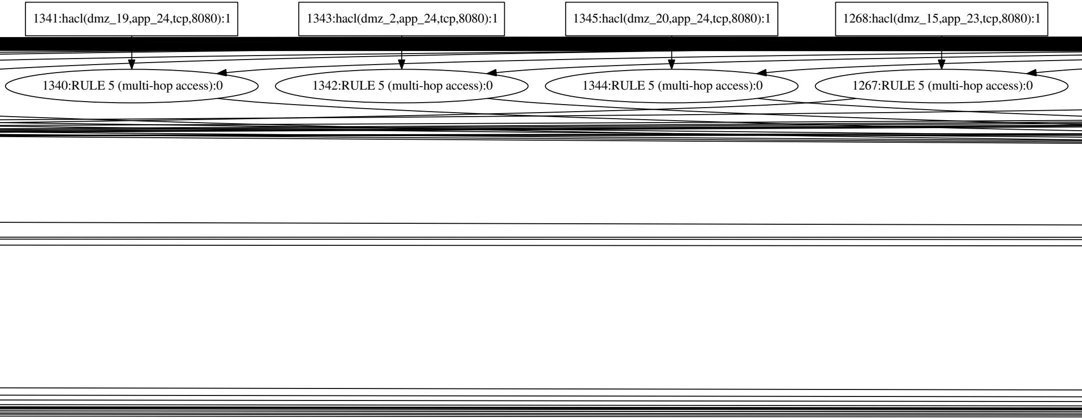

1178:hacl(dmz_8,app_21,tcp,8080):1
1258:hacl(dmz_10,app_23,tcp,8080):1
1260:hacl(dmz_11,app_23,tcp,8080):1
1235:hacl(dmz_4,app_22,tcp,8080):1
1177:RULE 5 (multi-hop access):0
1257:RULE 5 (multi-hop access):0
1259:RULE 5 (multi-hop access):0
1234:RULE 5 (multi-hop access):0
1252:netAccess(

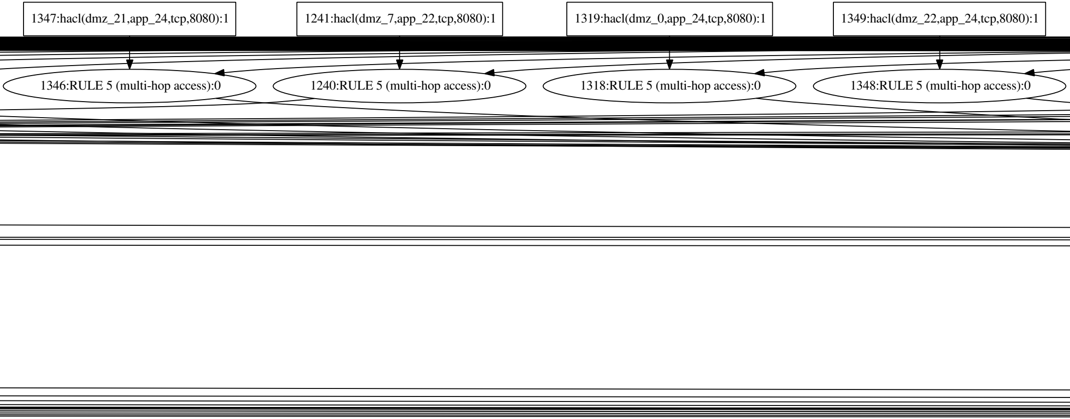

1199:hacl(dmz_13,app_22,tcp,8080):1
1262:hacl(dmz_12,app_23,tcp,8080):1
1227:hacl(dmz_26,app_22,tcp,8080):1
1286:hacl(dmz_23,app_23,tcp,8080):1
1198:RULE 5 (multi-hop access):0
1261:RULE 5 (multi-hop access):0
1226:RULE 5 (multi-hop access):0
1285:RULE 5 (multi-hop access):0
app_23,tcp,8080):0

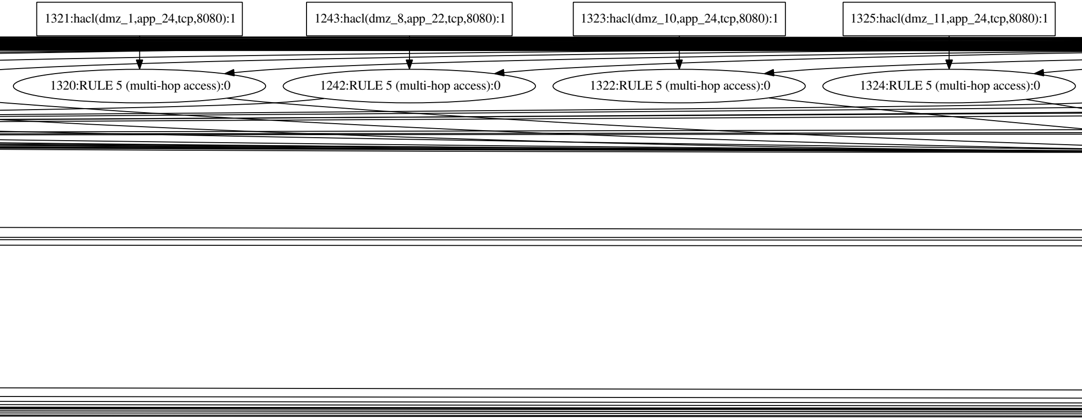

1205:hacl(dmz_16,app_22,tcp,8080):1
1288:hacl(dmz_24,app_23,tcp,8080):1
1237:hacl(dmz_5,app_22,tcp,8080):1
1359:hacl(dmz_27,app_24,tcp,8080):1
1204:RULE 5 (multi-hop access):0
1287:RULE 5 (multi-hop access):0
1236:RULE 5 (multi-hop access):0
1358:RULE 5 (multi-hop access):0


1361:hacl(dmz_28,app_24,tcp,8080):1
1363:hacl(dmz_3,app_24,tcp,8080):1
1290:hacl(dmz_25,app_23,tcp,8080):1
1331:hacl(dmz_14,app_24,tcp,8080):1
1360:RULE 5 (multi-hop access):0
1362:RULE 5 (multi-hop access):0
1289:RULE 5 (multi-hop access):0
1330:RULE 5 (multi-hop access):0

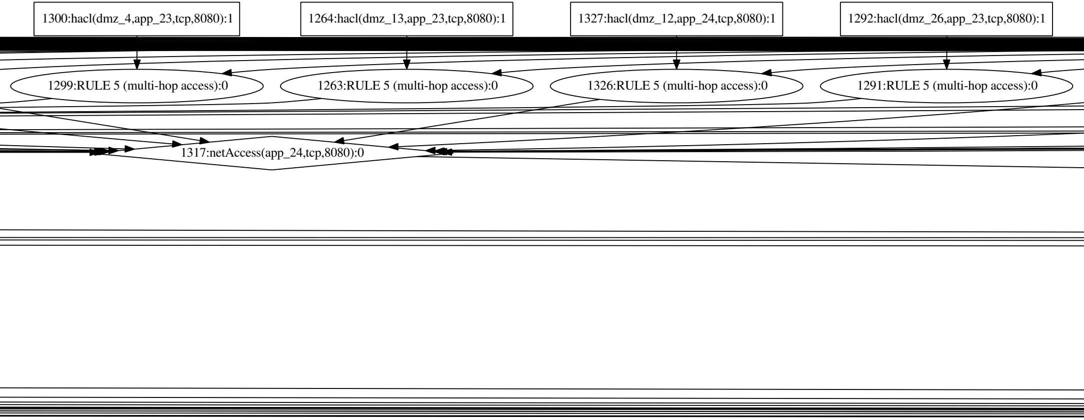

1337:hacl(dmz_17,app_24,tcp,8080):1
1310:hacl(dmz_9,app_23,tcp,8080):1
1339:hacl(dmz_18,app_24,tcp,8080):1
1239:hacl(dmz_6,app_22,tcp,8080):1
1336:RULE 5 (multi-hop access):0
1309:RULE 5 (multi-hop access):0
1338:RULE 5 (multi-hop access):0
1238:RULE 5 (multi-hop access):0
1311:networkServiceInfo(app_23,tomcat,tcp,8080,root):1
1312:vulExists(app_23,vul_app,tomcat,remoteExploit,privEscalation):1
1251:RULE 2 (remote exploit of a server program):0

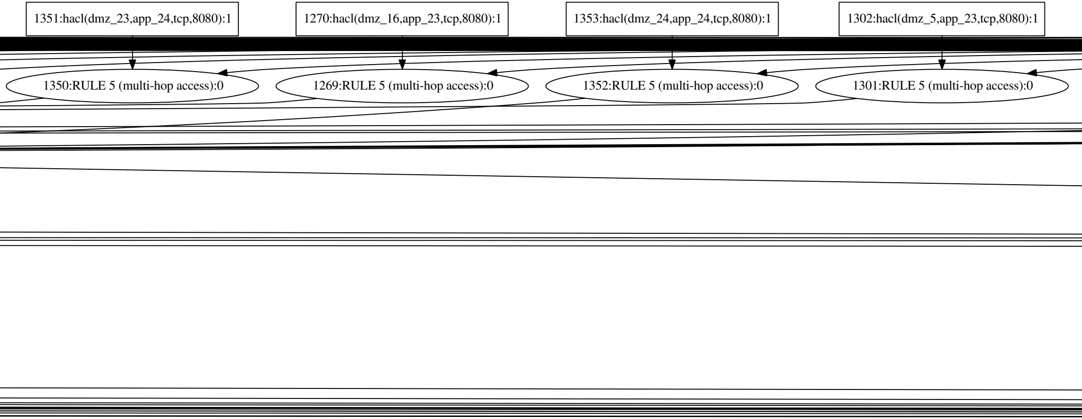

1341:hacl(dmz_19,app_24,tcp,8080):1
1343:hacl(dmz_2,app_24,tcp,8080):1
1345:hacl(dmz_20,app_24,tcp,8080):1
1268:hacl(dmz_15,app_23,tcp,8080):1
1340:RULE 5 (multi-hop access):0
1342:RULE 5 (multi-hop access):0
1344:RULE 5 (multi-hop access):0
1267:RULE 5 (multi-hop access):0

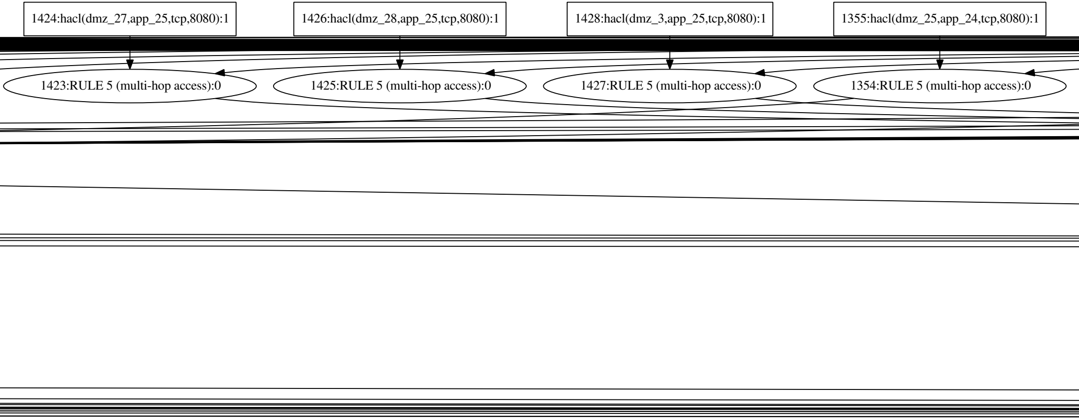

1347:hacl(dmz_21,app_24,tcp,8080):1
1241:hacl(dmz_7,app_22,tcp,8080):1
1319:hacl(dmz_0,app_24,tcp,8080):1
1349:hacl(dmz_22,app_24,tcp,8080):1
1346:RULE 5 (multi-hop access):0
1240:RULE 5 (multi-hop access):0
1318:RULE 5 (multi-hop access):0
1348:RULE 5 (multi-hop access):0

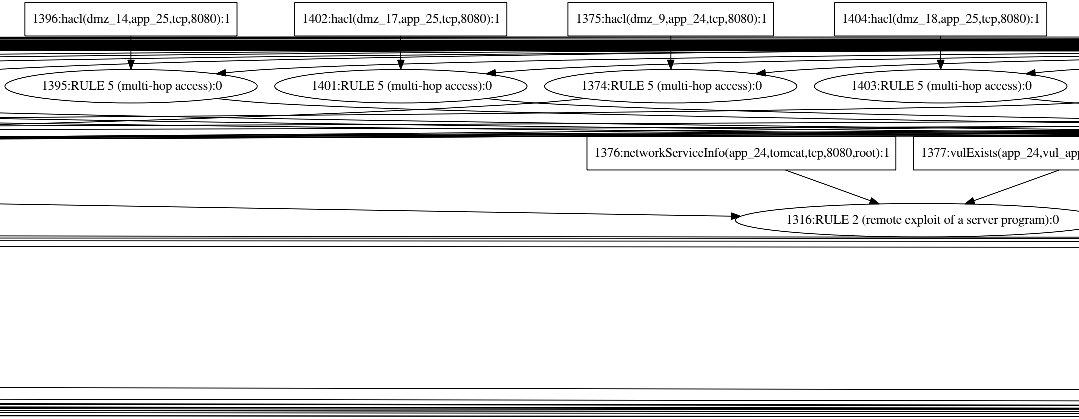


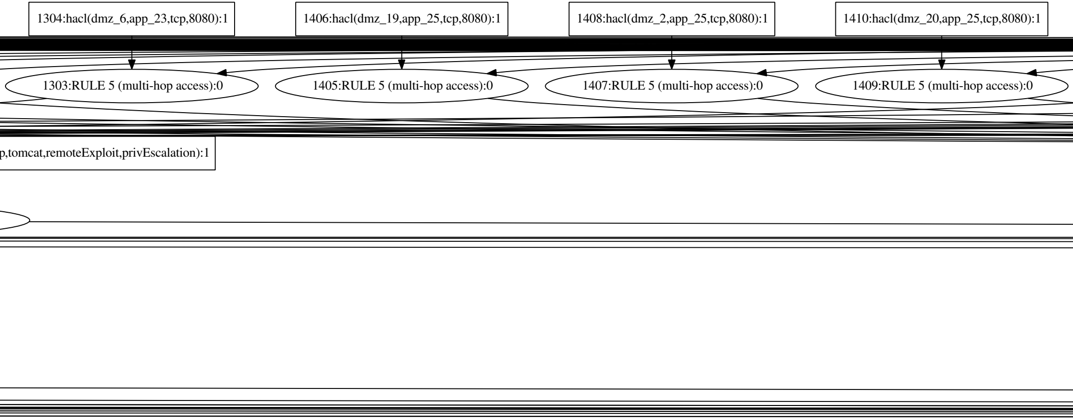
1321:hacl(dmz_1,app_24,tcp,8080):1
1243:hacl(dmz_8,app_22,tcp,8080):1
1323:hacl(dmz_10,app_24,tcp,8080):1
1325:hacl(dmz_11,app_24,tcp,8080):1
1320:RULE 5 (multi-hop access):0
1242:RULE 5 (multi-hop access):0
1322:RULE 5 (multi-hop access):0
1324:RULE 5 (multi-hop access):0

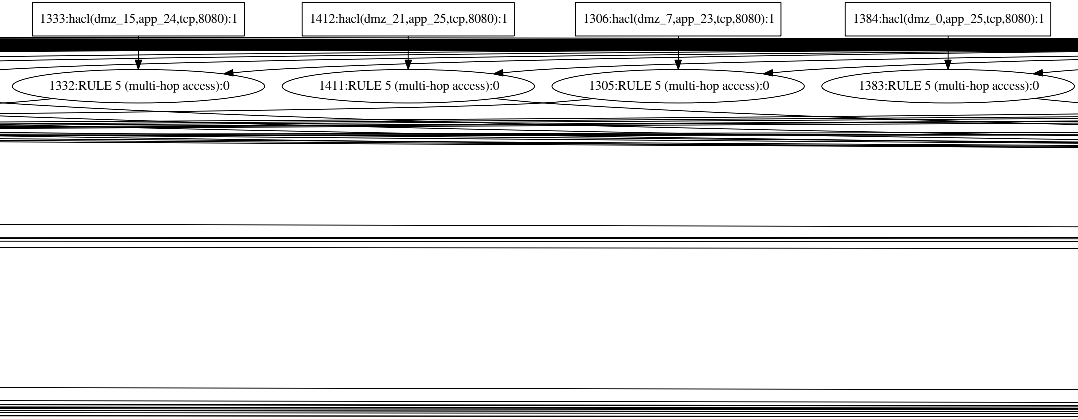

1300:hacl(dmz_4,app_23,tcp,8080):1
1264:hacl(dmz_13,app_23,tcp,8080):1
1327:hacl(dmz_12,app_24,tcp,8080):1
1292:hacl(dmz_26,app_23,tcp,8080):1
1299:RULE 5 (multi-hop access):0
1263:RULE 5 (multi-hop access):0
1326:RULE 5 (multi-hop access):0
1291:RULE 5 (multi-hop access):0
1317:netAccess(app_24,tcp,8080):0

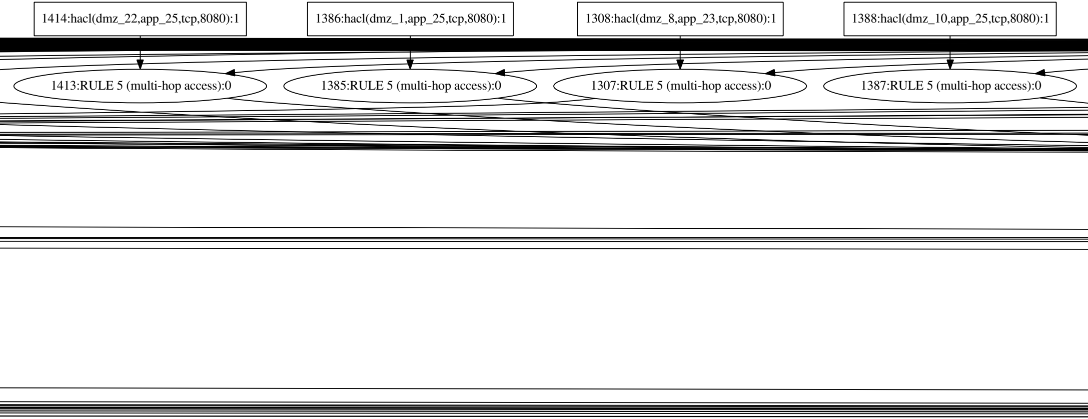

1351:hacl(dmz_23,app_24,tcp,8080):1
1270:hacl(dmz_16,app_23,tcp,8080):1
1353:hacl(dmz_24,app_24,tcp,8080):1
1302:hacl(dmz_5,app_23,tcp,8080):1
1350:RULE 5 (multi-hop access):0
1269:RULE 5 (multi-hop access):0
1352:RULE 5 (multi-hop access):0
1301:RULE 5 (multi-hop access):0

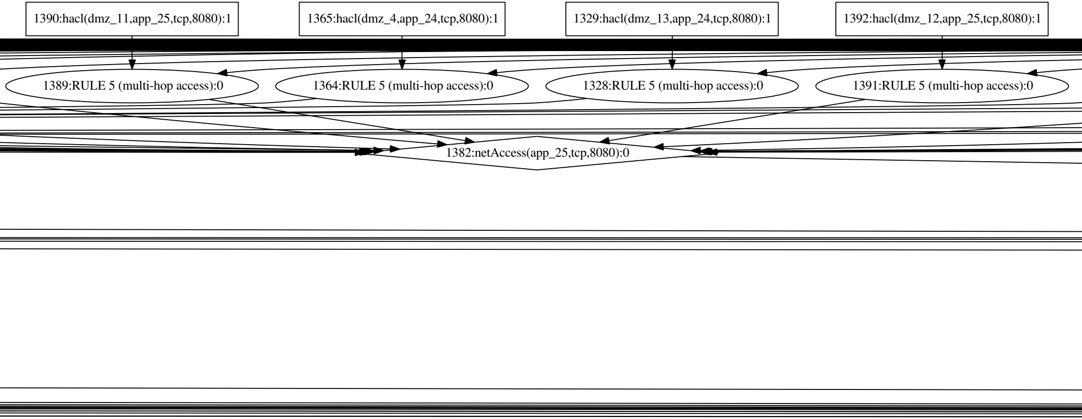

1424:hacl(dmz_27,app_25,tcp,8080):1
1426:hacl(dmz_28,app_25,tcp,8080):1
1428:hacl(dmz_3,app_25,tcp,8080):1
1355:hacl(dmz_25,app_24,tcp,8080):1
1423:RULE 5 (multi-hop access):0
1425:RULE 5 (multi-hop access):0
1427:RULE 5 (multi-hop access):0
1354:RULE 5 (multi-hop access):0

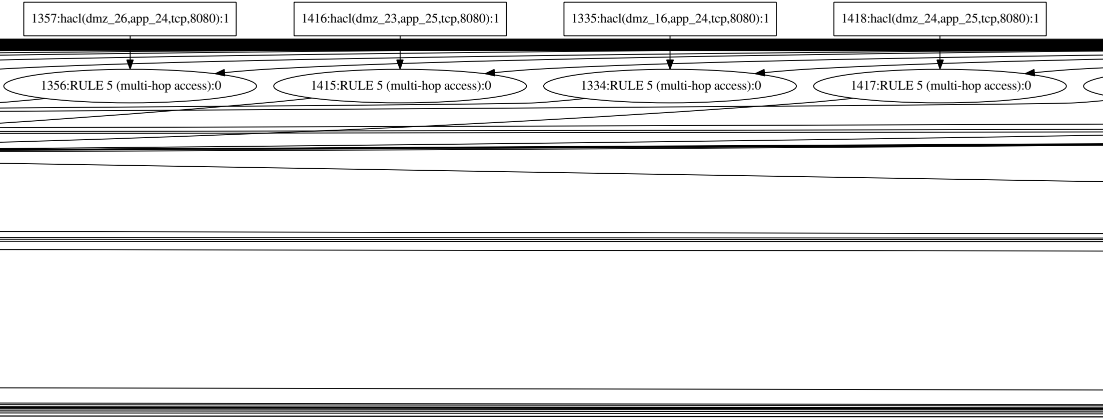

1396:hacl(dmz_14,app_25,tcp,8080):1
1402:hacl(dmz_17,app_25,tcp,8080):1
1375:hacl(dmz_9,app_24,tcp,8080):1
1404:hacl(dmz_18,app_25,tcp,8080):1
1395:RULE 5 (multi-hop access):0
1401:RULE 5 (multi-hop access):0
1374:RULE 5 (multi-hop access):0
1403:RULE 5 (multi-hop access):0
1376:networkServiceInfo(app_24,tomcat,tcp,8080,root):1
1377:vulExists(app_24,vul_app
1316:RULE 2 (remote exploit of a server program):0

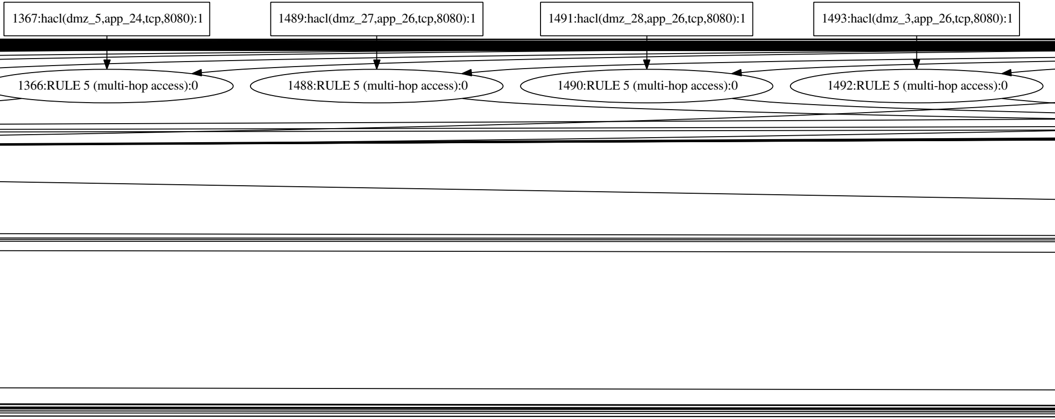

1304:hacl(dmz_6,app_23,tcp,8080):1
1406:hacl(dmz_19,app_25,tcp,8080):1
1408:hacl(dmz_2,app_25,tcp,8080):1
1410:hacl(dmz_20,app_25,tcp,8080):1
1303:RULE 5 (multi-hop access):0
1405:RULE 5 (multi-hop access):0
1407:RULE 5 (multi-hop access):0
1409:RULE 5 (multi-hop access):0
p,tomcat,remoteExploit,privEscalation):1

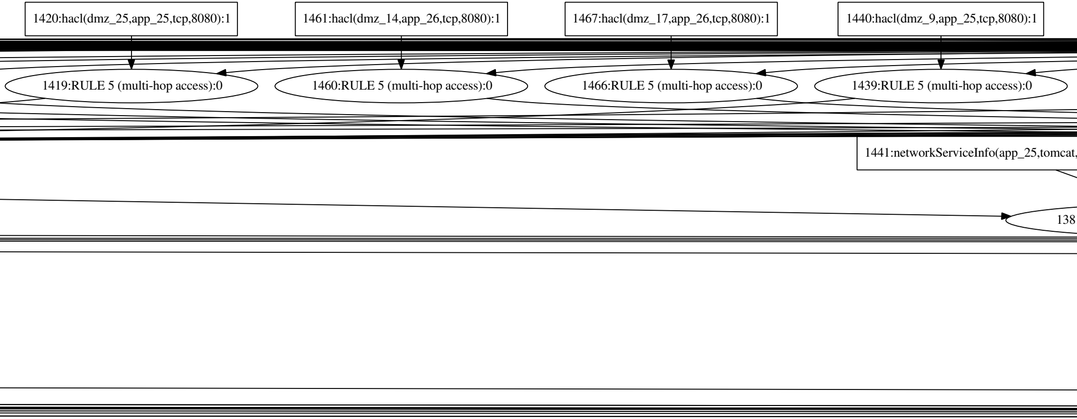

1333:hacl(dmz_15,app_24,tcp,8080):1
1412:hacl(dmz_21,app_25,tcp,8080):1
1306:hacl(dmz_7,app_23,tcp,8080):1
1384:hacl(dmz_0,app_25,tcp,8080):1
1332:RULE 5 (multi-hop access):0
1411:RULE 5 (multi-hop access):0
1305:RULE 5 (multi-hop access):0
1383:RULE 5 (multi-hop access):0

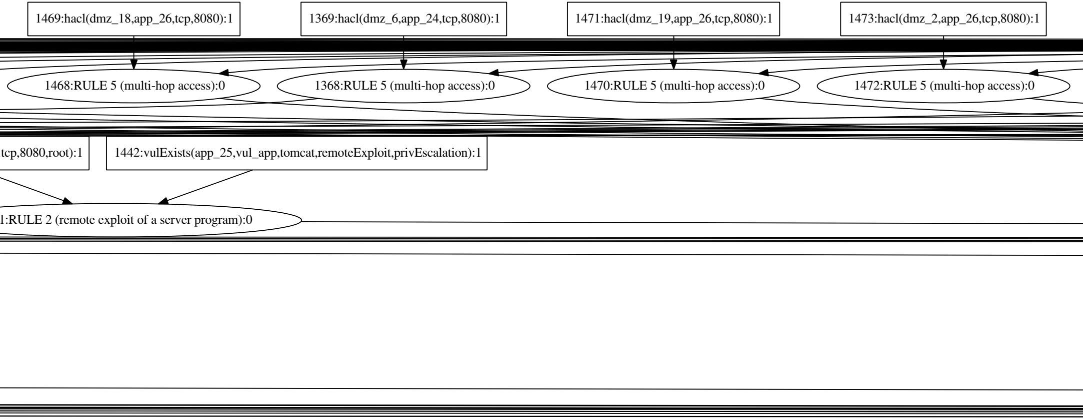

1414:hacl(dmz_22,app_25,tcp,8080):1
1386:hacl(dmz_1,app_25,tcp,8080):1
1308:hacl(dmz_8,app_23,tcp,8080):1
1388:hacl(dmz_10,app_25,tcp,8080):1
1413:RULE 5 (multi-hop access):0
1385:RULE 5 (multi-hop access):0
1307:RULE 5 (multi-hop access):0
1387:RULE 5 (multi-hop access):0

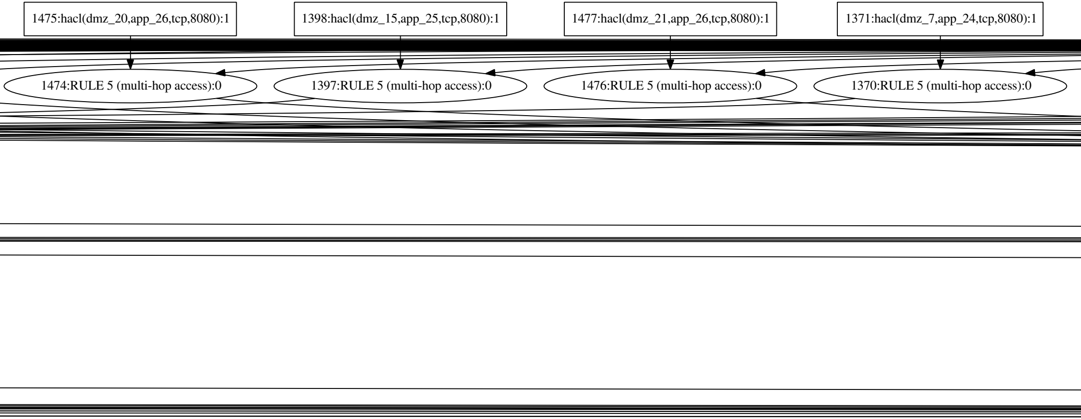

1390:hacl(dmz_11,app_25,tcp,8080):1
1365:hacl(dmz_4,app_24,tcp,8080):1
1329:hacl(dmz_13,app_24,tcp,8080):1
1392:hacl(dmz_12,app_25,tcp,8080):1
1389:RULE 5 (multi-hop access):0
1364:RULE 5 (multi-hop access):0
1328:RULE 5 (multi-hop access):0
1391:RULE 5 (multi-hop access):0
1382:netAccess(app_25,tcp,8080):0

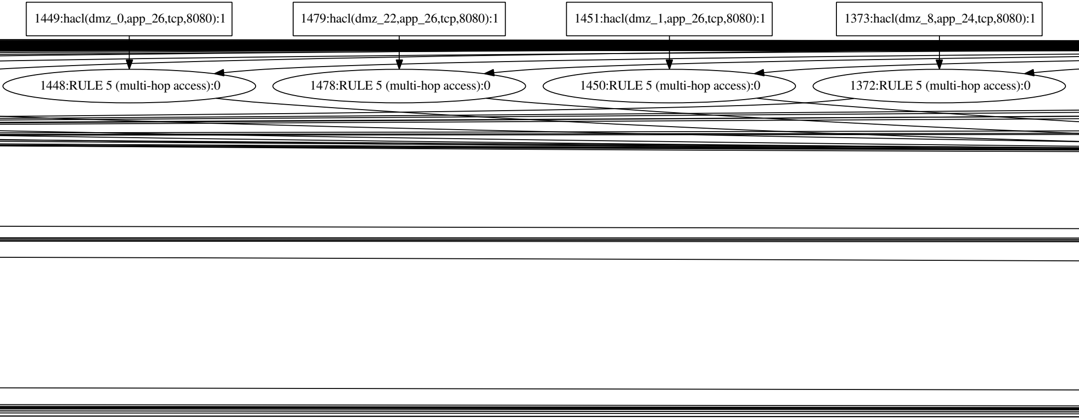

1357:hacl(dmz_26,app_24,tcp,8080):1
1416:hacl(dmz_23,app_25,tcp,8080):1
1335:hacl(dmz_16,app_24,tcp,8080):1
1418:hacl(dmz_24,app_25,tcp,8080):1
1356:RULE 5 (multi-hop access):0
1415:RULE 5 (multi-hop access):0
1334:RULE 5 (multi-hop access):0
1417:RULE 5 (multi-hop access):0

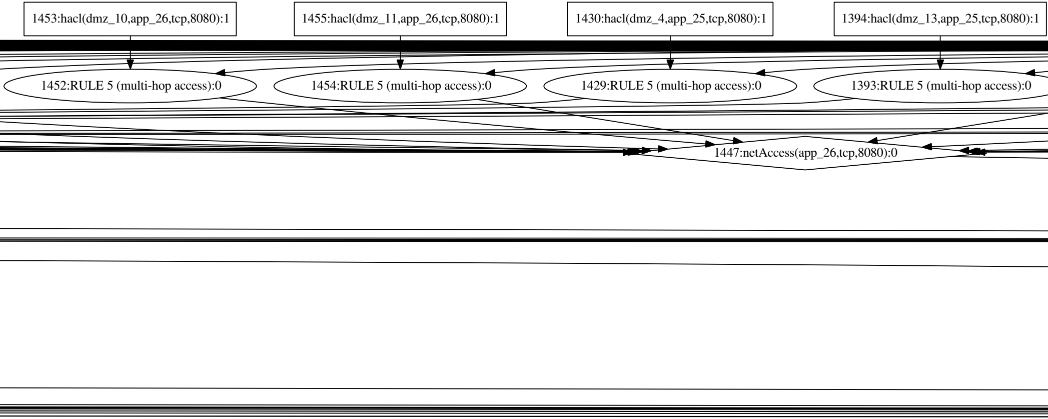

1367:hacl(dmz_5,app_24,tcp,8080):1
1489:hacl(dmz_27,app_26,tcp,8080):1
1491:hacl(dmz_28,app_26,tcp,8080):1
1493:hacl(dmz_3,app_26,tcp,8080):1
1366:RULE 5 (multi-hop access):0
1488:RULE 5 (multi-hop access):0
1490:RULE 5 (multi-hop access):0
1492:RULE 5 (multi-hop access):0

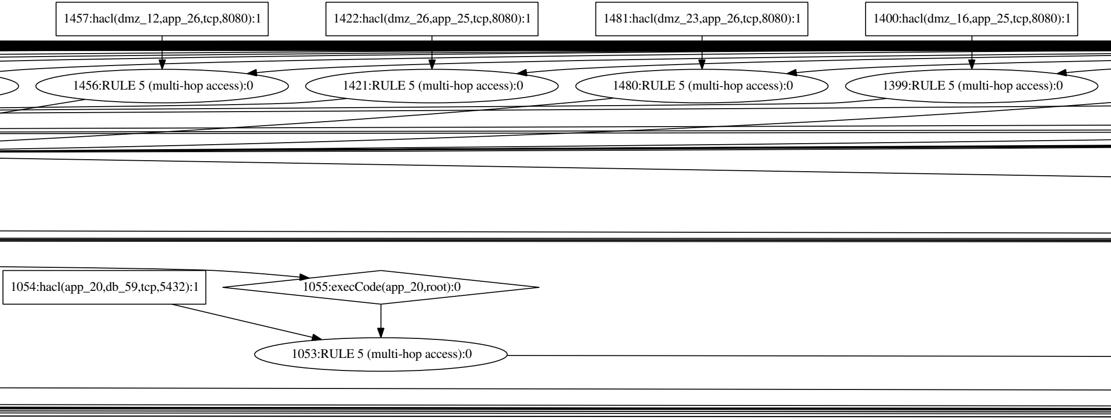

1420:hacl(dmz_25,app_25,tcp,8080):1
1461:hacl(dmz_14,app_26,tcp,8080):1
1467:hacl(dmz_17,app_26,tcp,8080):1
1440:hacl(dmz_9,app_25,tcp,8080):1
1419:RULE 5 (multi-hop access):0
1460:RULE 5 (multi-hop access):0
1466:RULE 5 (multi-hop access):0
1439:RULE 5 (multi-hop access):0
1441:networkServiceInfo(app_25,tomcat,
138

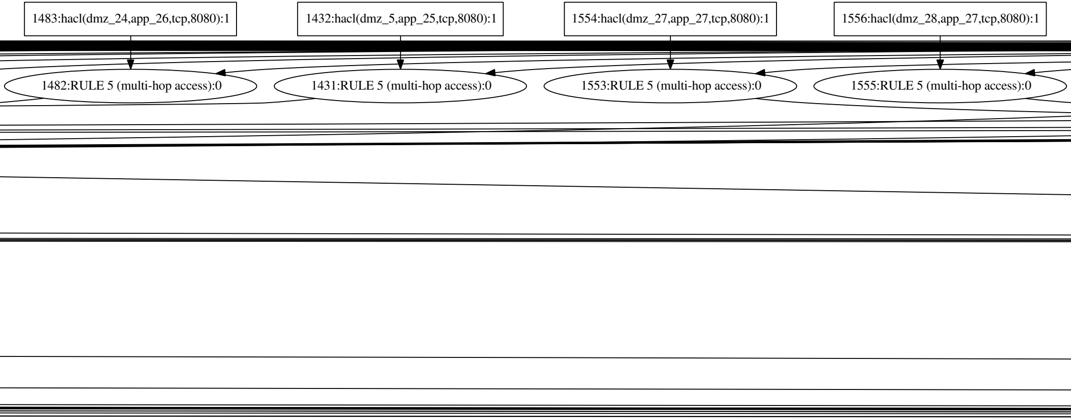

1469:hacl(dmz_18,app_26,tcp,8080):1
1369:hacl(dmz_6,app_24,tcp,8080):1
1471:hacl(dmz_19,app_26,tcp,8080):1
1473:hacl(dmz_2,app_26,tcp,8080):1
1468:RULE 5 (multi-hop access):0
1368:RULE 5 (multi-hop access):0
1470:RULE 5 (multi-hop access):0
1472:RULE 5 (multi-hop access):0
tcp,8080,root):1
1442:vulExists(app_25,vul_app,tomcat,remoteExploit,privEscalation):1
1:RULE 2 (remote exploit of a server program):0

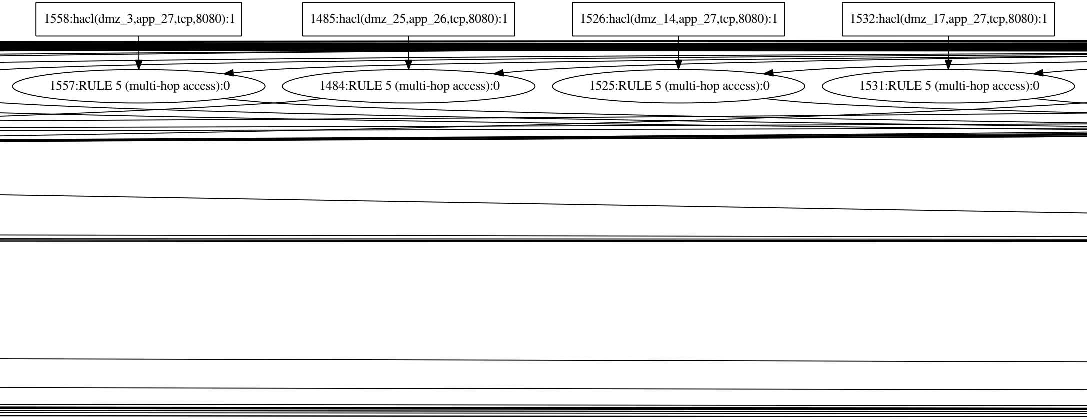

1475:hacl(dmz_20,app_26,tcp,8080):1
1398:hacl(dmz_15,app_25,tcp,8080):1
1477:hacl(dmz_21,app_26,tcp,8080):1
1371:hacl(dmz_7,app_24,tcp,8080):1
1474:RULE 5 (multi-hop access):0
1397:RULE 5 (multi-hop access):0
1476:RULE 5 (multi-hop access):0
1370:RULE 5 (multi-hop access):0

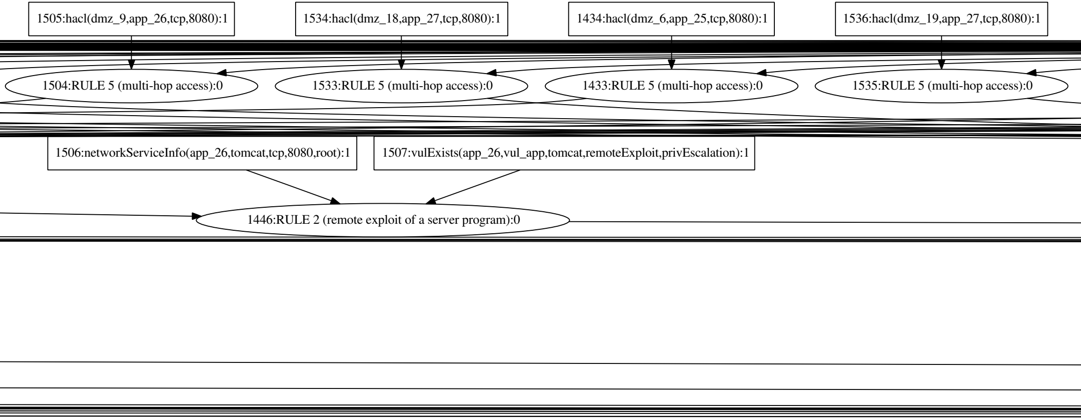

1449:hacl(dmz_0,app_26,tcp,8080):1
1479:hacl(dmz_22,app_26,tcp,8080):1
1451:hacl(dmz_1,app_26,tcp,8080):1
1373:hacl(dmz_8,app_24,tcp,8080):1
1448:RULE 5 (multi-hop access):0
1478:RULE 5 (multi-hop access):0
1450:RULE 5 (multi-hop access):0
1372:RULE 5 (multi-hop access):0

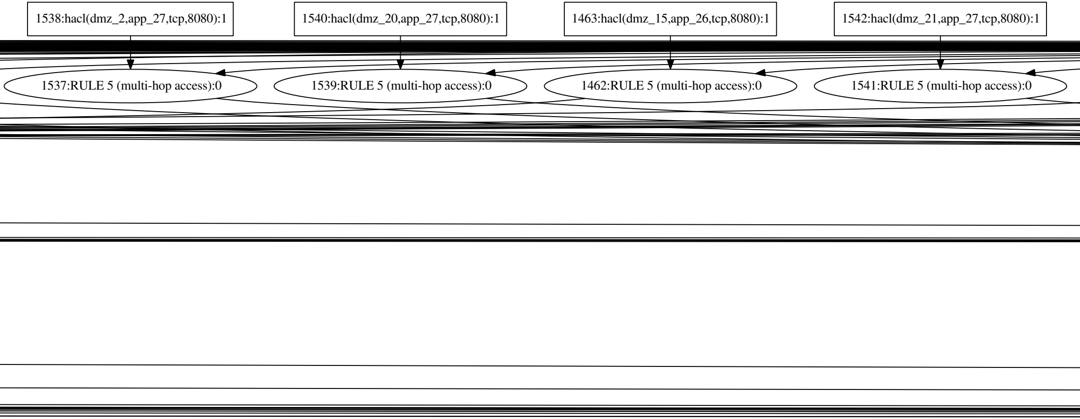

1453:hacl(dmz_10,app_26,tcp,8080):1
1455:hacl(dmz_11,app_26,tcp,8080):1
1430:hacl(dmz_4,app_25,tcp,8080):1
1394:hacl(dmz_13,app_25,tcp,8080):1
1452:RULE 5 (multi-hop access):0
1454:RULE 5 (multi-hop access):0
1429:RULE 5 (multi-hop access):0
1393:RULE 5 (multi-hop access):0
1447:netAccess(app_26,tcp,8080):0

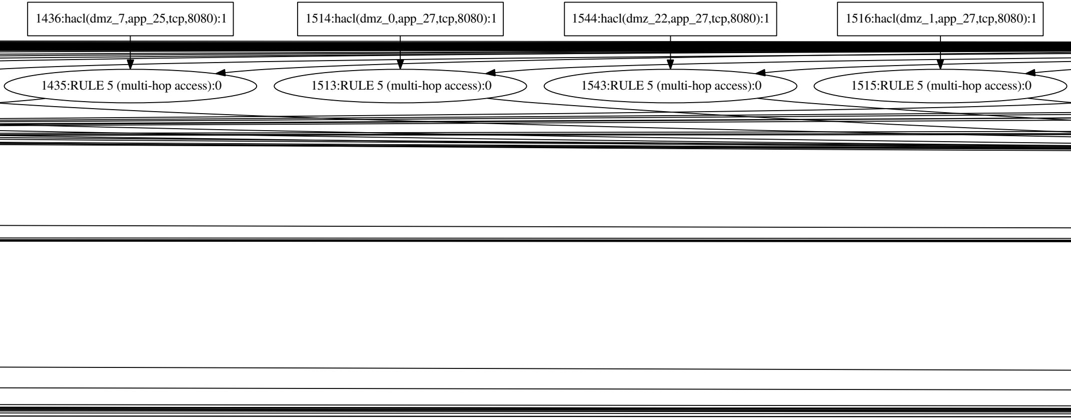

1457:hacl(dmz_12,app_26,tcp,8080):1
1422:hacl(dmz_26,app_25,tcp,8080):1
1481:hacl(dmz_23,app_26,tcp,8080):1
1400:hacl(dmz_16,app_25,tcp,8080):1
1456:RULE 5 (multi-hop access):0
1421:RULE 5 (multi-hop access):0
1480:RULE 5 (multi-hop access):0
1399:RULE 5 (multi-hop access):0
1054:hacl(app_20,db_59,tcp,5432):1
1055:execCode(app_20,root):0
1053:RULE 5 (multi-hop access):0

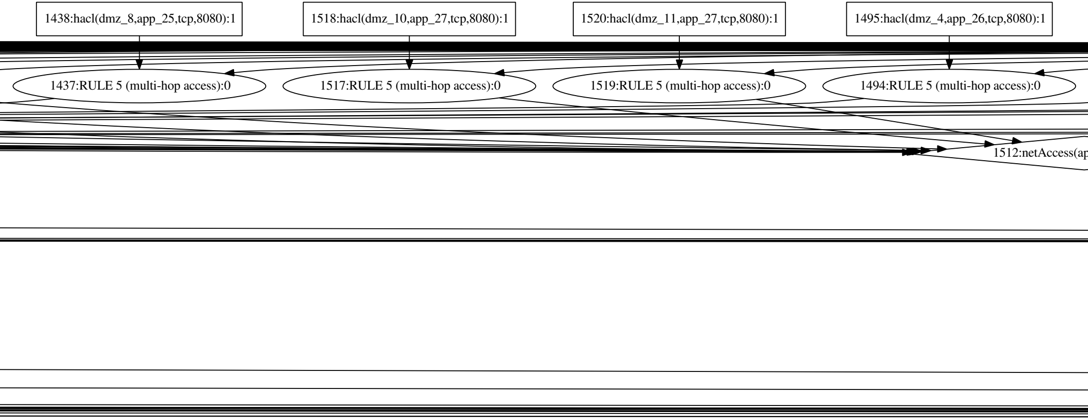

1483:hacl(dmz_24,app_26,tcp,8080):1
1432:hacl(dmz_5,app_25,tcp,8080):1
1554:hacl(dmz_27,app_27,tcp,8080):1
1556:hacl(dmz_28,app_27,tcp,8080):1
1482:RULE 5 (multi-hop access):0
1431:RULE 5 (multi-hop access):0
1553:RULE 5 (multi-hop access):0
1555:RULE 5 (multi-hop access):0

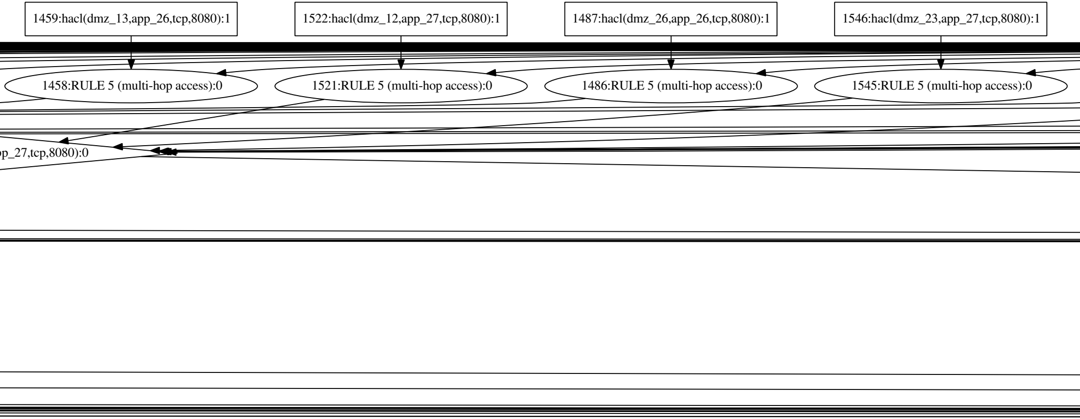

1558:hacl(dmz_3,app_27,tcp,8080):1
1485:hacl(dmz_25,app_26,tcp,8080):1
1526:hacl(dmz_14,app_27,tcp,8080):1
1532:hacl(dmz_17,app_27,tcp,8080):1
1557:RULE 5 (multi-hop access):0
1484:RULE 5 (multi-hop access):0
1525:RULE 5 (multi-hop access):0
1531:RULE 5 (multi-hop access):0

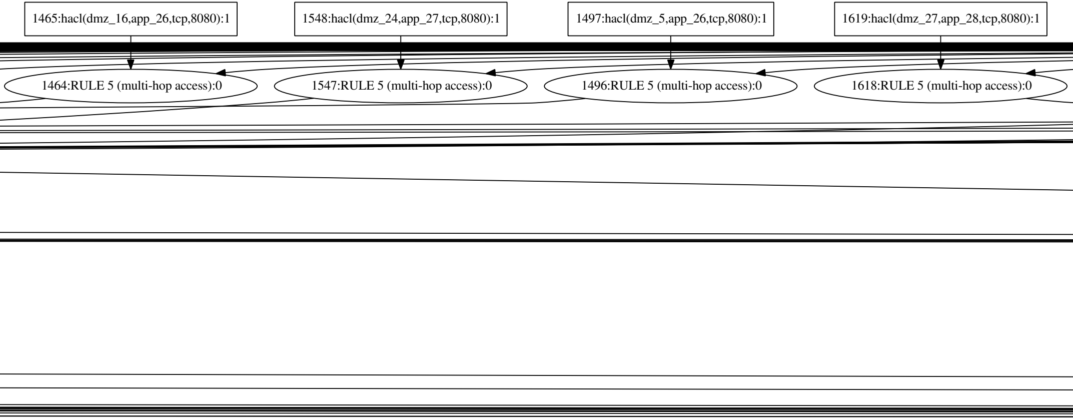

1505:hacl(dmz_9,app_26,tcp,8080):1
1534:hacl(dmz_18,app_27,tcp,8080):1
1434:hacl(dmz_6,app_25,tcp,8080):1
1536:hacl(dmz_19,app_27,tcp,8080):1
1504:RULE 5 (multi-hop access):0
1533:RULE 5 (multi-hop access):0
1433:RULE 5 (multi-hop access):0
1535:RULE 5 (multi-hop access):0
1506:networkServiceInfo(app_26,tomcat,tcp,8080,root):1
1507:vulExists(app_26,vul_app,tomcat,remoteExploit,privEscalation):1
1446:RULE 2 (remote exploit of a server program):0

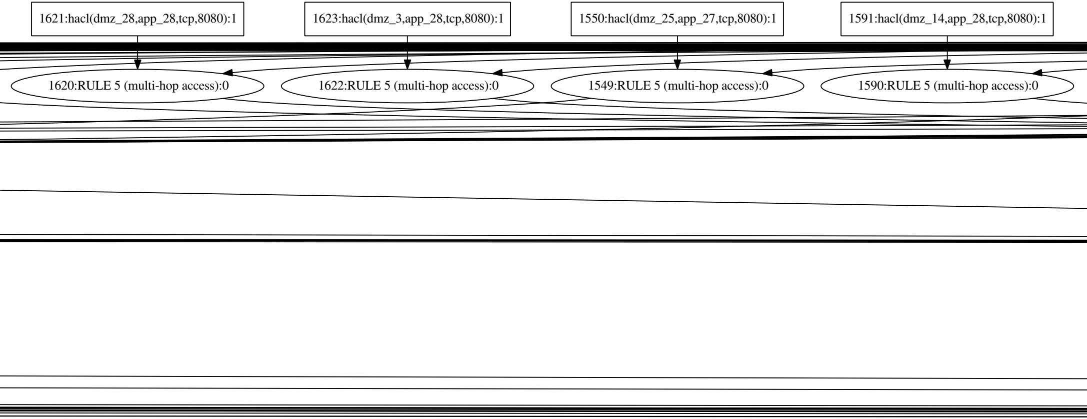

1538:hacl(dmz_2,app_27,tcp,8080):1
1540:hacl(dmz_20,app_27,tcp,8080):1
1463:hacl(dmz_15,app_26,tcp,8080):1
1542:hacl(dmz_21,app_27,tcp,8080):1
1537:RULE 5 (multi-hop access):0
1539:RULE 5 (multi-hop access):0
1462:RULE 5 (multi-hop access):0
1541:RULE 5 (multi-hop access):0

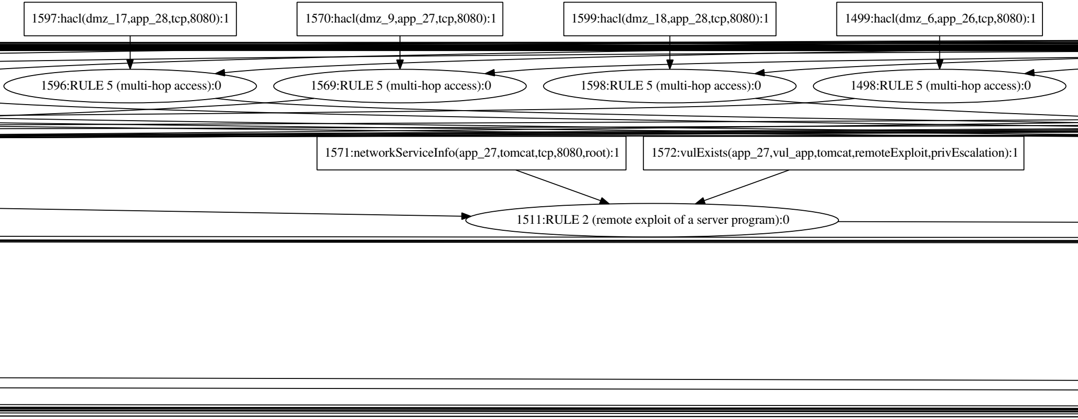

1436:hacl(dmz_7,app_25,tcp,8080):1
1514:hacl(dmz_0,app_27,tcp,8080):1
1544:hacl(dmz_22,app_27,tcp,8080):1
1516:hacl(dmz_1,app_27,tcp,8080):1
1435:RULE 5 (multi-hop access):0
1513:RULE 5 (multi-hop access):0
1543:RULE 5 (multi-hop access):0
1515:RULE 5 (multi-hop access):0

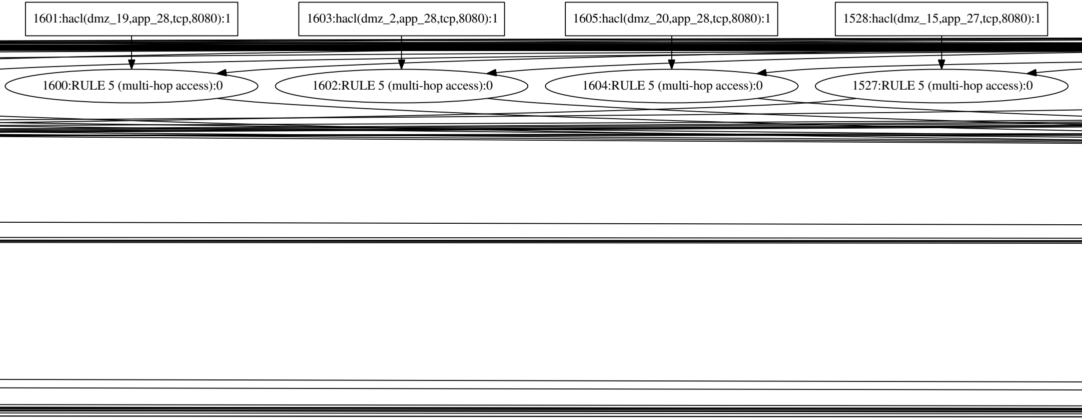

1438:hacl(dmz_8,app_25,tcp,8080):1
1518:hacl(dmz_10,app_27,tcp,8080):1
1520:hacl(dmz_11,app_27,tcp,8080):1
1495:hacl(dmz_4,app_26,tcp,8080):1
1437:RULE 5 (multi-hop access):0
1517:RULE 5 (multi-hop access):0
1519:RULE 5 (multi-hop access):0
1494:RULE 5 (multi-hop access):0
1512:netAccess(ap

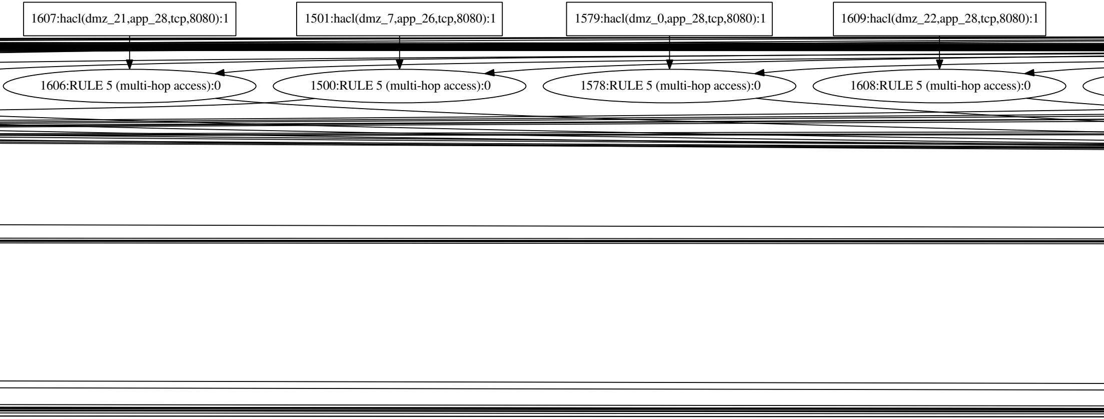

1459:hacl(dmz_13,app_26,tcp,8080):1
1522:hacl(dmz_12,app_27,tcp,8080):1
1487:hacl(dmz_26,app_26,tcp,8080):1
1546:hacl(dmz_23,app_27,tcp,8080):1
1458:RULE 5 (multi-hop access):0
1521:RULE 5 (multi-hop access):0
1486:RULE 5 (multi-hop access):0
1545:RULE 5 (multi-hop access):0
p_27,tcp,8080):0

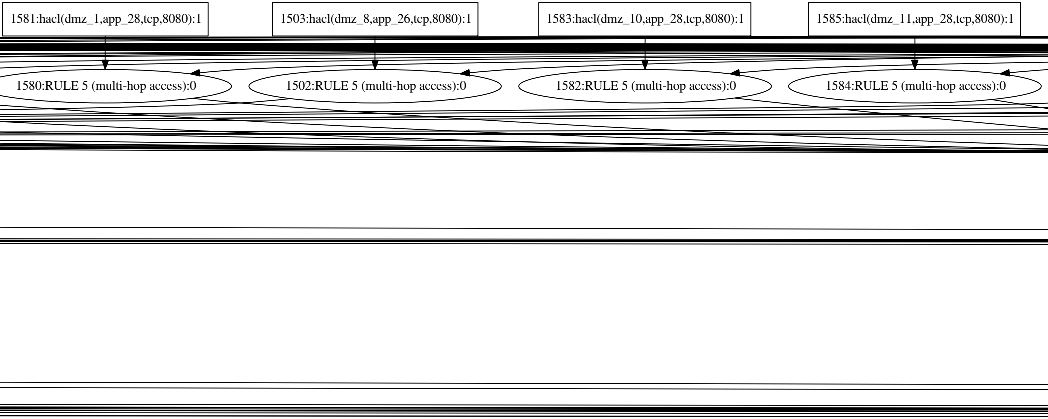

1465:hacl(dmz_16,app_26,tcp,8080):1
1548:hacl(dmz_24,app_27,tcp,8080):1
1497:hacl(dmz_5,app_26,tcp,8080):1
1619:hacl(dmz_27,app_28,tcp,8080):1
1464:RULE 5 (multi-hop access):0
1547:RULE 5 (multi-hop access):0
1496:RULE 5 (multi-hop access):0
1618:RULE 5 (multi-hop access):0

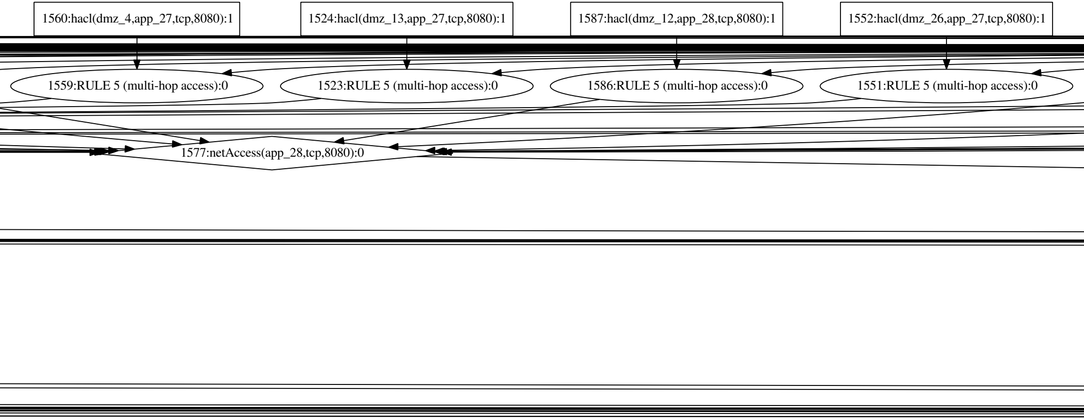

1621:hacl(dmz_28,app_28,tcp,8080):1
1623:hacl(dmz_3,app_28,tcp,8080):1
1550:hacl(dmz_25,app_27,tcp,8080):1
1591:hacl(dmz_14,app_28,tcp,8080):1
1620:RULE 5 (multi-hop access):0
1622:RULE 5 (multi-hop access):0
1549:RULE 5 (multi-hop access):0
1590:RULE 5 (multi-hop access):0

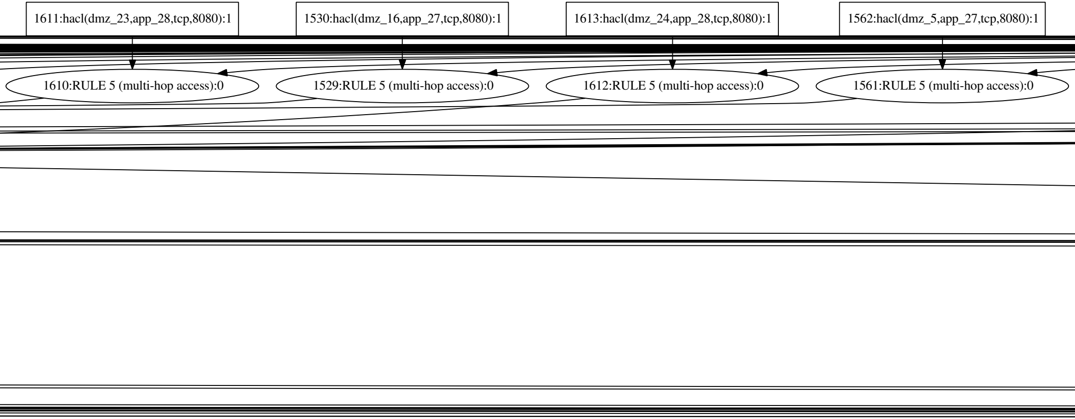

1597:hacl(dmz_17,app_28,tcp,8080):1
1570:hacl(dmz_9,app_27,tcp,8080):1
1599:hacl(dmz_18,app_28,tcp,8080):1
1499:hacl(dmz_6,app_26,tcp,8080):1
1596:RULE 5 (multi-hop access):0
1569:RULE 5 (multi-hop access):0
1598:RULE 5 (multi-hop access):0
1498:RULE 5 (multi-hop access):0
1571:networkServiceInfo(app_27,tomcat,tcp,8080,root):1
1572:vulExists(app_27,vul_app,tomcat,remoteExploit,privEscalation):1
1511:RULE 2 (remote exploit of a server program):0

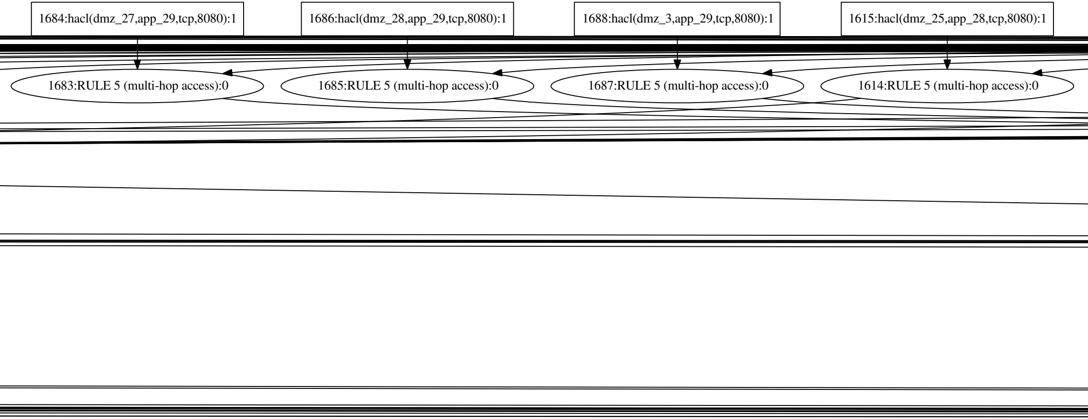

1601:hacl(dmz_19,app_28,tcp,8080):1
1603:hacl(dmz_2,app_28,tcp,8080):1
1605:hacl(dmz_20,app_28,tcp,8080):1
1528:hacl(dmz_15,app_27,tcp,8080):1
1600:RULE 5 (multi-hop access):0
1602:RULE 5 (multi-hop access):0
1604:RULE 5 (multi-hop access):0
1527:RULE 5 (multi-hop access):0

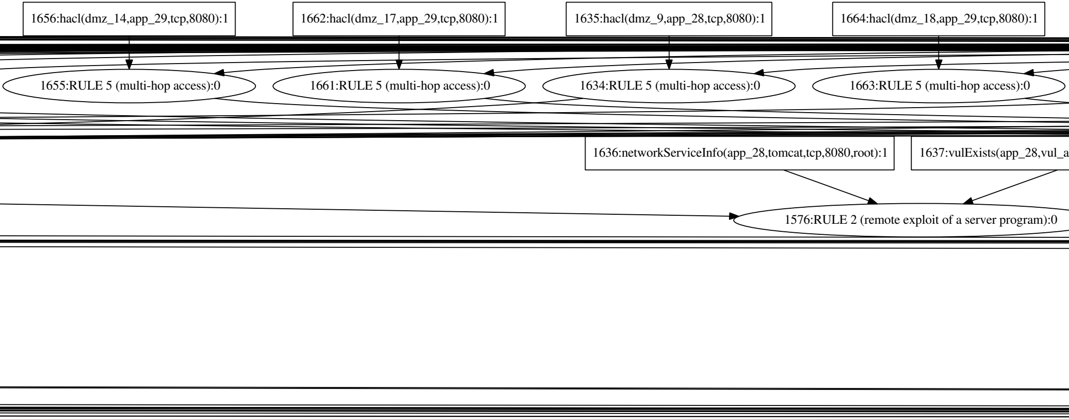

1607:hacl(dmz_21,app_28,tcp,8080):1
1501:hacl(dmz_7,app_26,tcp,8080):1
1579:hacl(dmz_0,app_28,tcp,8080):1
1609:hacl(dmz_22,app_28,tcp,8080):1
1606:RULE 5 (multi-hop access):0
1500:RULE 5 (multi-hop access):0
1578:RULE 5 (multi-hop access):0
1608:RULE 5 (multi-hop access):0

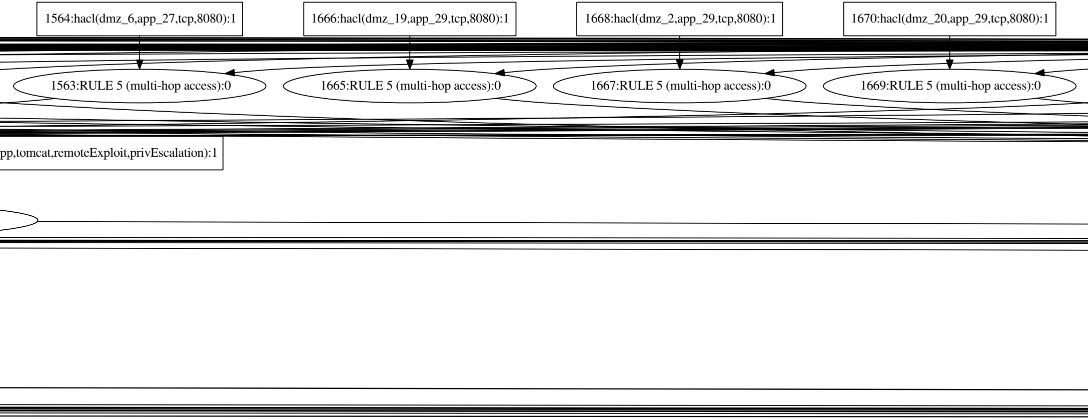

1581:hacl(dmz_1,app_28,tcp,8080):1
1503:hacl(dmz_8,app_26,tcp,8080):1
1583:hacl(dmz_10,app_28,tcp,8080):1
1585:hacl(dmz_11,app_28,tcp,8080):1
1580:RULE 5 (multi-hop access):0
1502:RULE 5 (multi-hop access):0
1582:RULE 5 (multi-hop access):0
1584:RULE 5 (multi-hop access):0

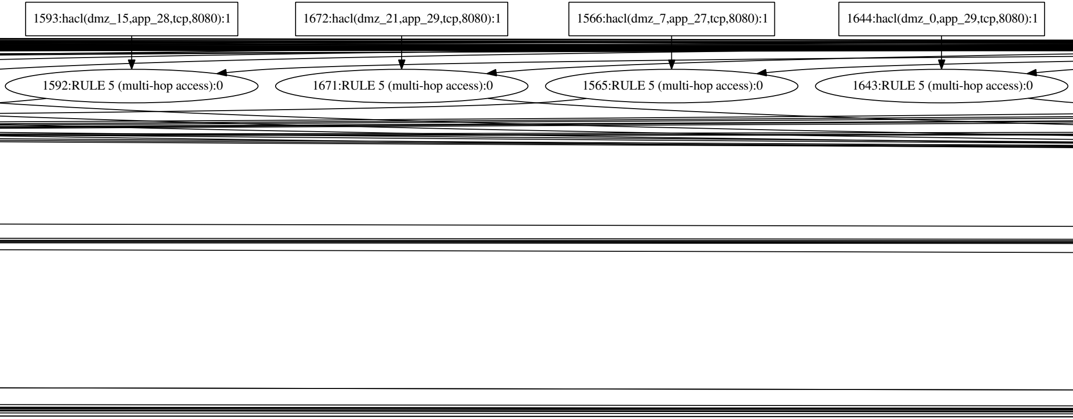

1560:hacl(dmz_4,app_27,tcp,8080):1
1524:hacl(dmz_13,app_27,tcp,8080):1
1587:hacl(dmz_12,app_28,tcp,8080):1
1552:hacl(dmz_26,app_27,tcp,8080):1
1559:RULE 5 (multi-hop access):0
1523:RULE 5 (multi-hop access):0
1586:RULE 5 (multi-hop access):0
1551:RULE 5 (multi-hop access):0
1577:netAccess(app_28,tcp,8080):0

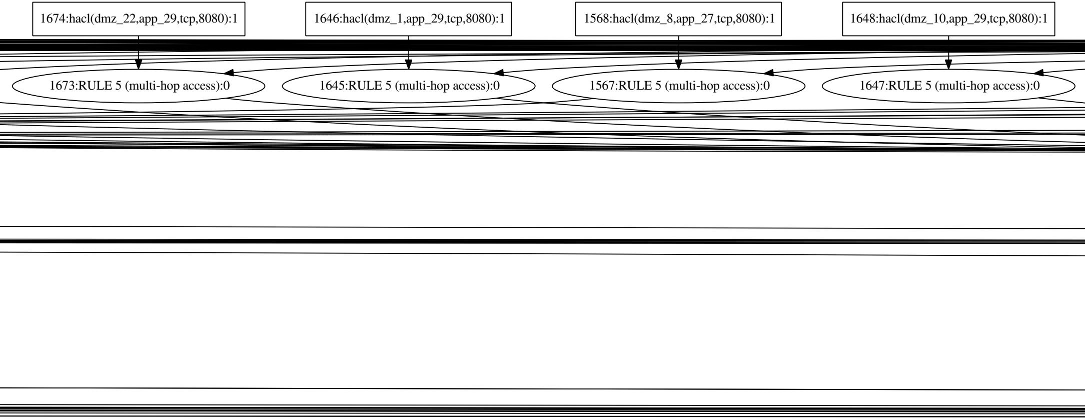

1611:hacl(dmz_23,app_28,tcp,8080):1
1530:hacl(dmz_16,app_27,tcp,8080):1
1613:hacl(dmz_24,app_28,tcp,8080):1
1562:hacl(dmz_5,app_27,tcp,8080):1
1610:RULE 5 (multi-hop access):0
1529:RULE 5 (multi-hop access):0
1612:RULE 5 (multi-hop access):0
1561:RULE 5 (multi-hop access):0

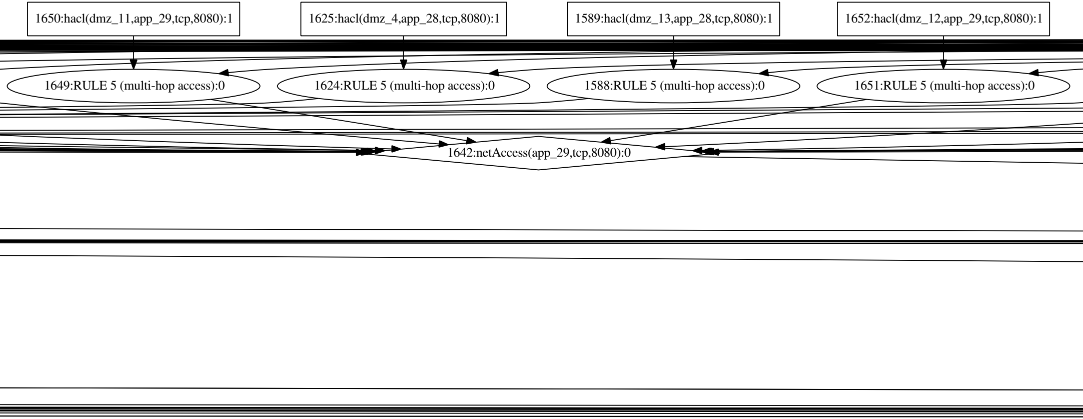

1684:hacl(dmz_27,app_29,tcp,8080):1
1686:hacl(dmz_28,app_29,tcp,8080):1
1688:hacl(dmz_3,app_29,tcp,8080):1
1615:hacl(dmz_25,app_28,tcp,8080):1
1683:RULE 5 (multi-hop access):0
1685:RULE 5 (multi-hop access):0
1687:RULE 5 (multi-hop access):0
1614:RULE 5 (multi-hop access):0

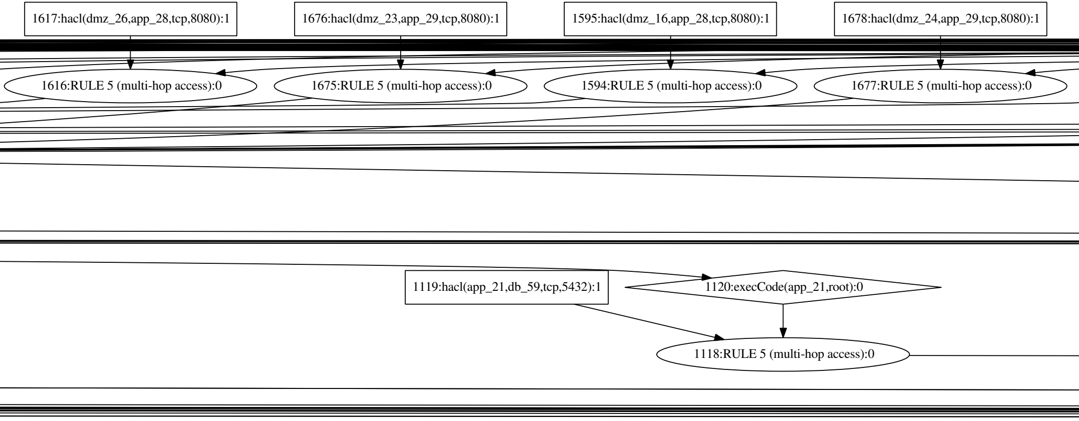

1656:hacl(dmz_14,app_29,tcp,8080):1
1662:hacl(dmz_17,app_29,tcp,8080):1
1635:hacl(dmz_9,app_28,tcp,8080):1
1664:hacl(dmz_18,app_29,tcp,8080):1
1655:RULE 5 (multi-hop access):0
1661:RULE 5 (multi-hop access):0
1634:RULE 5 (multi-hop access):0
1663:RULE 5 (multi-hop access):0
1636:networkServiceInfo(app_28,tomcat,tcp,8080,root):1
1637:vulExists(app_28,vul_a
1576:RULE 2 (remote exploit of a server program):0

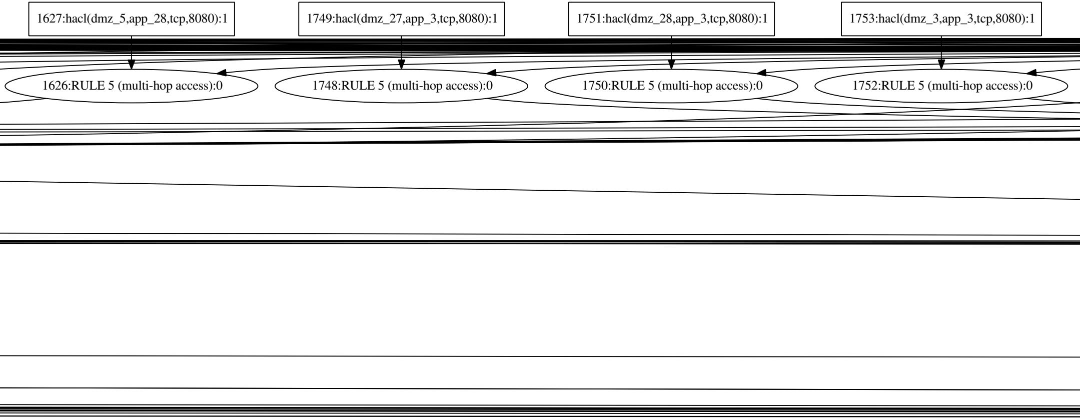

1564:hacl(dmz_6,app_27,tcp,8080):1
1666:hacl(dmz_19,app_29,tcp,8080):1
1668:hacl(dmz_2,app_29,tcp,8080):1
1670:hacl(dmz_20,app_29,tcp,8080):1
1563:RULE 5 (multi-hop access):0
1665:RULE 5 (multi-hop access):0
1667:RULE 5 (multi-hop access):0
1669:RULE 5 (multi-hop access):0
pp,tomcat,remoteExploit,privEscalation):1

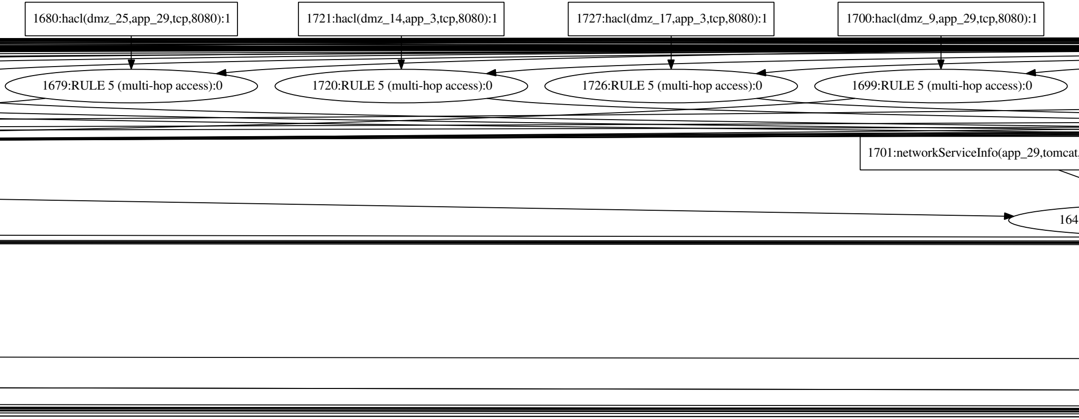

1593:hacl(dmz_15,app_28,tcp,8080):1
1672:hacl(dmz_21,app_29,tcp,8080):1
1566:hacl(dmz_7,app_27,tcp,8080):1
1644:hacl(dmz_0,app_29,tcp,8080):1
1592:RULE 5 (multi-hop access):0
1671:RULE 5 (multi-hop access):0
1565:RULE 5 (multi-hop access):0
1643:RULE 5 (multi-hop access):0

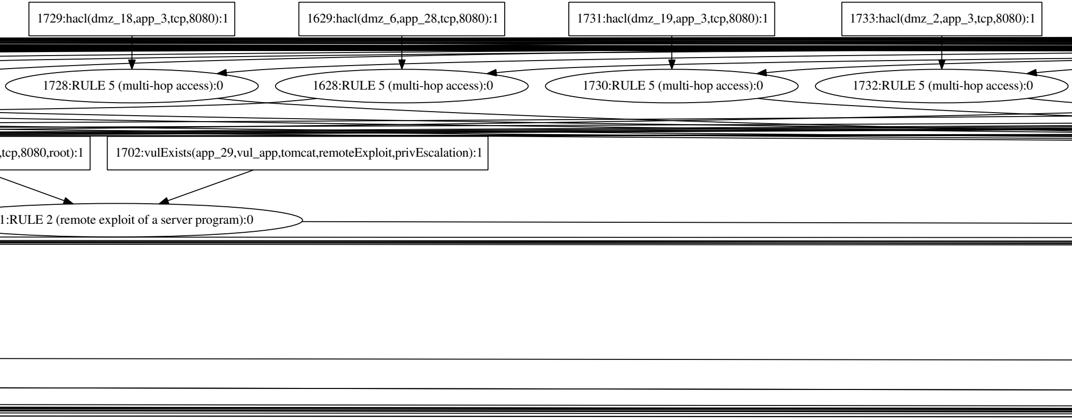

1674:hacl(dmz_22,app_29,tcp,8080):1
1646:hacl(dmz_1,app_29,tcp,8080):1
1568:hacl(dmz_8,app_27,tcp,8080):1
1648:hacl(dmz_10,app_29,tcp,8080):1
1673:RULE 5 (multi-hop access):0
1645:RULE 5 (multi-hop access):0
1567:RULE 5 (multi-hop access):0
1647:RULE 5 (multi-hop access):0

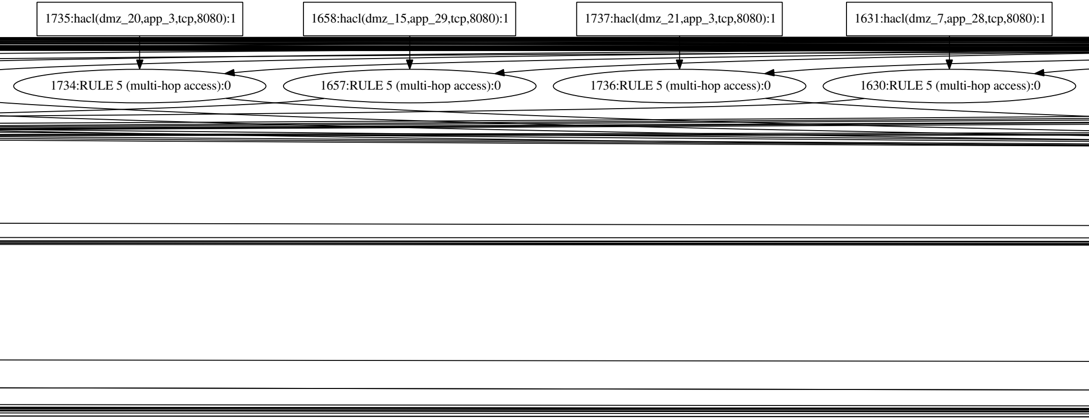

1650:hacl(dmz_11,app_29,tcp,8080):1
1625:hacl(dmz_4,app_28,tcp,8080):1
1589:hacl(dmz_13,app_28,tcp,8080):1
1652:hacl(dmz_12,app_29,tcp,8080):1
1649:RULE 5 (multi-hop access):0
1624:RULE 5 (multi-hop access):0
1588:RULE 5 (multi-hop access):0
1651:RULE 5 (multi-hop access):0
1642:netAccess(app_29,tcp,8080):0

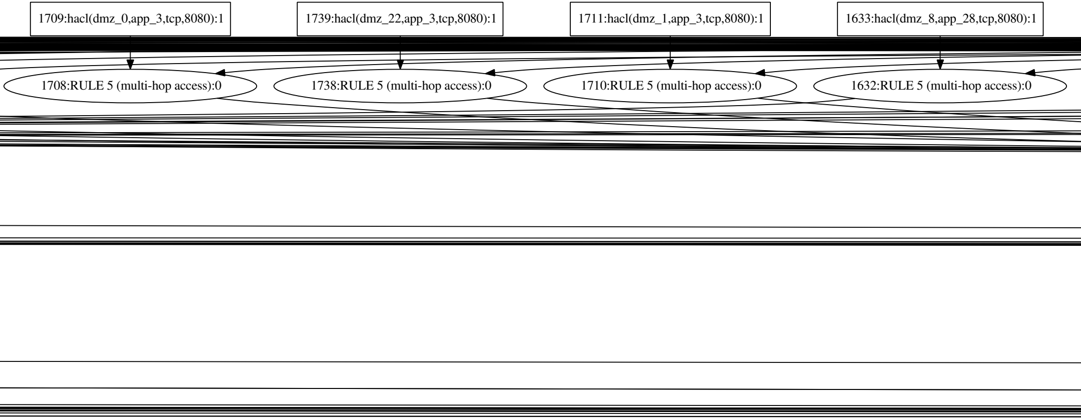

1617:hacl(dmz_26,app_28,tcp,8080):1
1676:hacl(dmz_23,app_29,tcp,8080):1
1595:hacl(dmz_16,app_28,tcp,8080):1
1678:hacl(dmz_24,app_29,tcp,8080):1
1616:RULE 5 (multi-hop access):0
1675:RULE 5 (multi-hop access):0
1594:RULE 5 (multi-hop access):0
1677:RULE 5 (multi-hop access):0
1119:hacl(app_21,db_59,tcp,5432):1
1120:execCode(app_21,root):0
1118:RULE 5 (multi-hop access):0

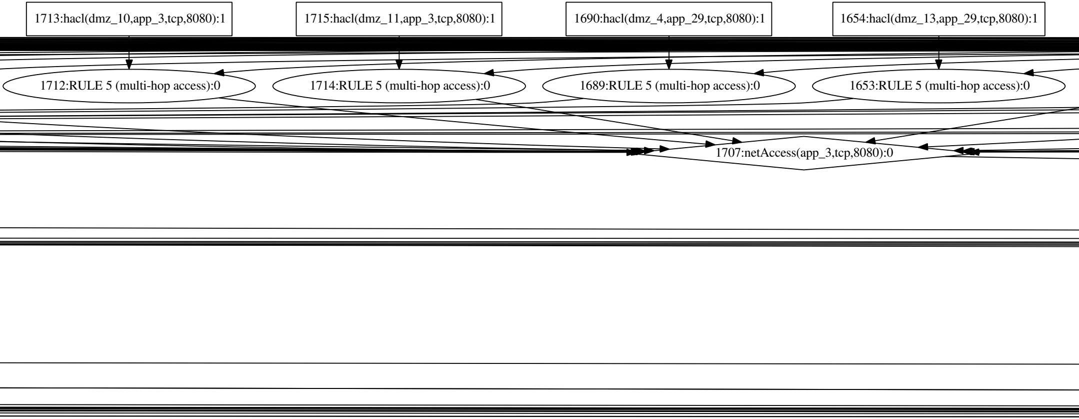

1627:hacl(dmz_5,app_28,tcp,8080):1
1749:hacl(dmz_27,app_3,tcp,8080):1
1751:hacl(dmz_28,app_3,tcp,8080):1
1753:hacl(dmz_3,app_3,tcp,8080):1
1626:RULE 5 (multi-hop access):0
1748:RULE 5 (multi-hop access):0
1750:RULE 5 (multi-hop access):0
1752:RULE 5 (multi-hop access):0

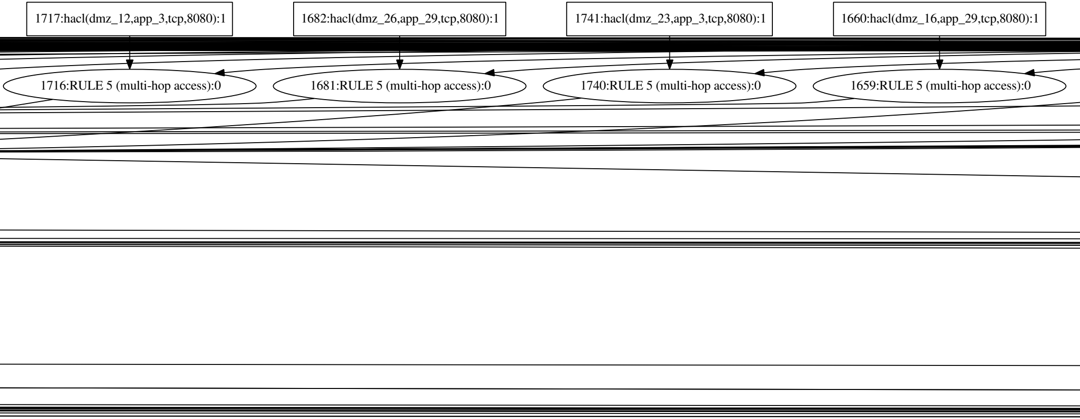

1680:hacl(dmz_25,app_29,tcp,8080):1
1721:hacl(dmz_14,app_3,tcp,8080):1
1727:hacl(dmz_17,app_3,tcp,8080):1
1700:hacl(dmz_9,app_29,tcp,8080):1
1679:RULE 5 (multi-hop access):0
1720:RULE 5 (multi-hop access):0
1726:RULE 5 (multi-hop access):0
1699:RULE 5 (multi-hop access):0
1701:networkServiceInfo(app_29,tomcat,
164

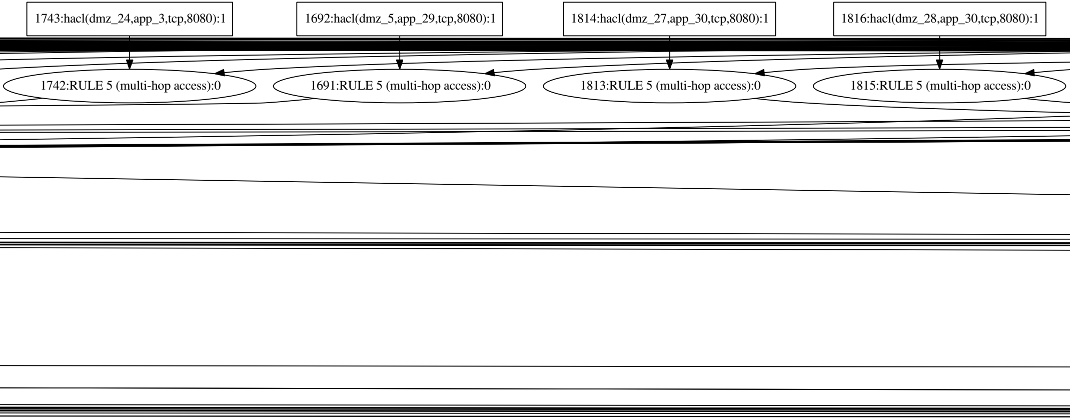

1729:hacl(dmz_18,app_3,tcp,8080):1
1629:hacl(dmz_6,app_28,tcp,8080):1
1731:hacl(dmz_19,app_3,tcp,8080):1
1733:hacl(dmz_2,app_3,tcp,8080):1
1728:RULE 5 (multi-hop access):0
1628:RULE 5 (multi-hop access):0
1730:RULE 5 (multi-hop access):0
1732:RULE 5 (multi-hop access):0
,tcp,8080,root):1
1702:vulExists(app_29,vul_app,tomcat,remoteExploit,privEscalation):1
1:RULE 2 (remote exploit of a server program):0

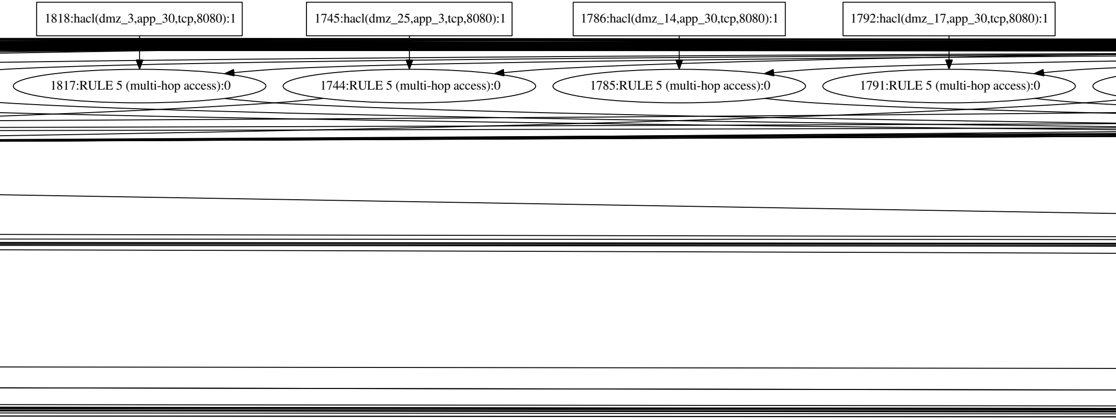

1735:hacl(dmz_20,app_3,tcp,8080):1
1658:hacl(dmz_15,app_29,tcp,8080):1
1737:hacl(dmz_21,app_3,tcp,8080):1
1631:hacl(dmz_7,app_28,tcp,8080):1
1734:RULE 5 (multi-hop access):0
1657:RULE 5 (multi-hop access):0
1736:RULE 5 (multi-hop access):0
1630:RULE 5 (multi-hop access):0

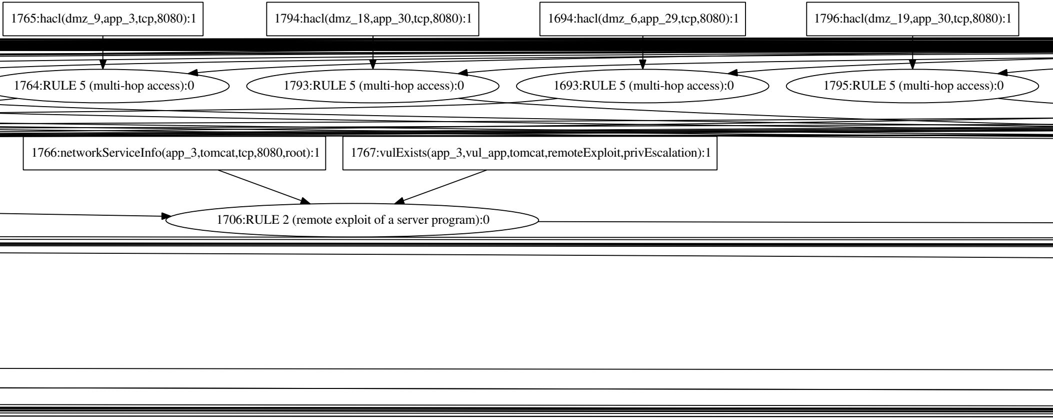

1709:hacl(dmz_0,app_3,tcp,8080):1
1739:hacl(dmz_22,app_3,tcp,8080):1
1711:hacl(dmz_1,app_3,tcp,8080):1
1633:hacl(dmz_8,app_28,tcp,8080):1
1708:RULE 5 (multi-hop access):0
1738:RULE 5 (multi-hop access):0
1710:RULE 5 (multi-hop access):0
1632:RULE 5 (multi-hop access):0

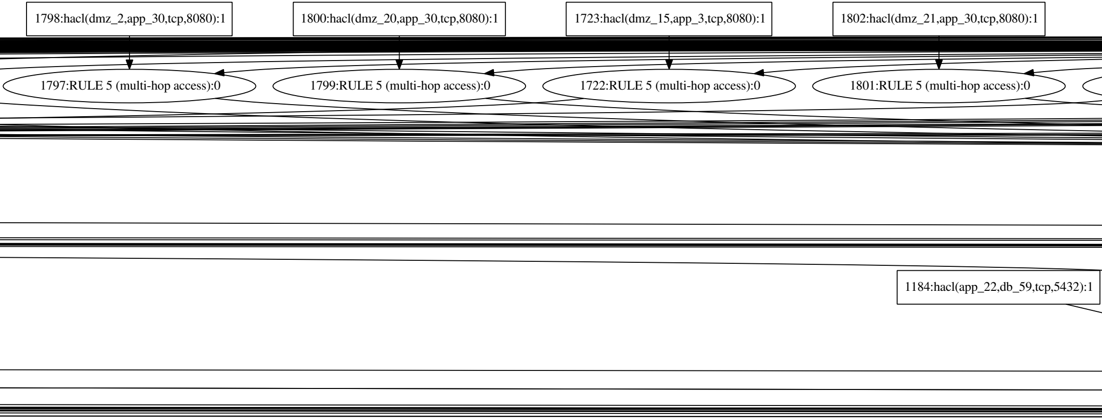

1713:hacl(dmz_10,app_3,tcp,8080):1
1715:hacl(dmz_11,app_3,tcp,8080):1
1690:hacl(dmz_4,app_29,tcp,8080):1
1654:hacl(dmz_13,app_29,tcp,8080):1
1712:RULE 5 (multi-hop access):0
1714:RULE 5 (multi-hop access):0
1689:RULE 5 (multi-hop access):0
1653:RULE 5 (multi-hop access):0
1707:netAccess(app_3,tcp,8080):0

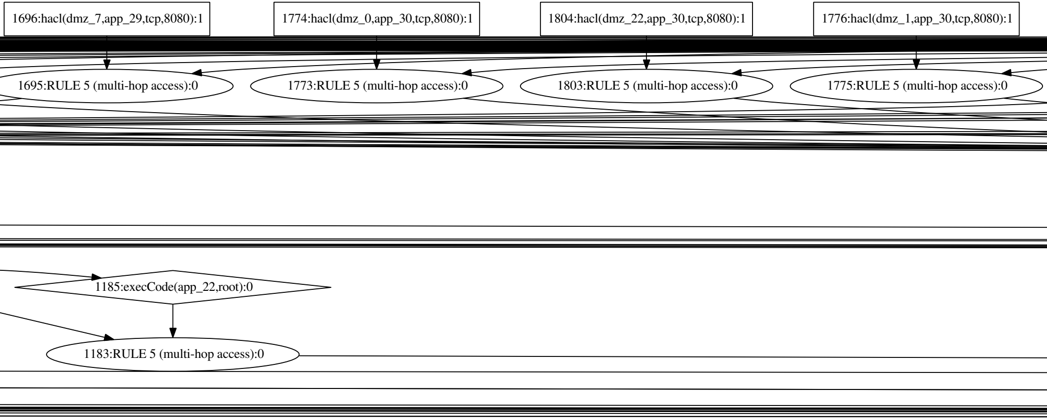

1717:hacl(dmz_12,app_3,tcp,8080):1
1682:hacl(dmz_26,app_29,tcp,8080):1
1741:hacl(dmz_23,app_3,tcp,8080):1
1660:hacl(dmz_16,app_29,tcp,8080):1
1716:RULE 5 (multi-hop access):0
1681:RULE 5 (multi-hop access):0
1740:RULE 5 (multi-hop access):0
1659:RULE 5 (multi-hop access):0

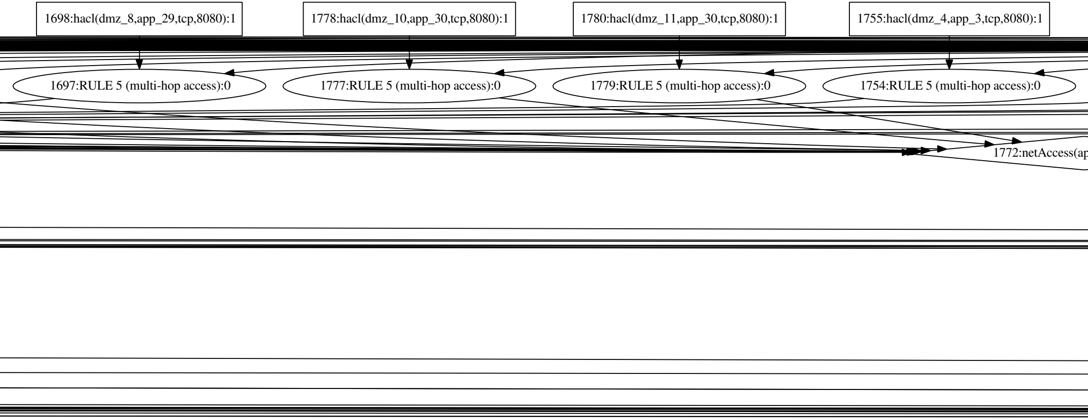

1743:hacl(dmz_24,app_3,tcp,8080):1
1692:hacl(dmz_5,app_29,tcp,8080):1
1814:hacl(dmz_27,app_30,tcp,8080):1
1816:hacl(dmz_28,app_30,tcp,8080):1
1742:RULE 5 (multi-hop access):0
1691:RULE 5 (multi-hop access):0
1813:RULE 5 (multi-hop access):0
1815:RULE 5 (multi-hop access):0

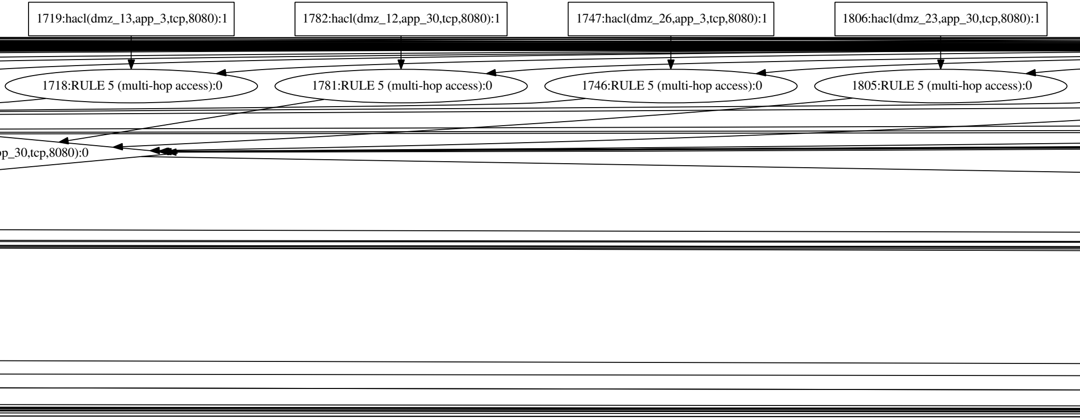

1818:hacl(dmz_3,app_30,tcp,8080):1
1745:hacl(dmz_25,app_3,tcp,8080):1
1786:hacl(dmz_14,app_30,tcp,8080):1
1792:hacl(dmz_17,app_30,tcp,8080):1
1817:RULE 5 (multi-hop access):0
1744:RULE 5 (multi-hop access):0
1785:RULE 5 (multi-hop access):0
1791:RULE 5 (multi-hop access):0

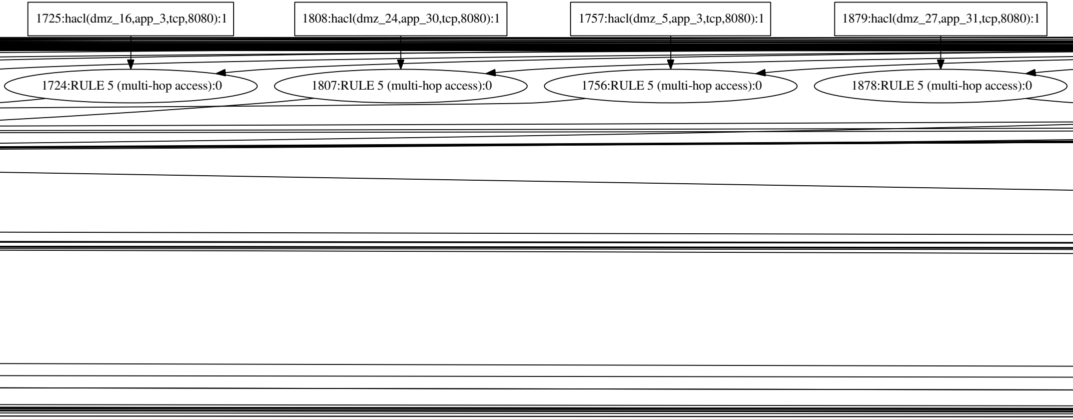

1765:hacl(dmz_9,app_3,tcp,8080):1
1794:hacl(dmz_18,app_30,tcp,8080):1
1694:hacl(dmz_6,app_29,tcp,8080):1
1796:hacl(dmz_19,app_30,tcp,8080):1
1764:RULE 5 (multi-hop access):0
1793:RULE 5 (multi-hop access):0
1693:RULE 5 (multi-hop access):0
1795:RULE 5 (multi-hop access):0
1766:networkServiceInfo(app_3,tomcat,tcp,8080,root):1
1767:vulExists(app_3,vul_app,tomcat,remoteExploit,privEscalation):1
1706:RULE 2 (remote exploit of a server program):0

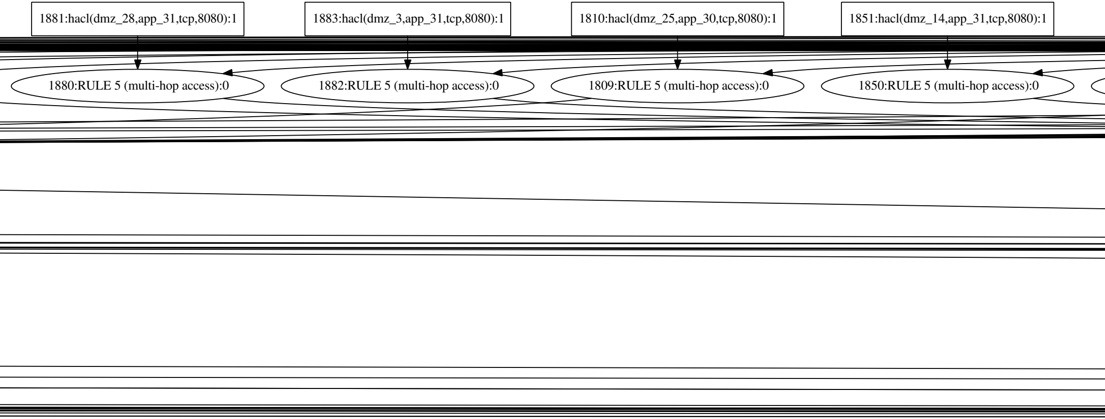

1798:hacl(dmz_2,app_30,tcp,8080):1
1800:hacl(dmz_20,app_30,tcp,8080):1
1723:hacl(dmz_15,app_3,tcp,8080):1
1802:hacl(dmz_21,app_30,tcp,8080):1
1797:RULE 5 (multi-hop access):0
1799:RULE 5 (multi-hop access):0
1722:RULE 5 (multi-hop access):0
1801:RULE 5 (multi-hop access):0
1184:hacl(app_22,db_59,tcp,5432):1

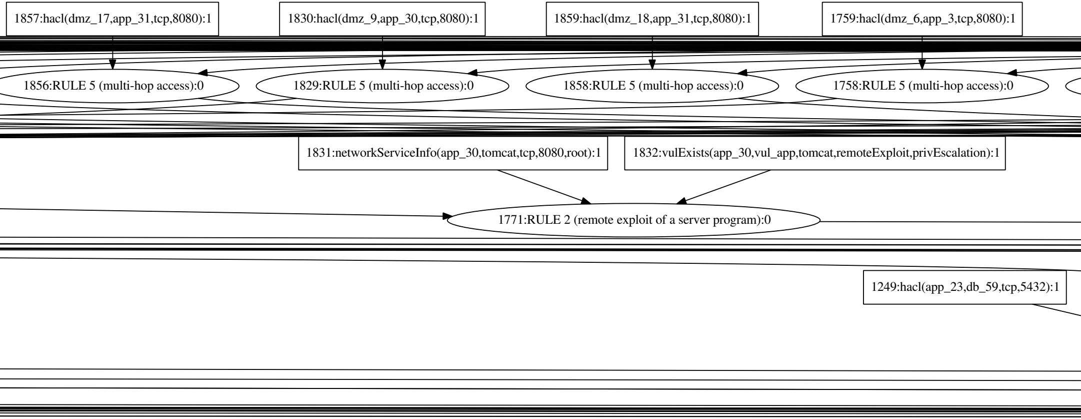

1696:hacl(dmz_7,app_29,tcp,8080):1
1774:hacl(dmz_0,app_30,tcp,8080):1
1804:hacl(dmz_22,app_30,tcp,8080):1
1776:hacl(dmz_1,app_30,tcp,8080):1
1695:RULE 5 (multi-hop access):0
1773:RULE 5 (multi-hop access):0
1803:RULE 5 (multi-hop access):0
1775:RULE 5 (multi-hop access):0
1185:execCode(app_22,root):0
1183:RULE 5 (multi-hop access):0

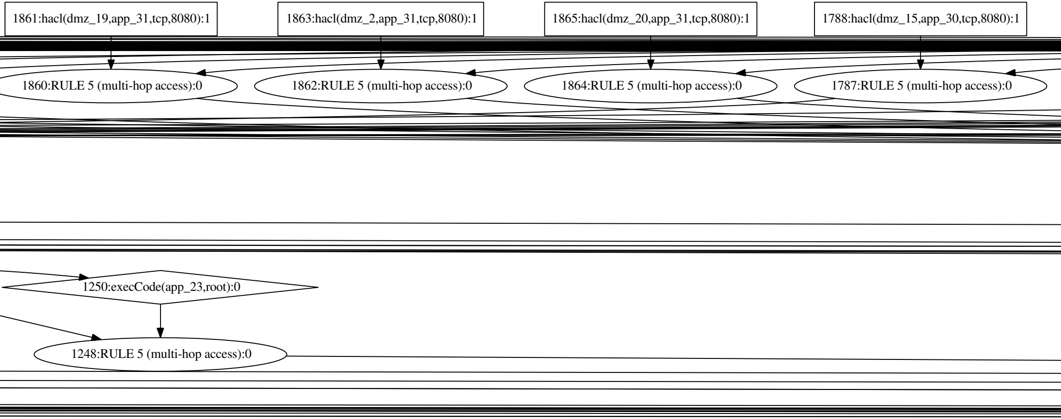

1698:hacl(dmz_8,app_29,tcp,8080):1
1778:hacl(dmz_10,app_30,tcp,8080):1
1780:hacl(dmz_11,app_30,tcp,8080):1
1755:hacl(dmz_4,app_3,tcp,8080):1
1697:RULE 5 (multi-hop access):0
1777:RULE 5 (multi-hop access):0
1779:RULE 5 (multi-hop access):0
1754:RULE 5 (multi-hop access):0
1772:netAccess(ap

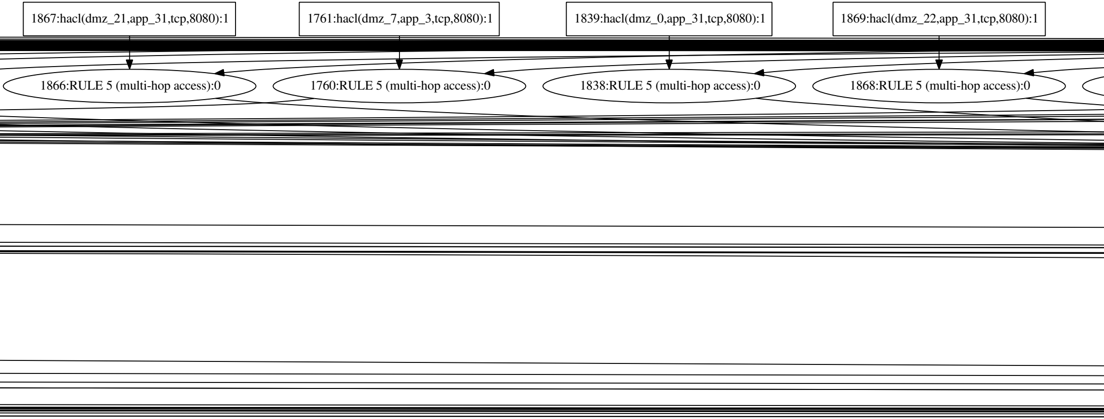

1719:hacl(dmz_13,app_3,tcp,8080):1
1782:hacl(dmz_12,app_30,tcp,8080):1
1747:hacl(dmz_26,app_3,tcp,8080):1
1806:hacl(dmz_23,app_30,tcp,8080):1
1718:RULE 5 (multi-hop access):0
1781:RULE 5 (multi-hop access):0
1746:RULE 5 (multi-hop access):0
1805:RULE 5 (multi-hop access):0
p_30,tcp,8080):0

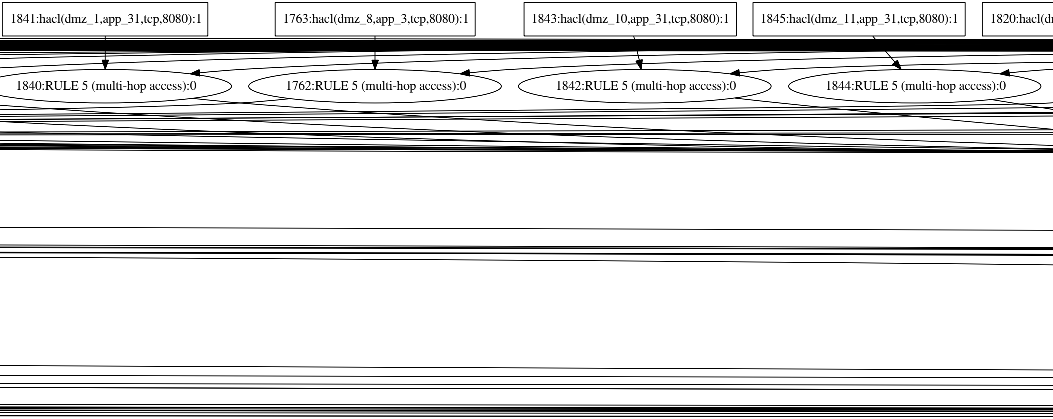

1725:hacl(dmz_16,app_3,tcp,8080):1
1808:hacl(dmz_24,app_30,tcp,8080):1
1757:hacl(dmz_5,app_3,tcp,8080):1
1879:hacl(dmz_27,app_31,tcp,8080):1
1724:RULE 5 (multi-hop access):0
1807:RULE 5 (multi-hop access):0
1756:RULE 5 (multi-hop access):0
1878:RULE 5 (multi-hop access):0

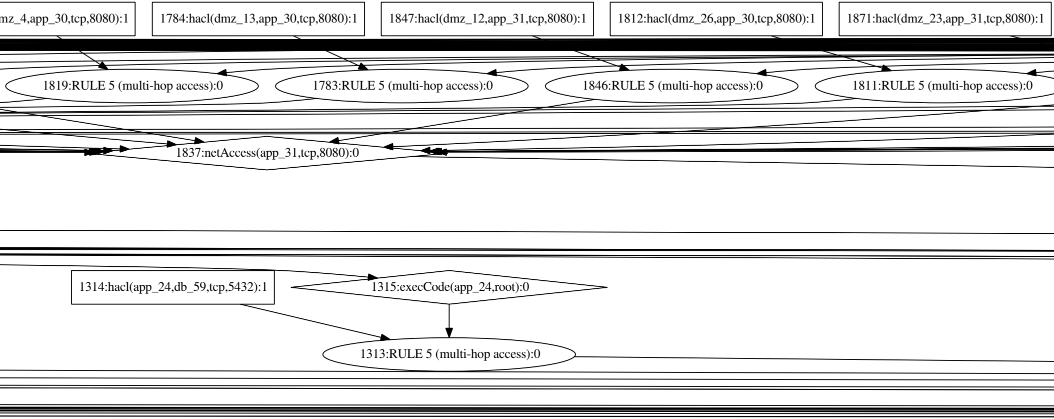

1881:hacl(dmz_28,app_31,tcp,8080):1
1883:hacl(dmz_3,app_31,tcp,8080):1
1810:hacl(dmz_25,app_30,tcp,8080):1
1851:hacl(dmz_14,app_31,tcp,8080):1
1880:RULE 5 (multi-hop access):0
1882:RULE 5 (multi-hop access):0
1809:RULE 5 (multi-hop access):0
1850:RULE 5 (multi-hop access):0

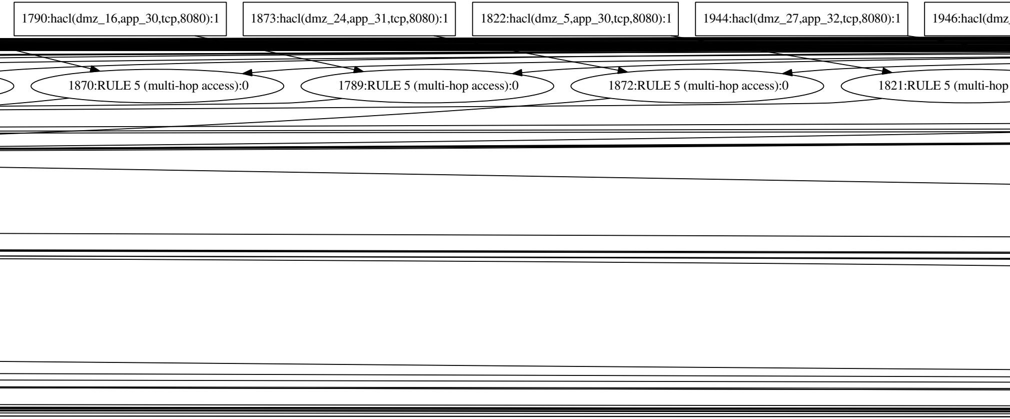

1857:hacl(dmz_17,app_31,tcp,8080):1
1830:hacl(dmz_9,app_30,tcp,8080):1
1859:hacl(dmz_18,app_31,tcp,8080):1
1759:hacl(dmz_6,app_3,tcp,8080):1
1856:RULE 5 (multi-hop access):0
1829:RULE 5 (multi-hop access):0
1858:RULE 5 (multi-hop access):0
1758:RULE 5 (multi-hop access):0
1831:networkServiceInfo(app_30,tomcat,tcp,8080,root):1
1832:vulExists(app_30,vul_app,tomcat,remoteExploit,privEscalation):1
1771:RULE 2 (remote exploit of a server program):0
1249:hacl(app_23,db_59,tcp,5432):1

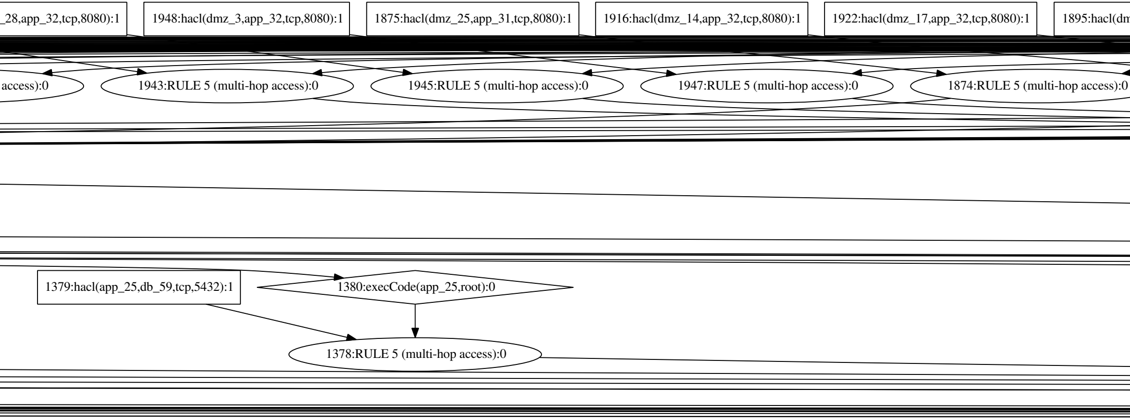

1861:hacl(dmz_19,app_31,tcp,8080):1
1863:hacl(dmz_2,app_31,tcp,8080):1
1865:hacl(dmz_20,app_31,tcp,8080):1
1788:hacl(dmz_15,app_30,tcp,8080):1
1860:RULE 5 (multi-hop access):0
1862:RULE 5 (multi-hop access):0
1864:RULE 5 (multi-hop access):0
1787:RULE 5 (multi-hop access):0
1250:execCode(app_23,root):0
1248:RULE 5 (multi-hop access):0

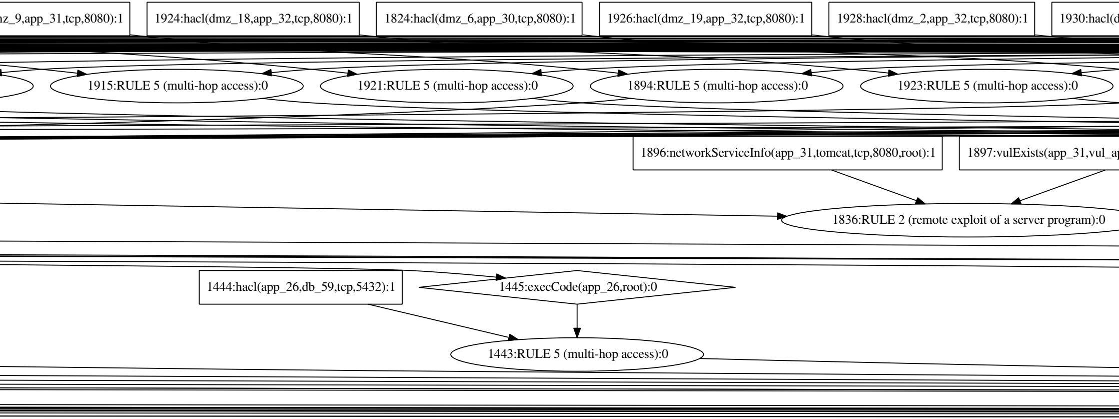

1867:hacl(dmz_21,app_31,tcp,8080):1
1761:hacl(dmz_7,app_3,tcp,8080):1
1839:hacl(dmz_0,app_31,tcp,8080):1
1869:hacl(dmz_22,app_31,tcp,8080):1
1866:RULE 5 (multi-hop access):0
1760:RULE 5 (multi-hop access):0
1838:RULE 5 (multi-hop access):0
1868:RULE 5 (multi-hop access):0

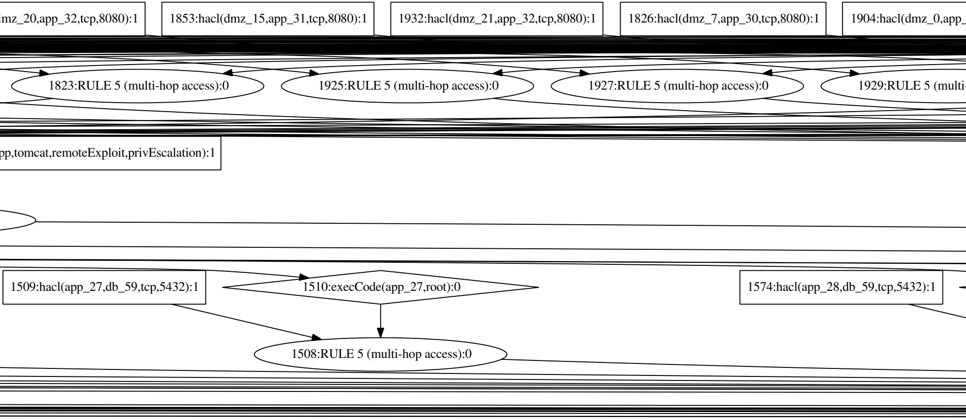

1841:hacl(dmz_1,app_31,tcp,8080):1
1763:hacl(dmz_8,app_3,tcp,8080):1
1843:hacl(dmz_10,app_31,tcp,8080):1
1845:hacl(dmz_11,app_31,tcp,8080):1
1820:hacl(d
1840:RULE 5 (multi-hop access):0
1762:RULE 5 (multi-hop access):0
1842:RULE 5 (multi-hop access):0
1844:RULE 5 (multi-hop access):0

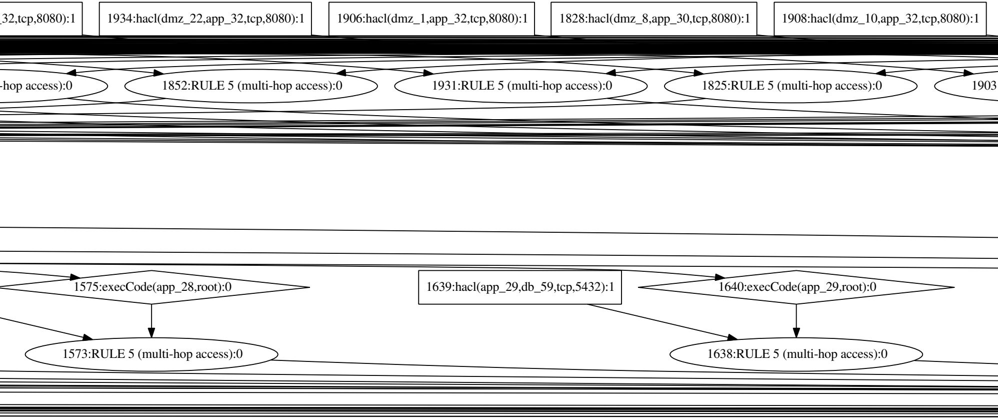

mz_4,app_30,tcp,8080):1
1784:hacl(dmz_13,app_30,tcp,8080):1
1847:hacl(dmz_12,app_31,tcp,8080):1
1812:hacl(dmz_26,app_30,tcp,8080):1
1871:hacl(dmz_23,app_31,tcp,8080):1
1819:RULE 5 (multi-hop access):0
1783:RULE 5 (multi-hop access):0
1846:RULE 5 (multi-hop access):0
1811:RULE 5 (multi-hop access):0
1837:netAccess(app_31,tcp,8080):0
1314:hacl(app_24,db_59,tcp,5432):1
1315:execCode(app_24,root):0
1313:RULE 5 (multi-hop access):0

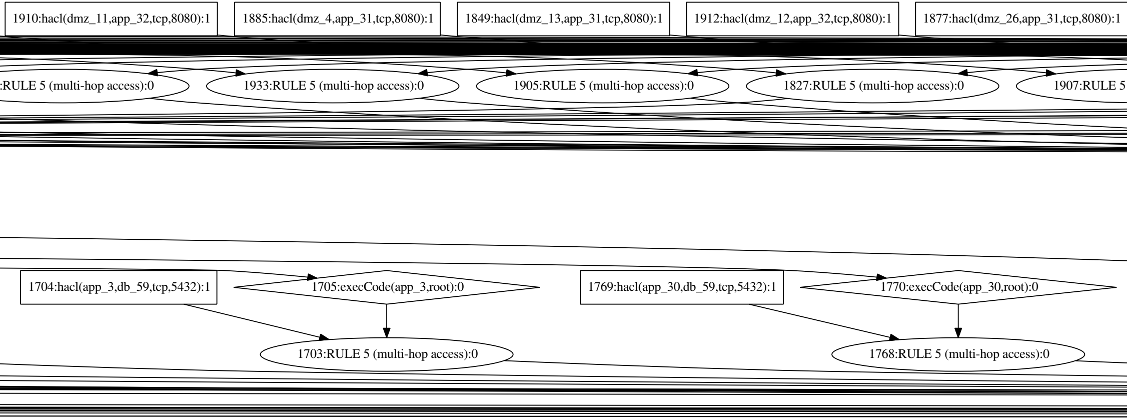

1790:hacl(dmz_16,app_30,tcp,8080):1
1873:hacl(dmz_24,app_31,tcp,8080):1
1822:hacl(dmz_5,app_30,tcp,8080):1
1944:hacl(dmz_27,app_32,tcp,8080):1
1946:hacl(dmz_
1870:RULE 5 (multi-hop access):0
1789:RULE 5 (multi-hop access):0
1872:RULE 5 (multi-hop access):0
1821:RULE 5 (multi-hop

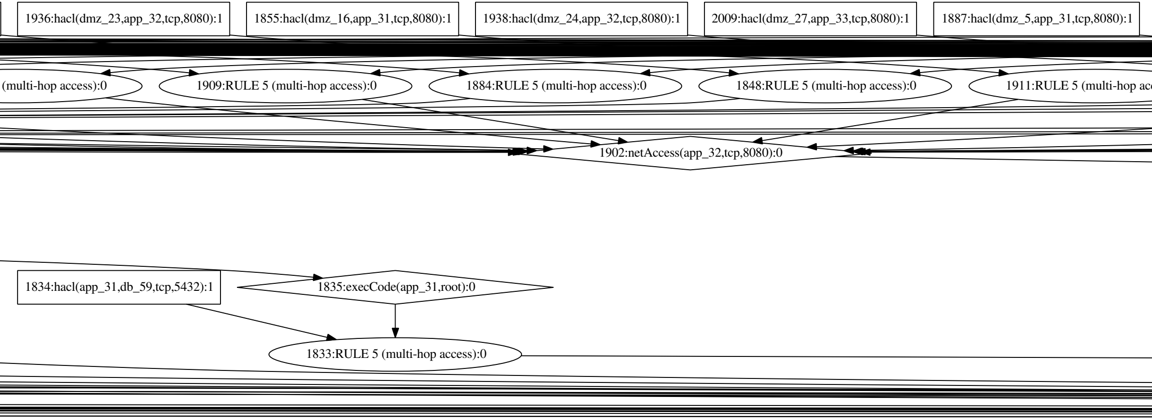

_28,app_32,tcp,8080):1
1948:hacl(dmz_3,app_32,tcp,8080):1
1875:hacl(dmz_25,app_31,tcp,8080):1
1916:hacl(dmz_14,app_32,tcp,8080):1
1922:hacl(dmz_17,app_32,tcp,8080):1
1895:hacl(dm
access):0
1943:RULE 5 (multi-hop access):0
1945:RULE 5 (multi-hop access):0
1947:RULE 5 (multi-hop access):0
1874:RULE 5 (multi-hop access):0
1379:hacl(app_25,db_59,tcp,5432):1
1380:execCode(app_25,root):0
1378:RULE 5 (multi-hop access):0

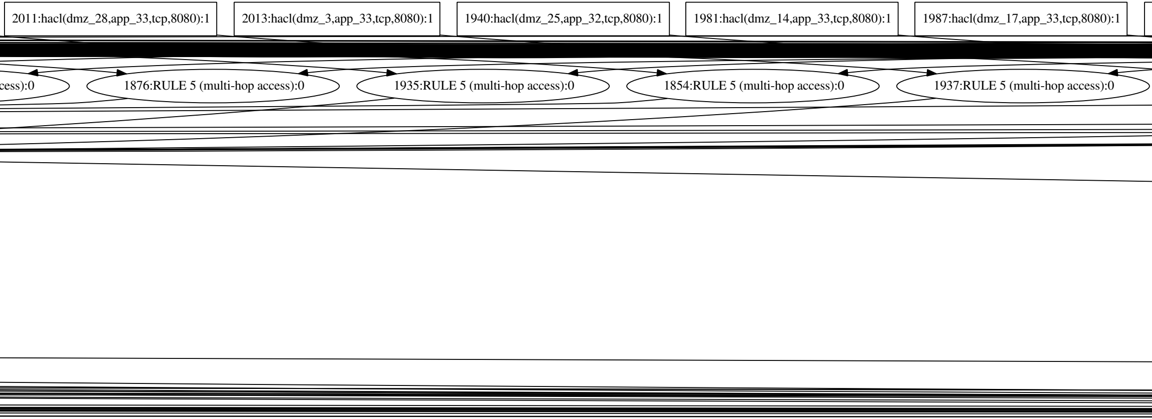

nz_9,app_31,tcp,8080):1
1924:hacl(dmz_18,app_32,tcp,8080):1
1824:hacl(dmz_6,app_30,tcp,8080):1
1926:hacl(dmz_19,app_32,tcp,8080):1
1928:hacl(dmz_2,app_32,tcp,8080):1
1930:hacl(d
1915:RULE 5 (multi-hop access):0
1921:RULE 5 (multi-hop access):0
1894:RULE 5 (multi-hop access):0
1923:RULE 5 (multi-hop access):0
1896:networkServiceInfo(app_31,tomcat,tcp,8080,root):1
1897:vulExists(app_31,vul_ap
1836:RULE 2 (remote exploit of a server program):0
1444:hacl(app_26,db_59,tcp,5432):1
1445:execCode(app_26,root):0
1443:RULE 5 (multi-hop access):0

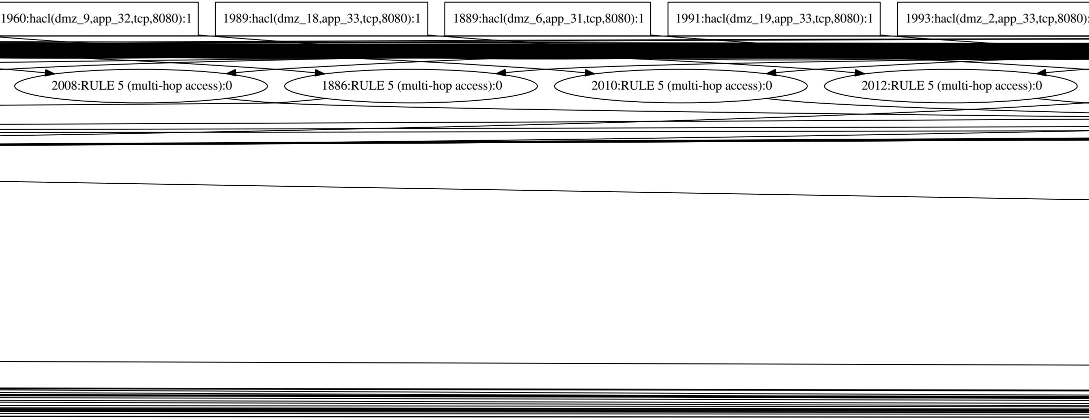

mz_20,app_32,tcp,8080):1
1853:hacl(dmz_15,app_31,tcp,8080):1
1932:hacl(dmz_21,app_32,tcp,8080):1
1826:hacl(dmz_7,app_30,tcp,8080):1
1904:hacl(dmz_0,app_
1823:RULE 5 (multi-hop access):0
1925:RULE 5 (multi-hop access):0
1927:RULE 5 (multi-hop access):0
1929:RULE 5 (multi-
pp,tomcat,remoteExploit,privEscalation):1
1509:hacl(app_27,db_59,tcp,5432):1
1510:execCode(app_27,root):0
1574:hacl(app_28,db_59,tcp,5432):1
1508:RULE 5 (multi-hop access):0

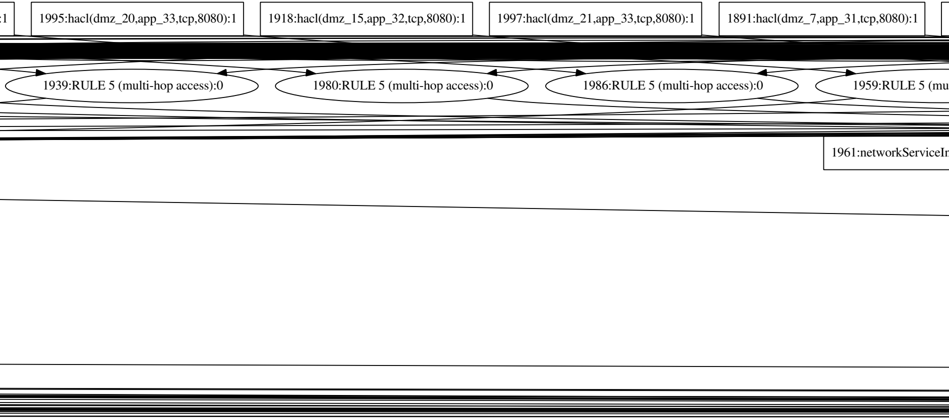

_32,tcp,8080):1
1934:hacl(dmz_22,app_32,tcp,8080):1
1906:hacl(dmz_1,app_32,tcp,8080):1
1828:hacl(dmz_8,app_30,tcp,8080):1
1908:hacl(dmz_10,app_32,tcp,8080):1
-hop access):0
1852:RULE 5 (multi-hop access):0
1931:RULE 5 (multi-hop access):0
1825:RULE 5 (multi-hop access):0
1903
1575:execCode(app_28,root):0
1639:hacl(app_29,db_59,tcp,5432):1
1640:execCode(app_29,root):0
1573:RULE 5 (multi-hop access):0
1638:RULE 5 (multi-hop access):0

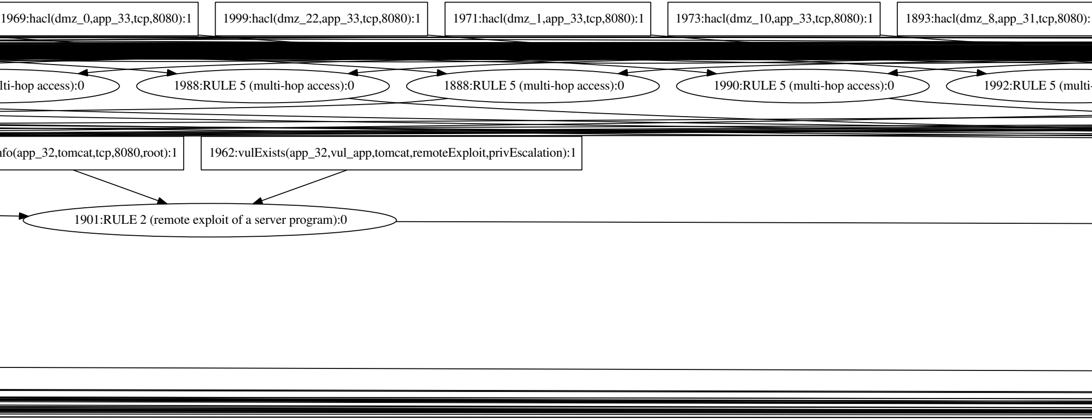

1910:hacl(dmz_11,app_32,tcp,8080):1
1885:hacl(dmz_4,app_31,tcp,8080):1
1849:hacl(dmz_13,app_31,tcp,8080):1
1912:hacl(dmz_12,app_32,tcp,8080):1
1877:hacl(dmz_26,app_31,tcp,8080):1
:RULE 5 (multi-hop access):0
1933:RULE 5 (multi-hop access):0
1905:RULE 5 (multi-hop access):0
1827:RULE 5 (multi-hop access):0
1907:RULE 5
1704:hacl(app_3,db_59,tcp,5432):1
1705:execCode(app_3,root):0
1769:hacl(app_30,db_59,tcp,5432):1
1770:execCode(app_30,root):0
1703:RULE 5 (multi-hop access):0
1768:RULE 5 (multi-hop access):0

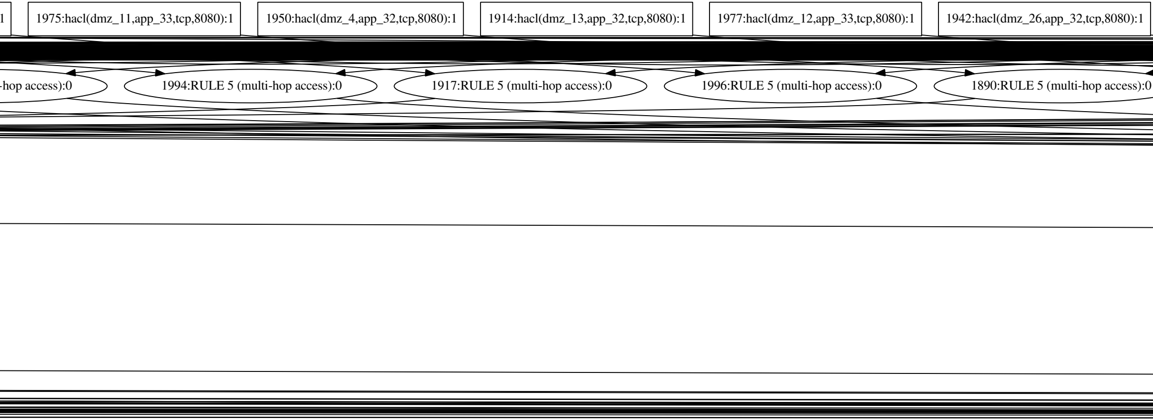

1936:hacl(dmz_23,app_32,tcp,8080):1
1855:hacl(dmz_16,app_31,tcp,8080):1
1938:hacl(dmz_24,app_32,tcp,8080):1
2009:hacl(dmz_27,app_33,tcp,8080):1
1887:hacl(dmz_5,app_31,tcp,8080):1
(multi-hop access):0
1909:RULE 5 (multi-hop access):0
1884:RULE 5 (multi-hop access):0
1848:RULE 5 (multi-hop access):0
1911:RULE 5 (multi-hop ac
1902:netAccess(app_32,tcp,8080):0
1834:hacl(app_31,db_59,tcp,5432):1
1835:execCode(app_31,root):0
1833:RULE 5 (multi-hop access):0

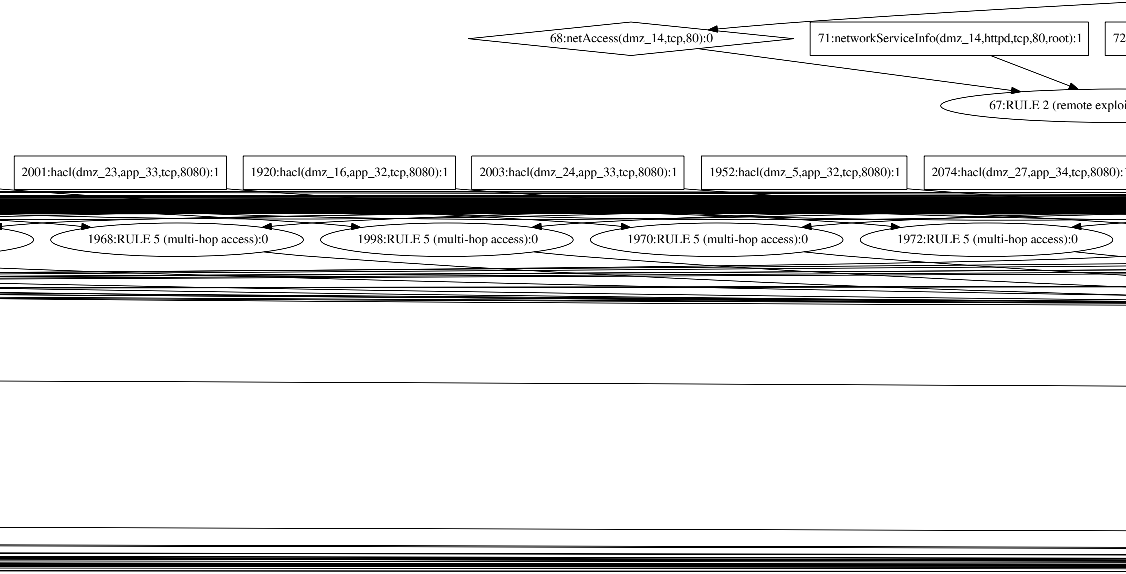

2011:hacl(dmz_28,app_33,tcp,8080):1
2013:hacl(dmz_3,app_33,tcp,8080):1
1940:hacl(dmz_25,app_32,tcp,8080):1
1981:hacl(dmz_14,app_33,tcp,8080):1
1987:hacl(dmz_17,app_33,tcp,8080):1
cess):0
1876:RULE 5 (multi-hop access):0
1935:RULE 5 (multi-hop access):0
1854:RULE 5 (multi-hop access):0
1937:RULE 5 (multi-hop access):0

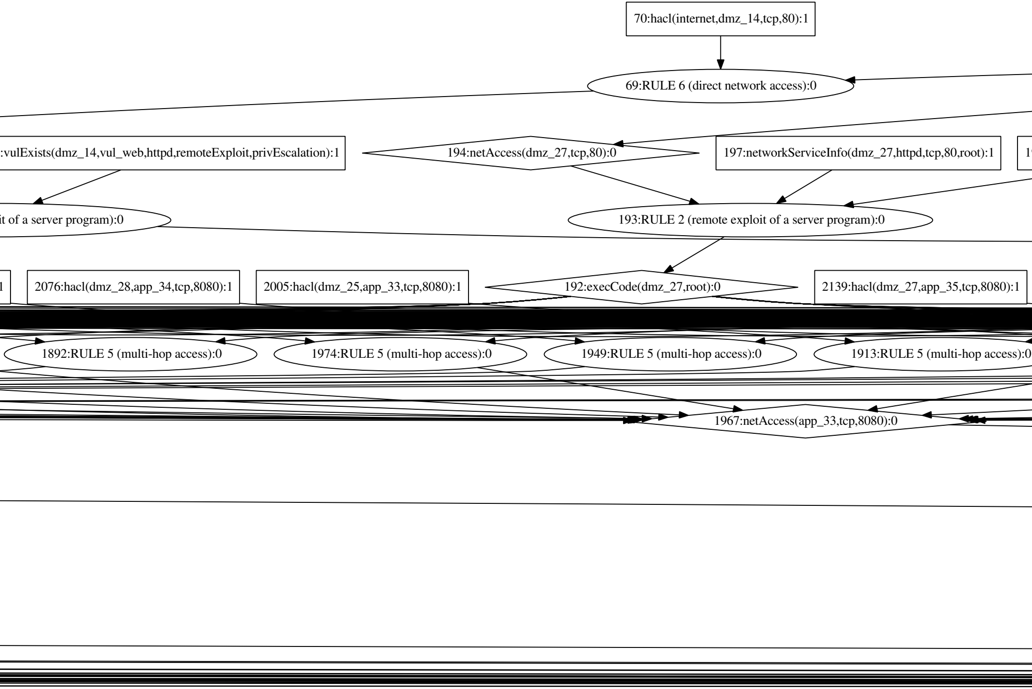

1960:hacl(dmz_9,app_32,tcp,8080):1
1989:hacl(dmz_18,app_33,tcp,8080):1
1889:hacl(dmz_6,app_31,tcp,8080):1
1991:hacl(dmz_19,app_33,tcp,8080):1
1993:hacl(dmz_2,app_33,tcp,8080):
2008:RULE 5 (multi-hop access):0
1886:RULE 5 (multi-hop access):0
2010:RULE 5 (multi-hop access):0
2012:RULE 5 (multi-hop access):0

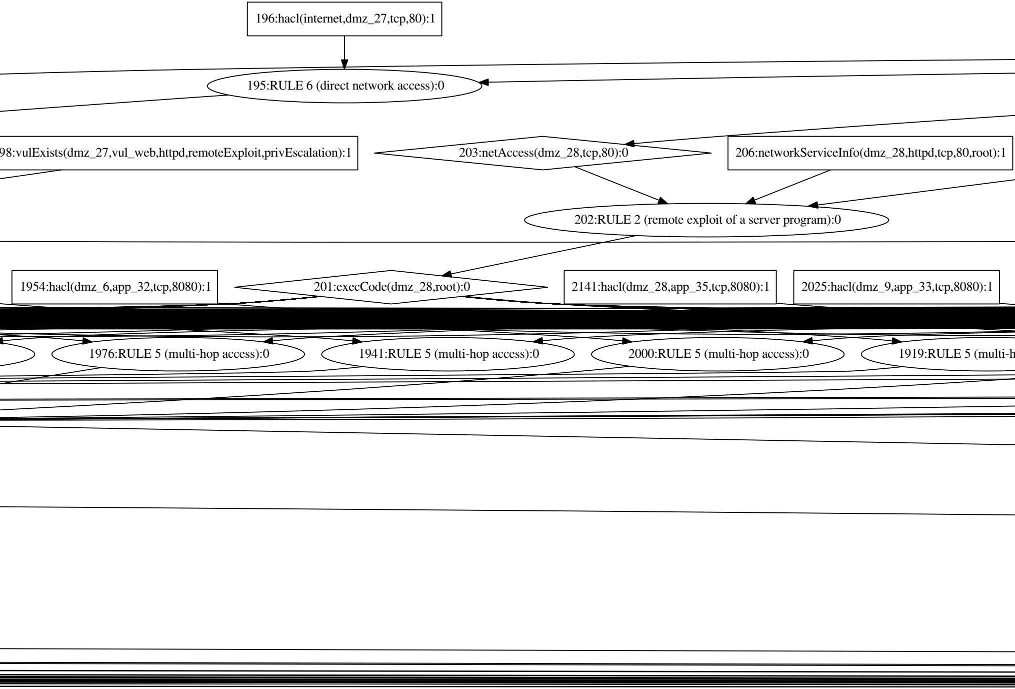

:1
1995:hacl(dmz_20,app_33,tcp,8080):1
1918:hacl(dmz_15,app_32,tcp,8080):1
1997:hacl(dmz_21,app_33,tcp,8080):1
1891:hacl(dmz_7,app_31,tcp,8080):1
1939:RULE 5 (multi-hop access):0
1980:RULE 5 (multi-hop access):0
1986:RULE 5 (multi-hop access):0
1959:RULE 5 (mu
1961:networkServiceIn

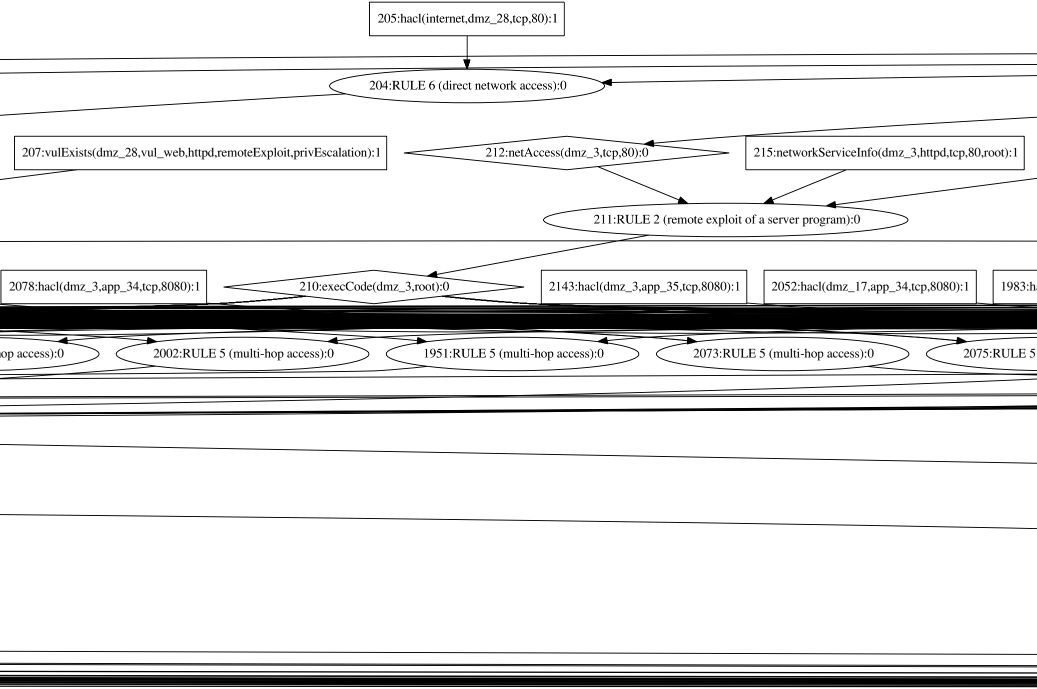

1969:hacl(dmz_0,app_33,tcp,8080):1
1999:hacl(dmz_22,app_33,tcp,8080):1
1971:hacl(dmz_1,app_33,tcp,8080):1
1973:hacl(dmz_10,app_33,tcp,8080):1
1893:hacl(dmz_8,app_31,tcp,8080):
lti-hop access):0
1988:RULE 5 (multi-hop access):0
1888:RULE 5 (multi-hop access):0
1990:RULE 5 (multi-hop access):0
1992:RULE 5 (multi-
nfo(app_32,tomcat,tcp,8080,root):1
1962:vulExists(app_32,vul_app,tomcat,remoteExploit,privEscalation):1
1901:RULE 2 (remote exploit of a server program):0

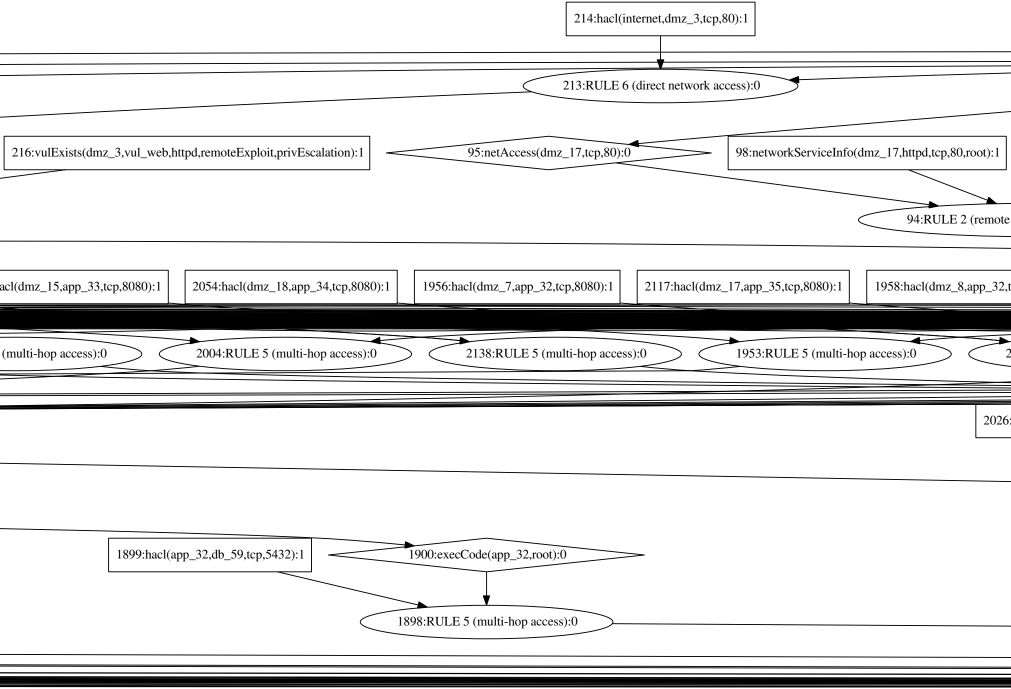

1975:hacl(dmz_11,app_33,tcp,8080):1
1950:hacl(dmz_4,app_32,tcp,8080):1
1914:hacl(dmz_13,app_32,tcp,8080):1
1977:hacl(dmz_12,app_33,tcp,8080):1
1942:hacl(dmz_26,app_32,tcp,8080):1
-hop access):0
1994:RULE 5 (multi-hop access):0
1917:RULE 5 (multi-hop access):0
1996:RULE 5 (multi-hop access):0
1890:RULE 5 (multi-hop access):0

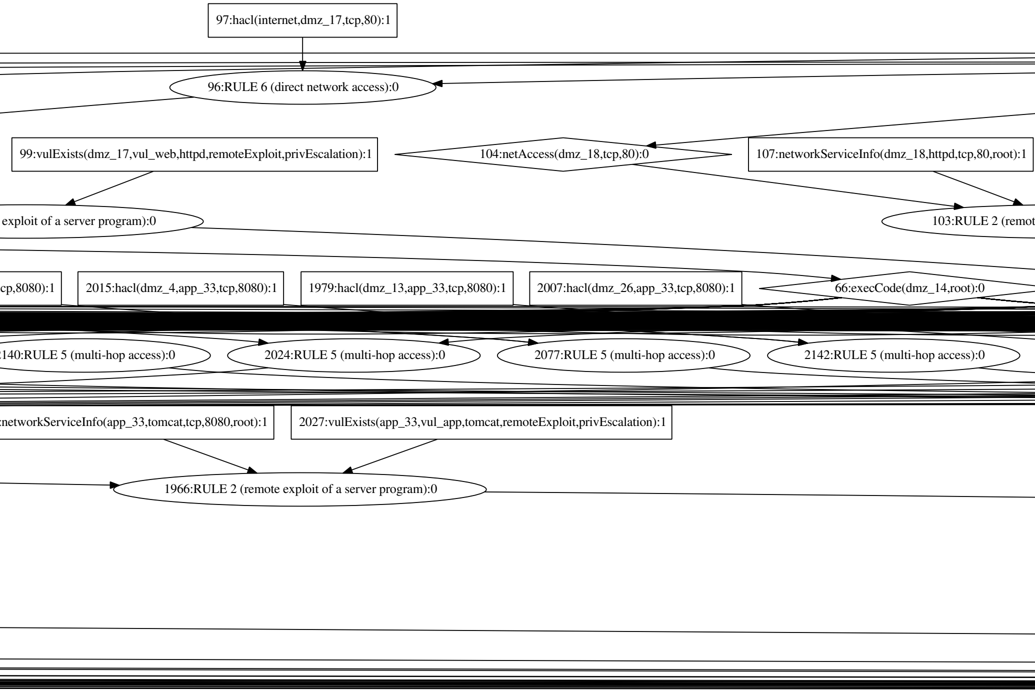

68:netAccess(dmz_14,tcp,80):0
71:networkServiceInfo(dmz_14,httpd,tcp,80,root):1
72
67:RULE 2 (remote exploi
2001:hacl(dmz_23,app_33,tcp,8080):1
1920:hacl(dmz_16,app_32,tcp,8080):1
2003:hacl(dmz_24,app_33,tcp,8080):1
1952:hacl(dmz_5,app_32,tcp,8080):1
2074:hacl(dmz_27,app_34,tcp,8080):
1968:RULE 5 (multi-hop access):0
1998:RULE 5 (multi-hop access):0
1970:RULE 5 (multi-hop access):0
1972:RULE 5 (multi-hop access):0

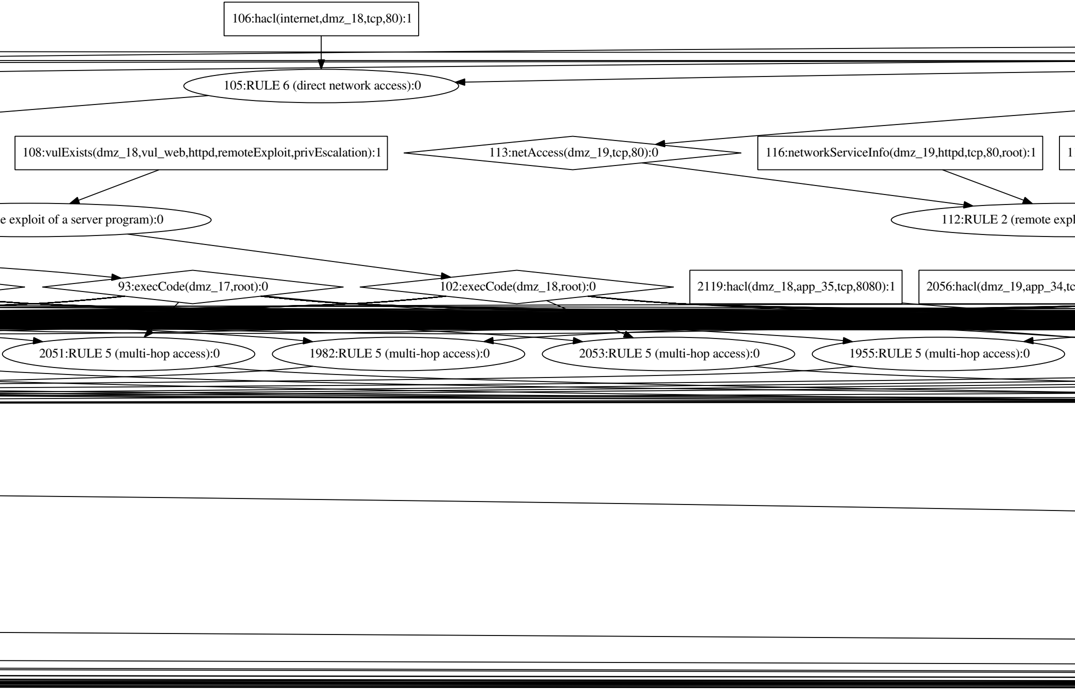

70:hacl(internet,dmz_14,tcp,80):1
69:RULE 6 (direct network access):0
:vulExists(dmz_14,vul_web,httpd,remoteExploit,privEscalation):1
194:netAccess(dmz_27,tcp,80):0
197:networkServiceInfo(dmz_27,httpd,tcp,80,root):1
it of a server program):0
193:RULE 2 (remote exploit of a server program):0
2076:hacl(dmz_28,app_34,tcp,8080):1
2005:hacl(dmz_25,app_33,tcp,8080):1
192:execCode(dmz_27,root):0
2139:hacl(dmz_27,app_35,tcp,8080):1
1892:RULE 5 (multi-hop access):0
1974:RULE 5 (multi-hop access):0
1949:RULE 5 (multi-hop access):0
1913:RULE 5 (multi-hop access):0
1967:netAccess(app_33,tcp,8080):0

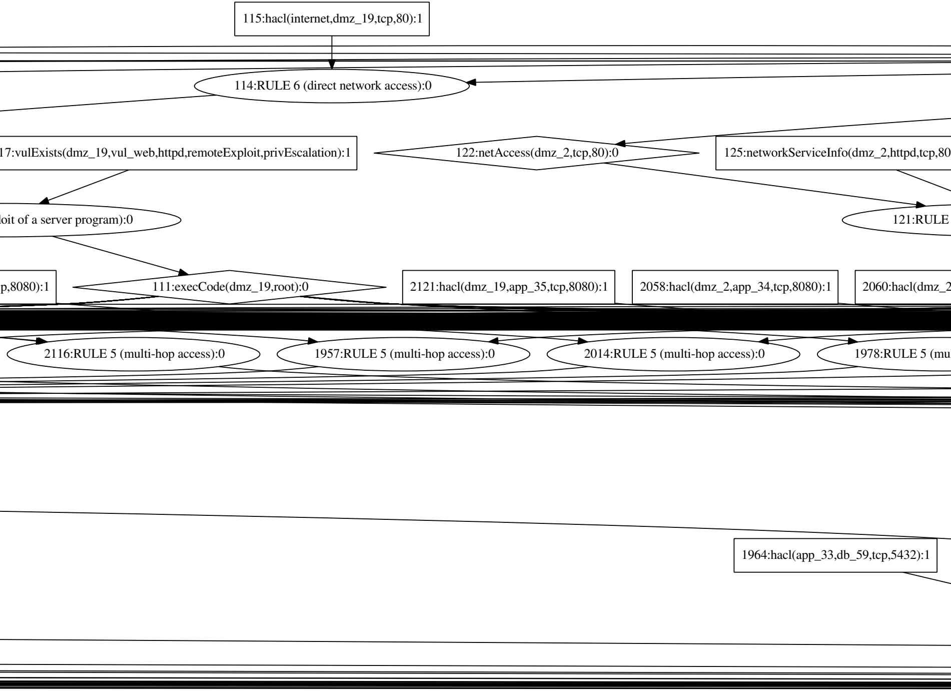

196:hacl(internet,dmz_27,tcp,80):1
195:RULE 6 (direct network access):0
98:vulExists(dmz_27,vul_web,httpd,remoteExploit,privEscalation):1
203:netAccess(dmz_28,tcp,80):0
206:networkServiceInfo(dmz_28,httpd,tcp,80,root):1
202:RULE 2 (remote exploit of a server program):0
1954:hacl(dmz_6,app_32,tcp,8080):1
201:execCode(dmz_28,root):0
2141:hacl(dmz_28,app_35,tcp,8080):1
2025:hacl(dmz_9,app_33,tcp,8080):1
1976:RULE 5 (multi-hop access):0
1941:RULE 5 (multi-hop access):0
2000:RULE 5 (multi-hop access):0
1919:RULE 5 (multi-h

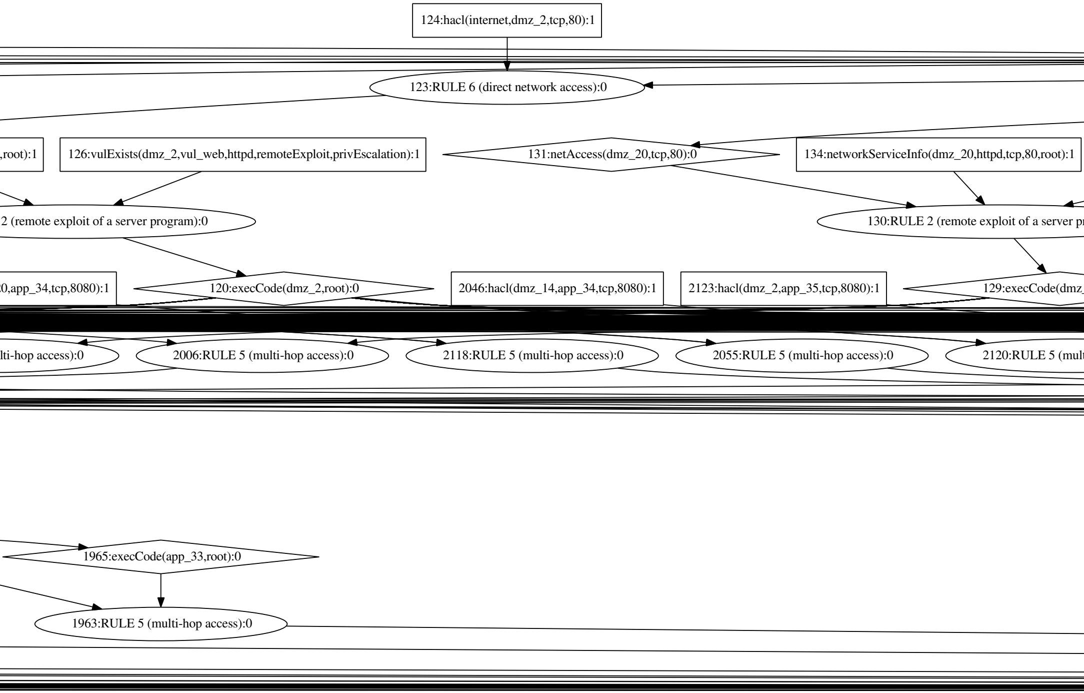

205:hacl(internet,dmz_28,tcp,80):1
204:RULE 6 (direct network access):0
207:vulExists(dmz_28,vul_web,httpd,remoteExploit,privEscalation):1
212:netAccess(dmz_3,tcp,80):0
215:networkServiceInfo(dmz_3,httpd,tcp,80,root):1
211:RULE 2 (remote exploit of a server program):0
2078:hacl(dmz_3,app_34,tcp,8080):1
210:execCode(dmz_3,root):0
2143:hacl(dmz_3,app_35,tcp,8080):1
2052:hacl(dmz_17,app_34,tcp,8080):1
1983:h
op access):0
2002:RULE 5 (multi-hop access):0
1951:RULE 5 (multi-hop access):0
2073:RULE 5 (multi-hop access):0
2075:RULE 5

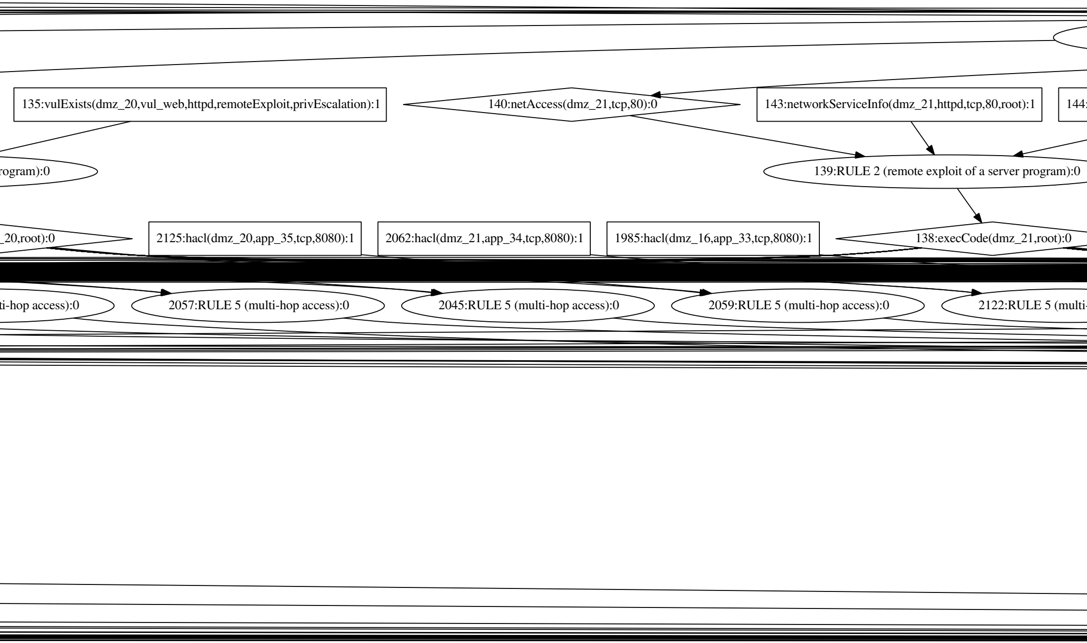

214:hacl(internet,dmz_3,tcp,80):1
213:RULE 6 (direct network access):0
216:vulExists(dmz_3,vul_web,httpd,remoteExploit,privEscalation):1
95:netAccess(dmz_17,tcp,80):0
98:networkServiceInfo(dmz_17,httpd,tcp,80,root):1
94:RULE 2 (remote
acl(dmz_15,app_33,tcp,8080):1
2054:hacl(dmz_18,app_34,tcp,8080):1
1956:hacl(dmz_7,app_32,tcp,8080):1
2117:hacl(dmz_17,app_35,tcp,8080):1
1958:hacl(dmz_8,app_32,t
(multi-hop access):0
2004:RULE 5 (multi-hop access):0
2138:RULE 5 (multi-hop access):0
1953:RULE 5 (multi-hop access):0
2
2026:
1899:hacl(app_32,db_59,tcp,5432):1
1900:execCode(app_32,root):0
1898:RULE 5 (multi-hop access):0

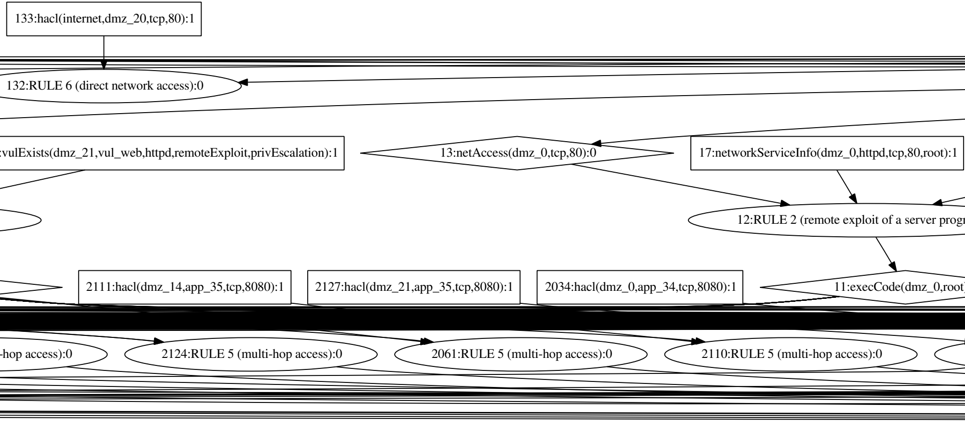

97:hacl(internet,dmz_17,tcp,80):1
96:RULE 6 (direct network access):0
99:vulExists(dmz_17,vul_web,httpd,remoteExploit,privEscalation):1
104:netAccess(dmz_18,tcp,80):0
107:networkServiceInfo(dmz_18,httpd,tcp,80,root):1
exploit of a server program):0
103:RULE 2 (remot
cp,8080):1
2015:hacl(dmz_4,app_33,tcp,8080):1
1979:hacl(dmz_13,app_33,tcp,8080):1
2007:hacl(dmz_26,app_33,tcp,8080):1
66:execCode(dmz_14,root):0
2140:RULE 5 (multi-hop access):0
2024:RULE 5 (multi-hop access):0
2077:RULE 5 (multi-hop access):0
2142:RULE 5 (multi-hop access):0
networkServiceInfo(app_33,tomcat,tcp,8080,root):1
2027:vulExists(app_33,vul_app,tomcat,remoteExploit,privEscalation):1
1966:RULE 2 (remote exploit of a server program):0

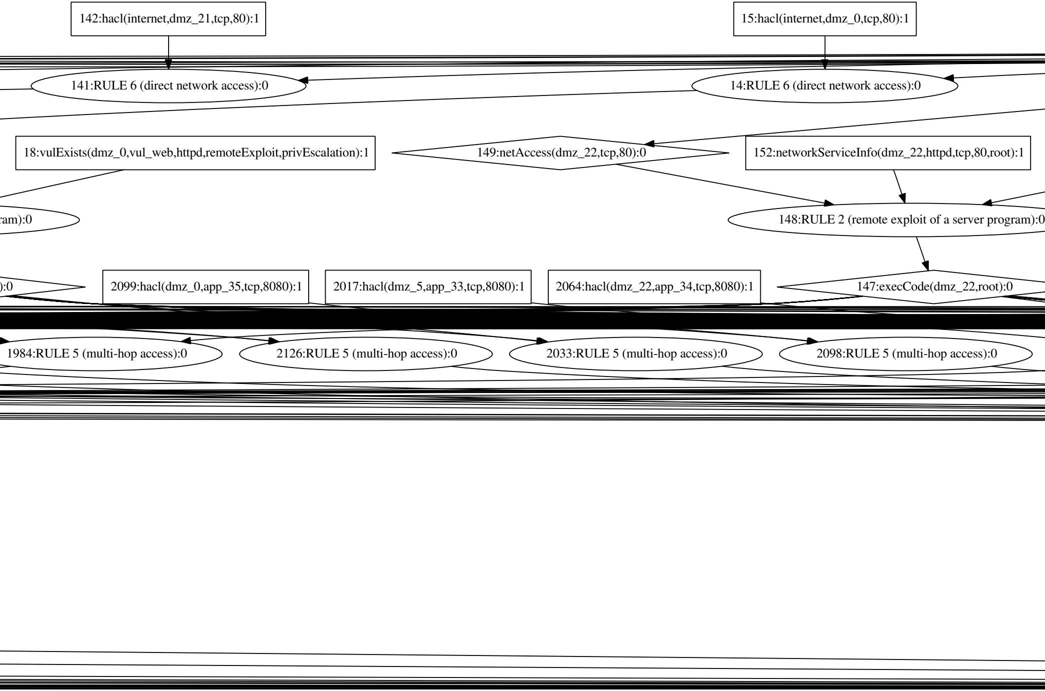

106:hacl(internet,dmz_18,tcp,80):1
105:RULE 6 (direct network access):0
108:vulExists(dmz_18,vul_web,httpd,remoteExploit,privEscalation):1
113:netAccess(dmz_19,tcp,80):0
116:networkServiceInfo(dmz_19,httpd,tcp,80,root):1
1
e exploit of a server program):0
112:RULE 2 (remote expl
93:execCode(dmz_17,root):0
102:execCode(dmz_18,root):0
2119:hacl(dmz_18,app_35,tcp,8080):1
2056:hacl(dmz_19,app_34,tc
2051:RULE 5 (multi-hop access):0
1982:RULE 5 (multi-hop access):0
2053:RULE 5 (multi-hop access):0
1955:RULE 5 (multi-hop access):0

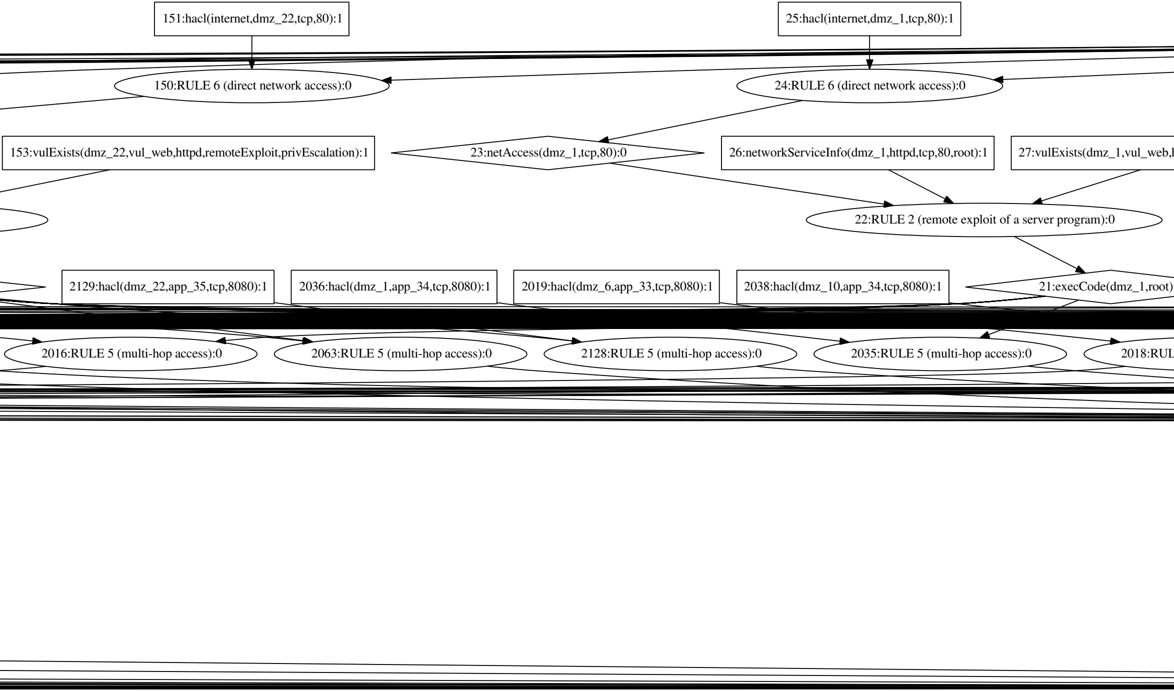

115:hacl(internet,dmz_19,tcp,80):1
114:RULE 6 (direct network access):0
17:vulExists(dmz_19,vul_web,httpd,remoteExploit,privEscalation):1
122:netAccess(dmz_2,tcp,80):0
125:networkServiceInfo(dmz_2,httpd,tcp,80
loit of a server program):0
121:RULE
p,8080):1
111:execCode(dmz_19,root):0
2121:hacl(dmz_19,app_35,tcp,8080):1
2058:hacl(dmz_2,app_34,tcp,8080):1
2060:hacl(dmz_2
2116:RULE 5 (multi-hop access):0
1957:RULE 5 (multi-hop access):0
2014:RULE 5 (multi-hop access):0
1978:RULE 5 (mu
1964:hacl(app_33,db_59,tcp,5432):1

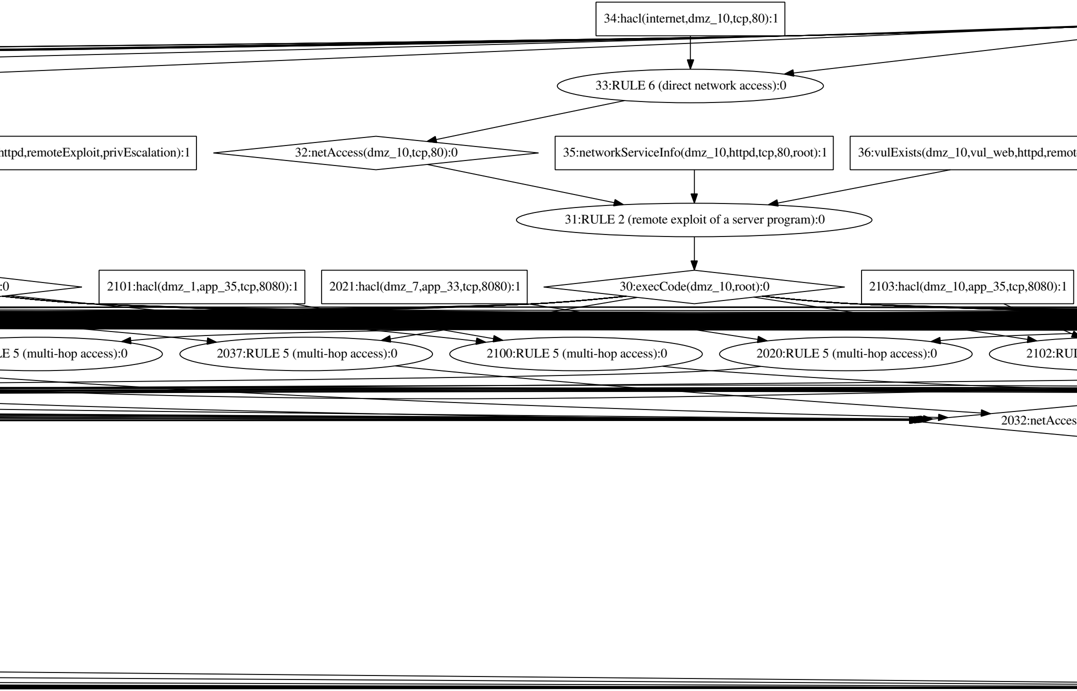

124:hacl(internet,dmz_2,tcp,80):1
123:RULE 6 (direct network access):0
,root):1
126:vulExists(dmz_2,vul_web,httpd,remoteExploit,privEscalation):1
131:netAccess(dmz_20,tcp,80):0
134:networkServiceInfo(dmz_20,httpd,tcp,80,root):1
2 (remote exploit of a server program):0
130:RULE 2 (remote exploit of a server pr
20,app_34,tcp,8080):1
120:execCode(dmz_2,root):0
2046:hacl(dmz_14,app_34,tcp,8080):1
2123:hacl(dmz_2,app_35,tcp,8080):1
129:execCode(dmz_
lti-hop access):0
2006:RULE 5 (multi-hop access):0
2118:RULE 5 (multi-hop access):0
2055:RULE 5 (multi-hop access):0
2120:RULE 5 (mult
1965:execCode(app_33,root):0
1963:RULE 5 (multi-hop access):0

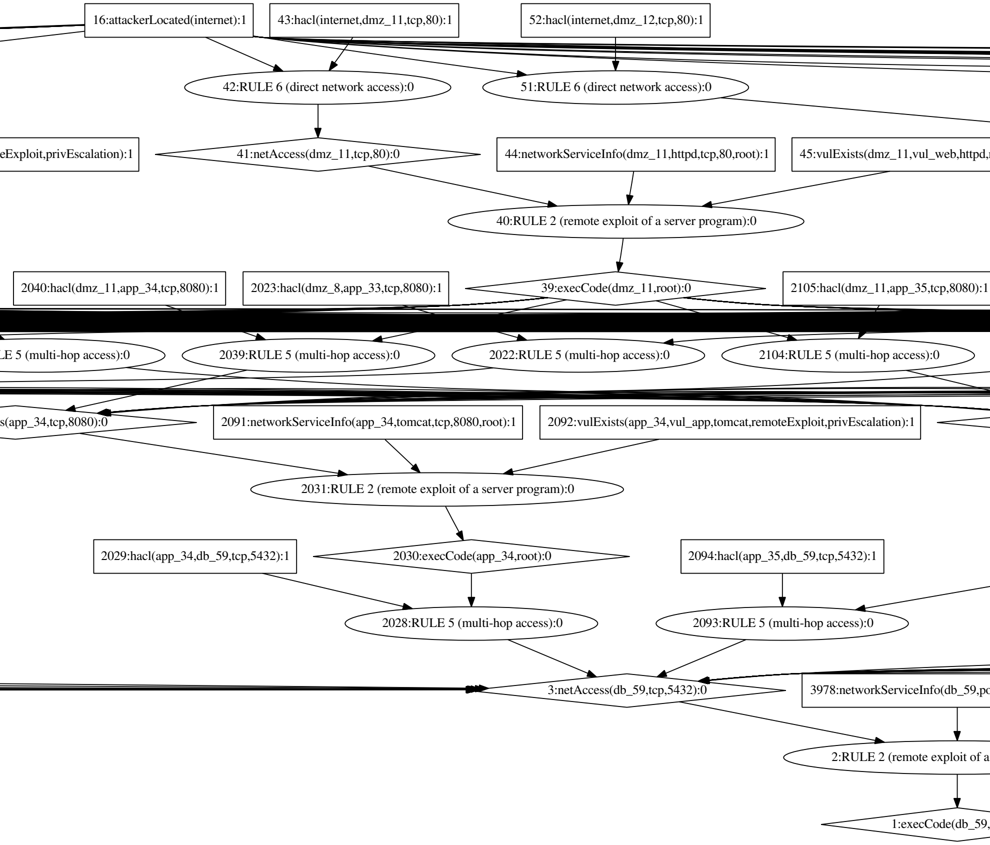

135:vulExists(dmz_20,vul_web,httpd,remoteExploit,privEscalation):1
140:netAccess(dmz_21,tcp,80):0
143:networkServiceInfo(dmz_21,httpd,tcp,80,root):1
144:
rogram):0
139:RULE 2 (remote exploit of a server program):0
_20,root):0
2125:hacl(dmz_20,app_35,tcp,8080):1
2062:hacl(dmz_21,app_34,tcp,8080):1
1985:hacl(dmz_16,app_33,tcp,8080):1
138:execCode(dmz_21,root):0
ti-hop access):0
2057:RULE 5 (multi-hop access):0
2045:RULE 5 (multi-hop access):0
2059:RULE 5 (multi-hop access):0
2122:RULE 5 (multi-

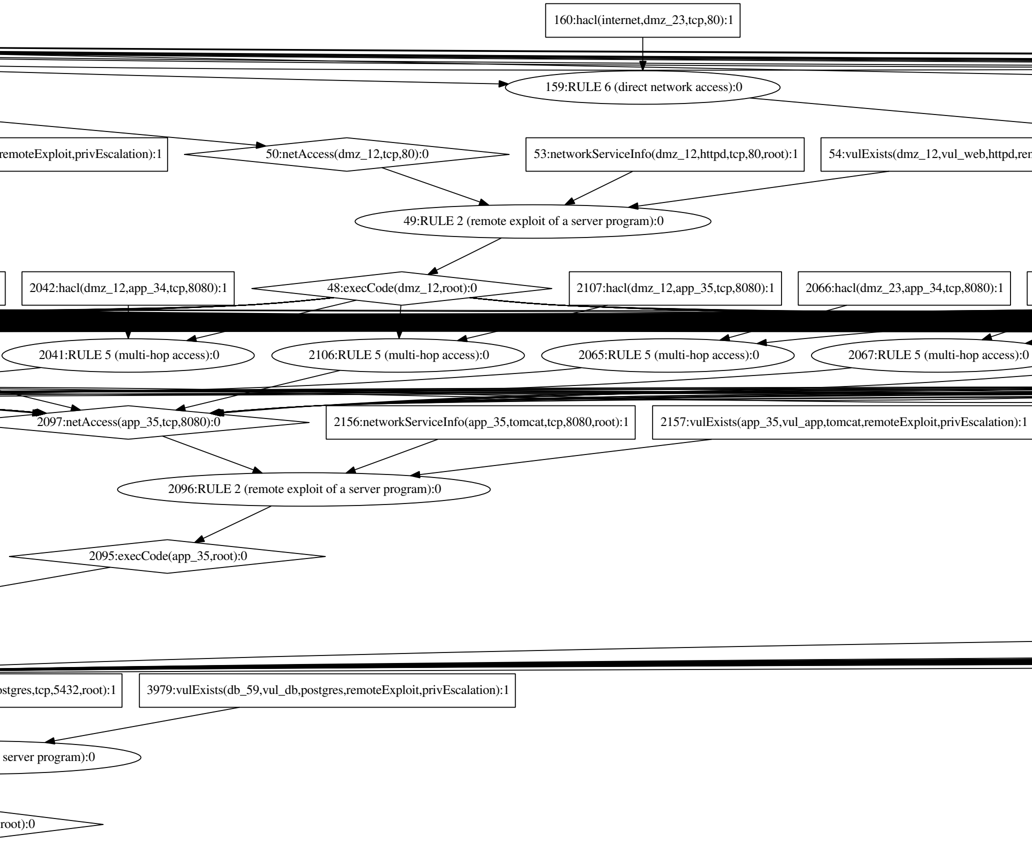

133:hacl(internet,dmz_20,tcp,80):1
132:RULE 6 (direct network access):0
:vulExists(dmz_21,vul_web,httpd,remoteExploit,privEscalation):1
13:netAccess(dmz_0,tcp,80):0
17:networkServiceInfo(dmz_0,httpd,tcp,80,root):1
12:RULE 2 (remote exploit of a server progr
2111:hacl(dmz_14,app_35,tcp,8080):1
2127:hacl(dmz_21,app_35,tcp,8080):1
2034:hacl(dmz_0,app_34,tcp,8080):1
11:execCode(dmz_0,root
-hop access):0
2124:RULE 5 (multi-hop access):0
2061:RULE 5 (multi-hop access):0
2110:RULE 5 (multi-hop access):0

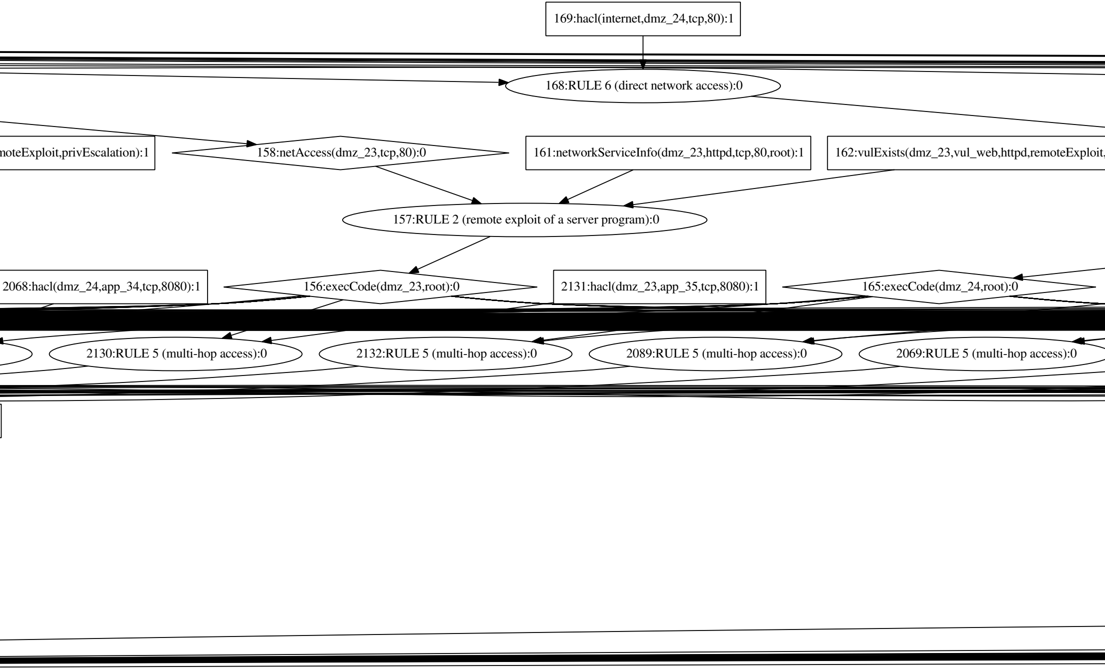

142:hacl(internet,dmz_21,tcp,80):1
15:hacl(internet,dmz_0,tcp,80):1
141:RULE 6 (direct network access):0
14:RULE 6 (direct network access):0
18:vulExists(dmz_0,vul_web,httpd,remoteExploit,privEscalation):1
149:netAccess(dmz_22,tcp,80):0
152:networkServiceInfo(dmz_22,httpd,tcp,80,root):1
ram):0
148:RULE 2 (remote exploit of a server program):0
):0
2099:hacl(dmz_0,app_35,tcp,8080):1
2017:hacl(dmz_5,app_33,tcp,8080):1
2064:hacl(dmz_22,app_34,tcp,8080):1
147:execCode(dmz_22,root):0
1984:RULE 5 (multi-hop access):0
2126:RULE 5 (multi-hop access):0
2033:RULE 5 (multi-hop access):0
2098:RULE 5 (multi-hop access):0

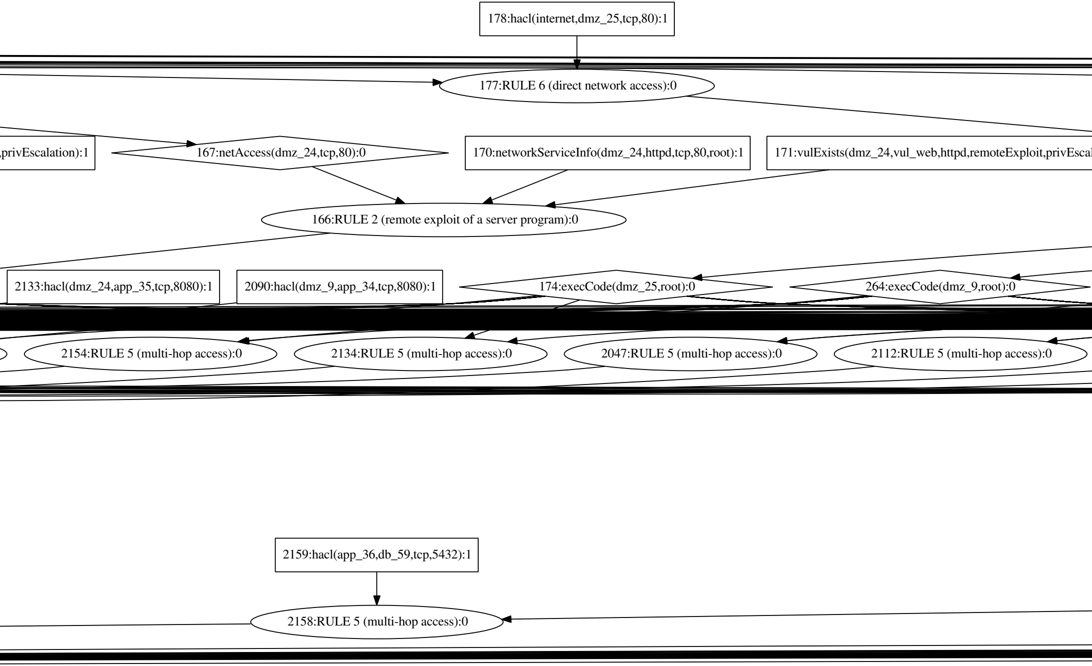

151:hacl(internet,dmz_22,tcp,80):1
25:hacl(internet,dmz_1,tcp,80):1
150:RULE 6 (direct network access):0
24:RULE 6 (direct network access):0
153:vulExists(dmz_22,vul_web,httpd,remoteExploit,privEscalation):1
23:netAccess(dmz_1,tcp,80):0
26:networkServiceInfo(dmz_1,httpd,tcp,80,root):1
27:vulExists(dmz_1,vul_web,
22:RULE 2 (remote exploit of a server program):0
2129:hacl(dmz_22,app_35,tcp,8080):1
2036:hacl(dmz_1,app_34,tcp,8080):1
2019:hacl(dmz_6,app_33,tcp,8080):1
2038:hacl(dmz_10,app_34,tcp,8080):1
21:execCode(dmz_1,root)
2016:RULE 5 (multi-hop access):0
2063:RULE 5 (multi-hop access):0
2128:RULE 5 (multi-hop access):0
2035:RULE 5 (multi-hop access):0
2018:RUL

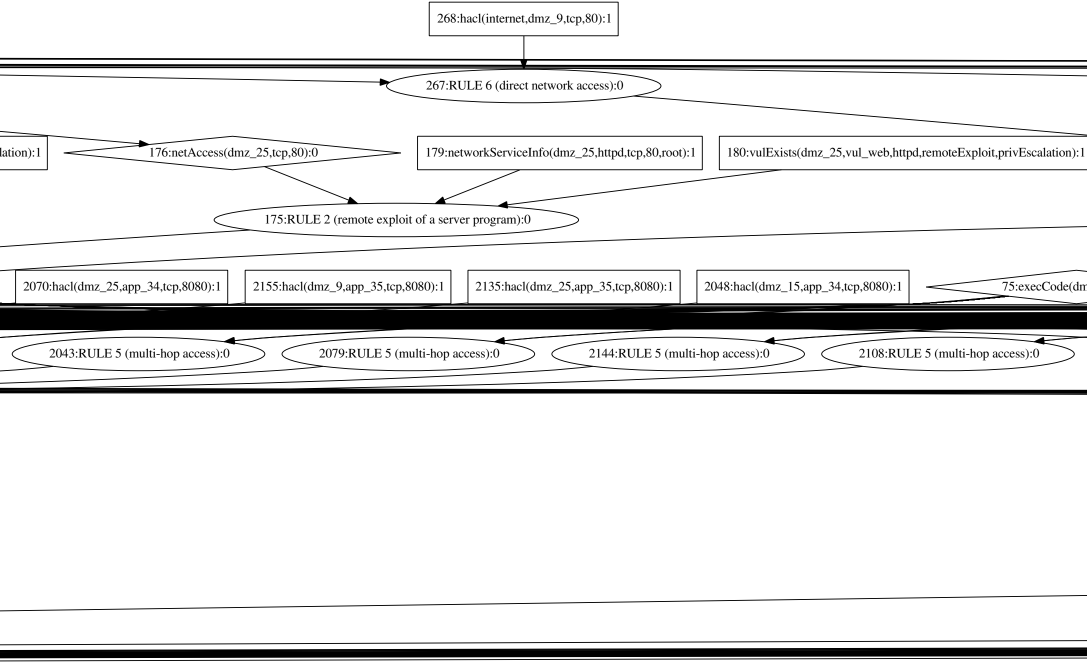

34:hacl(internet,dmz_10,tcp,80):1
33:RULE 6 (direct network access):0
httpd,remoteExploit,privEscalation):1
32:netAccess(dmz_10,tcp,80):0
35:networkServiceInfo(dmz_10,httpd,tcp,80,root):1
36:vulExists(dmz_10,vul_web,httpd,remot
31:RULE 2 (remote exploit of a server program):0
:0
2101:hacl(dmz_1,app_35,tcp,8080):1
2021:hacl(dmz_7,app_33,tcp,8080):1
30:execCode(dmz_10,root):0
2103:hacl(dmz_10,app_35,tcp,8080):1
E 5 (multi-hop access):0
2037:RULE 5 (multi-hop access):0
2100:RULE 5 (multi-hop access):0
2020:RULE 5 (multi-hop access):0
2102:RUL
2032:netAcces

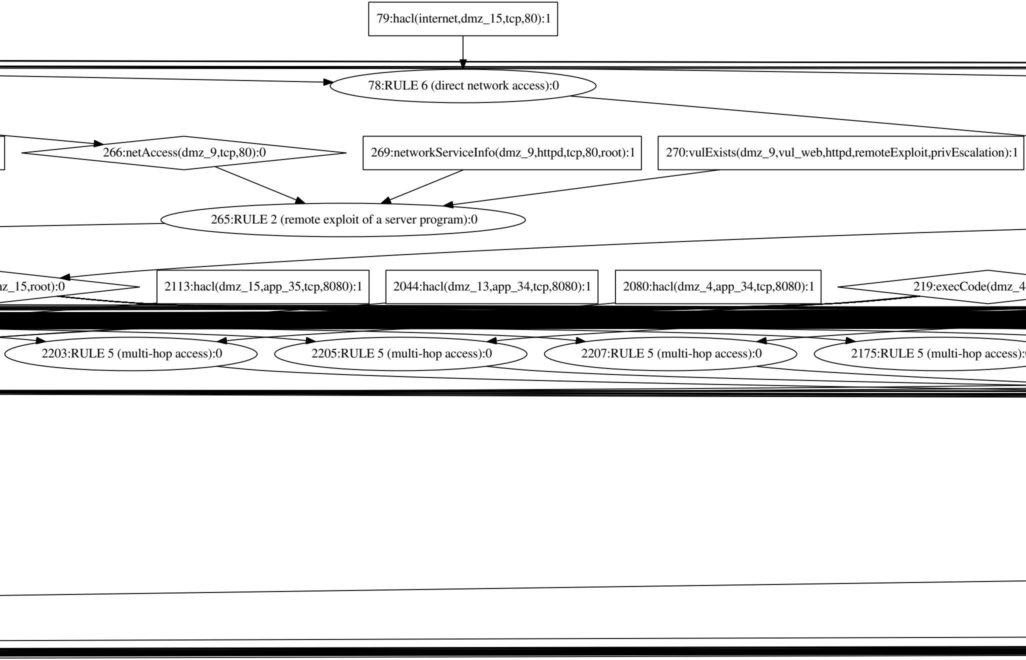

16:attackerLocated(internet):1
43:hacl(internet,dmz_11,tcp,80):1
52:hacl(internet,dmz_12,tcp,80):1
42:RULE 6 (direct network access):0
51:RULE 6 (direct network access):0
eExploit,privEscalation):1
41:netAccess(dmz_11,tcp,80):0
44:networkServiceInfo(dmz_11,httpd,tcp,80,root):1
45:vulExists(dmz_11,vul_web,httpd,
40:RULE 2 (remote exploit of a server program):0
2040:hacl(dmz_11,app_34,tcp,8080):1
2023:hacl(dmz_8,app_33,tcp,8080):1
39:execCode(dmz_11,root):0
2105:hacl(dmz_11,app_35,tcp,8080):1
LE 5 (multi-hop access):0
2039:RULE 5 (multi-hop access):0
2022:RULE 5 (multi-hop access):0
2104:RULE 5 (multi-hop access):0
s(app_34,tcp,8080):0
2091:networkServiceInfo(app_34,tomcat,tcp,8080,root):1
2092:vulExists(app_34,vul_app,tomcat,remoteExploit,privEscalation):1
2031:RULE 2 (remote exploit of a server program):0
2029:hacl(app_34,db_59,tcp,5432):1
2030:execCode(app_34,root):0
2094:hacl(app_35,db_59,tcp,5432):1
2028:RULE 5 (multi-hop access):0
2093:RULE 5 (multi-hop access):0
3:netAccess(db_59,tcp,5432):0
3978:networkServiceInfo(db_59,po
2:RULE 2 (remote exploit of a
1:execCode(db_59,

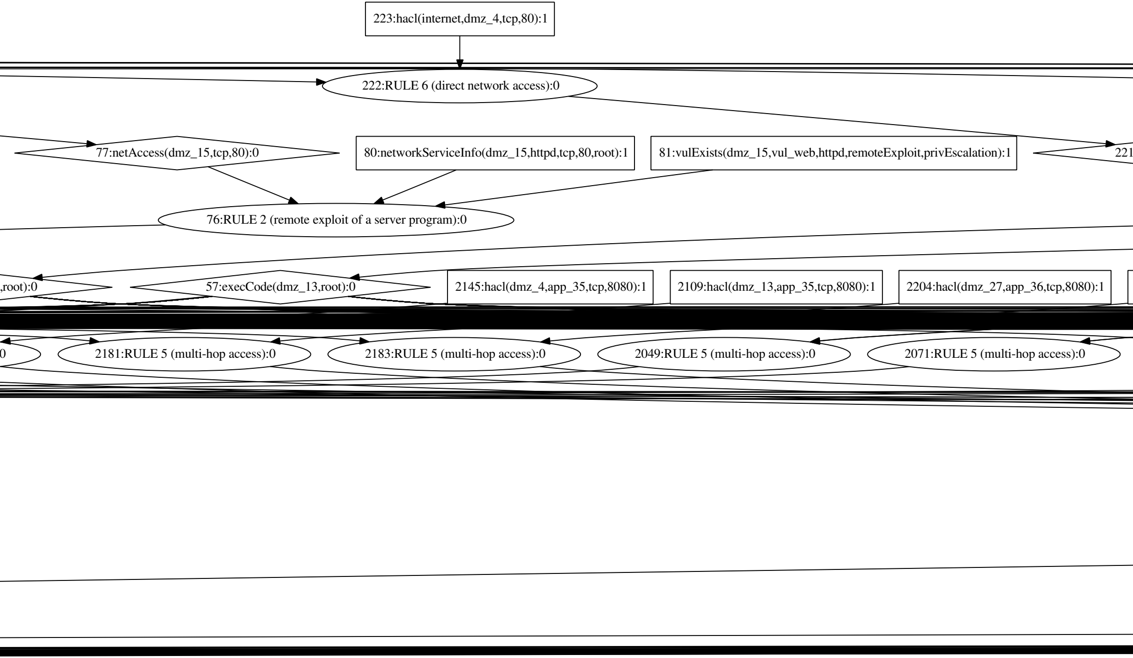

160:hacl(internet,dmz_23,tcp,80):1
159:RULE 6 (direct network access):0
remoteExploit,privEscalation):1
50:netAccess(dmz_12,tcp,80):0
53:networkServiceInfo(dmz_12,httpd,tcp,80,root):1
54:vulExists(dmz_12,vul_web,httpd,rer
49:RULE 2 (remote exploit of a server program):0
2042:hacl(dmz_12,app_34,tcp,8080):1
48:execCode(dmz_12,root):0
2107:hacl(dmz_12,app_35,tcp,8080):1
2066:hacl(dmz_23,app_34,tcp,8080):1
2041:RULE 5 (multi-hop access):0
2106:RULE 5 (multi-hop access):0
2065:RULE 5 (multi-hop access):0
2067:RULE 5 (multi-hop access):0
2097:netAccess(app_35,tcp,8080):0
2156:networkServiceInfo(app_35,tomcat,tcp,8080,root):1
2157:vulExists(app_35,vul_app,tomcat,remoteExploit,privEscalation):1
2096:RULE 2 (remote exploit of a server program):0
2095:execCode(app_35,root):0
stgres,tcp,5432,root):1
3979:vulExists(db_59,vul_db,postgres,remoteExploit,privEscalation):1
server program):0
root):0

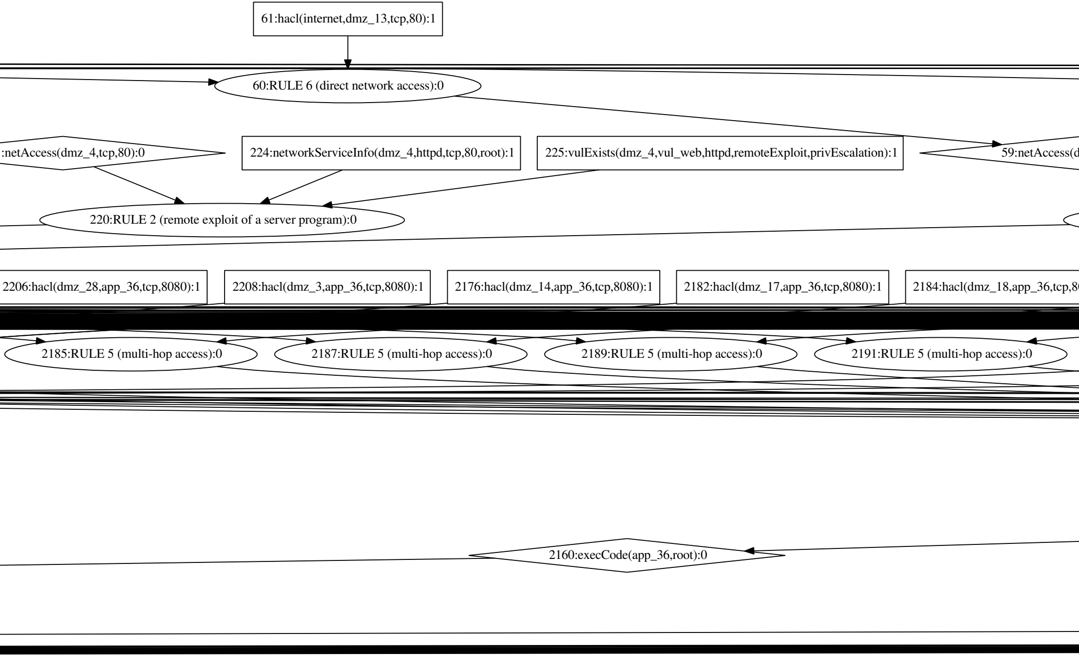

169:hacl(internet,dmz_24,tcp,80):1
168:RULE 6 (direct network access):0
moteExploit,privEscalation):1
158:netAccess(dmz_23,tcp,80):0
161:networkServiceInfo(dmz_23,httpd,tcp,80,root):1
162:vulExists(dmz_23,vul_web,httpd,remoteExploit,
157:RULE 2 (remote exploit of a server program):0
2068:hacl(dmz_24,app_34,tcp,8080):1
156:execCode(dmz_23,root):0
2131:hacl(dmz_23,app_35,tcp,8080):1
165:execCode(dmz_24,root):0
2130:RULE 5 (multi-hop access):0
2132:RULE 5 (multi-hop access):0
2089:RULE 5 (multi-hop access):0
2069:RULE 5 (multi-hop access):0

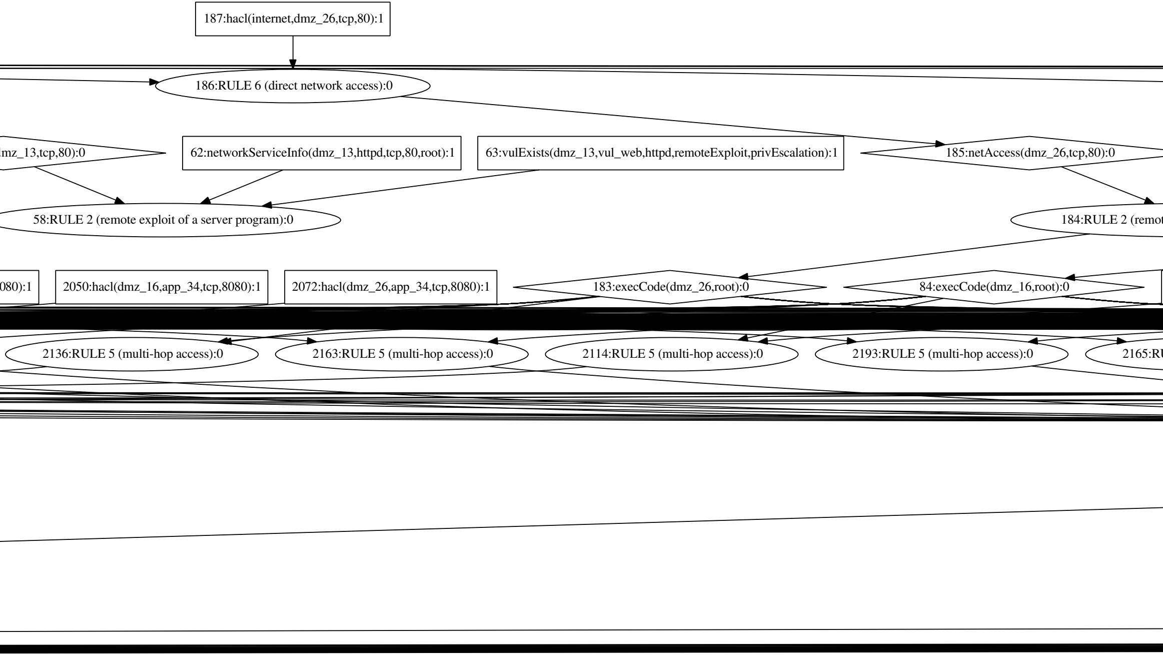

178:hacl(internet,dmz_25,tcp,80):1
177:RULE 6 (direct network access):0
,privEscalation):1
167:netAccess(dmz_24,tcp,80):0
170:networkServiceInfo(dmz_24,httpd,tcp,80,root):1
171:vulExists(dmz_24,vul_web,httpd,remoteExploit,privEscal
166:RULE 2 (remote exploit of a server program):0
2133:hacl(dmz_24,app_35,tcp,8080):1
2090:hacl(dmz_9,app_34,tcp,8080):1
174:execCode(dmz_25,root):0
264:execCode(dmz_9,root):0
2154:RULE 5 (multi-hop access):0
2134:RULE 5 (multi-hop access):0
2047:RULE 5 (multi-hop access):0
2112:RULE 5 (multi-hop access):0
2159:hacl(app_36,db_59,tcp,5432):1
2158:RULE 5 (multi-hop access):0

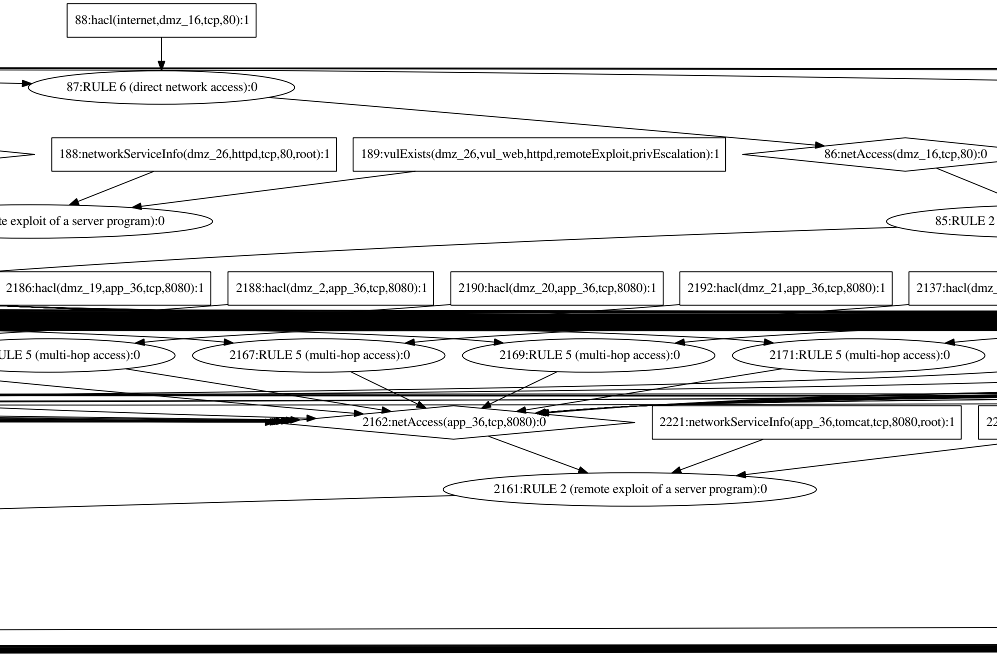

268:hacl(internet,dmz_9,tcp,80):1
267:RULE 6 (direct network access):0
lation):1
176:netAccess(dmz_25,tcp,80):0
179:networkServiceInfo(dmz_25,httpd,tcp,80,root):1
180:vulExists(dmz_25,vul_web,httpd,remoteExploit,privEscalation):1
175:RULE 2 (remote exploit of a server program):0
2070:hacl(dmz_25,app_34,tcp,8080):1
2155:hacl(dmz_9,app_35,tcp,8080):1
2135:hacl(dmz_25,app_35,tcp,8080):1
2048:hacl(dmz_15,app_34,tcp,8080):1
75:execCode(dm
2043:RULE 5 (multi-hop access):0
2079:RULE 5 (multi-hop access):0
2144:RULE 5 (multi-hop access):0
2108:RULE 5 (multi-hop access):0

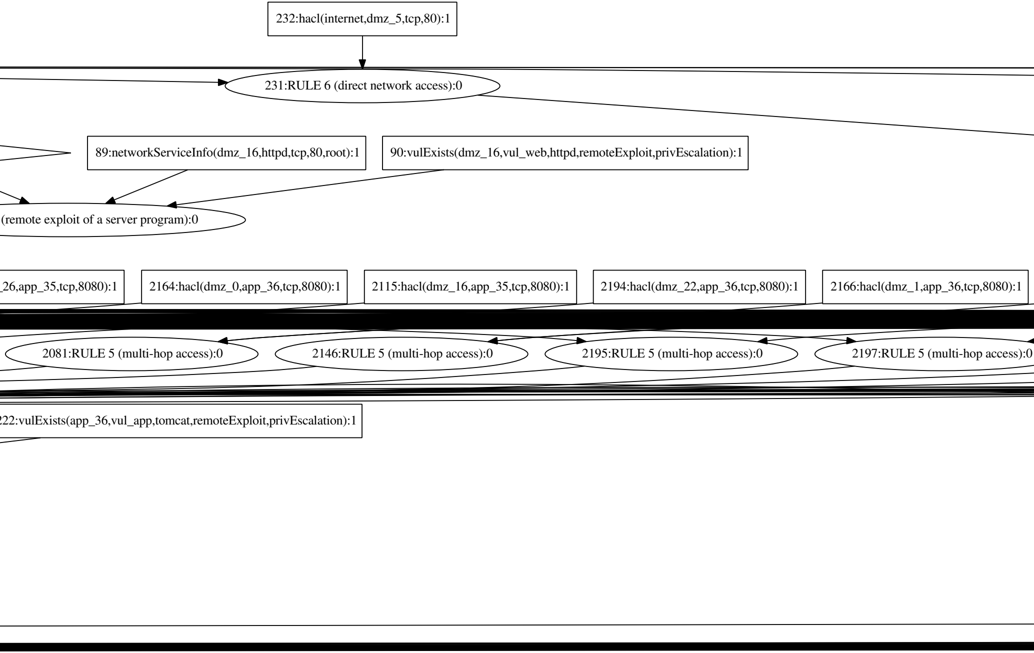

79:hacl(internet,dmz_15,tcp,80):1
78:RULE 6 (direct network access):0
266:netAccess(dmz_9,tcp,80):0
269:networkServiceInfo(dmz_9,httpd,tcp,80,root):1
270:vulExists(dmz_9,vul_web,httpd,remoteExploit,privEscalation):1
265:RULE 2 (remote exploit of a server program):0
z_15,root):0
2113:hacl(dmz_15,app_35,tcp,8080):1
2044:hacl(dmz_13,app_34,tcp,8080):1
2080:hacl(dmz_4,app_34,tcp,8080):1
219:execCode(dmz_4
2203:RULE 5 (multi-hop access):0
2205:RULE 5 (multi-hop access):0
2207:RULE 5 (multi-hop access):0
2175:RULE 5 (multi-hop access):

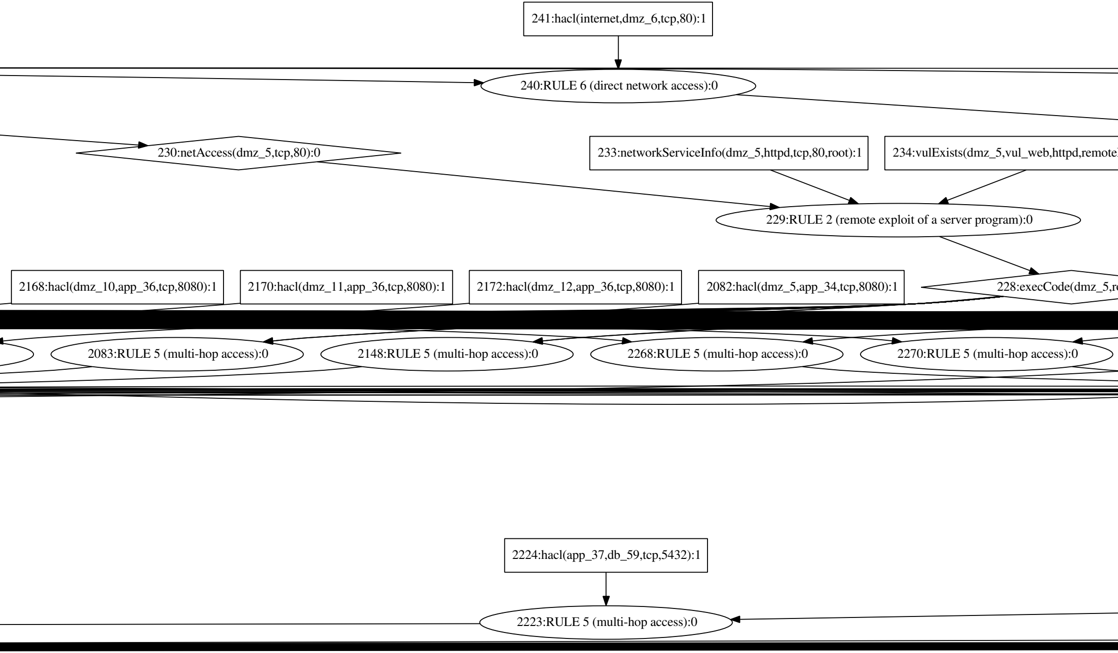

223:hacl(internet,dmz_4,tcp,80):1
222:RULE 6 (direct network access):0
77:netAccess(dmz_15,tcp,80):0
80:networkServiceInfo(dmz_15,httpd,tcp,80,root):1
81:vulExists(dmz_15,vul_web,httpd,remoteExploit,privEscalation):1
221
76:RULE 2 (remote exploit of a server program):0
,root):0
57:execCode(dmz_13,root):0
2145:hacl(dmz_4,app_35,tcp,8080):1
2109:hacl(dmz_13,app_35,tcp,8080):1
2204:hacl(dmz_27,app_36,tcp,8080):1
0
2181:RULE 5 (multi-hop access):0
2183:RULE 5 (multi-hop access):0
2049:RULE 5 (multi-hop access):0
2071:RULE 5 (multi-hop access):0

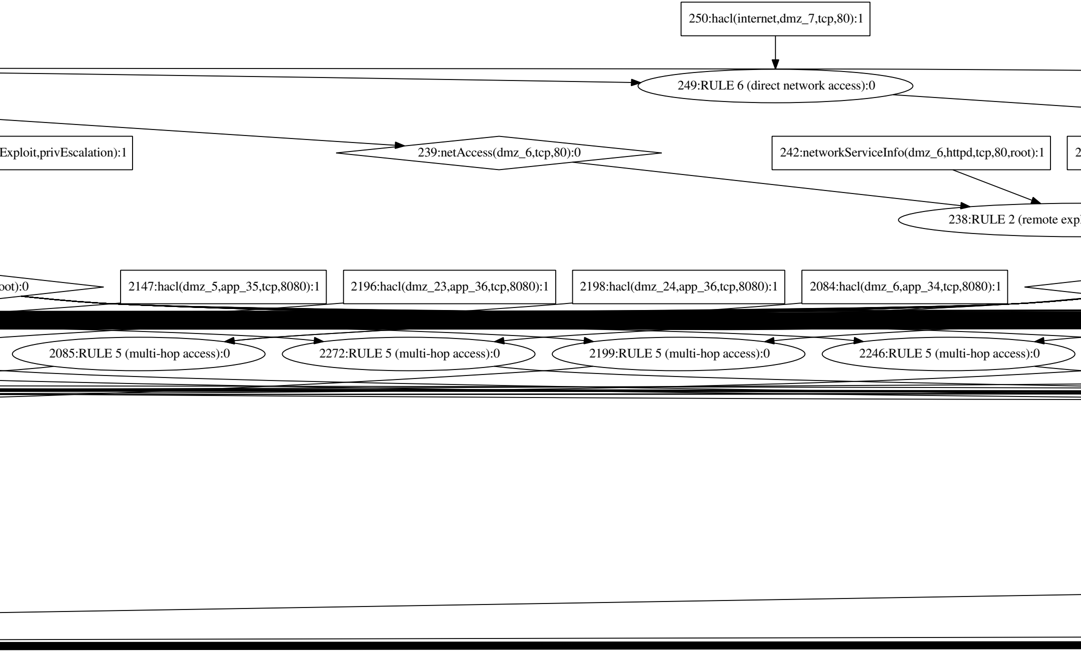

61:hacl(internet,dmz_13,tcp,80):1
60:RULE 6 (direct network access):0
:netAccess(dmz_4,tcp,80):0
224:networkServiceInfo(dmz_4,httpd,tcp,80,root):1
225:vulExists(dmz_4,vul_web,httpd,remoteExploit,privEscalation):1
59:netAccess(d
220:RULE 2 (remote exploit of a server program):0
2206:hacl(dmz_28,app_36,tcp,8080):1
2208:hacl(dmz_3,app_36,tcp,8080):1
2176:hacl(dmz_14,app_36,tcp,8080):1
2182:hacl(dmz_17,app_36,tcp,8080):1
2184:hacl(dmz_18,app_36,tcp,8
2185:RULE 5 (multi-hop access):0
2187:RULE 5 (multi-hop access):0
2189:RULE 5 (multi-hop access):0
2191:RULE 5 (multi-hop access):0
2160:execCode(app_36,root):0

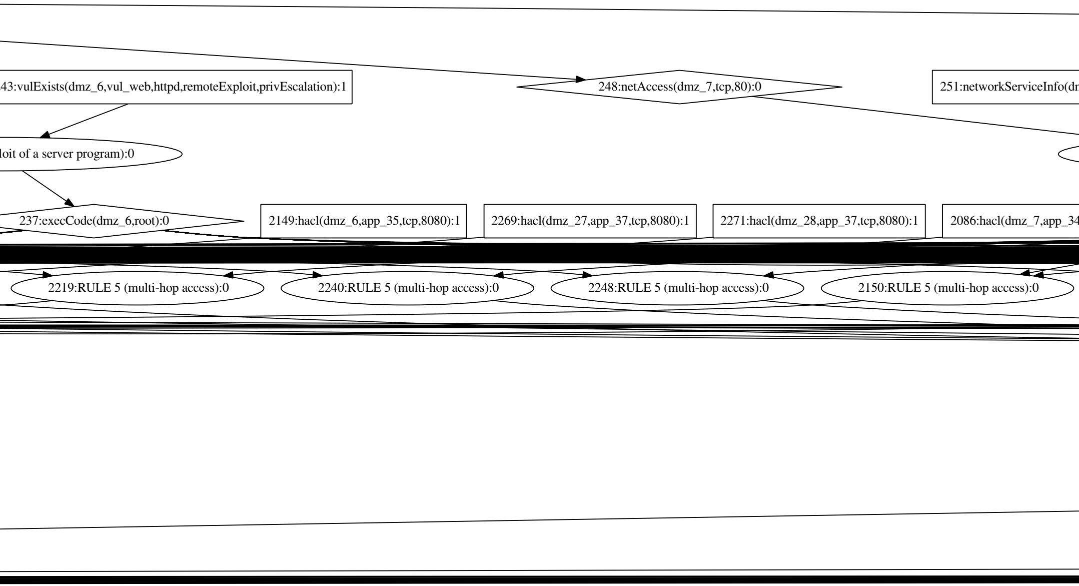

187:hacl(internet,dmz_26,tcp,80):1
186:RULE 6 (direct network access):0
lmz_13,tcp,80):0
62:networkServiceInfo(dmz_13,httpd,tcp,80,root):1
63:vulExists(dmz_13,vul_web,httpd,remoteExploit,privEscalation):1
185:netAccess(dmz_26,tcp,80):0
58:RULE 2 (remote exploit of a server program):0
184:RULE 2 (remot
080):1
2050:hacl(dmz_16,app_34,tcp,8080):1
2072:hacl(dmz_26,app_34,tcp,8080):1
183:execCode(dmz_26,root):0
84:execCode(dmz_16,root):0
2136:RULE 5 (multi-hop access):0
2163:RULE 5 (multi-hop access):0
2114:RULE 5 (multi-hop access):0
2193:RULE 5 (multi-hop access):0
2165:RU

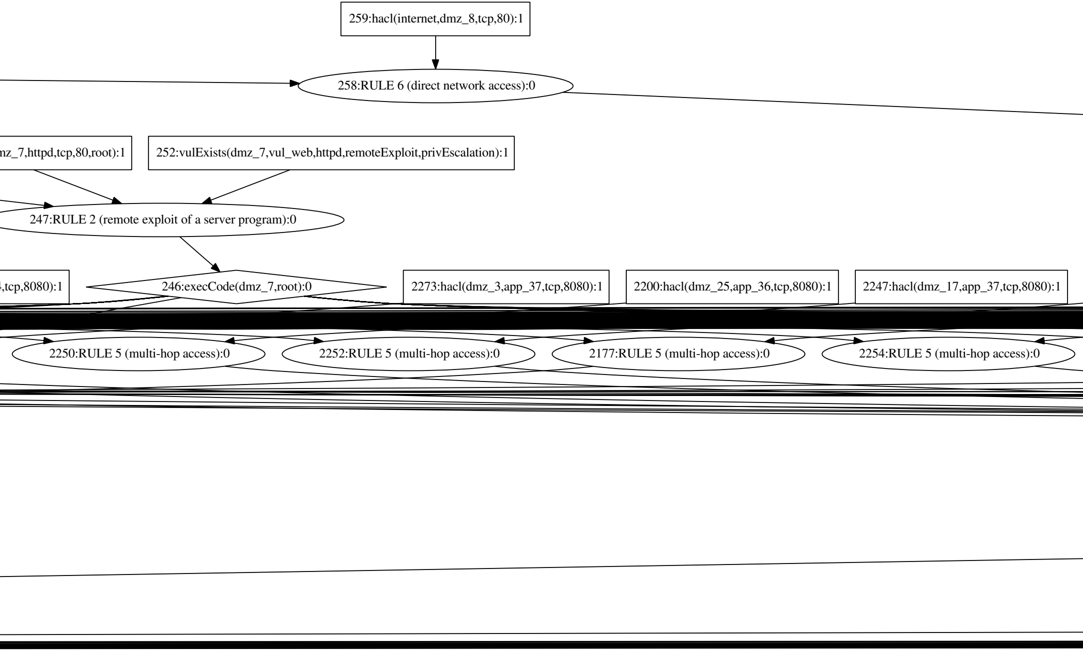

88:hacl(internet,dmz_16,tcp,80):1
87:RULE 6 (direct network access):0
188:networkServiceInfo(dmz_26,httpd,tcp,80,root):1
189:vulExists(dmz_26,vul_web,httpd,remoteExploit,privEscalation):1
86:netAccess(dmz_16,tcp,80):0
te exploit of a server program):0
85:RULE 2
2186:hacl(dmz_19,app_36,tcp,8080):1
2188:hacl(dmz_2,app_36,tcp,8080):1
2190:hacl(dmz_20,app_36,tcp,8080):1
2192:hacl(dmz_21,app_36,tcp,8080):1
2137:hacl(dmz_
ULE 5 (multi-hop access):0
2167:RULE 5 (multi-hop access):0
2169:RULE 5 (multi-hop access):0
2171:RULE 5 (multi-hop access):0
2162:netAccess(app_36,tcp,8080):0
2221:networkServiceInfo(app_36,tomcat,tcp,8080,root):1
22
2161:RULE 2 (remote exploit of a server program):0

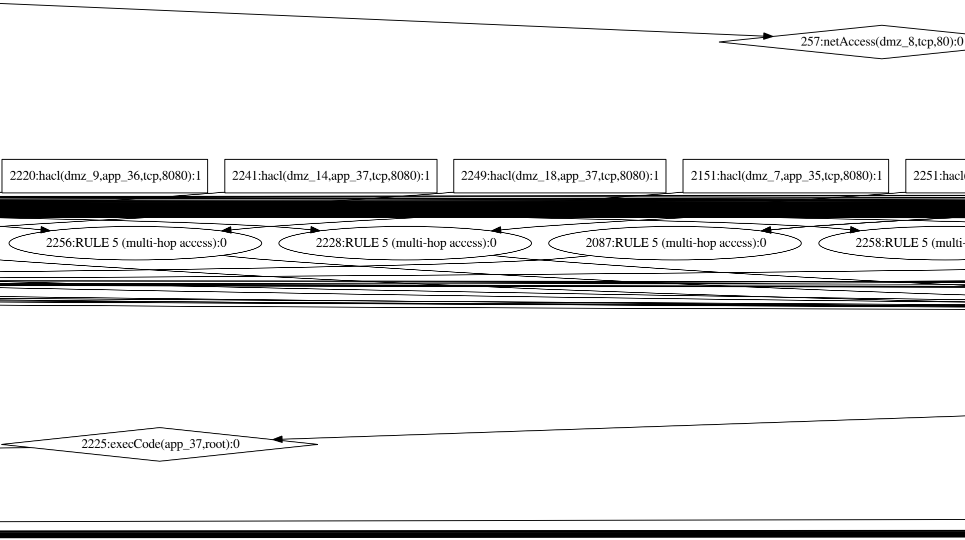

232:hacl(internet,dmz_5,tcp,80):1
231:RULE 6 (direct network access):0
89:networkServiceInfo(dmz_16,httpd,tcp,80,root):1
90:vulExists(dmz_16,vul_web,httpd,remoteExploit,privEscalation):1
(remote exploit of a server program):0
26,app_35,tcp,8080):1
2164:hacl(dmz_0,app_36,tcp,8080):1
2115:hacl(dmz_16,app_35,tcp,8080):1
2194:hacl(dmz_22,app_36,tcp,8080):1
2166:hacl(dmz_1,app_36,tcp,8080):1
2081:RULE 5 (multi-hop access):0
2146:RULE 5 (multi-hop access):0
2195:RULE 5 (multi-hop access):0
2197:RULE 5 (multi-hop access):0
22:vulExists(app_36,vul_app,tomcat,remoteExploit,privEscalation):1

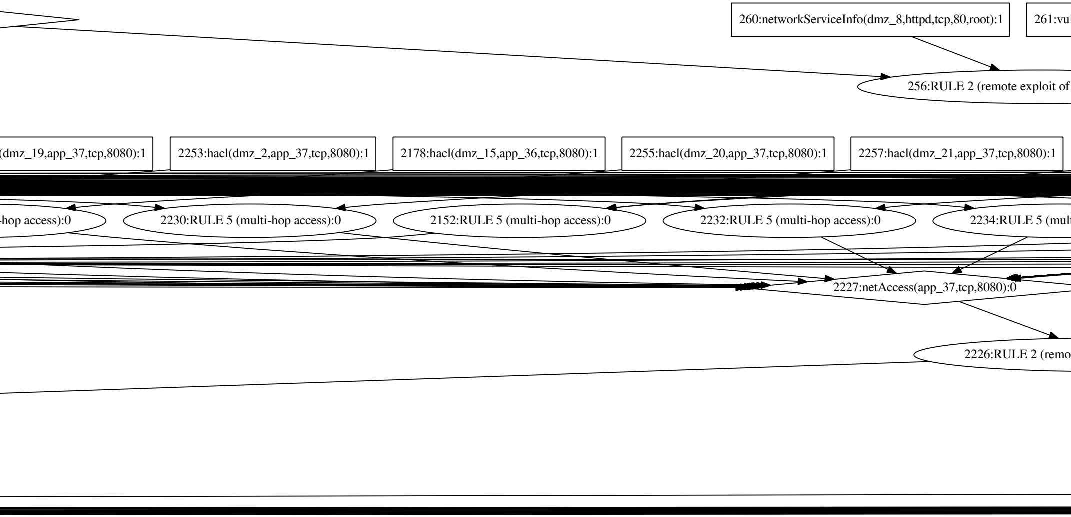

241:hacl(internet,dmz_6,tcp,80):1
240:RULE 6 (direct network access):0
230:netAccess(dmz_5,tcp,80):0
233:networkServiceInfo(dmz_5,httpd,tcp,80,root):1
234:vulExists(dmz_5,vul_web,httpd,remote
229:RULE 2 (remote exploit of a server program):0
2168:hacl(dmz_10,app_36,tcp,8080):1
2170:hacl(dmz_11,app_36,tcp,8080):1
2172:hacl(dmz_12,app_36,tcp,8080):1
2082:hacl(dmz_5,app_34,tcp,8080):1
228:execCode(dmz_5,r
2083:RULE 5 (multi-hop access):0
2148:RULE 5 (multi-hop access):0
2268:RULE 5 (multi-hop access):0
2270:RULE 5 (multi-hop access):0
2224:hacl(app_37,db_59,tcp,5432):1
2223:RULE 5 (multi-hop access):0

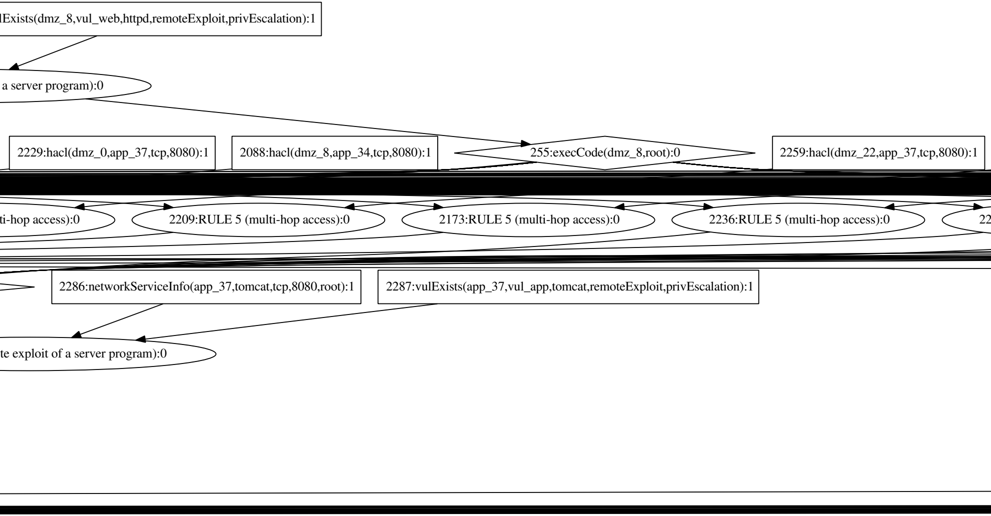

250:hacl(internet,dmz_7,tcp,80):1
249:RULE 6 (direct network access):0
Exploit,privEscalation):1
239:netAccess(dmz_6,tcp,80):0
242:networkServiceInfo(dmz_6,httpd,tcp,80,root):1
238:RULE 2 (remote exp
oot):0
2147:hacl(dmz_5,app_35,tcp,8080):1
2196:hacl(dmz_23,app_36,tcp,8080):1
2198:hacl(dmz_24,app_36,tcp,8080):1
2084:hacl(dmz_6,app_34,tcp,8080):1
2085:RULE 5 (multi-hop access):0
2272:RULE 5 (multi-hop access):0
2199:RULE 5 (multi-hop access):0
2246:RULE 5 (multi-hop access):0

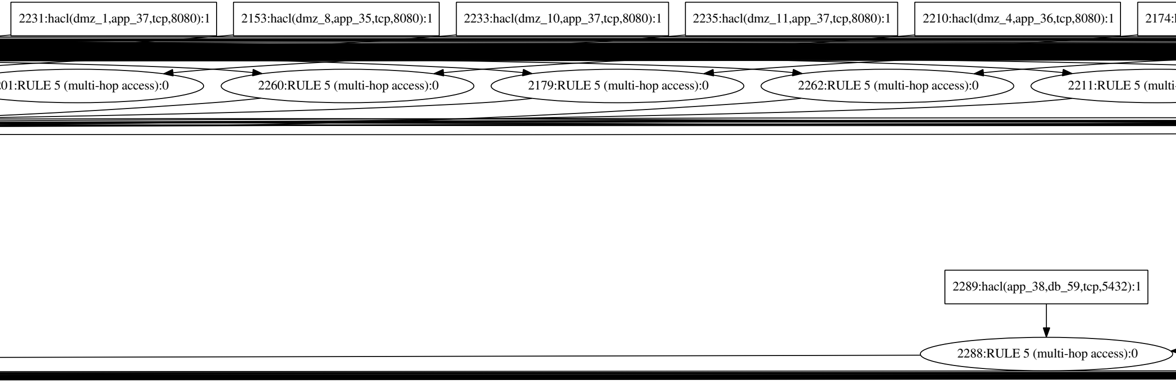

43:vulExists(dmz_6,vul_web,httpd,remoteExploit,privEscalation):1
248:netAccess(dmz_7,tcp,80):0
251:networkServiceInfo(dr
loit of a server program):0
237:execCode(dmz_6,root):0
2149:hacl(dmz_6,app_35,tcp,8080):1
2269:hacl(dmz_27,app_37,tcp,8080):1
2271:hacl(dmz_28,app_37,tcp,8080):1
2086:hacl(dmz_7,app_34
2219:RULE 5 (multi-hop access):0
2240:RULE 5 (multi-hop access):0
2248:RULE 5 (multi-hop access):0
2150:RULE 5 (multi-hop access):0

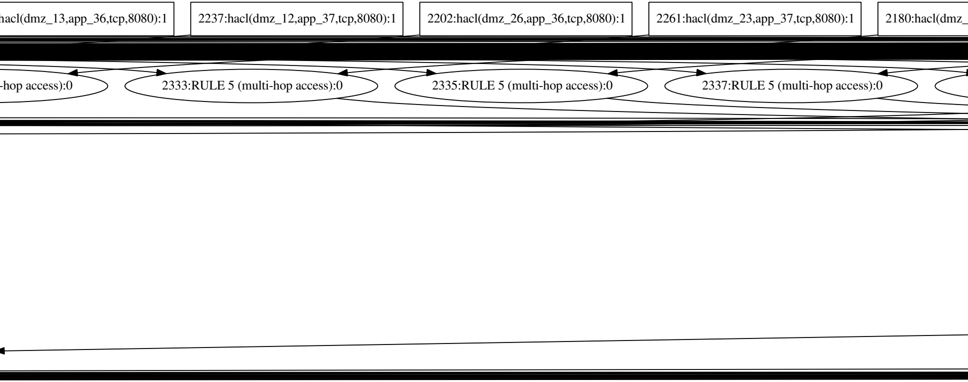

259:hacl(internet,dmz_8,tcp,80):1
258:RULE 6 (direct network access):0
mz_7,httpd,tcp,80,root):1
252:vulExists(dmz_7,vul_web,httpd,remoteExploit,privEscalation):1
247:RULE 2 (remote exploit of a server program):0
,tcp,8080):1
246:execCode(dmz_7,root):0
2273:hacl(dmz_3,app_37,tcp,8080):1
2200:hacl(dmz_25,app_36,tcp,8080):1
2247:hacl(dmz_17,app_37,tcp,8080):1
2250:RULE 5 (multi-hop access):0
2252:RULE 5 (multi-hop access):0
2177:RULE 5 (multi-hop access):0
2254:RULE 5 (multi-hop access):0

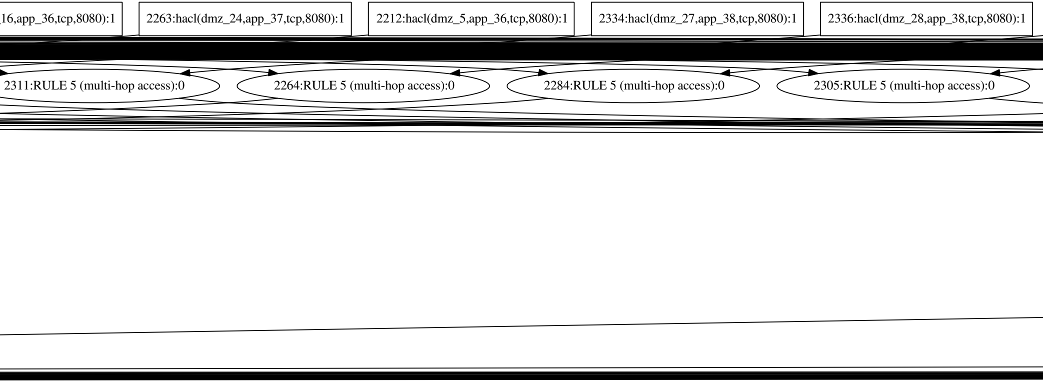

257:netAccess(dmz_8,tcp,80):0
2220:hacl(dmz_9,app_36,tcp,8080):1
2241:hacl(dmz_14,app_37,tcp,8080):1
2249:hacl(dmz_18,app_37,tcp,8080):1
2151:hacl(dmz_7,app_35,tcp,8080):1
2251:hacl
2256:RULE 5 (multi-hop access):0
2228:RULE 5 (multi-hop access):0
2087:RULE 5 (multi-hop access):0
2258:RULE 5 (multi-
2225:execCode(app_37,root):0

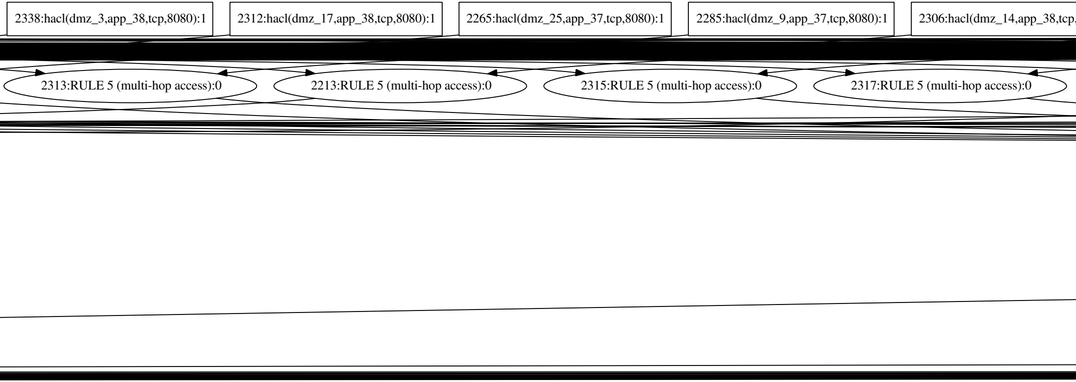

260:networkServiceInfo(dmz_8,httpd,tcp,80,root):1
261:vu
256:RULE 2 (remote exploit of
(dmz_19,app_37,tcp,8080):1
2253:hacl(dmz_2,app_37,tcp,8080):1
2178:hacl(dmz_15,app_36,tcp,8080):1
2255:hacl(dmz_20,app_37,tcp,8080):1
2257:hacl(dmz_21,app_37,tcp,8080):1
-hop access):0
2230:RULE 5 (multi-hop access):0
2152:RULE 5 (multi-hop access):0
2232:RULE 5 (multi-hop access):0
2234:RULE 5 (mult
2227:netAccess(app_37,tcp,8080):0
2226:RULE 2 (remo

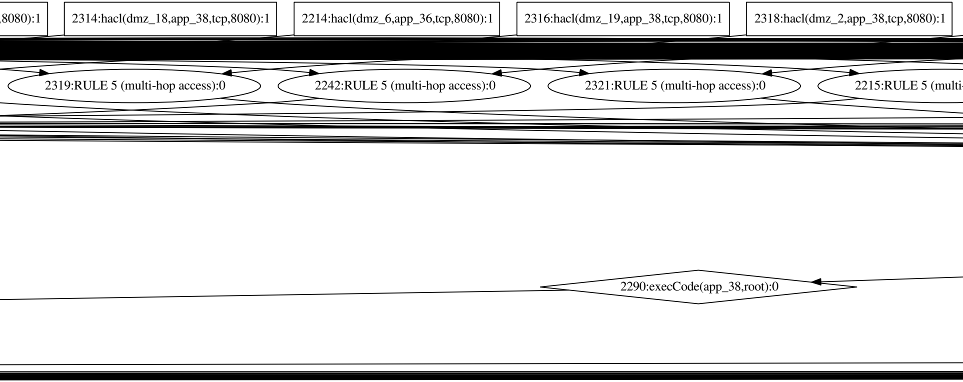

lExists(dmz_8,vul_web,httpd,remoteExploit,privEscalation):1
a server program):0
2229:hacl(dmz_0,app_37,tcp,8080):1
2088:hacl(dmz_8,app_34,tcp,8080):1
255:execCode(dmz_8,root):0
2259:hacl(dmz_22,app_37,tcp,8080):1
ti-hop access):0
2209:RULE 5 (multi-hop access):0
2173:RULE 5 (multi-hop access):0
2236:RULE 5 (multi-hop access):0
22
2286:networkServiceInfo(app_37,tomcat,tcp,8080,root):1
2287:vulExists(app_37,vul_app,tomcat,remoteExploit,privEscalation):1
te exploit of a server program):0

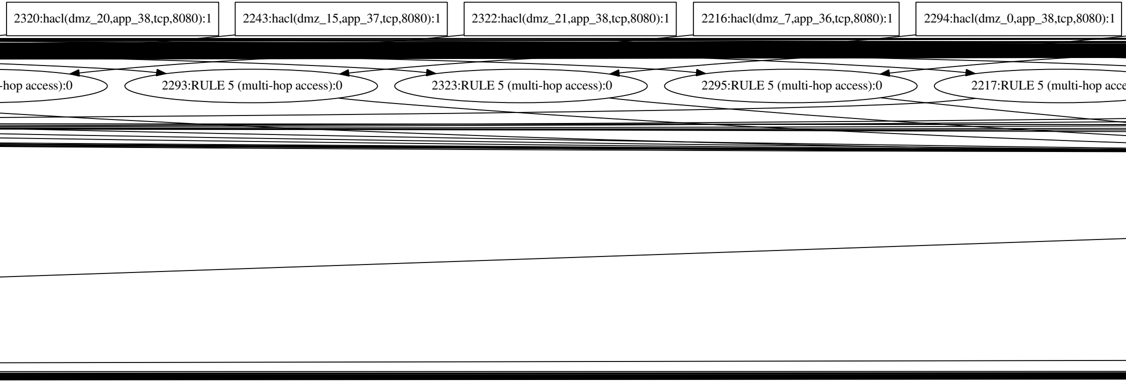

2231:hacl(dmz_1,app_37,tcp,8080):1
2153:hacl(dmz_8,app_35,tcp,8080):1
2233:hacl(dmz_10,app_37,tcp,8080):1
2235:hacl(dmz_11,app_37,tcp,8080):1
2210:hacl(dmz_4,app_36,tcp,8080):1
2174:
01:RULE 5 (multi-hop access):0
2260:RULE 5 (multi-hop access):0
2179:RULE 5 (multi-hop access):0
2262:RULE 5 (multi-hop access):0
2211:RULE 5 (multi-
2289:hacl(app_38,db_59,tcp,5432):1
2288:RULE 5 (multi-hop access):0

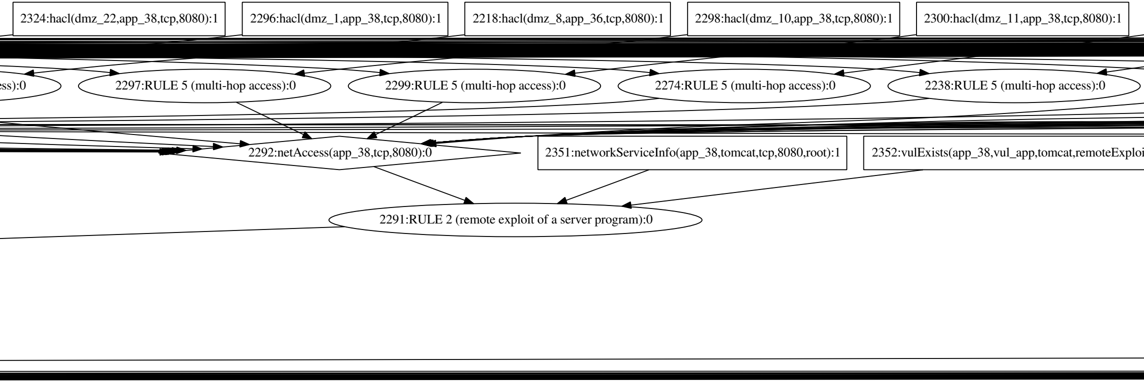

hacl(dmz_13,app_36,tcp,8080):1
2237:hacl(dmz_12,app_37,tcp,8080):1
2202:hacl(dmz_26,app_36,tcp,8080):1
2261:hacl(dmz_23,app_37,tcp,8080):1
2180:hacl(dmz_
-hop access):0
2333:RULE 5 (multi-hop access):0
2335:RULE 5 (multi-hop access):0
2337:RULE 5 (multi-hop access):0

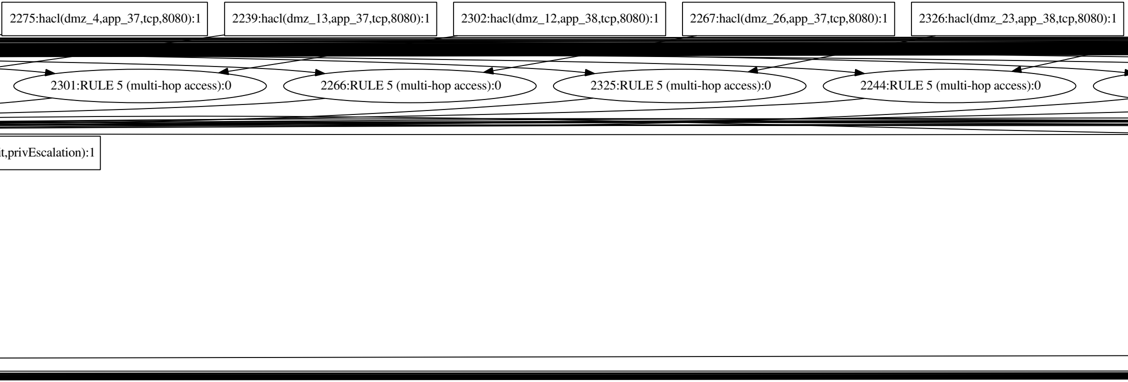

16,app_36,tcp,8080):1
2263:hacl(dmz_24,app_37,tcp,8080):1
2212:hacl(dmz_5,app_36,tcp,8080):1
2334:hacl(dmz_27,app_38,tcp,8080):1
2336:hacl(dmz_28,app_38,tcp,8080):1
2311:RULE 5 (multi-hop access):0
2264:RULE 5 (multi-hop access):0
2284:RULE 5 (multi-hop access):0
2305:RULE 5 (multi-hop access):0

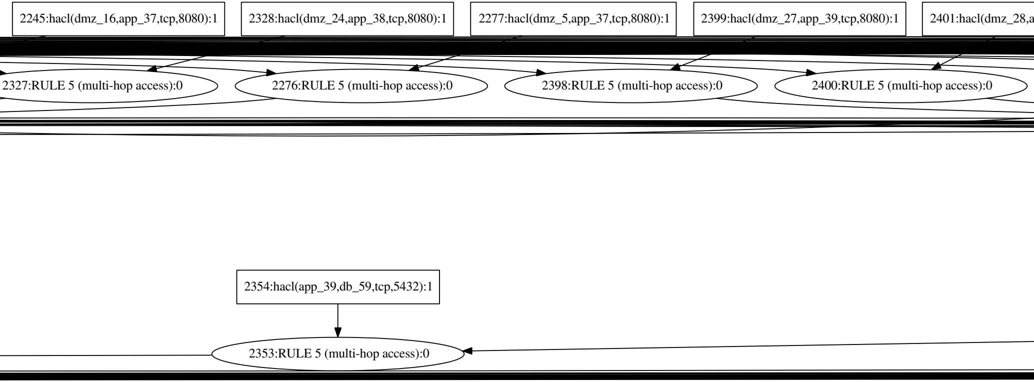

2338:hacl(dmz_3,app_38,tcp,8080):1
2312:hacl(dmz_17,app_38,tcp,8080):1
2265:hacl(dmz_25,app_37,tcp,8080):1
2285:hacl(dmz_9,app_37,tcp,8080):1
2306:hacl(dmz_14,app_38,tcp,
2313:RULE 5 (multi-hop access):0
2213:RULE 5 (multi-hop access):0
2315:RULE 5 (multi-hop access):0
2317:RULE 5 (multi-hop access):0

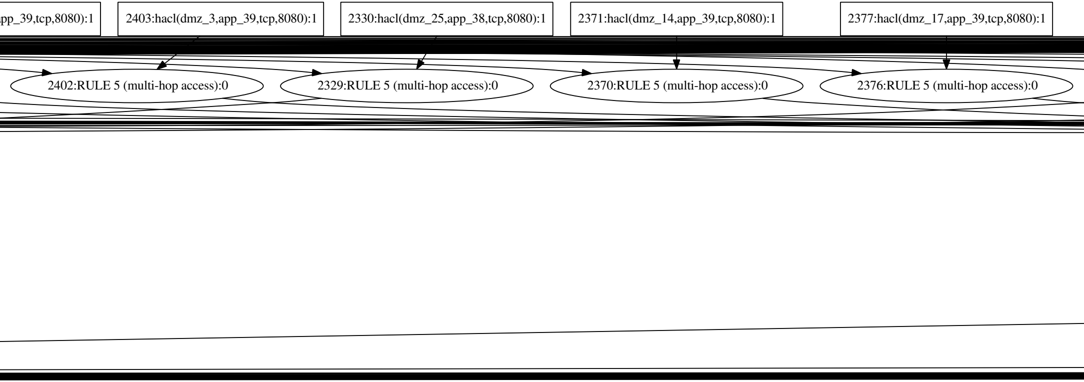

8080):1
2314:hacl(dmz_18,app_38,tcp,8080):1
2214:hacl(dmz_6,app_36,tcp,8080):1
2316:hacl(dmz_19,app_38,tcp,8080):1
2318:hacl(dmz_2,app_38,tcp,8080):1
2319:RULE 5 (multi-hop access):0
2242:RULE 5 (multi-hop access):0
2321:RULE 5 (multi-hop access):0
2215:RULE 5 (multi-
2290:execCode(app_38,root):0

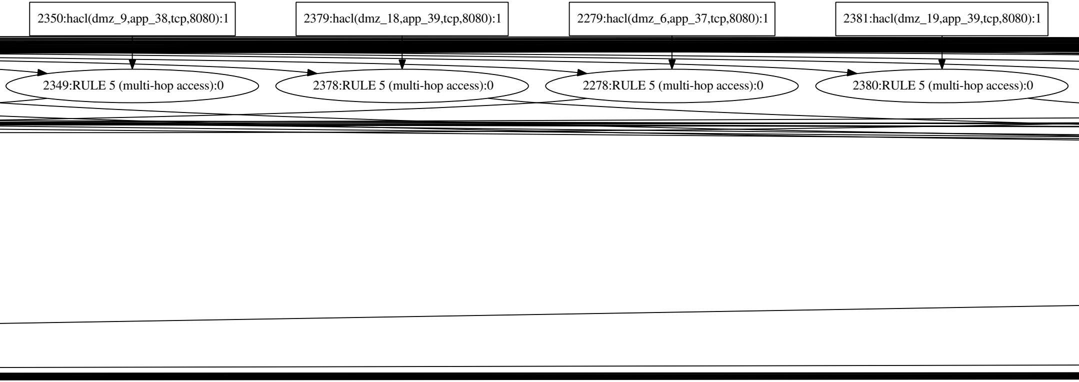

2320:hacl(dmz_20,app_38,tcp,8080):1
2243:hacl(dmz_15,app_37,tcp,8080):1
2322:hacl(dmz_21,app_38,tcp,8080):1
2216:hacl(dmz_7,app_36,tcp,8080):1
2294:hacl(dmz_0,app_38,tcp,8080):1
-hop access):0
2293:RULE 5 (multi-hop access):0
2323:RULE 5 (multi-hop access):0
2295:RULE 5 (multi-hop access):0
2217:RULE 5 (multi-hop acce

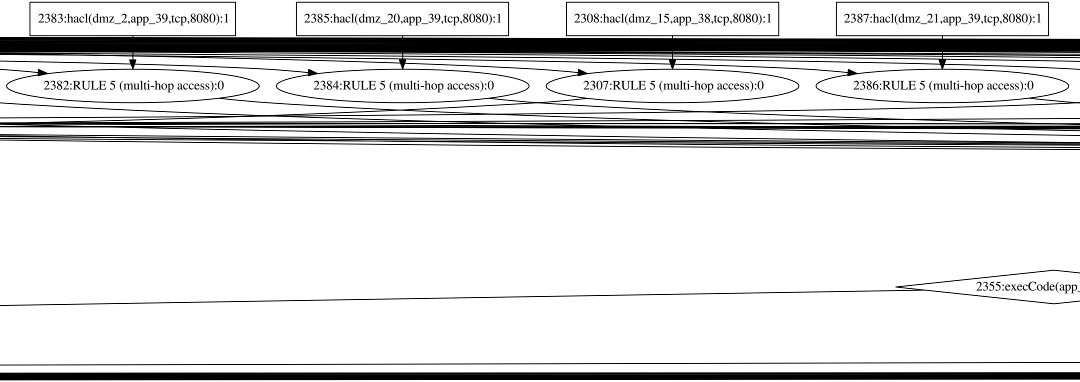

2324:hacl(dmz_22,app_38,tcp,8080):1
2296:hacl(dmz_1,app_38,tcp,8080):1
2218:hacl(dmz_8,app_36,tcp,8080):1
2298:hacl(dmz_10,app_38,tcp,8080):1
2300:hacl(dmz_11,app_38,tcp,8080):1
ss):0
2297:RULE 5 (multi-hop access):0
2299:RULE 5 (multi-hop access):0
2274:RULE 5 (multi-hop access):0
2238:RULE 5 (multi-hop access):0
2292:netAccess(app_38,tcp,8080):0
2351:networkServiceInfo(app_38,tomcat,tcp,8080,root):1
2352:vulExists(app_38,vul_app,tomcat,remoteExploi
2291:RULE 2 (remote exploit of a server program):0

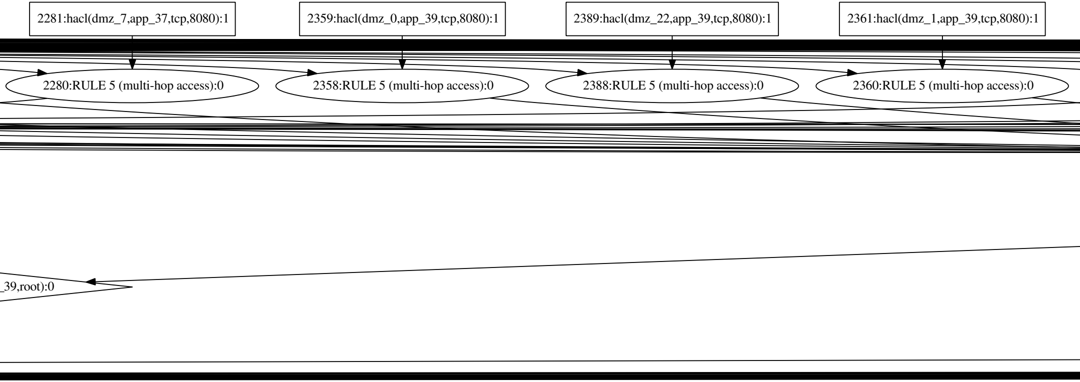

2275:hacl(dmz_4,app_37,tcp,8080):1
2239:hacl(dmz_13,app_37,tcp,8080):1
2302:hacl(dmz_12,app_38,tcp,8080):1
2267:hacl(dmz_26,app_37,tcp,8080):1
2326:hacl(dmz_23,app_38,tcp,8080):1
2301:RULE 5 (multi-hop access):0
2266:RULE 5 (multi-hop access):0
2325:RULE 5 (multi-hop access):0
2244:RULE 5 (multi-hop access):0
t,privEscalation):1

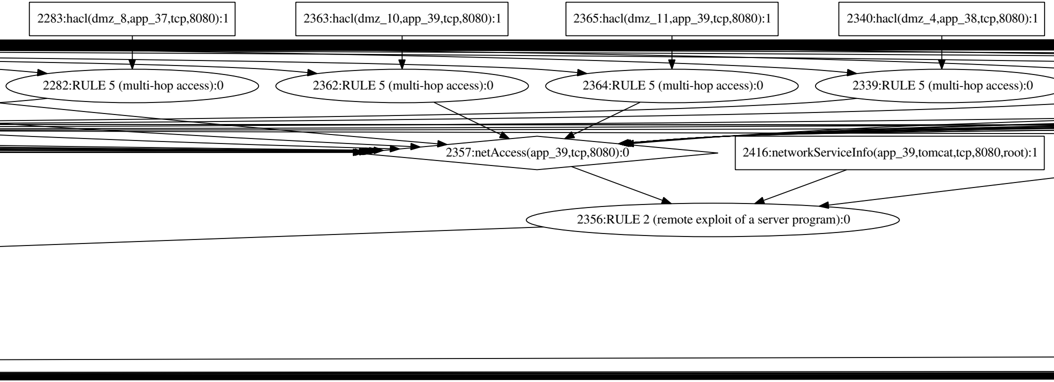

2245:hacl(dmz_16,app_37,tcp,8080):1
2328:hacl(dmz_24,app_38,tcp,8080):1
2277:hacl(dmz_5,app_37,tcp,8080):1
2399:hacl(dmz_27,app_39,tcp,8080):1
2401:hacl(dmz_28,a
2327:RULE 5 (multi-hop access):0
2276:RULE 5 (multi-hop access):0
2398:RULE 5 (multi-hop access):0
2400:RULE 5 (multi-hop access):0
2354:hacl(app_39,db_59,tcp,5432):1
2353:RULE 5 (multi-hop access):0

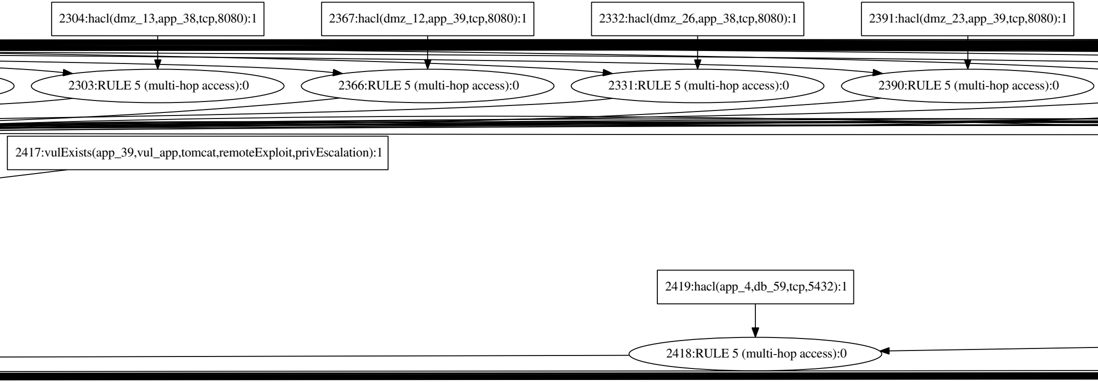

pp_39,tcp,8080):1
2403:hacl(dmz_3,app_39,tcp,8080):1
2330:hacl(dmz_25,app_38,tcp,8080):1
2371:hacl(dmz_14,app_39,tcp,8080):1
2377:hacl(dmz_17,app_39,tcp,8080):1
2402:RULE 5 (multi-hop access):0
2329:RULE 5 (multi-hop access):0
2370:RULE 5 (multi-hop access):0
2376:RULE 5 (multi-hop access):0

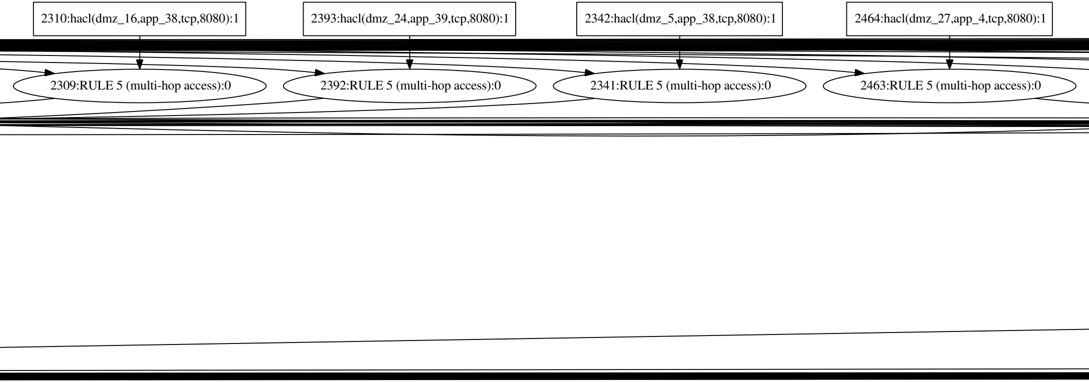

2350:hacl(dmz_9,app_38,tcp,8080):1
2379:hacl(dmz_18,app_39,tcp,8080):1
2279:hacl(dmz_6,app_37,tcp,8080):1
2381:hacl(dmz_19,app_39,tcp,8080):1
2349:RULE 5 (multi-hop access):0
2378:RULE 5 (multi-hop access):0
2278:RULE 5 (multi-hop access):0
2380:RULE 5 (multi-hop access):0

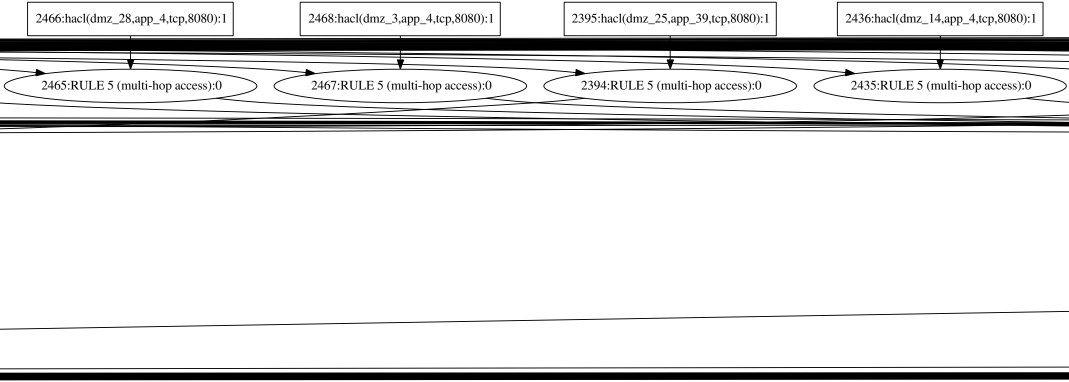

2383:hacl(dmz_2,app_39,tcp,8080):1
2385:hacl(dmz_20,app_39,tcp,8080):1
2308:hacl(dmz_15,app_38,tcp,8080):1
2387:hacl(dmz_21,app_39,tcp,8080):1
2382:RULE 5 (multi-hop access):0
2384:RULE 5 (multi-hop access):0
2307:RULE 5 (multi-hop access):0
2386:RULE 5 (multi-hop access):0
2355:execCode(app_

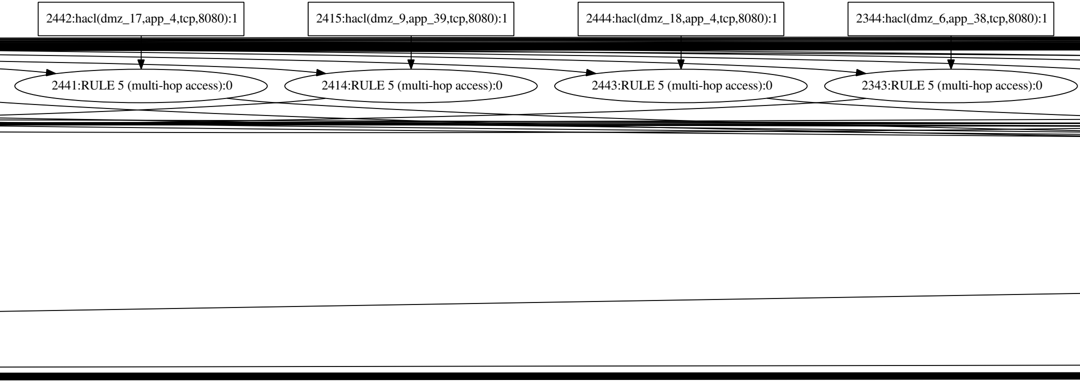

2281:hacl(dmz_7,app_37,tcp,8080):1
2359:hacl(dmz_0,app_39,tcp,8080):1
2389:hacl(dmz_22,app_39,tcp,8080):1
2361:hacl(dmz_1,app_39,tcp,8080):1
2280:RULE 5 (multi-hop access):0
2358:RULE 5 (multi-hop access):0
2388:RULE 5 (multi-hop access):0
2360:RULE 5 (multi-hop access):0
_39,root):0

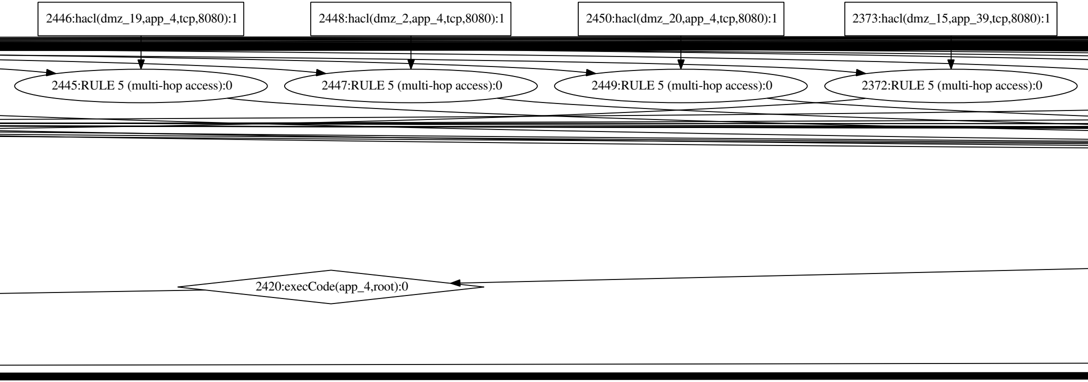

2283:hacl(dmz_8,app_37,tcp,8080):1
2363:hacl(dmz_10,app_39,tcp,8080):1
2365:hacl(dmz_11,app_39,tcp,8080):1
2340:hacl(dmz_4,app_38,tcp,8080):1
2282:RULE 5 (multi-hop access):0
2362:RULE 5 (multi-hop access):0
2364:RULE 5 (multi-hop access):0
2339:RULE 5 (multi-hop access):0
2357:netAccess(app_39,tcp,8080):0
2416:networkServiceInfo(app_39,tomcat,tcp,8080,root):1
2356:RULE 2 (remote exploit of a server program):0

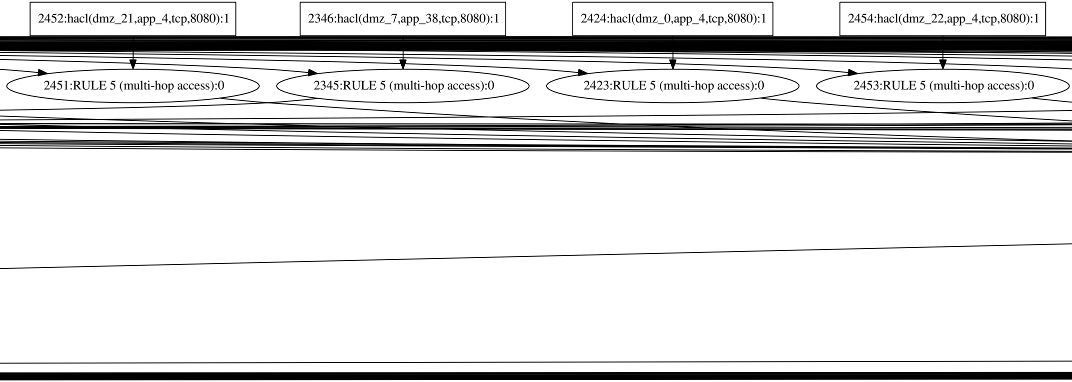

2304:hacl(dmz_13,app_38,tcp,8080):1
2367:hacl(dmz_12,app_39,tcp,8080):1
2332:hacl(dmz_26,app_38,tcp,8080):1
2391:hacl(dmz_23,app_39,tcp,8080):1
2303:RULE 5 (multi-hop access):0
2366:RULE 5 (multi-hop access):0
2331:RULE 5 (multi-hop access):0
2390:RULE 5 (multi-hop access):0
2417:vulExists(app_39,vul_app,tomcat,remoteExploit,privEscalation):1
2419:hacl(app_4,db_59,tcp,5432):1
2418:RULE 5 (multi-hop access):0

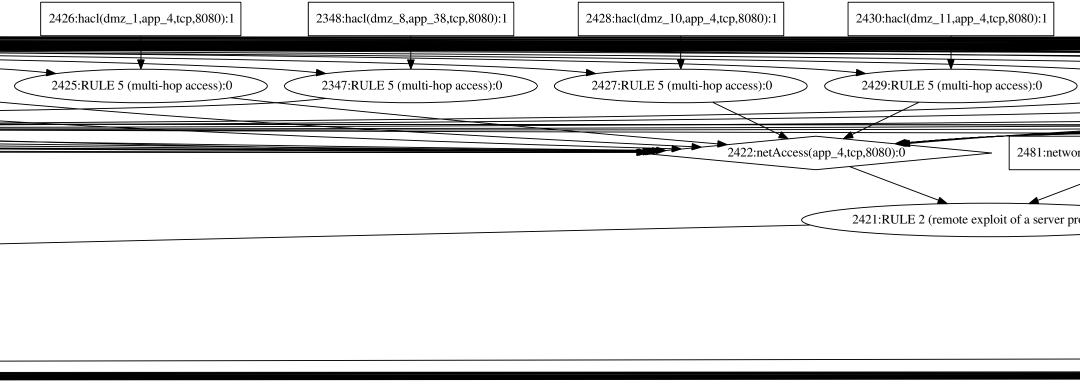

2310:hacl(dmz_16,app_38,tcp,8080):1
2393:hacl(dmz_24,app_39,tcp,8080):1
2342:hacl(dmz_5,app_38,tcp,8080):1
2464:hacl(dmz_27,app_4,tcp,8080):1
2309:RULE 5 (multi-hop access):0
2392:RULE 5 (multi-hop access):0
2341:RULE 5 (multi-hop access):0
2463:RULE 5 (multi-hop access):0

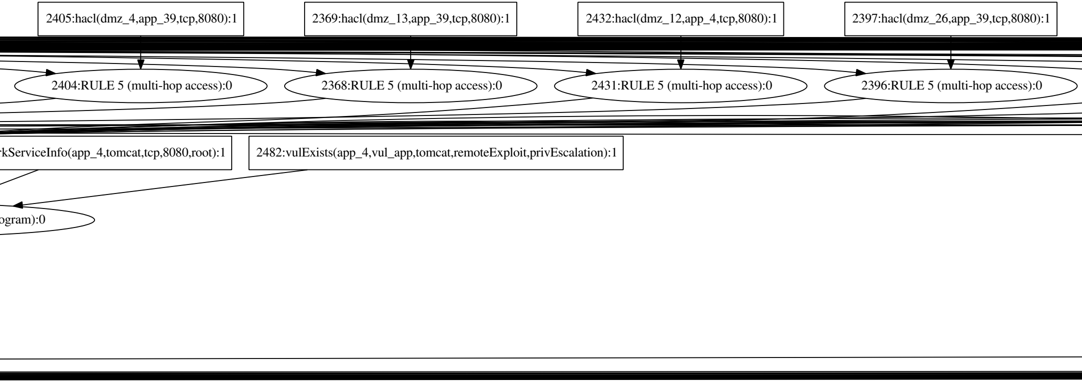

2466:hacl(dmz_28,app_4,tcp,8080):1
2468:hacl(dmz_3,app_4,tcp,8080):1
2395:hacl(dmz_25,app_39,tcp,8080):1
2436:hacl(dmz_14,app_4,tcp,8080):1
2465:RULE 5 (multi-hop access):0
2467:RULE 5 (multi-hop access):0
2394:RULE 5 (multi-hop access):0
2435:RULE 5 (multi-hop access):0

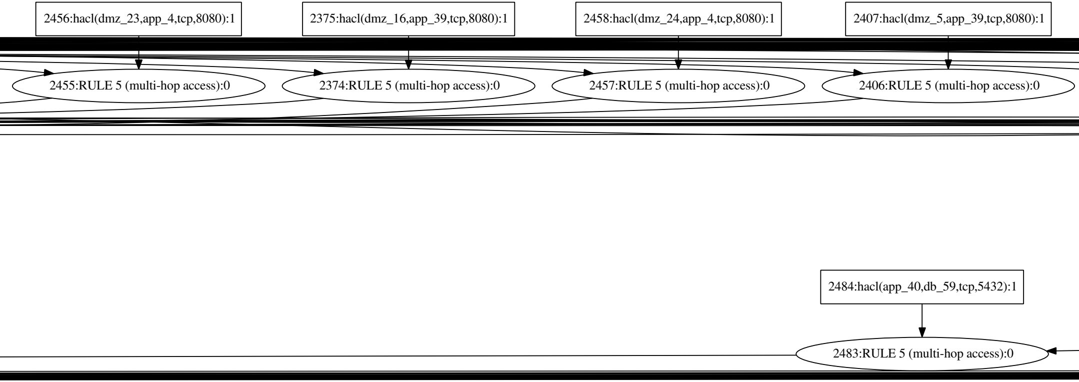

2442:hacl(dmz_17,app_4,tcp,8080):1
2415:hacl(dmz_9,app_39,tcp,8080):1
2444:hacl(dmz_18,app_4,tcp,8080):1
2344:hacl(dmz_6,app_38,tcp,8080):1
2441:RULE 5 (multi-hop access):0
2414:RULE 5 (multi-hop access):0
2443:RULE 5 (multi-hop access):0
2343:RULE 5 (multi-hop access):0

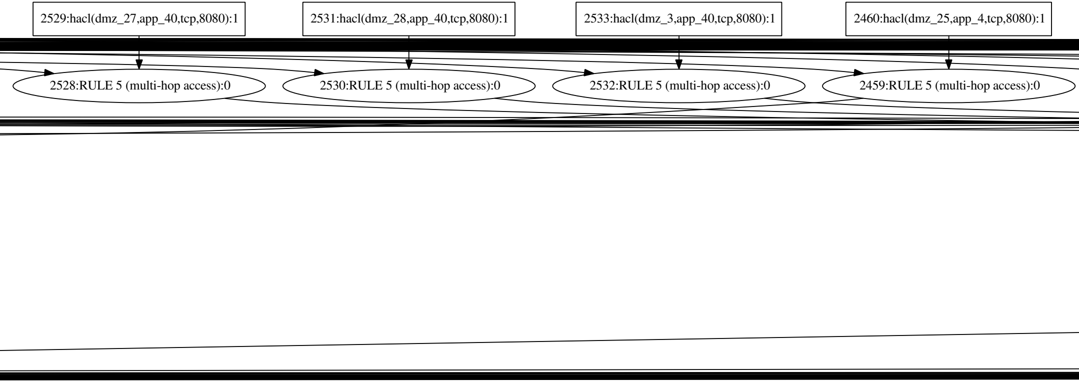

2446:hacl(dmz_19,app_4,tcp,8080):1
2448:hacl(dmz_2,app_4,tcp,8080):1
2450:hacl(dmz_20,app_4,tcp,8080):1
2373:hacl(dmz_15,app_39,tcp,8080):1
2445:RULE 5 (multi-hop access):0
2447:RULE 5 (multi-hop access):0
2449:RULE 5 (multi-hop access):0
2372:RULE 5 (multi-hop access):0
2420:execCode(app_4,root):0

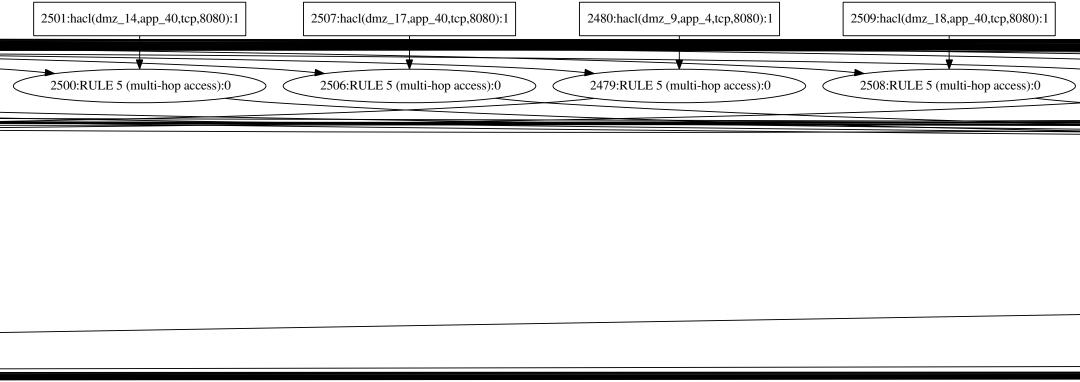

2452:hacl(dmz_21,app_4,tcp,8080):1
2346:hacl(dmz_7,app_38,tcp,8080):1
2424:hacl(dmz_0,app_4,tcp,8080):1
2454:hacl(dmz_22,app_4,tcp,8080):1
2451:RULE 5 (multi-hop access):0
2345:RULE 5 (multi-hop access):0
2423:RULE 5 (multi-hop access):0
2453:RULE 5 (multi-hop access):0

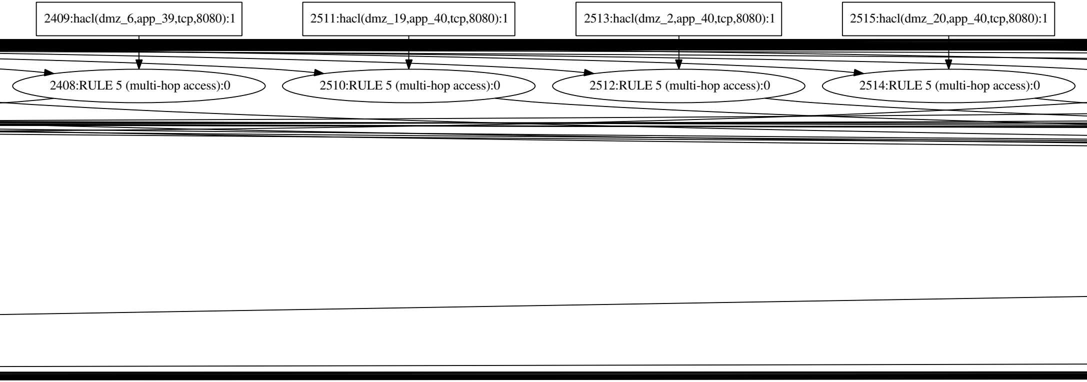

2426:hacl(dmz_1,app_4,tcp,8080):1
2348:hacl(dmz_8,app_38,tcp,8080):1
2428:hacl(dmz_10,app_4,tcp,8080):1
2430:hacl(dmz_11,app_4,tcp,8080):1
2425:RULE 5 (multi-hop access):0
2347:RULE 5 (multi-hop access):0
2427:RULE 5 (multi-hop access):0
2429:RULE 5 (multi-hop access):0
2422:netAccess(app_4,tcp,8080):0
2481:networ
2421:RULE 2 (remote exploit of a server pro

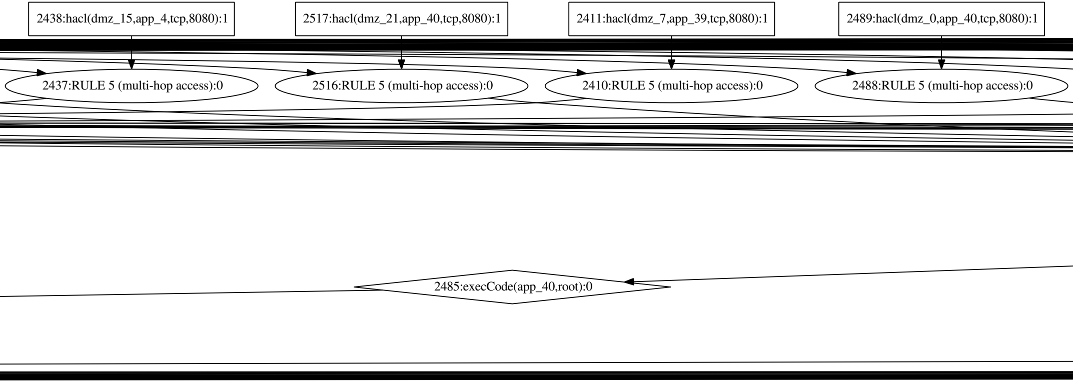

2405:hacl(dmz_4,app_39,tcp,8080):1
2369:hacl(dmz_13,app_39,tcp,8080):1
2432:hacl(dmz_12,app_4,tcp,8080):1
2397:hacl(dmz_26,app_39,tcp,8080):1
2404:RULE 5 (multi-hop access):0
2368:RULE 5 (multi-hop access):0
2431:RULE 5 (multi-hop access):0
2396:RULE 5 (multi-hop access):0
rkServiceInfo(app_4,tomcat,tcp,8080,root):1
2482:vulExists(app_4,vul_app,tomcat,remoteExploit,privEscalation):1
ogram):0

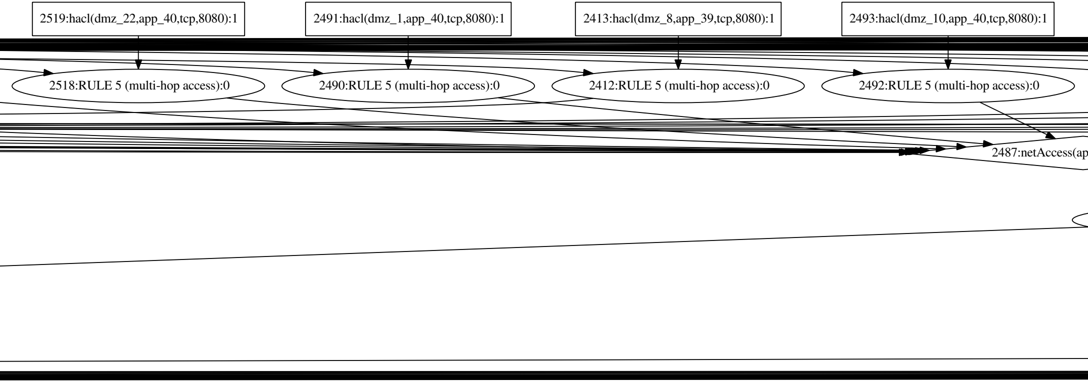

2456:hacl(dmz_23,app_4,tcp,8080):1
2375:hacl(dmz_16,app_39,tcp,8080):1
2458:hacl(dmz_24,app_4,tcp,8080):1
2407:hacl(dmz_5,app_39,tcp,8080):1
2455:RULE 5 (multi-hop access):0
2374:RULE 5 (multi-hop access):0
2457:RULE 5 (multi-hop access):0
2406:RULE 5 (multi-hop access):0
2484:hacl(app_40,db_59,tcp,5432):1
2483:RULE 5 (multi-hop access):0

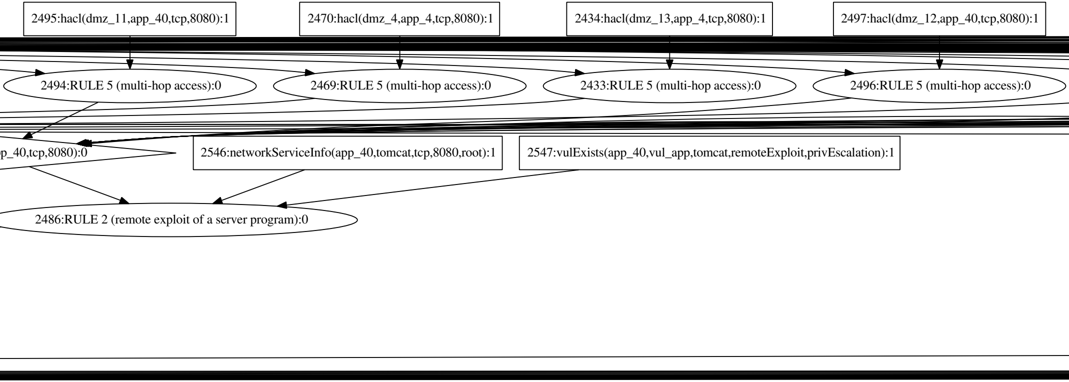

2529:hacl(dmz_27,app_40,tcp,8080):1
2531:hacl(dmz_28,app_40,tcp,8080):1
2533:hacl(dmz_3,app_40,tcp,8080):1
2460:hacl(dmz_25,app_4,tcp,8080):1
2528:RULE 5 (multi-hop access):0
2530:RULE 5 (multi-hop access):0
2532:RULE 5 (multi-hop access):0
2459:RULE 5 (multi-hop access):0

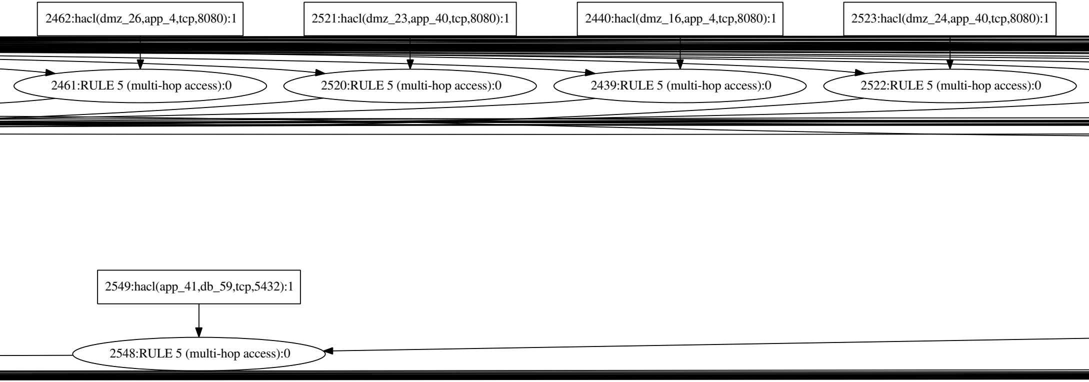

2501:hacl(dmz_14,app_40,tcp,8080):1
2507:hacl(dmz_17,app_40,tcp,8080):1
2480:hacl(dmz_9,app_4,tcp,8080):1
2509:hacl(dmz_18,app_40,tcp,8080):1
2500:RULE 5 (multi-hop access):0
2506:RULE 5 (multi-hop access):0
2479:RULE 5 (multi-hop access):0
2508:RULE 5 (multi-hop access):0

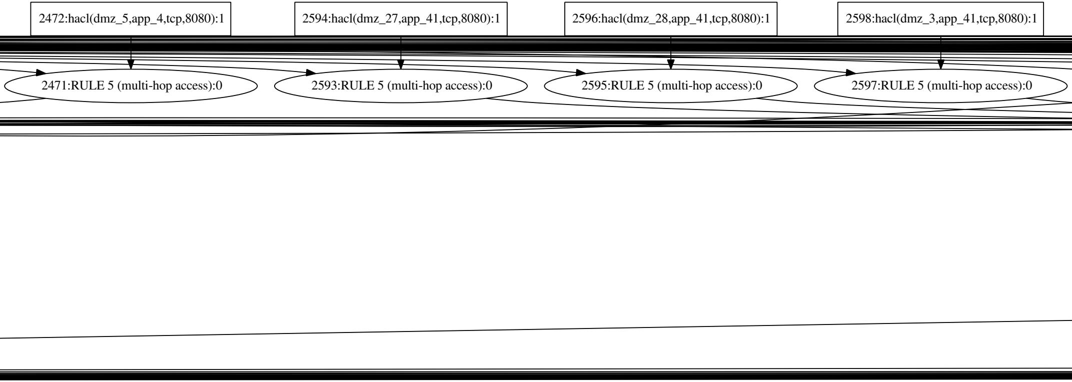

2409:hacl(dmz_6,app_39,tcp,8080):1
2511:hacl(dmz_19,app_40,tcp,8080):1
2513:hacl(dmz_2,app_40,tcp,8080):1
2515:hacl(dmz_20,app_40,tcp,8080):1
2408:RULE 5 (multi-hop access):0
2510:RULE 5 (multi-hop access):0
2512:RULE 5 (multi-hop access):0
2514:RULE 5 (multi-hop access):0

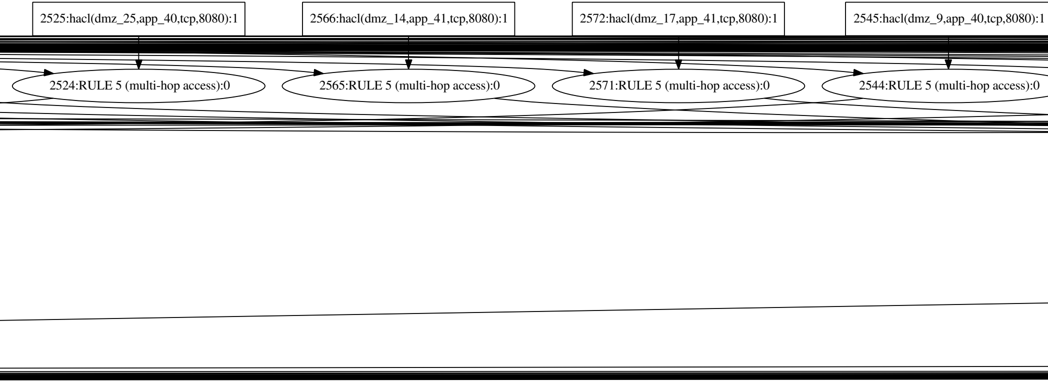

2438:hacl(dmz_15,app_4,tcp,8080):1
2517:hacl(dmz_21,app_40,tcp,8080):1
2411:hacl(dmz_7,app_39,tcp,8080):1
2489:hacl(dmz_0,app_40,tcp,8080):1
2437:RULE 5 (multi-hop access):0
2516:RULE 5 (multi-hop access):0
2410:RULE 5 (multi-hop access):0
2488:RULE 5 (multi-hop access):0
2485:execCode(app_40,root):0

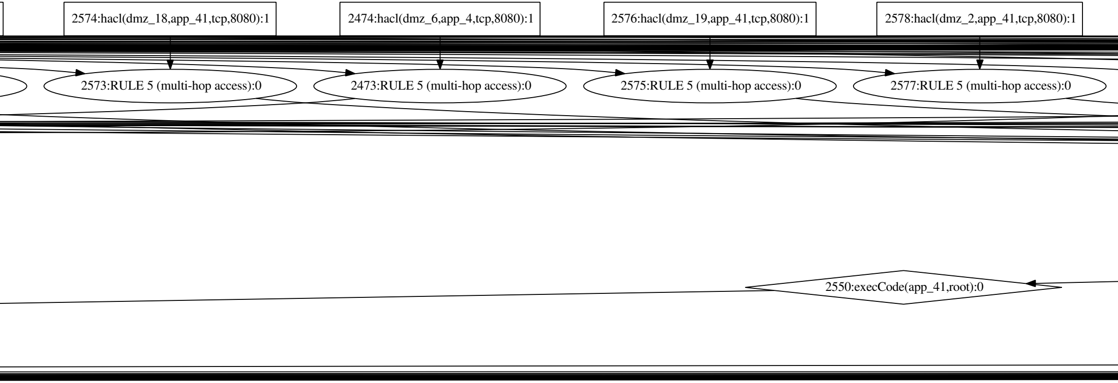

2519:hacl(dmz_22,app_40,tcp,8080):1
2491:hacl(dmz_1,app_40,tcp,8080):1
2413:hacl(dmz_8,app_39,tcp,8080):1
2493:hacl(dmz_10,app_40,tcp,8080):1
2518:RULE 5 (multi-hop access):0
2490:RULE 5 (multi-hop access):0
2412:RULE 5 (multi-hop access):0
2492:RULE 5 (multi-hop access):0
2487:netAccess(ap

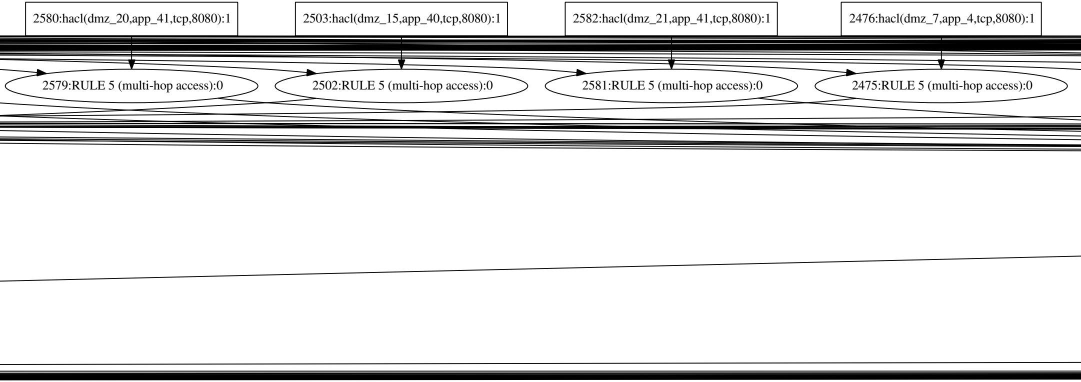

2495:hacl(dmz_11,app_40,tcp,8080):1
2470:hacl(dmz_4,app_4,tcp,8080):1
2434:hacl(dmz_13,app_4,tcp,8080):1
2497:hacl(dmz_12,app_40,tcp,8080):1
2494:RULE 5 (multi-hop access):0
2469:RULE 5 (multi-hop access):0
2433:RULE 5 (multi-hop access):0
2496:RULE 5 (multi-hop access):0
p_40,tcp,8080):0
2546:networkServiceInfo(app_40,tomcat,tcp,8080,root):1
2547:vulExists(app_40,vul_app,tomcat,remoteExploit,privEscalation):1
2486:RULE 2 (remote exploit of a server program):0

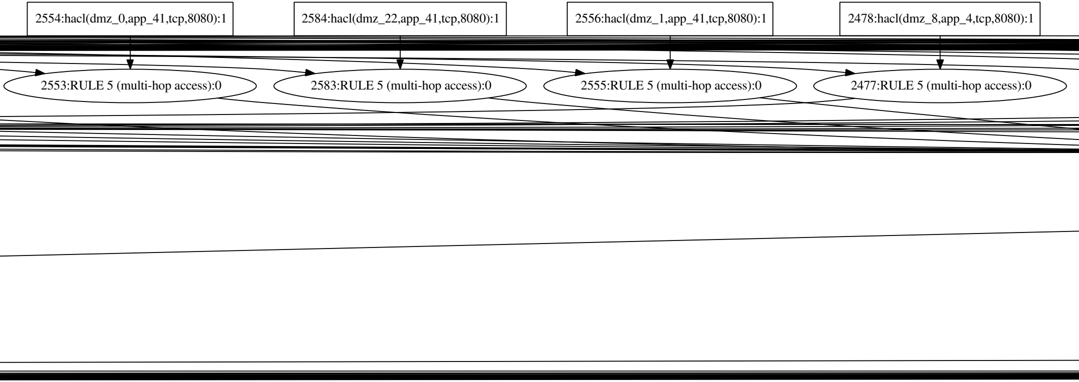

2462:hacl(dmz_26,app_4,tcp,8080):1
2521:hacl(dmz_23,app_40,tcp,8080):1
2440:hacl(dmz_16,app_4,tcp,8080):1
2523:hacl(dmz_24,app_40,tcp,8080):1
2461:RULE 5 (multi-hop access):0
2520:RULE 5 (multi-hop access):0
2439:RULE 5 (multi-hop access):0
2522:RULE 5 (multi-hop access):0
2549:hacl(app_41,db_59,tcp,5432):1
2548:RULE 5 (multi-hop access):0

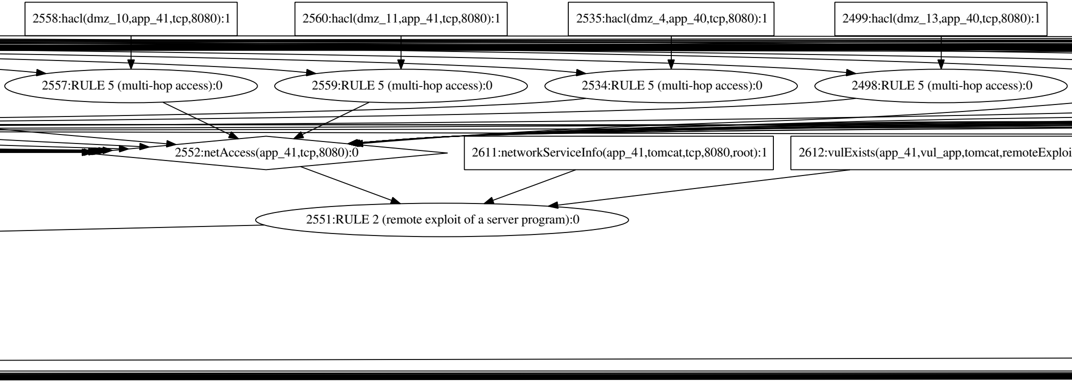

2472:hacl(dmz_5,app_4,tcp,8080):1
2594:hacl(dmz_27,app_41,tcp,8080):1
2596:hacl(dmz_28,app_41,tcp,8080):1
2598:hacl(dmz_3,app_41,tcp,8080):1
2471:RULE 5 (multi-hop access):0
2593:RULE 5 (multi-hop access):0
2595:RULE 5 (multi-hop access):0
2597:RULE 5 (multi-hop access):0

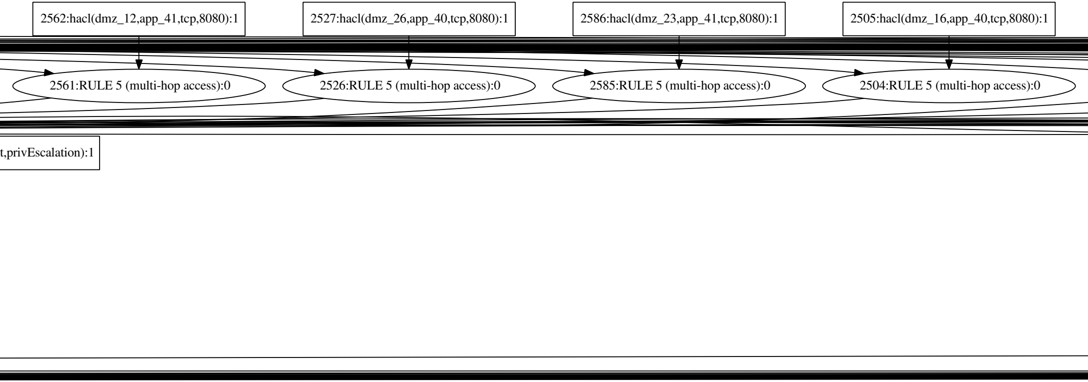

2525:hacl(dmz_25,app_40,tcp,8080):1
2566:hacl(dmz_14,app_41,tcp,8080):1
2572:hacl(dmz_17,app_41,tcp,8080):1
2545:hacl(dmz_9,app_40,tcp,8080):1
2524:RULE 5 (multi-hop access):0
2565:RULE 5 (multi-hop access):0
2571:RULE 5 (multi-hop access):0
2544:RULE 5 (multi-hop access):0

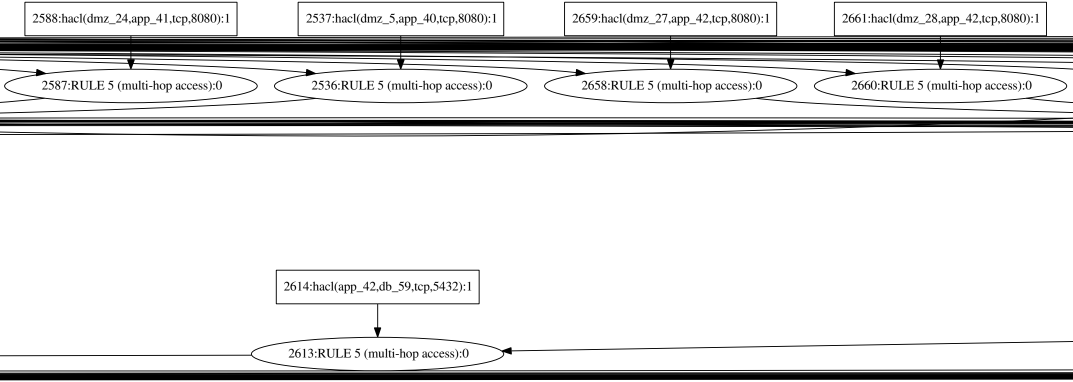

2574:hacl(dmz_18,app_41,tcp,8080):1
2474:hacl(dmz_6,app_4,tcp,8080):1
2576:hacl(dmz_19,app_41,tcp,8080):1
2578:hacl(dmz_2,app_41,tcp,8080):1
2573:RULE 5 (multi-hop access):0
2473:RULE 5 (multi-hop access):0
2575:RULE 5 (multi-hop access):0
2577:RULE 5 (multi-hop access):0
2550:execCode(app_41,root):0

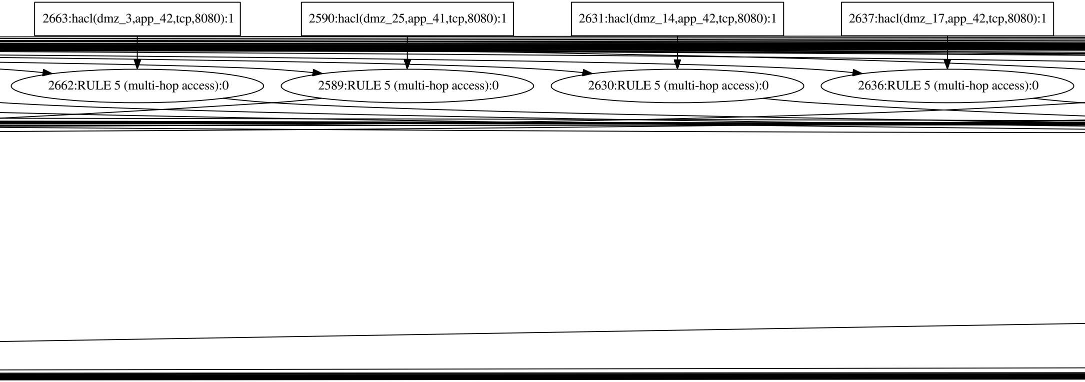

2580:hacl(dmz_20,app_41,tcp,8080):1
2503:hacl(dmz_15,app_40,tcp,8080):1
2582:hacl(dmz_21,app_41,tcp,8080):1
2476:hacl(dmz_7,app_4,tcp,8080):1
2579:RULE 5 (multi-hop access):0
2502:RULE 5 (multi-hop access):0
2581:RULE 5 (multi-hop access):0
2475:RULE 5 (multi-hop access):0

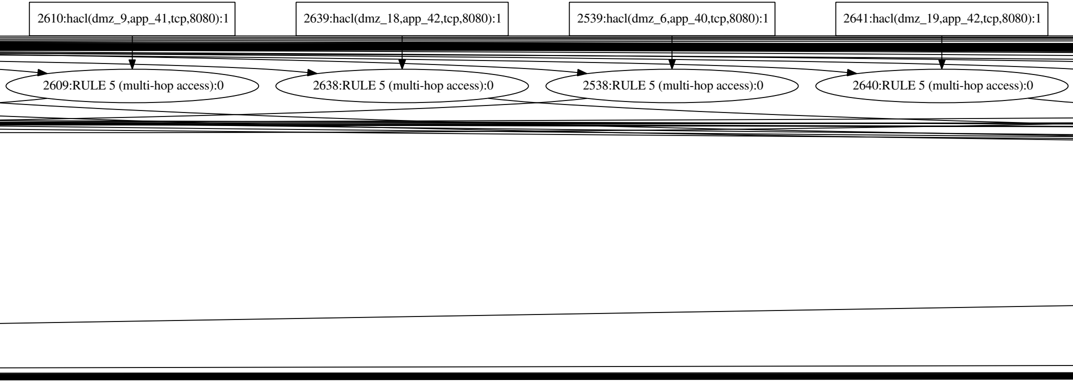

2554:hacl(dmz_0,app_41,tcp,8080):1
2584:hacl(dmz_22,app_41,tcp,8080):1
2556:hacl(dmz_1,app_41,tcp,8080):1
2478:hacl(dmz_8,app_4,tcp,8080):1
2553:RULE 5 (multi-hop access):0
2583:RULE 5 (multi-hop access):0
2555:RULE 5 (multi-hop access):0
2477:RULE 5 (multi-hop access):0

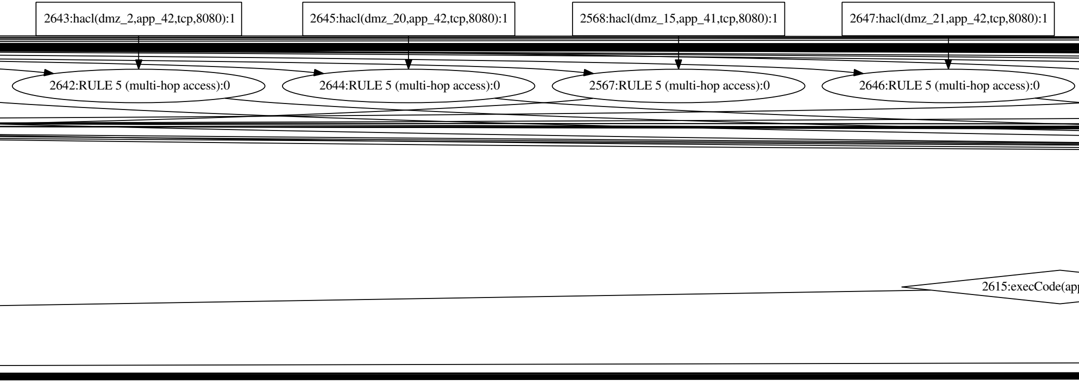

2558:hacl(dmz_10,app_41,tcp,8080):1
2560:hacl(dmz_11,app_41,tcp,8080):1
2535:hacl(dmz_4,app_40,tcp,8080):1
2499:hacl(dmz_13,app_40,tcp,8080):1
2557:RULE 5 (multi-hop access):0
2559:RULE 5 (multi-hop access):0
2534:RULE 5 (multi-hop access):0
2498:RULE 5 (multi-hop access):0
2552:netAccess(app_41,tcp,8080):0
2611:networkServiceInfo(app_41,tomcat,tcp,8080,root):1
2612:vulExists(app_41,vul_app,tomcat,remoteExploi
2551:RULE 2 (remote exploit of a server program):0

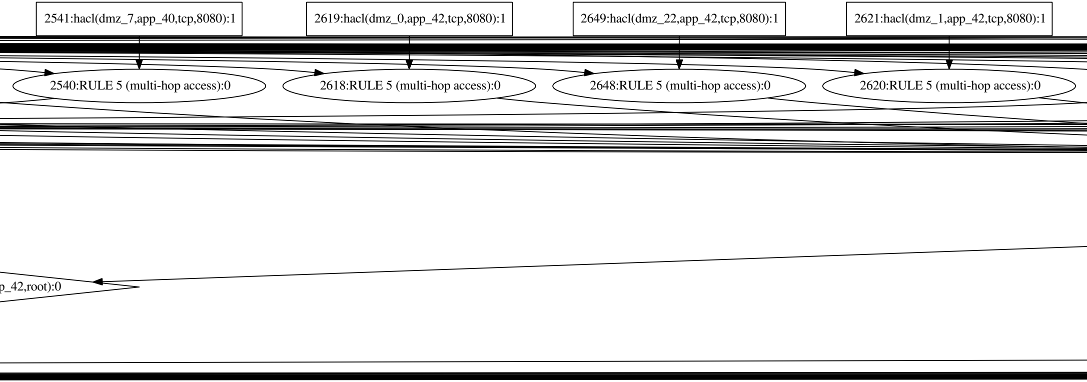

2562:hacl(dmz_12,app_41,tcp,8080):1
2527:hacl(dmz_26,app_40,tcp,8080):1
2586:hacl(dmz_23,app_41,tcp,8080):1
2505:hacl(dmz_16,app_40,tcp,8080):1
2561:RULE 5 (multi-hop access):0
2526:RULE 5 (multi-hop access):0
2585:RULE 5 (multi-hop access):0
2504:RULE 5 (multi-hop access):0
t,privEscalation):1

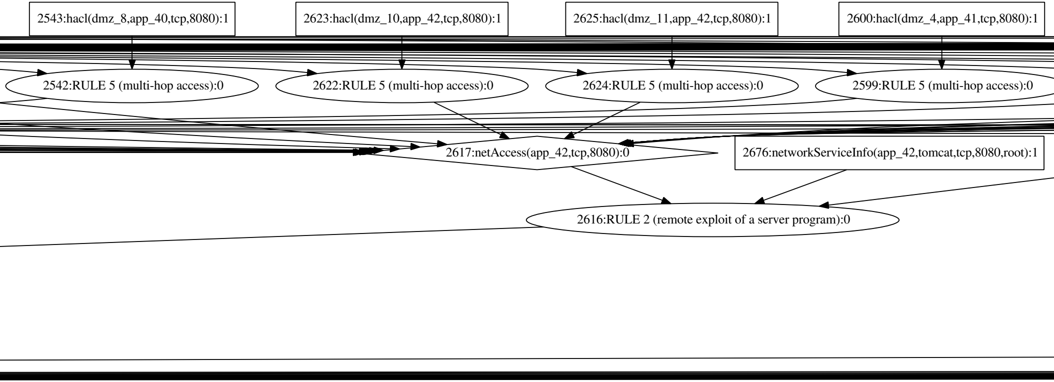

2588:hacl(dmz_24,app_41,tcp,8080):1
2537:hacl(dmz_5,app_40,tcp,8080):1
2659:hacl(dmz_27,app_42,tcp,8080):1
2661:hacl(dmz_28,app_42,tcp,8080):1
2587:RULE 5 (multi-hop access):0
2536:RULE 5 (multi-hop access):0
2658:RULE 5 (multi-hop access):0
2660:RULE 5 (multi-hop access):0
2614:hacl(app_42,db_59,tcp,5432):1
2613:RULE 5 (multi-hop access):0

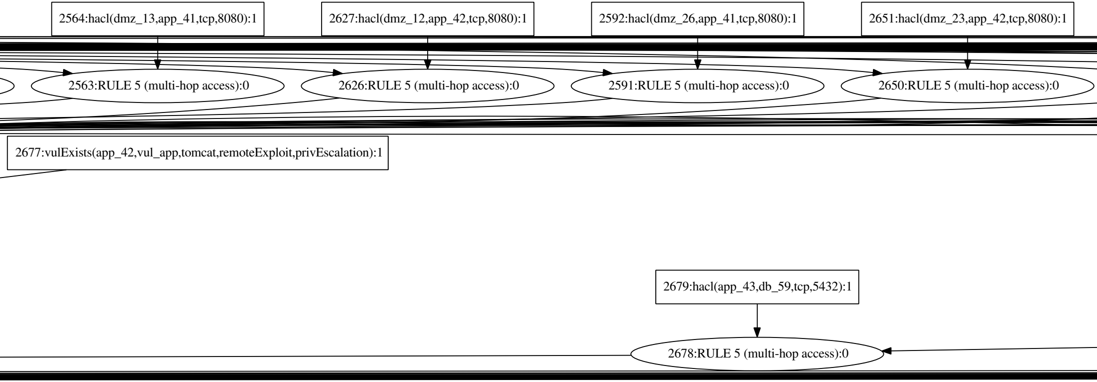

2663:hacl(dmz_3,app_42,tcp,8080):1
2590:hacl(dmz_25,app_41,tcp,8080):1
2631:hacl(dmz_14,app_42,tcp,8080):1
2637:hacl(dmz_17,app_42,tcp,8080):1
2662:RULE 5 (multi-hop access):0
2589:RULE 5 (multi-hop access):0
2630:RULE 5 (multi-hop access):0
2636:RULE 5 (multi-hop access):0

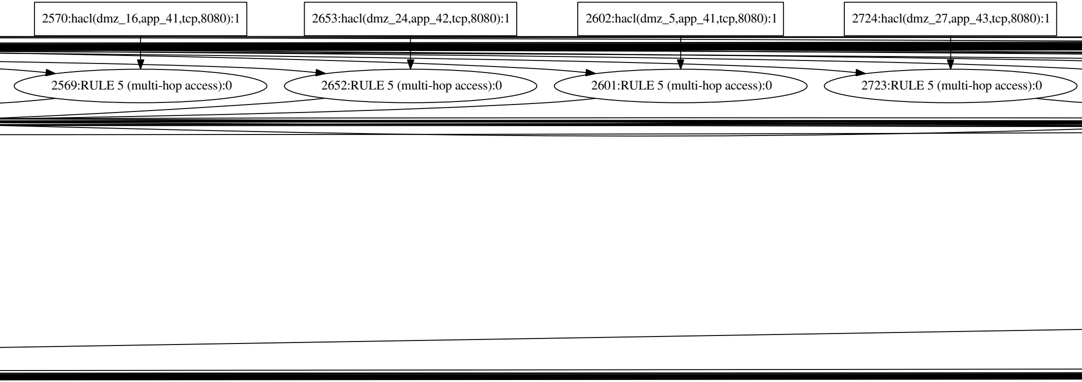

2610:hacl(dmz_9,app_41,tcp,8080):1
2639:hacl(dmz_18,app_42,tcp,8080):1
2539:hacl(dmz_6,app_40,tcp,8080):1
2641:hacl(dmz_19,app_42,tcp,8080):1
2609:RULE 5 (multi-hop access):0
2638:RULE 5 (multi-hop access):0
2538:RULE 5 (multi-hop access):0
2640:RULE 5 (multi-hop access):0

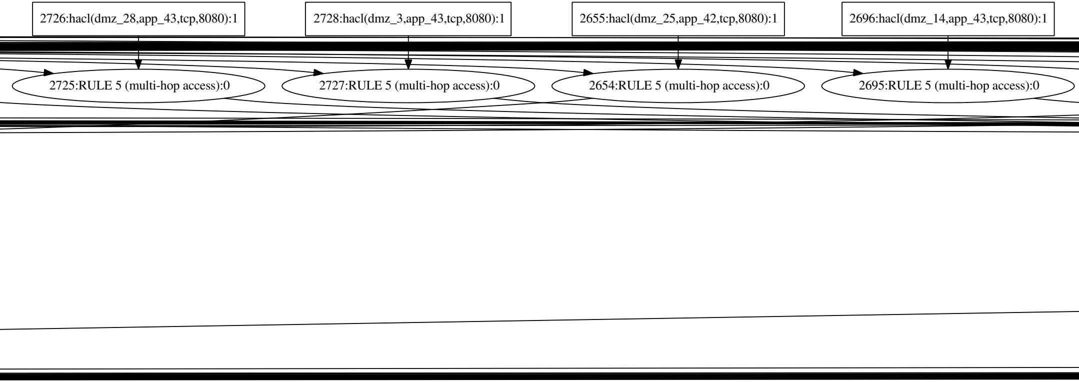

2643:hacl(dmz_2,app_42,tcp,8080):1
2645:hacl(dmz_20,app_42,tcp,8080):1
2568:hacl(dmz_15,app_41,tcp,8080):1
2647:hacl(dmz_21,app_42,tcp,8080):1
2642:RULE 5 (multi-hop access):0
2644:RULE 5 (multi-hop access):0
2567:RULE 5 (multi-hop access):0
2646:RULE 5 (multi-hop access):0
2615:execCode(app

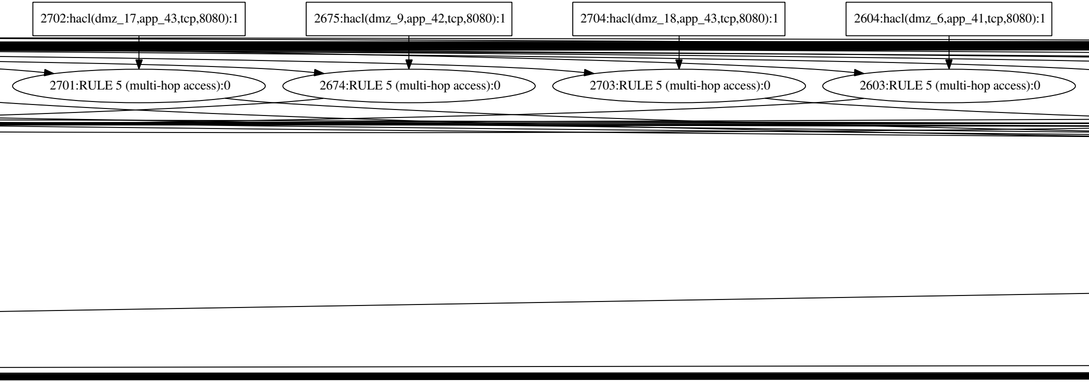

2541:hacl(dmz_7,app_40,tcp,8080):1
2619:hacl(dmz_0,app_42,tcp,8080):1
2649:hacl(dmz_22,app_42,tcp,8080):1
2621:hacl(dmz_1,app_42,tcp,8080):1
2540:RULE 5 (multi-hop access):0
2618:RULE 5 (multi-hop access):0
2648:RULE 5 (multi-hop access):0
2620:RULE 5 (multi-hop access):0
p_42,root):0

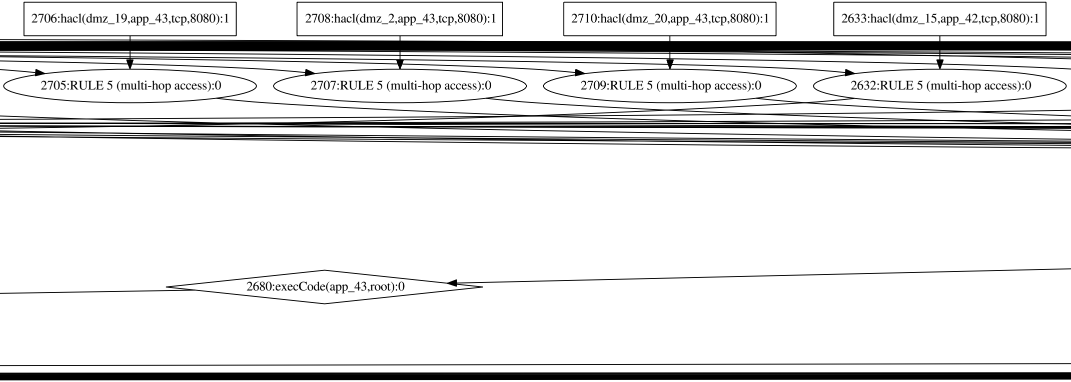

2543:hacl(dmz_8,app_40,tcp,8080):1
2623:hacl(dmz_10,app_42,tcp,8080):1
2625:hacl(dmz_11,app_42,tcp,8080):1
2600:hacl(dmz_4,app_41,tcp,8080):1
2542:RULE 5 (multi-hop access):0
2622:RULE 5 (multi-hop access):0
2624:RULE 5 (multi-hop access):0
2599:RULE 5 (multi-hop access):0
2617:netAccess(app_42,tcp,8080):0
2676:networkServiceInfo(app_42,tomcat,tcp,8080,root):1
2616:RULE 2 (remote exploit of a server program):0

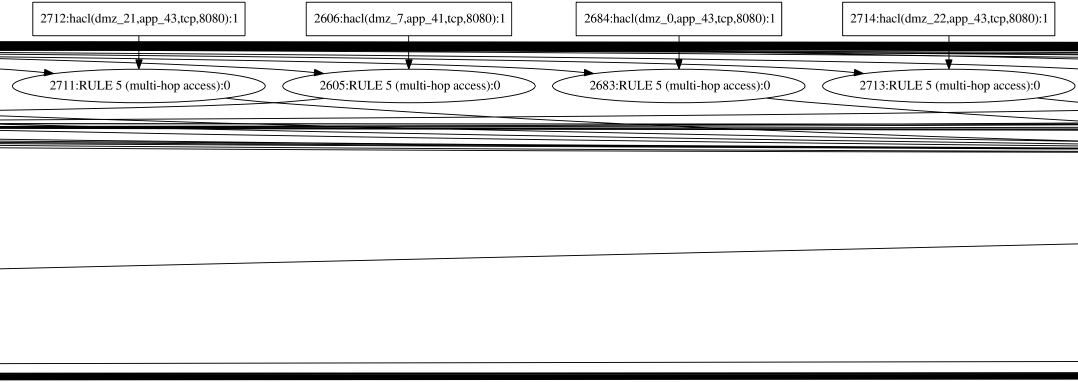

2564:hacl(dmz_13,app_41,tcp,8080):1
2627:hacl(dmz_12,app_42,tcp,8080):1
2592:hacl(dmz_26,app_41,tcp,8080):1
2651:hacl(dmz_23,app_42,tcp,8080):1
2563:RULE 5 (multi-hop access):0
2626:RULE 5 (multi-hop access):0
2591:RULE 5 (multi-hop access):0
2650:RULE 5 (multi-hop access):0
2677:vulExists(app_42,vul_app,tomcat,remoteExploit,privEscalation):1
2679:hacl(app_43,db_59,tcp,5432):1
2678:RULE 5 (multi-hop access):0

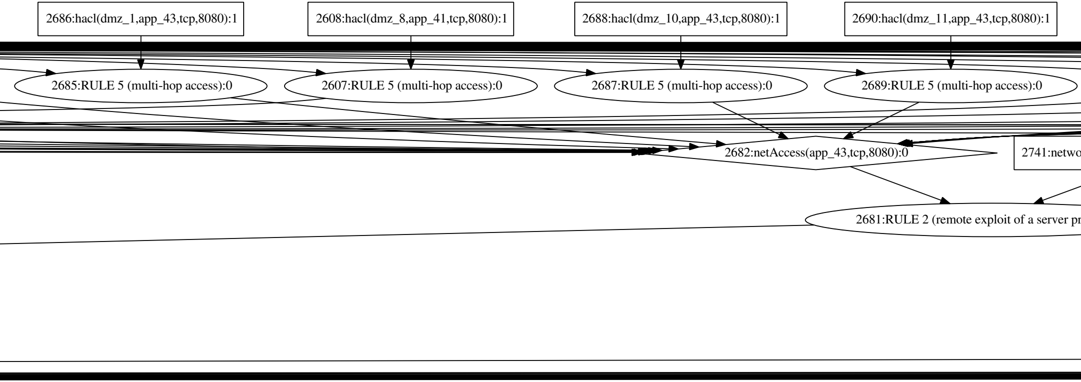

2570:hacl(dmz_16,app_41,tcp,8080):1
2653:hacl(dmz_24,app_42,tcp,8080):1
2602:hacl(dmz_5,app_41,tcp,8080):1
2724:hacl(dmz_27,app_43,tcp,8080):1
2569:RULE 5 (multi-hop access):0
2652:RULE 5 (multi-hop access):0
2601:RULE 5 (multi-hop access):0
2723:RULE 5 (multi-hop access):0

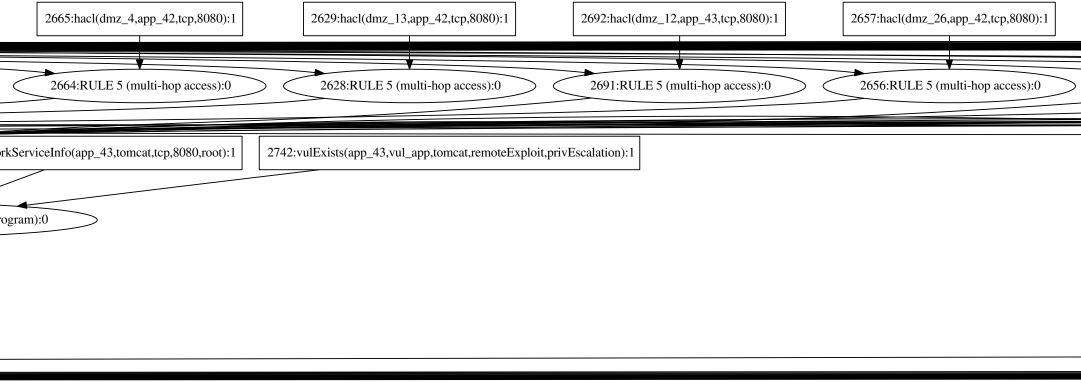

2726:hacl(dmz_28,app_43,tcp,8080):1
2728:hacl(dmz_3,app_43,tcp,8080):1
2655:hacl(dmz_25,app_42,tcp,8080):1
2696:hacl(dmz_14,app_43,tcp,8080):1
2725:RULE 5 (multi-hop access):0
2727:RULE 5 (multi-hop access):0
2654:RULE 5 (multi-hop access):0
2695:RULE 5 (multi-hop access):0

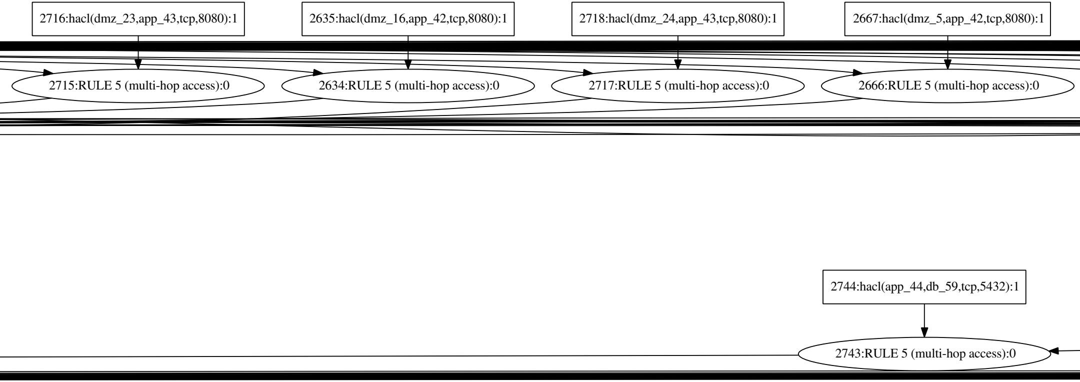

2702:hacl(dmz_17,app_43,tcp,8080):1
2675:hacl(dmz_9,app_42,tcp,8080):1
2704:hacl(dmz_18,app_43,tcp,8080):1
2604:hacl(dmz_6,app_41,tcp,8080):1
2701:RULE 5 (multi-hop access):0
2674:RULE 5 (multi-hop access):0
2703:RULE 5 (multi-hop access):0
2603:RULE 5 (multi-hop access):0

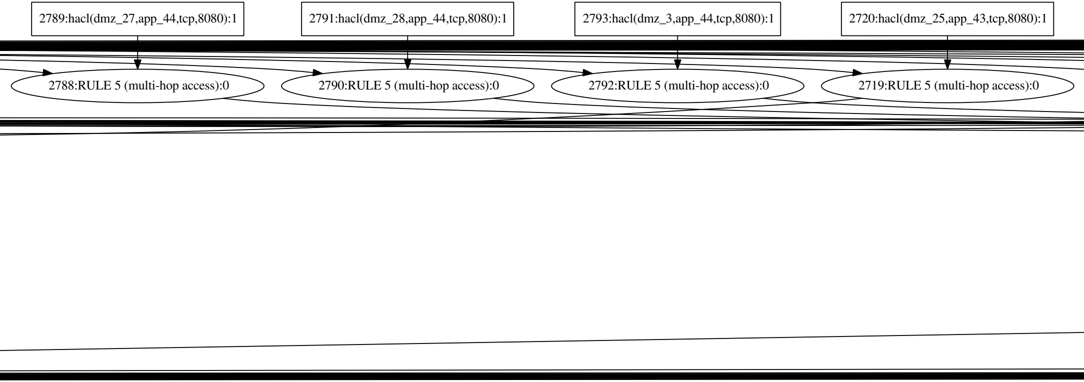

2706:hacl(dmz_19,app_43,tcp,8080):1
2708:hacl(dmz_2,app_43,tcp,8080):1
2710:hacl(dmz_20,app_43,tcp,8080):1
2633:hacl(dmz_15,app_42,tcp,8080):1
2705:RULE 5 (multi-hop access):0
2707:RULE 5 (multi-hop access):0
2709:RULE 5 (multi-hop access):0
2632:RULE 5 (multi-hop access):0
2680:execCode(app_43,root):0

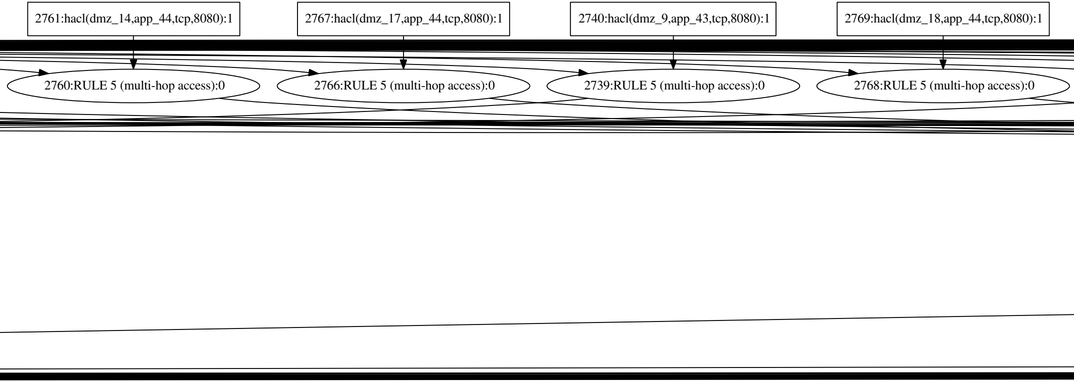

2712:hacl(dmz_21,app_43,tcp,8080):1
2606:hacl(dmz_7,app_41,tcp,8080):1
2684:hacl(dmz_0,app_43,tcp,8080):1
2714:hacl(dmz_22,app_43,tcp,8080):1
2711:RULE 5 (multi-hop access):0
2605:RULE 5 (multi-hop access):0
2683:RULE 5 (multi-hop access):0
2713:RULE 5 (multi-hop access):0

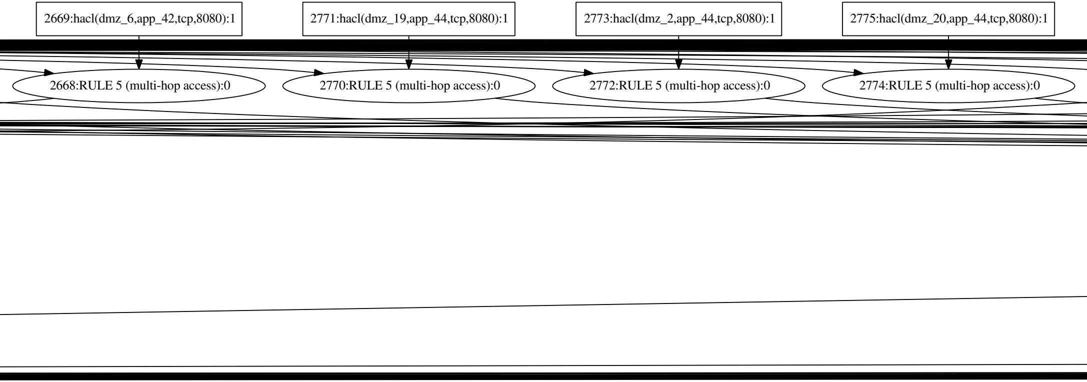

2686:hacl(dmz_1,app_43,tcp,8080):1
2608:hacl(dmz_8,app_41,tcp,8080):1
2688:hacl(dmz_10,app_43,tcp,8080):1
2690:hacl(dmz_11,app_43,tcp,8080):1
2685:RULE 5 (multi-hop access):0
2607:RULE 5 (multi-hop access):0
2687:RULE 5 (multi-hop access):0
2689:RULE 5 (multi-hop access):0
2682:netAccess(app_43,tcp,8080):0
2741:netwo
2681:RULE 2 (remote exploit of a server pr

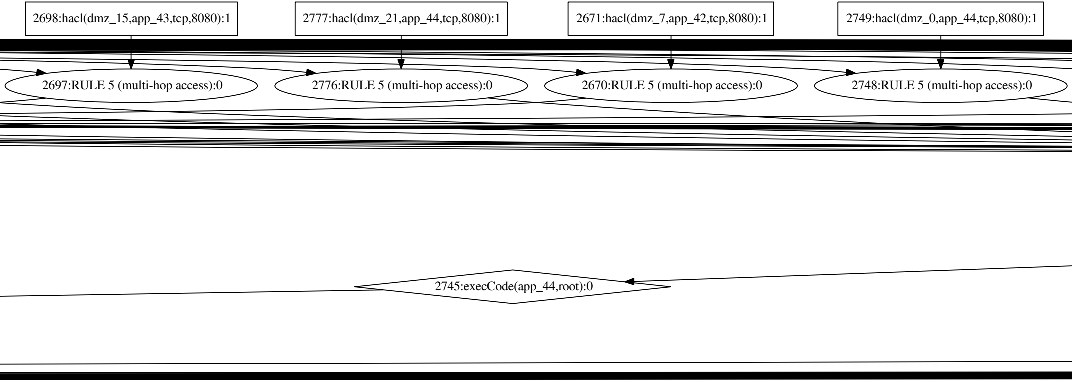

2665:hacl(dmz_4,app_42,tcp,8080):1
2629:hacl(dmz_13,app_42,tcp,8080):1
2692:hacl(dmz_12,app_43,tcp,8080):1
2657:hacl(dmz_26,app_42,tcp,8080):1
2664:RULE 5 (multi-hop access):0
2628:RULE 5 (multi-hop access):0
2691:RULE 5 (multi-hop access):0
2656:RULE 5 (multi-hop access):0
rkServiceInfo(app_43,tomcat,tcp,8080,root):1
2742:vulExists(app_43,vul_app,tomcat,remoteExploit,privEscalation):1
rogram):0

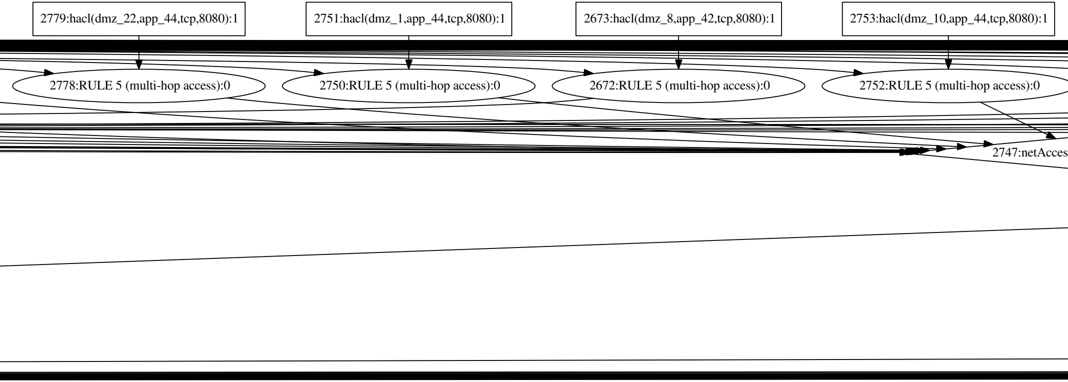

2716:hacl(dmz_23,app_43,tcp,8080):1
2635:hacl(dmz_16,app_42,tcp,8080):1
2718:hacl(dmz_24,app_43,tcp,8080):1
2667:hacl(dmz_5,app_42,tcp,8080):1
2715:RULE 5 (multi-hop access):0
2634:RULE 5 (multi-hop access):0
2717:RULE 5 (multi-hop access):0
2666:RULE 5 (multi-hop access):0
2744:hacl(app_44,db_59,tcp,5432):1
2743:RULE 5 (multi-hop access):0

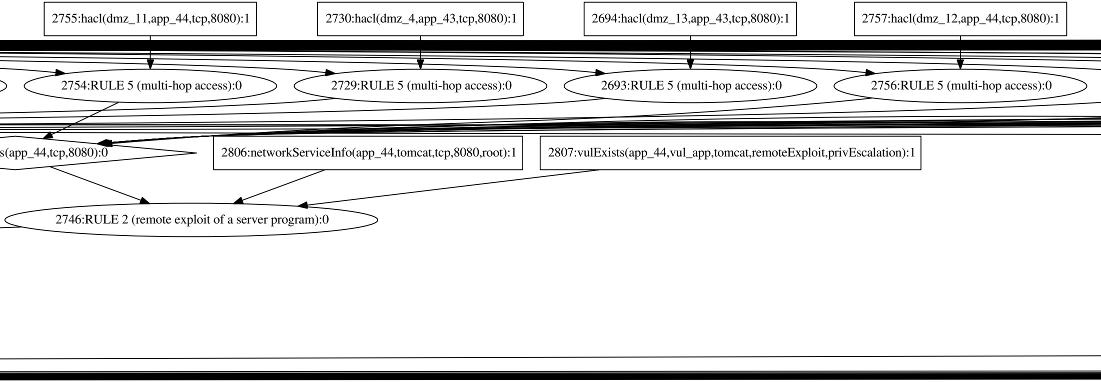

2789:hacl(dmz_27,app_44,tcp,8080):1
2791:hacl(dmz_28,app_44,tcp,8080):1
2793:hacl(dmz_3,app_44,tcp,8080):1
2720:hacl(dmz_25,app_43,tcp,8080):1
2788:RULE 5 (multi-hop access):0
2790:RULE 5 (multi-hop access):0
2792:RULE 5 (multi-hop access):0
2719:RULE 5 (multi-hop access):0

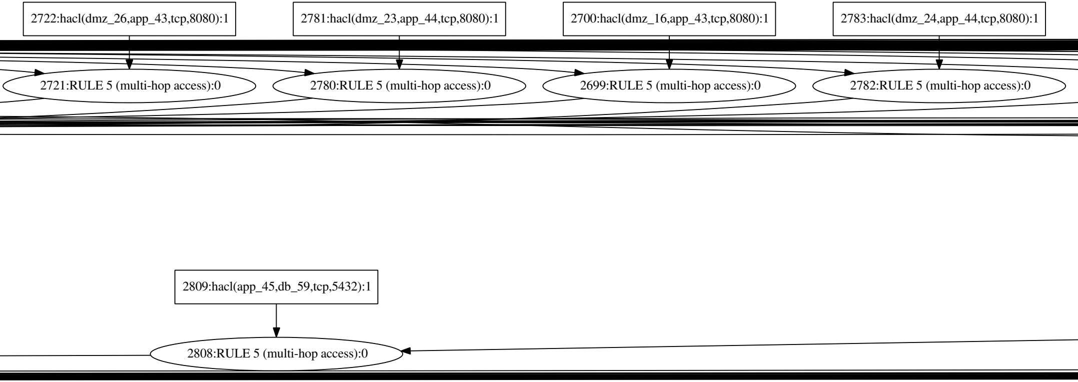

2761:hacl(dmz_14,app_44,tcp,8080):1
2767:hacl(dmz_17,app_44,tcp,8080):1
2740:hacl(dmz_9,app_43,tcp,8080):1
2769:hacl(dmz_18,app_44,tcp,8080):1
2760:RULE 5 (multi-hop access):0
2766:RULE 5 (multi-hop access):0
2739:RULE 5 (multi-hop access):0
2768:RULE 5 (multi-hop access):0

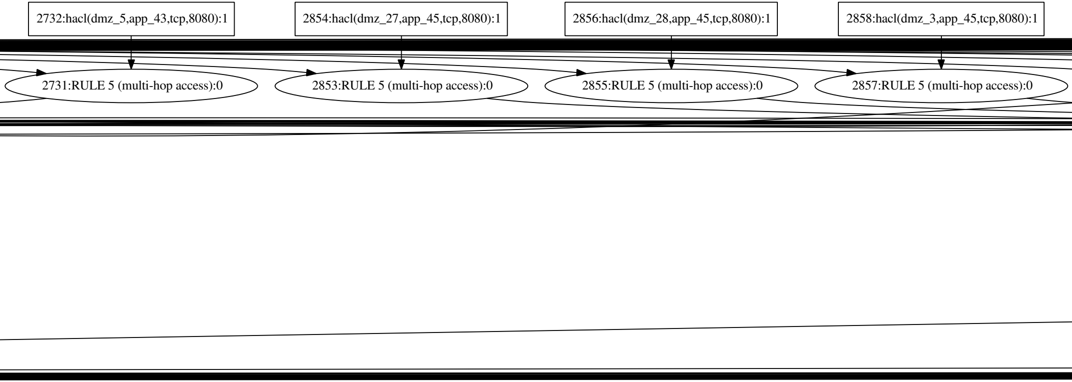

2669:hacl(dmz_6,app_42,tcp,8080):1
2771:hacl(dmz_19,app_44,tcp,8080):1
2773:hacl(dmz_2,app_44,tcp,8080):1
2775:hacl(dmz_20,app_44,tcp,8080):1
2668:RULE 5 (multi-hop access):0
2770:RULE 5 (multi-hop access):0
2772:RULE 5 (multi-hop access):0
2774:RULE 5 (multi-hop access):0

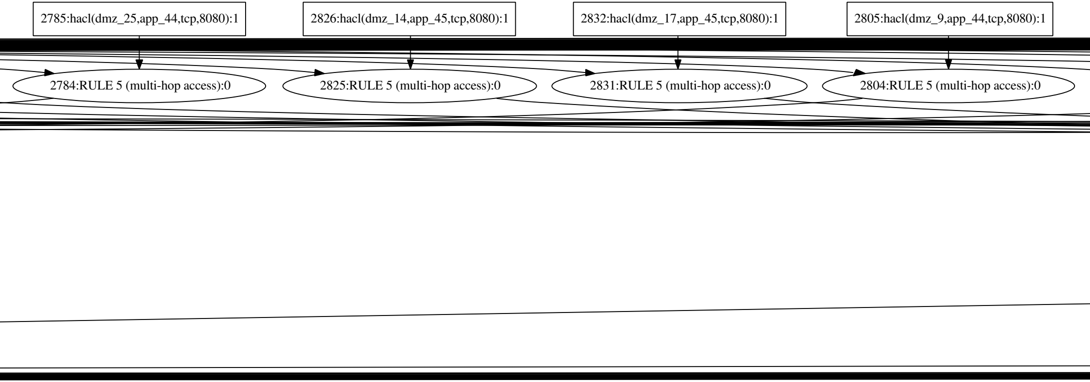

2698:hacl(dmz_15,app_43,tcp,8080):1
2777:hacl(dmz_21,app_44,tcp,8080):1
2671:hacl(dmz_7,app_42,tcp,8080):1
2749:hacl(dmz_0,app_44,tcp,8080):1
2697:RULE 5 (multi-hop access):0
2776:RULE 5 (multi-hop access):0
2670:RULE 5 (multi-hop access):0
2748:RULE 5 (multi-hop access):0
2745:execCode(app_44,root):0

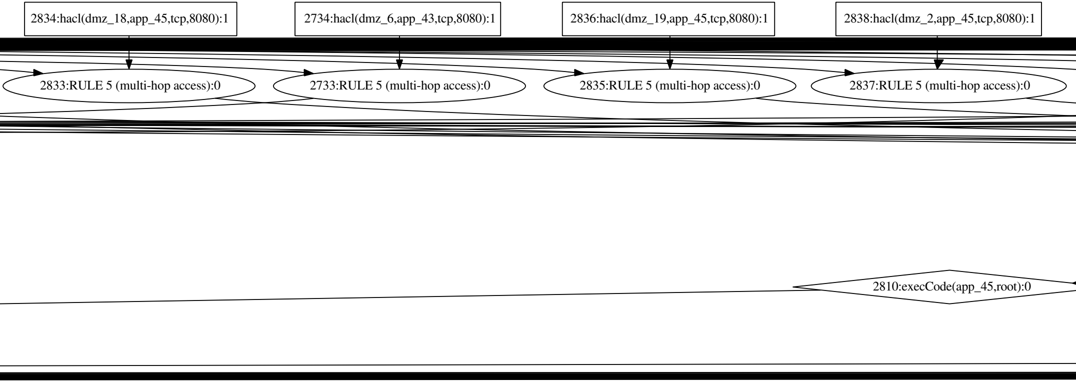

2779:hacl(dmz_22,app_44,tcp,8080):1
2751:hacl(dmz_1,app_44,tcp,8080):1
2673:hacl(dmz_8,app_42,tcp,8080):1
2753:hacl(dmz_10,app_44,tcp,8080):1
2778:RULE 5 (multi-hop access):0
2750:RULE 5 (multi-hop access):0
2672:RULE 5 (multi-hop access):0
2752:RULE 5 (multi-hop access):0
2747:netAcces

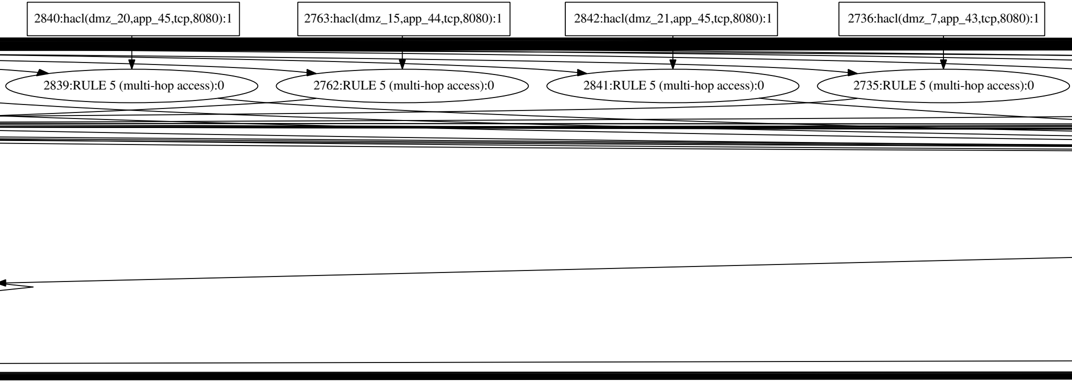

2755:hacl(dmz_11,app_44,tcp,8080):1
2730:hacl(dmz_4,app_43,tcp,8080):1
2694:hacl(dmz_13,app_43,tcp,8080):1
2757:hacl(dmz_12,app_44,tcp,8080):1
2754:RULE 5 (multi-hop access):0
2729:RULE 5 (multi-hop access):0
2693:RULE 5 (multi-hop access):0
2756:RULE 5 (multi-hop access):0
s(app_44,tcp,8080):0
2806:networkServiceInfo(app_44,tomcat,tcp,8080,root):1
2807:vulExists(app_44,vul_app,tomcat,remoteExploit,privEscalation):1
2746:RULE 2 (remote exploit of a server program):0

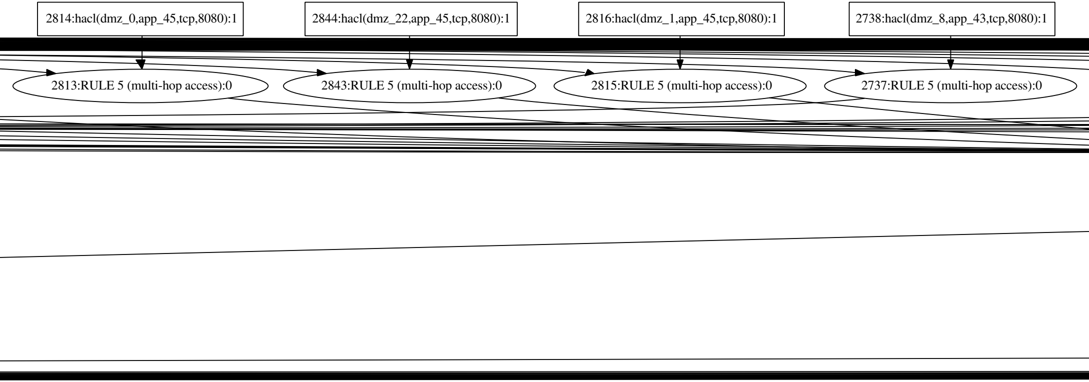

2722:hacl(dmz_26,app_43,tcp,8080):1
2781:hacl(dmz_23,app_44,tcp,8080):1
2700:hacl(dmz_16,app_43,tcp,8080):1
2783:hacl(dmz_24,app_44,tcp,8080):1
2721:RULE 5 (multi-hop access):0
2780:RULE 5 (multi-hop access):0
2699:RULE 5 (multi-hop access):0
2782:RULE 5 (multi-hop access):0
2809:hacl(app_45,db_59,tcp,5432):1
2808:RULE 5 (multi-hop access):0

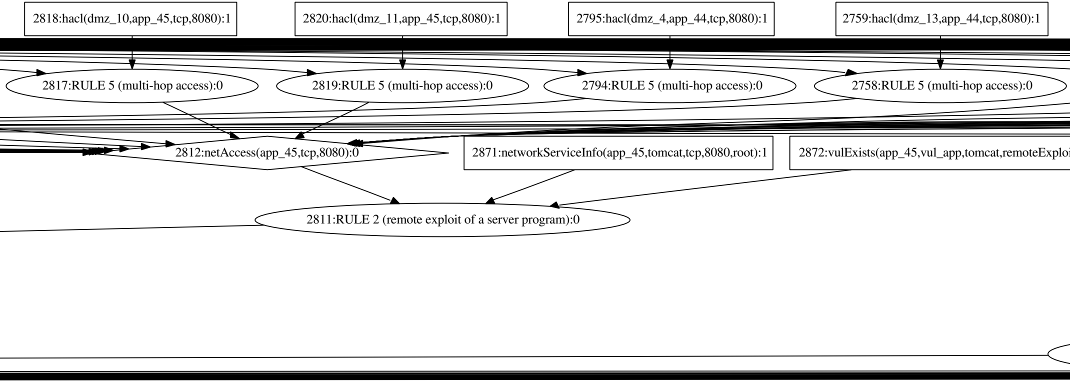

2732:hacl(dmz_5,app_43,tcp,8080):1
2854:hacl(dmz_27,app_45,tcp,8080):1
2856:hacl(dmz_28,app_45,tcp,8080):1
2858:hacl(dmz_3,app_45,tcp,8080):1
2731:RULE 5 (multi-hop access):0
2853:RULE 5 (multi-hop access):0
2855:RULE 5 (multi-hop access):0
2857:RULE 5 (multi-hop access):0

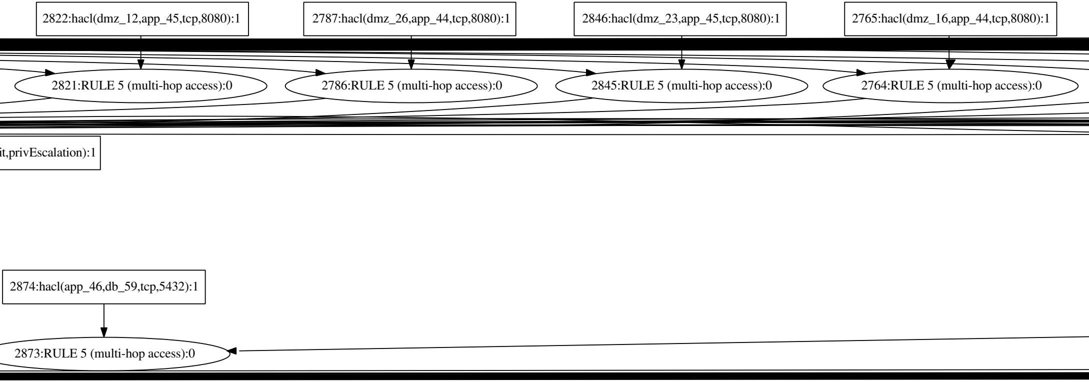

2785:hacl(dmz_25,app_44,tcp,8080):1
2826:hacl(dmz_14,app_45,tcp,8080):1
2832:hacl(dmz_17,app_45,tcp,8080):1
2805:hacl(dmz_9,app_44,tcp,8080):1
2784:RULE 5 (multi-hop access):0
2825:RULE 5 (multi-hop access):0
2831:RULE 5 (multi-hop access):0
2804:RULE 5 (multi-hop access):0

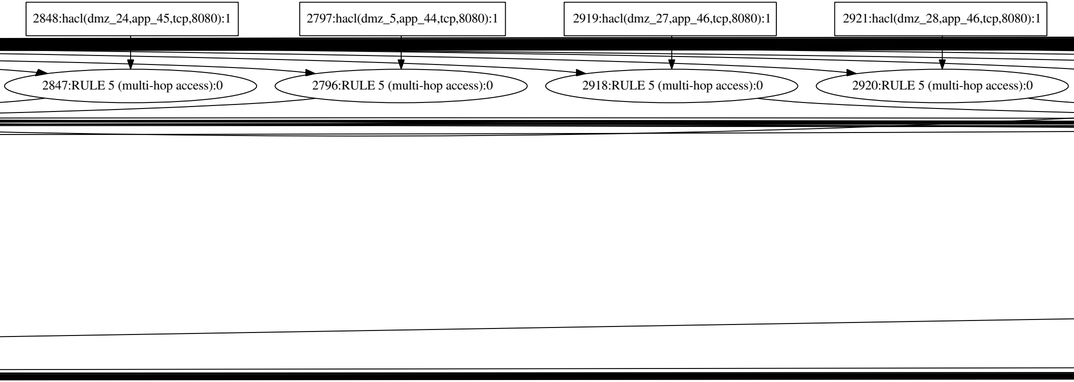

2834:hacl(dmz_18,app_45,tcp,8080):1
2734:hacl(dmz_6,app_43,tcp,8080):1
2836:hacl(dmz_19,app_45,tcp,8080):1
2838:hacl(dmz_2,app_45,tcp,8080):1
2833:RULE 5 (multi-hop access):0
2733:RULE 5 (multi-hop access):0
2835:RULE 5 (multi-hop access):0
2837:RULE 5 (multi-hop access):0
2810:execCode(app_45,root):0

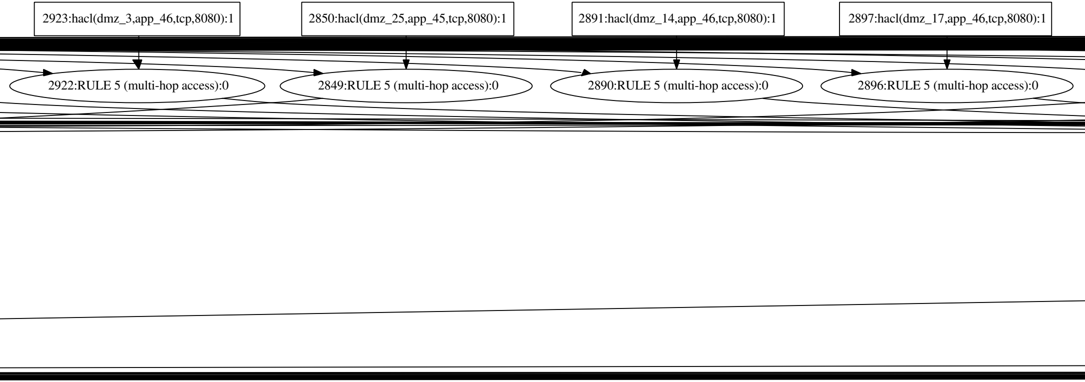

2840:hacl(dmz_20,app_45,tcp,8080):1
2763:hacl(dmz_15,app_44,tcp,8080):1
2842:hacl(dmz_21,app_45,tcp,8080):1
2736:hacl(dmz_7,app_43,tcp,8080):1
2839:RULE 5 (multi-hop access):0
2762:RULE 5 (multi-hop access):0
2841:RULE 5 (multi-hop access):0
2735:RULE 5 (multi-hop access):0

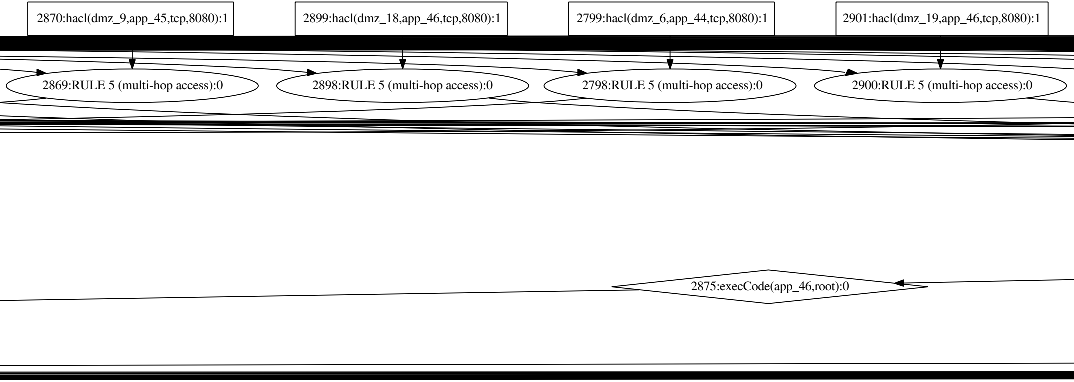

2814:hacl(dmz_0,app_45,tcp,8080):1
2844:hacl(dmz_22,app_45,tcp,8080):1
2816:hacl(dmz_1,app_45,tcp,8080):1
2738:hacl(dmz_8,app_43,tcp,8080):1
2813:RULE 5 (multi-hop access):0
2843:RULE 5 (multi-hop access):0
2815:RULE 5 (multi-hop access):0
2737:RULE 5 (multi-hop access):0

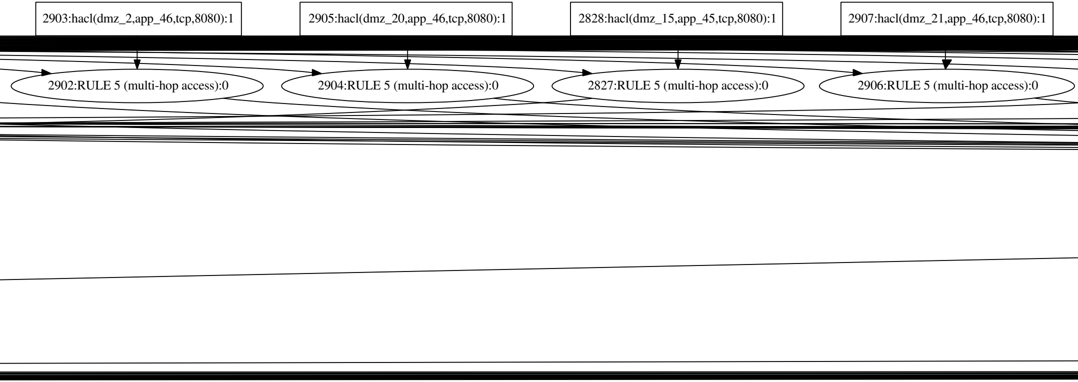

2818:hacl(dmz_10,app_45,tcp,8080):1
2820:hacl(dmz_11,app_45,tcp,8080):1
2795:hacl(dmz_4,app_44,tcp,8080):1
2759:hacl(dmz_13,app_44,tcp,8080):1
2817:RULE 5 (multi-hop access):0
2819:RULE 5 (multi-hop access):0
2794:RULE 5 (multi-hop access):0
2758:RULE 5 (multi-hop access):0
2812:netAccess(app_45,tcp,8080):0
2871:networkServiceInfo(app_45,tomcat,tcp,8080,root):1
2872:vulExists(app_45,vul_app,tomcat,remoteExploi
2811:RULE 2 (remote exploit of a server program):0

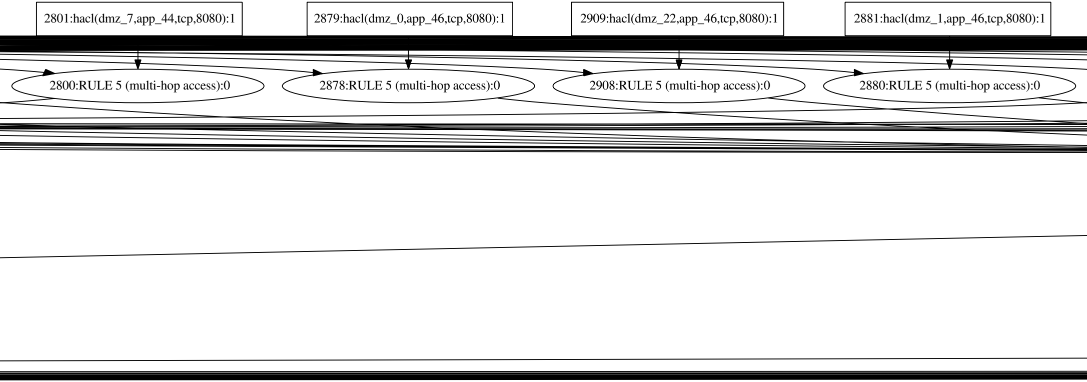

2822:hacl(dmz_12,app_45,tcp,8080):1
2787:hacl(dmz_26,app_44,tcp,8080):1
2846:hacl(dmz_23,app_45,tcp,8080):1
2765:hacl(dmz_16,app_44,tcp,8080):1
2821:RULE 5 (multi-hop access):0
2786:RULE 5 (multi-hop access):0
2845:RULE 5 (multi-hop access):0
2764:RULE 5 (multi-hop access):0
it,privEscalation):1
2874:hacl(app_46,db_59,tcp,5432):1
2873:RULE 5 (multi-hop access):0

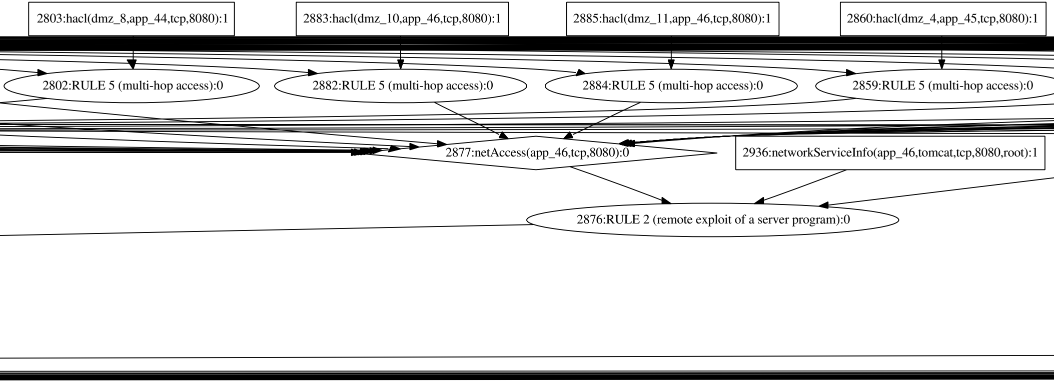

2848:hacl(dmz_24,app_45,tcp,8080):1
2797:hacl(dmz_5,app_44,tcp,8080):1
2919:hacl(dmz_27,app_46,tcp,8080):1
2921:hacl(dmz_28,app_46,tcp,8080):1
2847:RULE 5 (multi-hop access):0
2796:RULE 5 (multi-hop access):0
2918:RULE 5 (multi-hop access):0
2920:RULE 5 (multi-hop access):0

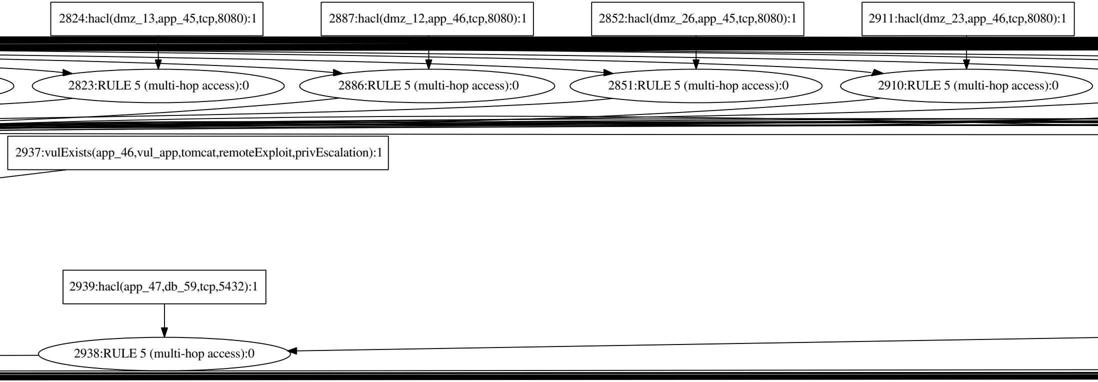

2923:hacl(dmz_3,app_46,tcp,8080):1
2850:hacl(dmz_25,app_45,tcp,8080):1
2891:hacl(dmz_14,app_46,tcp,8080):1
2897:hacl(dmz_17,app_46,tcp,8080):1
2922:RULE 5 (multi-hop access):0
2849:RULE 5 (multi-hop access):0
2890:RULE 5 (multi-hop access):0
2896:RULE 5 (multi-hop access):0

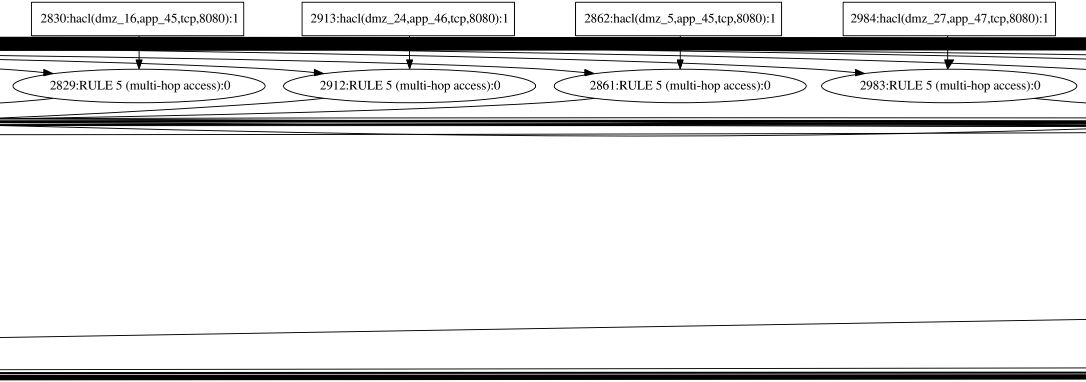

2870:hacl(dmz_9,app_45,tcp,8080):1
2899:hacl(dmz_18,app_46,tcp,8080):1
2799:hacl(dmz_6,app_44,tcp,8080):1
2901:hacl(dmz_19,app_46,tcp,8080):1
2869:RULE 5 (multi-hop access):0
2898:RULE 5 (multi-hop access):0
2798:RULE 5 (multi-hop access):0
2900:RULE 5 (multi-hop access):0
2875:execCode(app_46,root):0

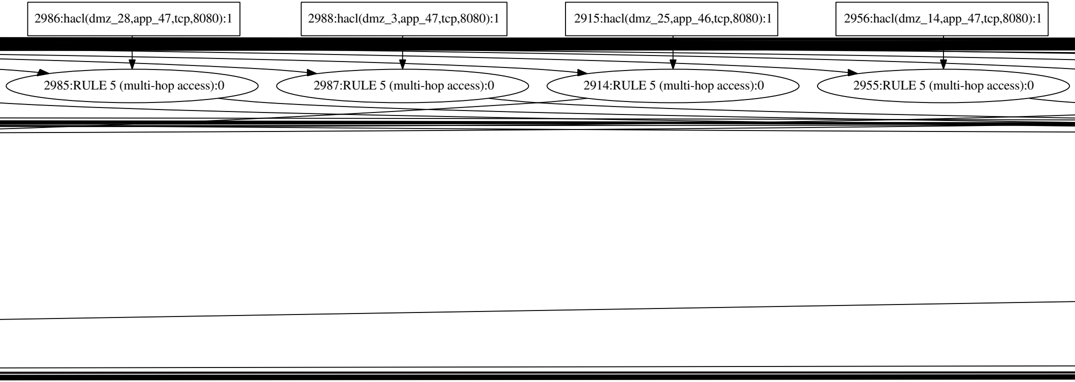

2903:hacl(dmz_2,app_46,tcp,8080):1
2905:hacl(dmz_20,app_46,tcp,8080):1
2828:hacl(dmz_15,app_45,tcp,8080):1
2907:hacl(dmz_21,app_46,tcp,8080):1
2902:RULE 5 (multi-hop access):0
2904:RULE 5 (multi-hop access):0
2827:RULE 5 (multi-hop access):0
2906:RULE 5 (multi-hop access):0

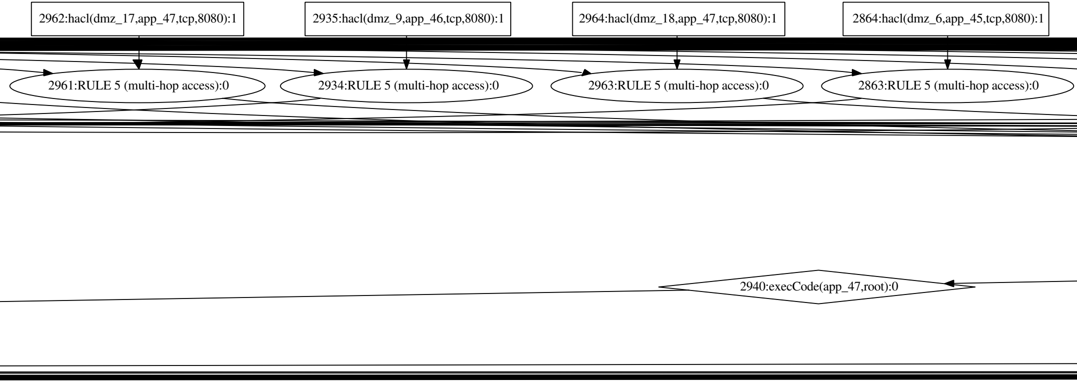

2801:hacl(dmz_7,app_44,tcp,8080):1
2879:hacl(dmz_0,app_46,tcp,8080):1
2909:hacl(dmz_22,app_46,tcp,8080):1
2881:hacl(dmz_1,app_46,tcp,8080):1
2800:RULE 5 (multi-hop access):0
2878:RULE 5 (multi-hop access):0
2908:RULE 5 (multi-hop access):0
2880:RULE 5 (multi-hop access):0

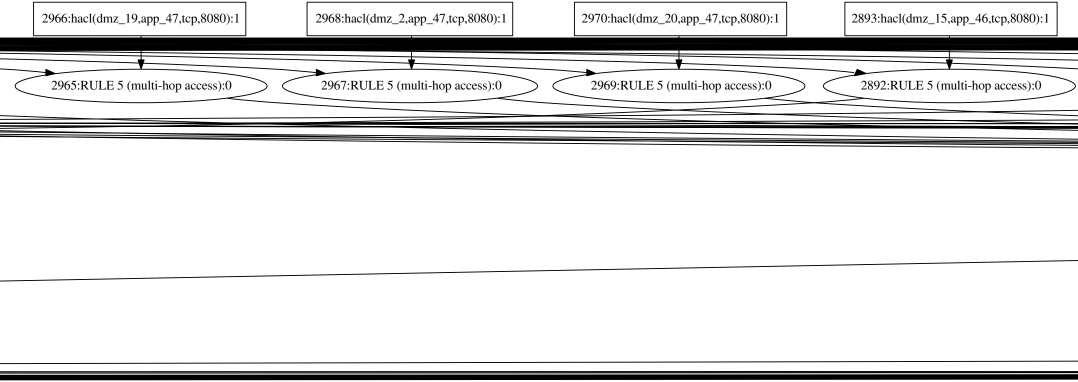

2803:hacl(dmz_8,app_44,tcp,8080):1
2883:hacl(dmz_10,app_46,tcp,8080):1
2885:hacl(dmz_11,app_46,tcp,8080):1
2860:hacl(dmz_4,app_45,tcp,8080):1
2802:RULE 5 (multi-hop access):0
2882:RULE 5 (multi-hop access):0
2884:RULE 5 (multi-hop access):0
2859:RULE 5 (multi-hop access):0
2877:netAccess(app_46,tcp,8080):0
2936:networkServiceInfo(app_46,tomcat,tcp,8080,root):1
2876:RULE 2 (remote exploit of a server program):0

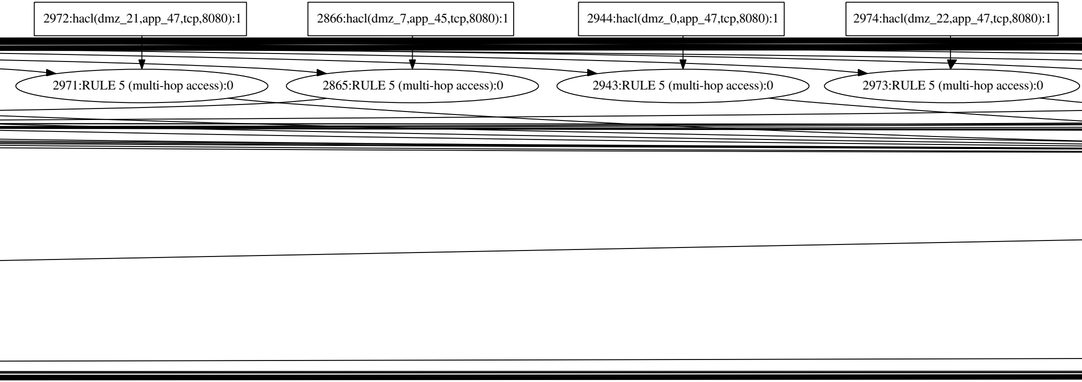

2824:hacl(dmz_13,app_45,tcp,8080):1
2887:hacl(dmz_12,app_46,tcp,8080):1
2852:hacl(dmz_26,app_45,tcp,8080):1
2911:hacl(dmz_23,app_46,tcp,8080):1
2823:RULE 5 (multi-hop access):0
2886:RULE 5 (multi-hop access):0
2851:RULE 5 (multi-hop access):0
2910:RULE 5 (multi-hop access):0
2937:vulExists(app_46,vul_app,tomcat,remoteExploit,privEscalation):1
2939:hacl(app_47,db_59,tcp,5432):1
2938:RULE 5 (multi-hop access):0

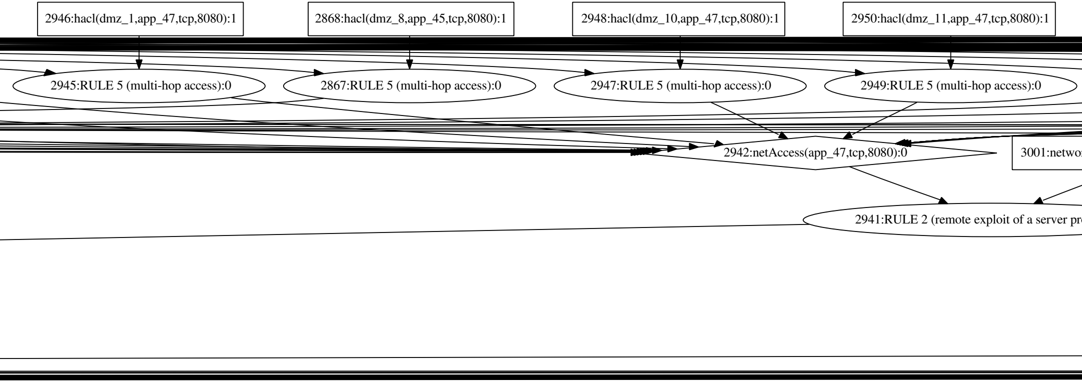

2830:hacl(dmz_16,app_45,tcp,8080):1
2913:hacl(dmz_24,app_46,tcp,8080):1
2862:hacl(dmz_5,app_45,tcp,8080):1
2984:hacl(dmz_27,app_47,tcp,8080):1
2829:RULE 5 (multi-hop access):0
2912:RULE 5 (multi-hop access):0
2861:RULE 5 (multi-hop access):0
2983:RULE 5 (multi-hop access):0

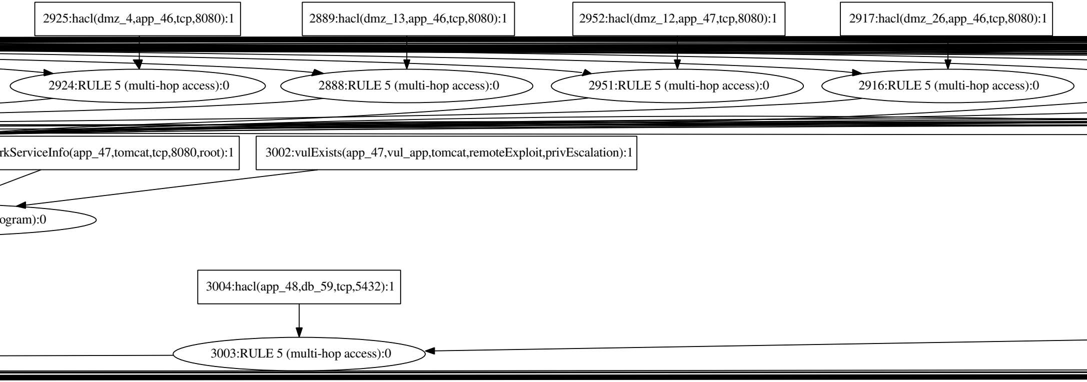

2986:hacl(dmz_28,app_47,tcp,8080):1
2988:hacl(dmz_3,app_47,tcp,8080):1
2915:hacl(dmz_25,app_46,tcp,8080):1
2956:hacl(dmz_14,app_47,tcp,8080):1
2985:RULE 5 (multi-hop access):0
2987:RULE 5 (multi-hop access):0
2914:RULE 5 (multi-hop access):0
2955:RULE 5 (multi-hop access):0

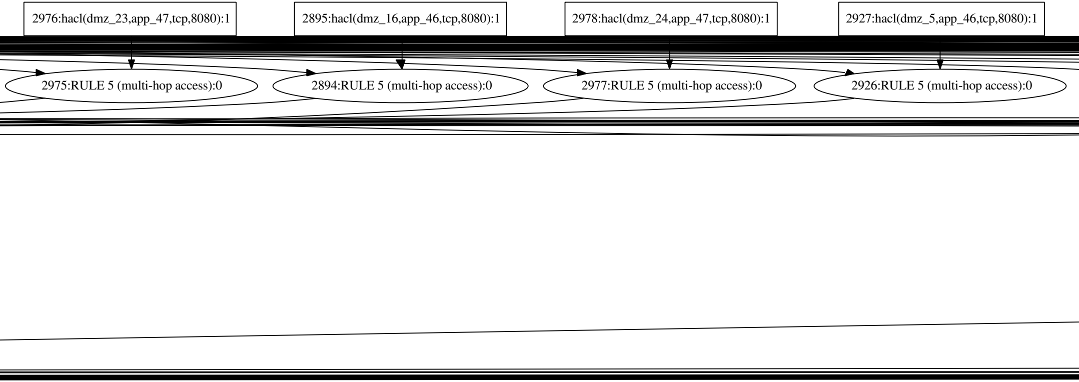

2962:hacl(dmz_17,app_47,tcp,8080):1
2935:hacl(dmz_9,app_46,tcp,8080):1
2964:hacl(dmz_18,app_47,tcp,8080):1
2864:hacl(dmz_6,app_45,tcp,8080):1
2961:RULE 5 (multi-hop access):0
2934:RULE 5 (multi-hop access):0
2963:RULE 5 (multi-hop access):0
2863:RULE 5 (multi-hop access):0
2940:execCode(app_47,root):0

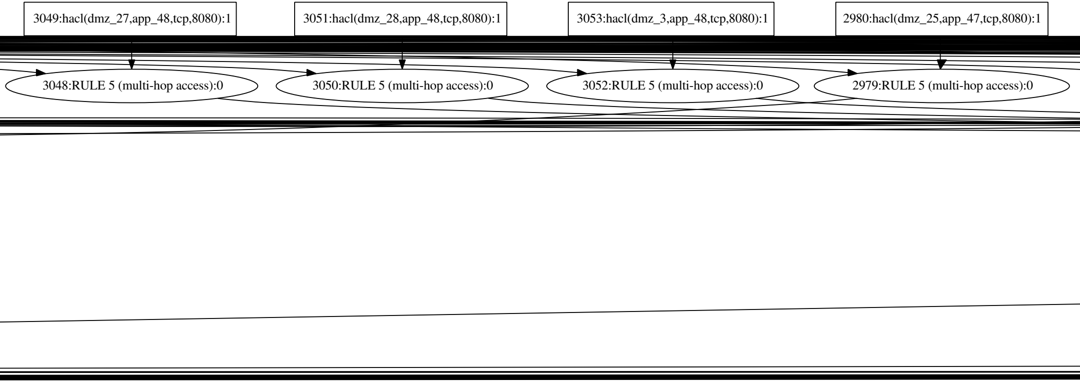

2966:hacl(dmz_19,app_47,tcp,8080):1
2968:hacl(dmz_2,app_47,tcp,8080):1
2970:hacl(dmz_20,app_47,tcp,8080):1
2893:hacl(dmz_15,app_46,tcp,8080):1
2965:RULE 5 (multi-hop access):0
2967:RULE 5 (multi-hop access):0
2969:RULE 5 (multi-hop access):0
2892:RULE 5 (multi-hop access):0

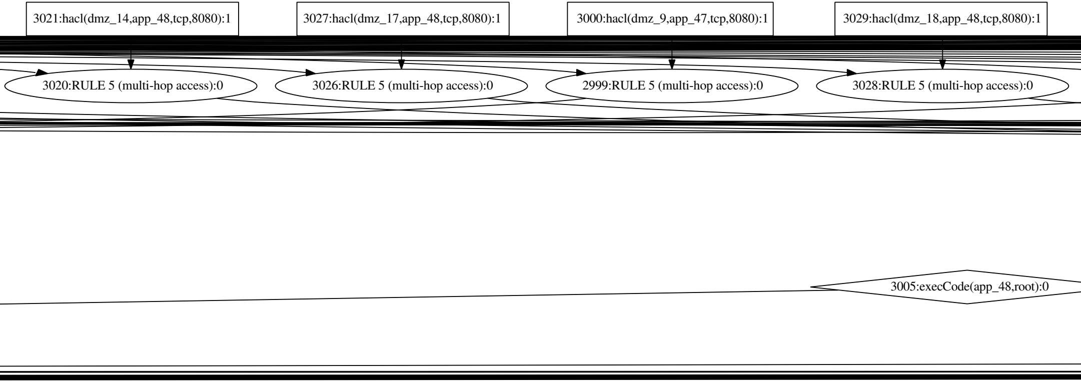

2972:hacl(dmz_21,app_47,tcp,8080):1
2866:hacl(dmz_7,app_45,tcp,8080):1
2944:hacl(dmz_0,app_47,tcp,8080):1
2974:hacl(dmz_22,app_47,tcp,8080):1
2971:RULE 5 (multi-hop access):0
2865:RULE 5 (multi-hop access):0
2943:RULE 5 (multi-hop access):0
2973:RULE 5 (multi-hop access):0

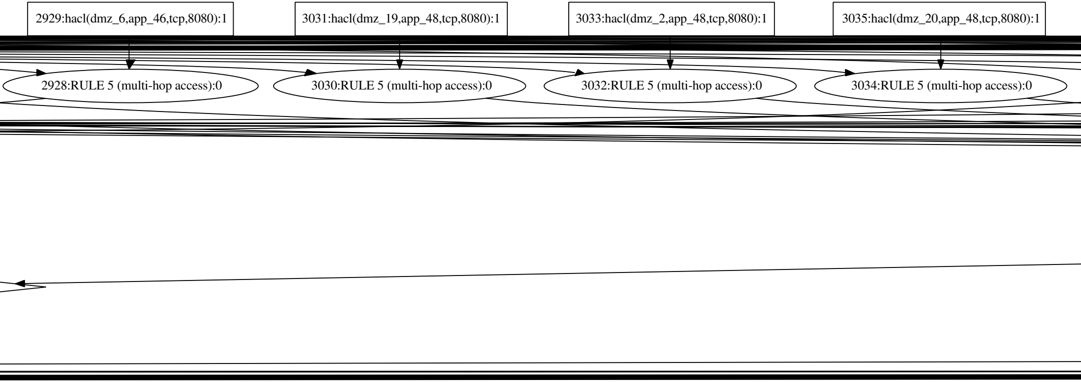

2946:hacl(dmz_1,app_47,tcp,8080):1
2868:hacl(dmz_8,app_45,tcp,8080):1
2948:hacl(dmz_10,app_47,tcp,8080):1
2950:hacl(dmz_11,app_47,tcp,8080):1
2945:RULE 5 (multi-hop access):0
2867:RULE 5 (multi-hop access):0
2947:RULE 5 (multi-hop access):0
2949:RULE 5 (multi-hop access):0
2942:netAccess(app_47,tcp,8080):0
3001:networ
2941:RULE 2 (remote exploit of a server pr

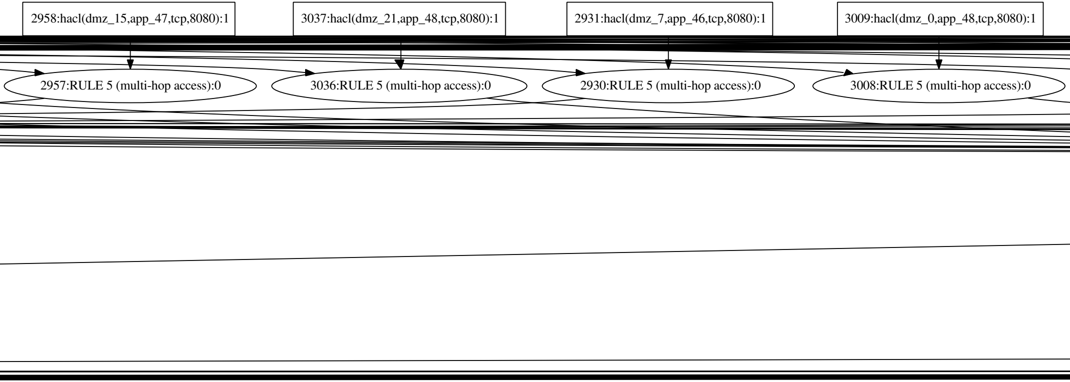

2925:hacl(dmz_4,app_46,tcp,8080):1
2889:hacl(dmz_13,app_46,tcp,8080):1
2952:hacl(dmz_12,app_47,tcp,8080):1
2917:hacl(dmz_26,app_46,tcp,8080):1
2924:RULE 5 (multi-hop access):0
2888:RULE 5 (multi-hop access):0
2951:RULE 5 (multi-hop access):0
2916:RULE 5 (multi-hop access):0
rkServiceInfo(app_47,tomcat,tcp,8080,root):1
3002:vulExists(app_47,vul_app,tomcat,remoteExploit,privEscalation):1
ogram):0
3004:hacl(app_48,db_59,tcp,5432):1
3003:RULE 5 (multi-hop access):0

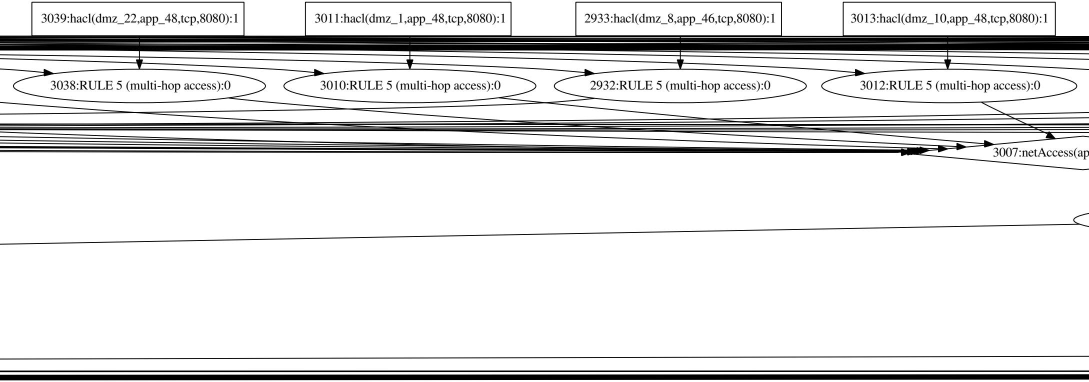

2976:hacl(dmz_23,app_47,tcp,8080):1
2895:hacl(dmz_16,app_46,tcp,8080):1
2978:hacl(dmz_24,app_47,tcp,8080):1
2927:hacl(dmz_5,app_46,tcp,8080):1
2975:RULE 5 (multi-hop access):0
2894:RULE 5 (multi-hop access):0
2977:RULE 5 (multi-hop access):0
2926:RULE 5 (multi-hop access):0

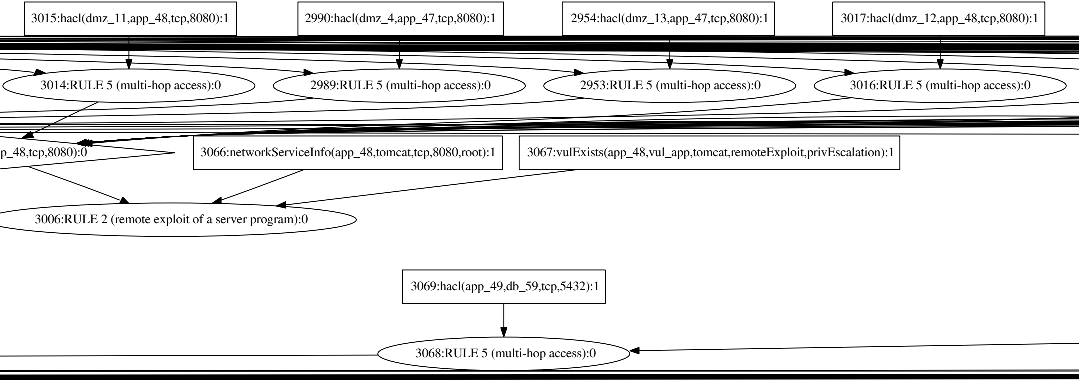

3049:hacl(dmz_27,app_48,tcp,8080):1
3051:hacl(dmz_28,app_48,tcp,8080):1
3053:hacl(dmz_3,app_48,tcp,8080):1
2980:hacl(dmz_25,app_47,tcp,8080):1
3048:RULE 5 (multi-hop access):0
3050:RULE 5 (multi-hop access):0
3052:RULE 5 (multi-hop access):0
2979:RULE 5 (multi-hop access):0

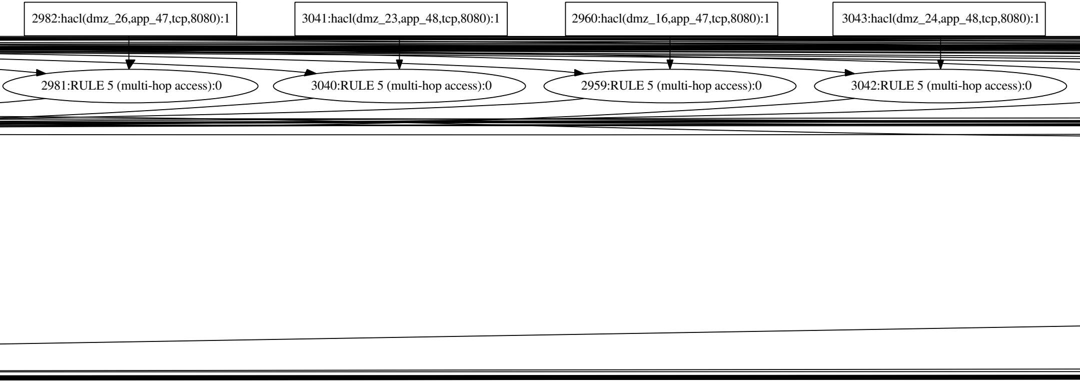

3021:hacl(dmz_14,app_48,tcp,8080):1
3027:hacl(dmz_17,app_48,tcp,8080):1
3000:hacl(dmz_9,app_47,tcp,8080):1
3029:hacl(dmz_18,app_48,tcp,8080):1
3020:RULE 5 (multi-hop access):0
3026:RULE 5 (multi-hop access):0
2999:RULE 5 (multi-hop access):0
3028:RULE 5 (multi-hop access):0
3005:execCode(app_48,root):0

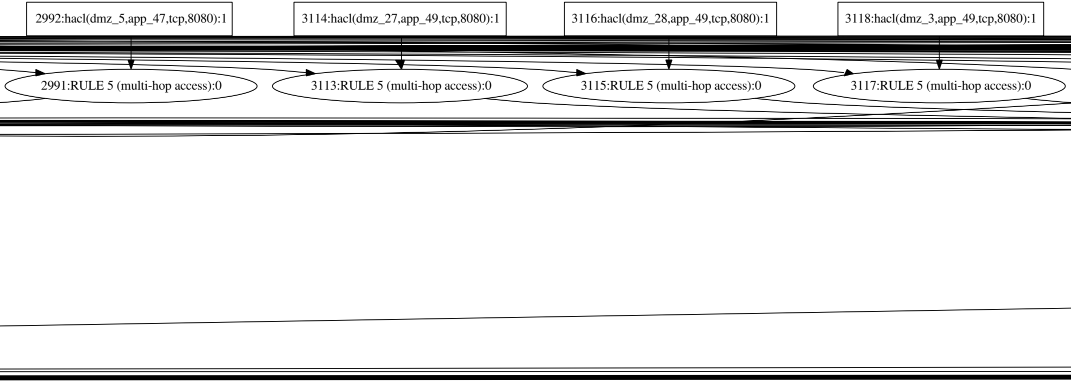

2929:hacl(dmz_6,app_46,tcp,8080):1
3031:hacl(dmz_19,app_48,tcp,8080):1
3033:hacl(dmz_2,app_48,tcp,8080):1
3035:hacl(dmz_20,app_48,tcp,8080):1
2928:RULE 5 (multi-hop access):0
3030:RULE 5 (multi-hop access):0
3032:RULE 5 (multi-hop access):0
3034:RULE 5 (multi-hop access):0

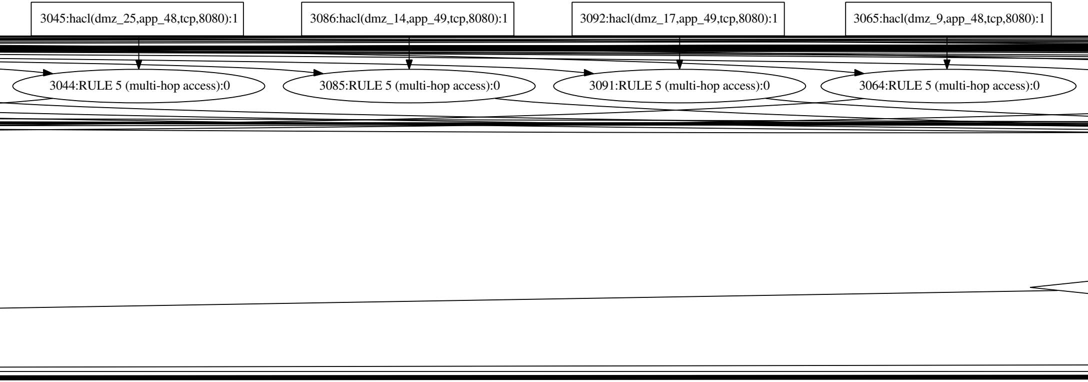

2958:hacl(dmz_15,app_47,tcp,8080):1
3037:hacl(dmz_21,app_48,tcp,8080):1
2931:hacl(dmz_7,app_46,tcp,8080):1
3009:hacl(dmz_0,app_48,tcp,8080):1
2957:RULE 5 (multi-hop access):0
3036:RULE 5 (multi-hop access):0
2930:RULE 5 (multi-hop access):0
3008:RULE 5 (multi-hop access):0

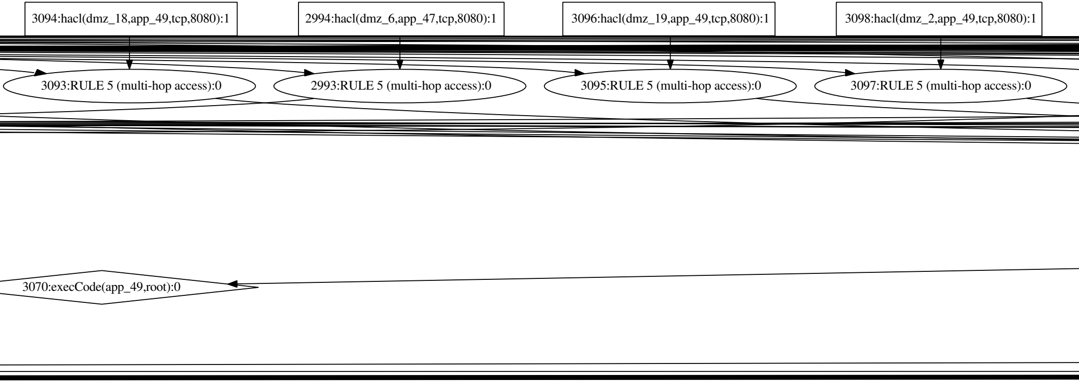

3039:hacl(dmz_22,app_48,tcp,8080):1
3011:hacl(dmz_1,app_48,tcp,8080):1
2933:hacl(dmz_8,app_46,tcp,8080):1
3013:hacl(dmz_10,app_48,tcp,8080):1
3038:RULE 5 (multi-hop access):0
3010:RULE 5 (multi-hop access):0
2932:RULE 5 (multi-hop access):0
3012:RULE 5 (multi-hop access):0
3007:netAccess(ap

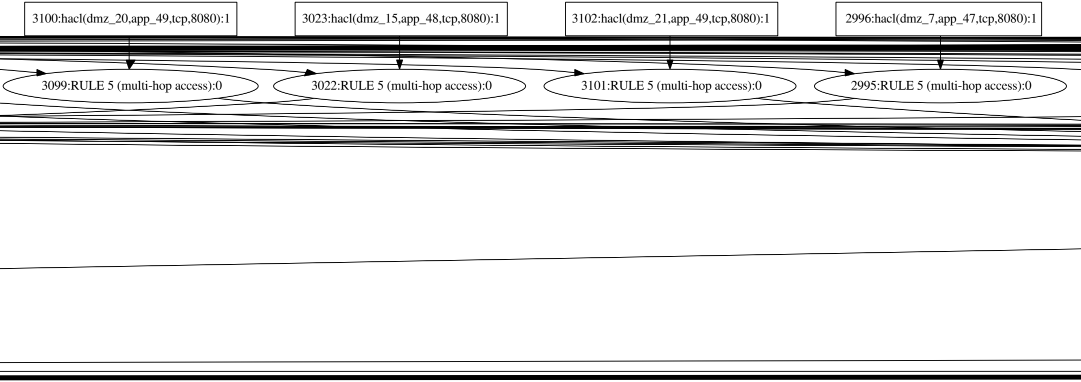

3015:hacl(dmz_11,app_48,tcp,8080):1
2990:hacl(dmz_4,app_47,tcp,8080):1
2954:hacl(dmz_13,app_47,tcp,8080):1
3017:hacl(dmz_12,app_48,tcp,8080):1
3014:RULE 5 (multi-hop access):0
2989:RULE 5 (multi-hop access):0
2953:RULE 5 (multi-hop access):0
3016:RULE 5 (multi-hop access):0
p_48,tcp,8080):0
3066:networkServiceInfo(app_48,tomcat,tcp,8080,root):1
3067:vulExists(app_48,vul_app,tomcat,remoteExploit,privEscalation):1
3006:RULE 2 (remote exploit of a server program):0
3069:hacl(app_49,db_59,tcp,5432):1
3068:RULE 5 (multi-hop access):0

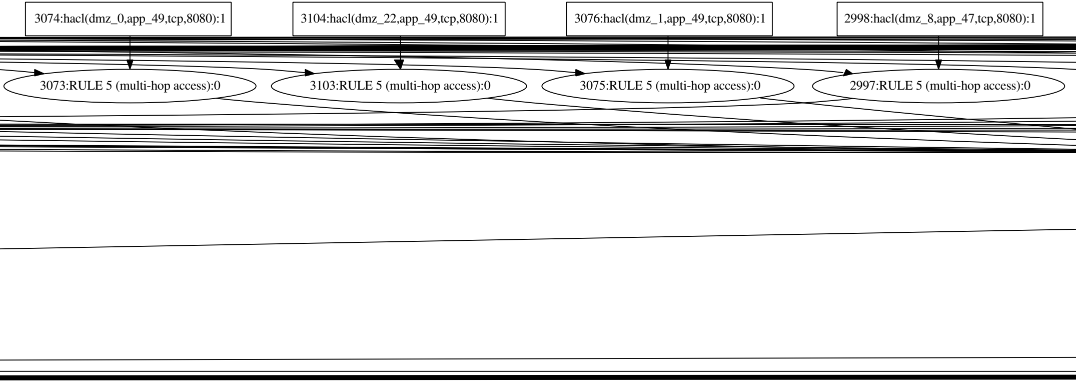

2982:hacl(dmz_26,app_47,tcp,8080):1
3041:hacl(dmz_23,app_48,tcp,8080):1
2960:hacl(dmz_16,app_47,tcp,8080):1
3043:hacl(dmz_24,app_48,tcp,8080):1
2981:RULE 5 (multi-hop access):0
3040:RULE 5 (multi-hop access):0
2959:RULE 5 (multi-hop access):0
3042:RULE 5 (multi-hop access):0

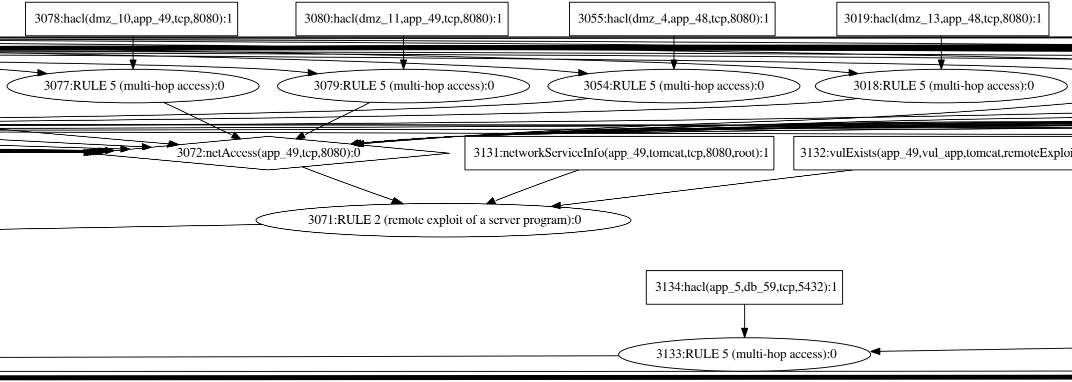

2992:hacl(dmz_5,app_47,tcp,8080):1
3114:hacl(dmz_27,app_49,tcp,8080):1
3116:hacl(dmz_28,app_49,tcp,8080):1
3118:hacl(dmz_3,app_49,tcp,8080):1
2991:RULE 5 (multi-hop access):0
3113:RULE 5 (multi-hop access):0
3115:RULE 5 (multi-hop access):0
3117:RULE 5 (multi-hop access):0

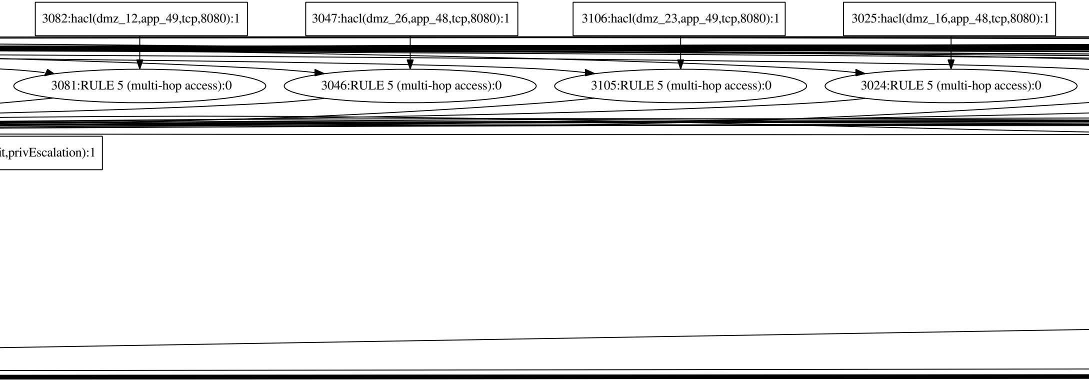

3045:hacl(dmz_25,app_48,tcp,8080):1
3086:hacl(dmz_14,app_49,tcp,8080):1
3092:hacl(dmz_17,app_49,tcp,8080):1
3065:hacl(dmz_9,app_48,tcp,8080):1
3044:RULE 5 (multi-hop access):0
3085:RULE 5 (multi-hop access):0
3091:RULE 5 (multi-hop access):0
3064:RULE 5 (multi-hop access):0

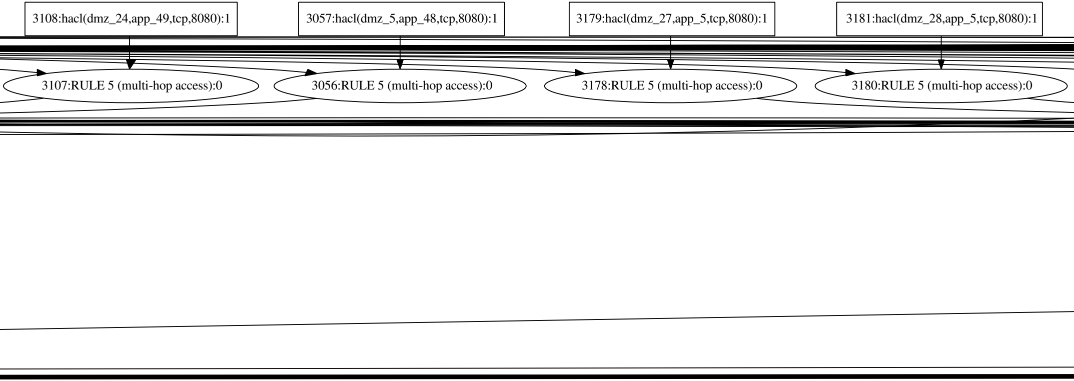

3094:hacl(dmz_18,app_49,tcp,8080):1
2994:hacl(dmz_6,app_47,tcp,8080):1
3096:hacl(dmz_19,app_49,tcp,8080):1
3098:hacl(dmz_2,app_49,tcp,8080):1
3093:RULE 5 (multi-hop access):0
2993:RULE 5 (multi-hop access):0
3095:RULE 5 (multi-hop access):0
3097:RULE 5 (multi-hop access):0
3070:execCode(app_49,root):0

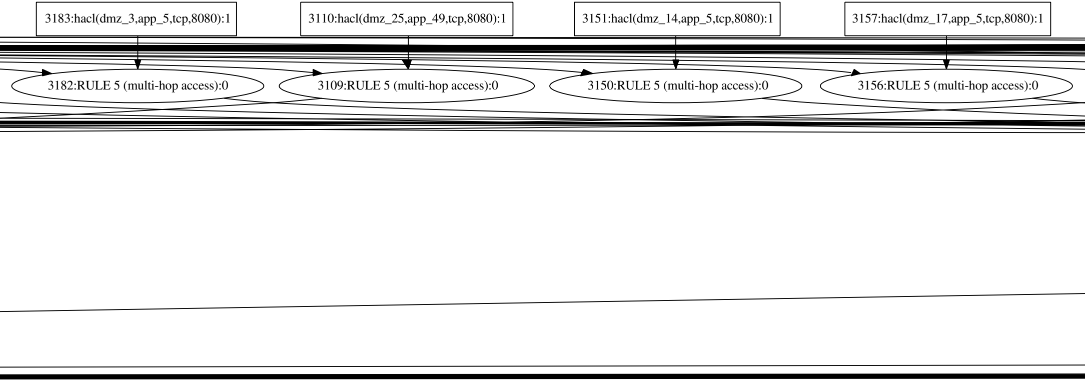

3100:hacl(dmz_20,app_49,tcp,8080):1
3023:hacl(dmz_15,app_48,tcp,8080):1
3102:hacl(dmz_21,app_49,tcp,8080):1
2996:hacl(dmz_7,app_47,tcp,8080):1
3099:RULE 5 (multi-hop access):0
3022:RULE 5 (multi-hop access):0
3101:RULE 5 (multi-hop access):0
2995:RULE 5 (multi-hop access):0

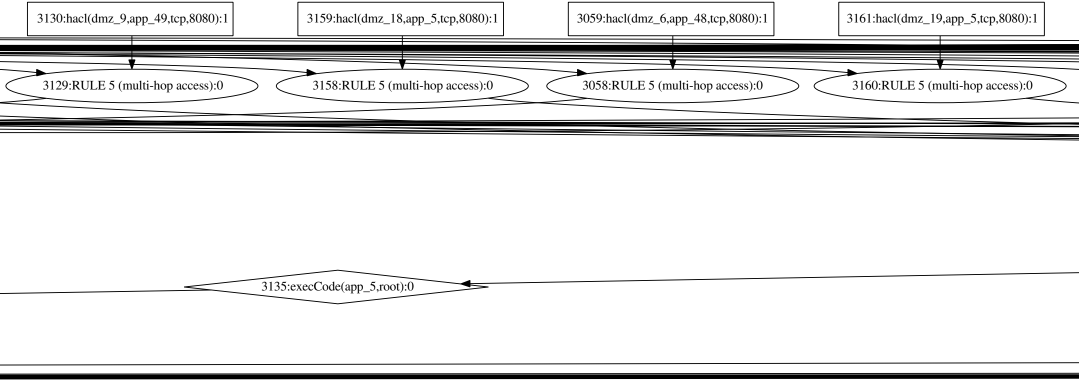

3074:hacl(dmz_0,app_49,tcp,8080):1
3104:hacl(dmz_22,app_49,tcp,8080):1
3076:hacl(dmz_1,app_49,tcp,8080):1
2998:hacl(dmz_8,app_47,tcp,8080):1
3073:RULE 5 (multi-hop access):0
3103:RULE 5 (multi-hop access):0
3075:RULE 5 (multi-hop access):0
2997:RULE 5 (multi-hop access):0

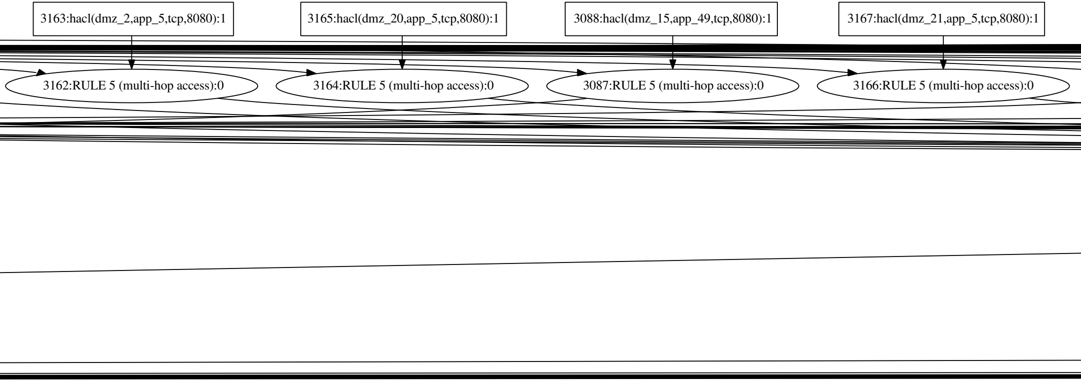

3078:hacl(dmz_10,app_49,tcp,8080):1
3080:hacl(dmz_11,app_49,tcp,8080):1
3055:hacl(dmz_4,app_48,tcp,8080):1
3019:hacl(dmz_13,app_48,tcp,8080):1
3077:RULE 5 (multi-hop access):0
3079:RULE 5 (multi-hop access):0
3054:RULE 5 (multi-hop access):0
3018:RULE 5 (multi-hop access):0
3072:netAccess(app_49,tcp,8080):0
3131:networkServiceInfo(app_49,tomcat,tcp,8080,root):1
3132:vulExists(app_49,vul_app,tomcat,remoteExploi
3071:RULE 2 (remote exploit of a server program):0
3134:hacl(app_5,db_59,tcp,5432):1
3133:RULE 5 (multi-hop access):0

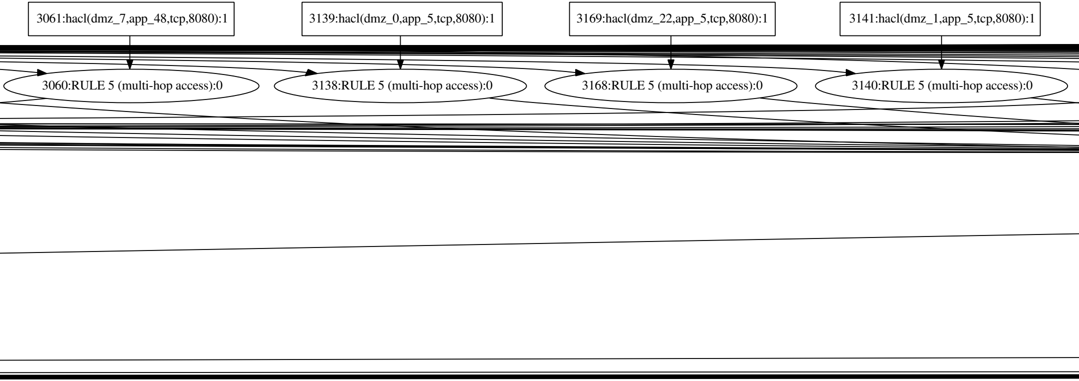

3082:hacl(dmz_12,app_49,tcp,8080):1
3047:hacl(dmz_26,app_48,tcp,8080):1
3106:hacl(dmz_23,app_49,tcp,8080):1
3025:hacl(dmz_16,app_48,tcp,8080):1
3081:RULE 5 (multi-hop access):0
3046:RULE 5 (multi-hop access):0
3105:RULE 5 (multi-hop access):0
3024:RULE 5 (multi-hop access):0
it,privEscalation):1

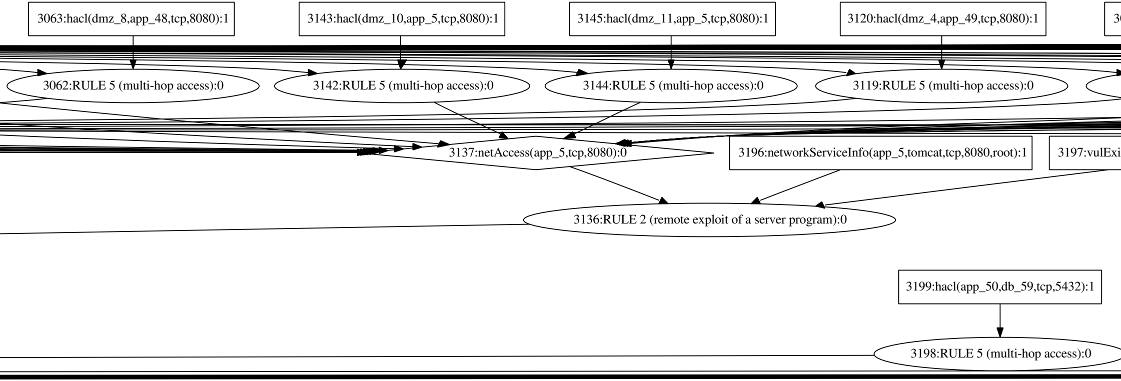

3108:hacl(dmz_24,app_49,tcp,8080):1
3057:hacl(dmz_5,app_48,tcp,8080):1
3179:hacl(dmz_27,app_5,tcp,8080):1
3181:hacl(dmz_28,app_5,tcp,8080):1
3107:RULE 5 (multi-hop access):0
3056:RULE 5 (multi-hop access):0
3178:RULE 5 (multi-hop access):0
3180:RULE 5 (multi-hop access):0

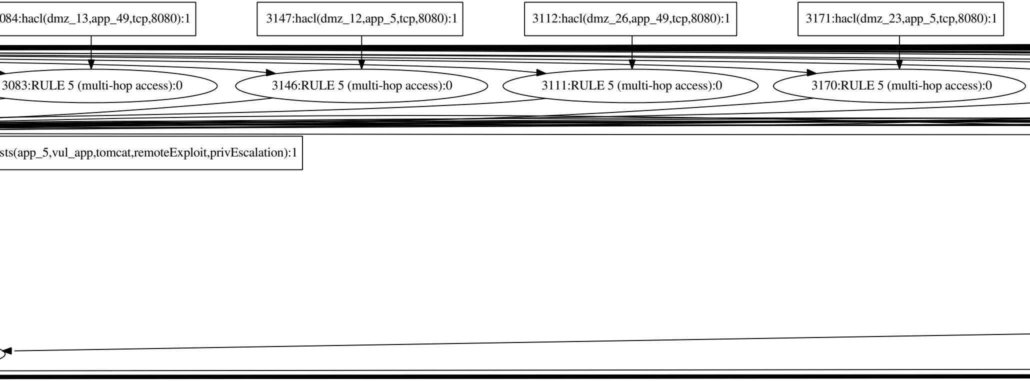

3183:hacl(dmz_3,app_5,tcp,8080):1
3110:hacl(dmz_25,app_49,tcp,8080):1
3151:hacl(dmz_14,app_5,tcp,8080):1
3157:hacl(dmz_17,app_5,tcp,8080):1
3182:RULE 5 (multi-hop access):0
3109:RULE 5 (multi-hop access):0
3150:RULE 5 (multi-hop access):0
3156:RULE 5 (multi-hop access):0

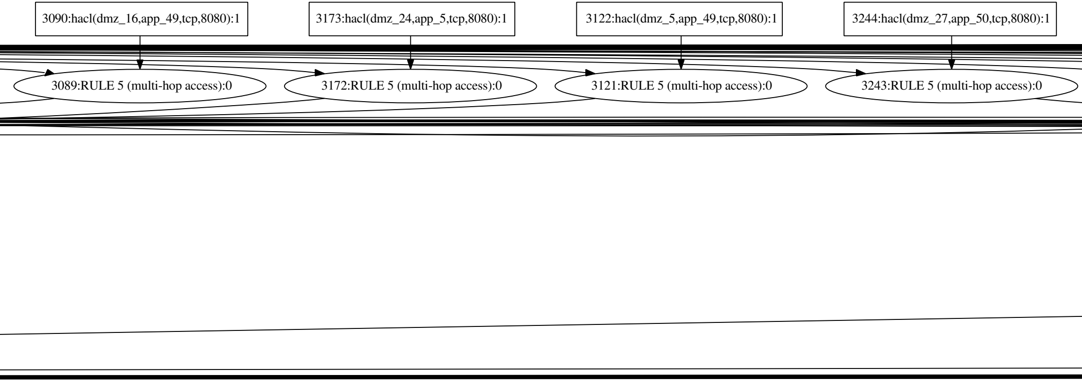

3130:hacl(dmz_9,app_49,tcp,8080):1
3159:hacl(dmz_18,app_5,tcp,8080):1
3059:hacl(dmz_6,app_48,tcp,8080):1
3161:hacl(dmz_19,app_5,tcp,8080):1
3129:RULE 5 (multi-hop access):0
3158:RULE 5 (multi-hop access):0
3058:RULE 5 (multi-hop access):0
3160:RULE 5 (multi-hop access):0
3135:execCode(app_5,root):0

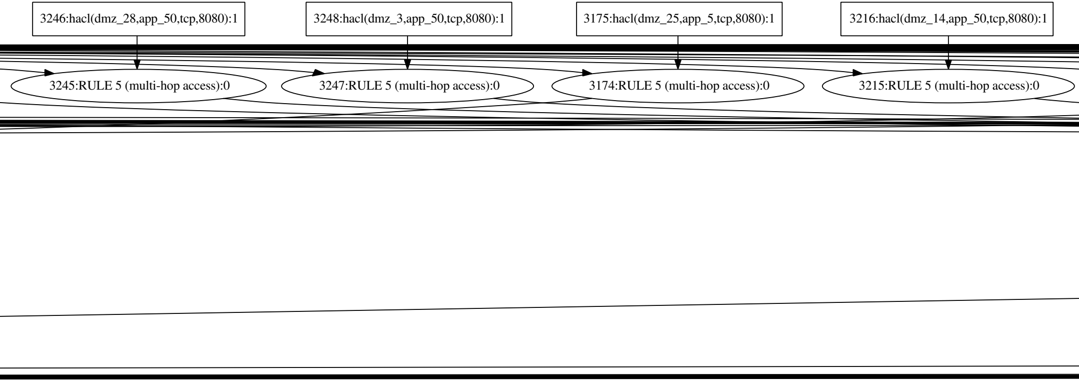

3163:hacl(dmz_2,app_5,tcp,8080):1
3165:hacl(dmz_20,app_5,tcp,8080):1
3088:hacl(dmz_15,app_49,tcp,8080):1
3167:hacl(dmz_21,app_5,tcp,8080):1
3162:RULE 5 (multi-hop access):0
3164:RULE 5 (multi-hop access):0
3087:RULE 5 (multi-hop access):0
3166:RULE 5 (multi-hop access):0

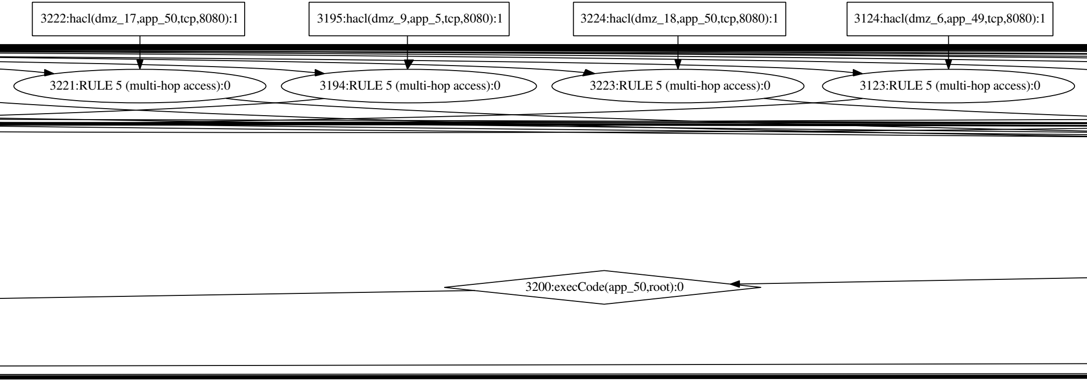

3061:hacl(dmz_7,app_48,tcp,8080):1
3139:hacl(dmz_0,app_5,tcp,8080):1
3169:hacl(dmz_22,app_5,tcp,8080):1
3141:hacl(dmz_1,app_5,tcp,8080):1
3060:RULE 5 (multi-hop access):0
3138:RULE 5 (multi-hop access):0
3168:RULE 5 (multi-hop access):0
3140:RULE 5 (multi-hop access):0

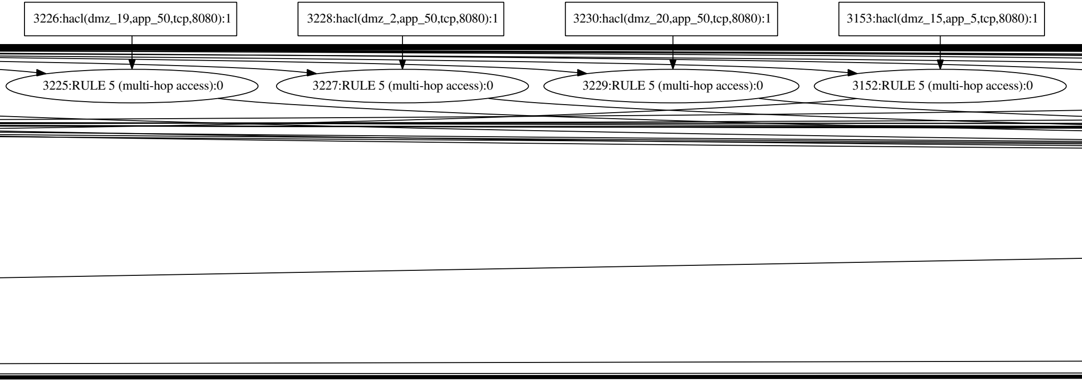

3063:hacl(dmz_8,app_48,tcp,8080):1
3143:hacl(dmz_10,app_5,tcp,8080):1
3145:hacl(dmz_11,app_5,tcp,8080):1
3120:hacl(dmz_4,app_49,tcp,8080):1
3062:RULE 5 (multi-hop access):0
3142:RULE 5 (multi-hop access):0
3144:RULE 5 (multi-hop access):0
3119:RULE 5 (multi-hop access):0
3137:netAccess(app_5,tcp,8080):0
3196:networkServiceInfo(app_5,tomcat,tcp,8080,root):1
3197:vulExi
3136:RULE 2 (remote exploit of a server program):0
3199:hacl(app_50,db_59,tcp,5432):1
3198:RULE 5 (multi-hop access):0

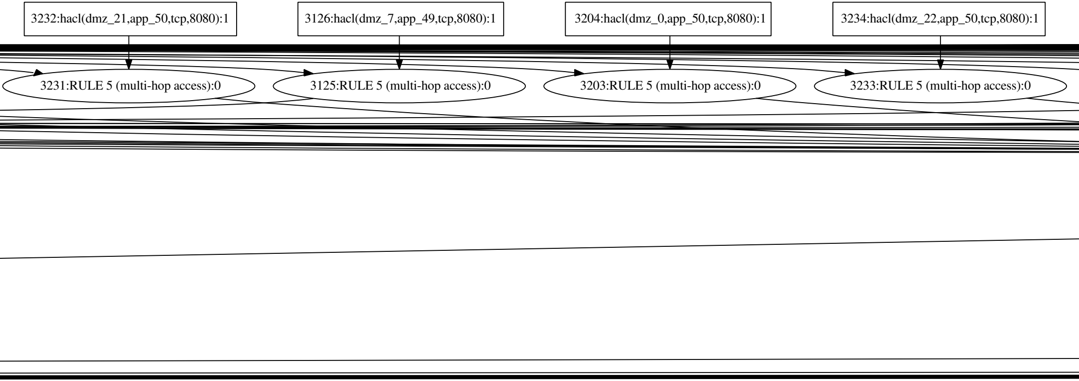

084:hacl(dmz_13,app_49,tcp,8080):1
3147:hacl(dmz_12,app_5,tcp,8080):1
3112:hacl(dmz_26,app_49,tcp,8080):1
3171:hacl(dmz_23,app_5,tcp,8080):1
3083:RULE 5 (multi-hop access):0
3146:RULE 5 (multi-hop access):0
3111:RULE 5 (multi-hop access):0
3170:RULE 5 (multi-hop access):0
sts(app_5,vul_app,tomcat,remoteExploit,privEscalation):1

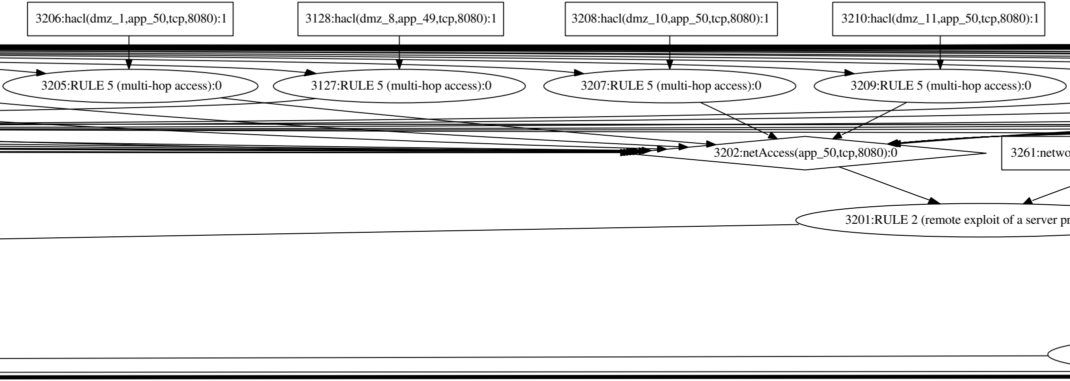

3090:hacl(dmz_16,app_49,tcp,8080):1
3173:hacl(dmz_24,app_5,tcp,8080):1
3122:hacl(dmz_5,app_49,tcp,8080):1
3244:hacl(dmz_27,app_50,tcp,8080):1
3089:RULE 5 (multi-hop access):0
3172:RULE 5 (multi-hop access):0
3121:RULE 5 (multi-hop access):0
3243:RULE 5 (multi-hop access):0

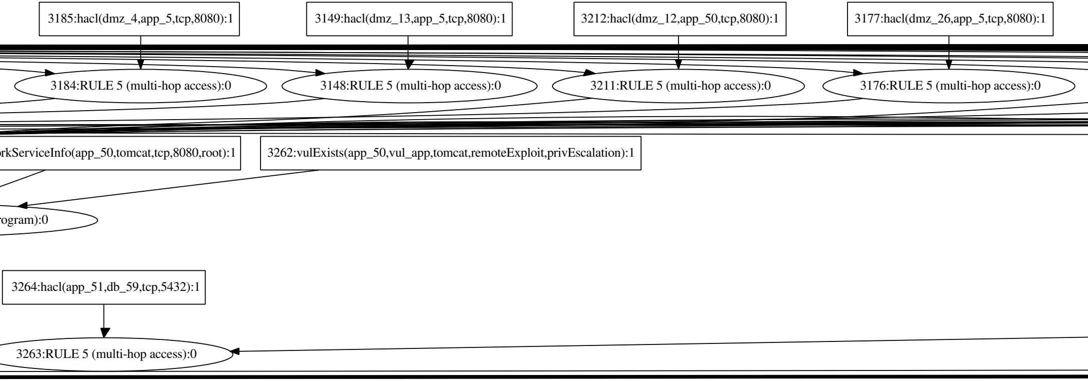

3246:hacl(dmz_28,app_50,tcp,8080):1
3248:hacl(dmz_3,app_50,tcp,8080):1
3175:hacl(dmz_25,app_5,tcp,8080):1
3216:hacl(dmz_14,app_50,tcp,8080):1
3245:RULE 5 (multi-hop access):0
3247:RULE 5 (multi-hop access):0
3174:RULE 5 (multi-hop access):0
3215:RULE 5 (multi-hop access):0

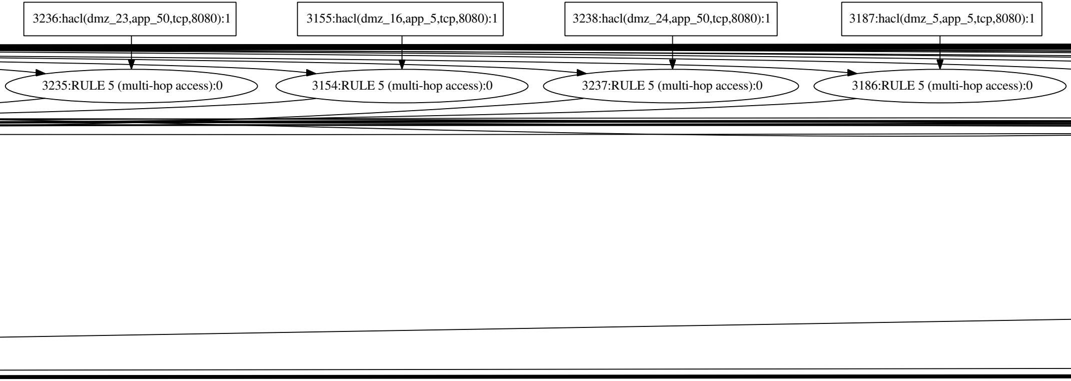

3222:hacl(dmz_17,app_50,tcp,8080):1
3195:hacl(dmz_9,app_5,tcp,8080):1
3224:hacl(dmz_18,app_50,tcp,8080):1
3124:hacl(dmz_6,app_49,tcp,8080):1
3221:RULE 5 (multi-hop access):0
3194:RULE 5 (multi-hop access):0
3223:RULE 5 (multi-hop access):0
3123:RULE 5 (multi-hop access):0
3200:execCode(app_50,root):0

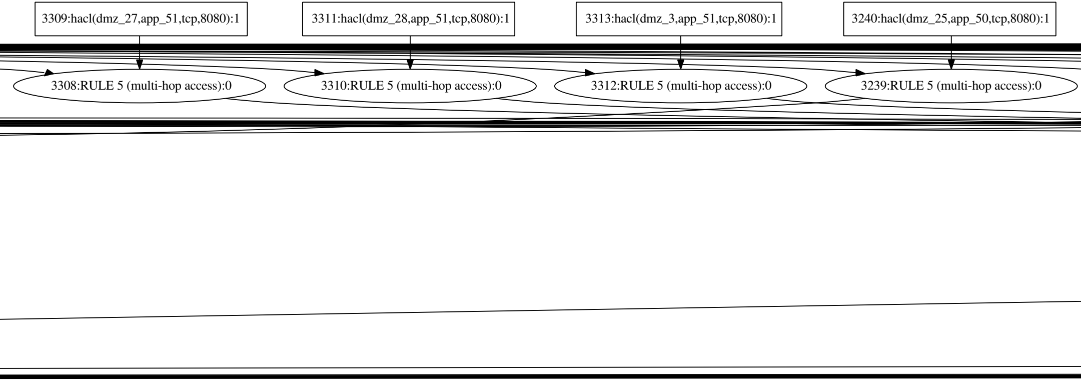

3226:hacl(dmz_19,app_50,tcp,8080):1
3228:hacl(dmz_2,app_50,tcp,8080):1
3230:hacl(dmz_20,app_50,tcp,8080):1
3153:hacl(dmz_15,app_5,tcp,8080):1
3225:RULE 5 (multi-hop access):0
3227:RULE 5 (multi-hop access):0
3229:RULE 5 (multi-hop access):0
3152:RULE 5 (multi-hop access):0

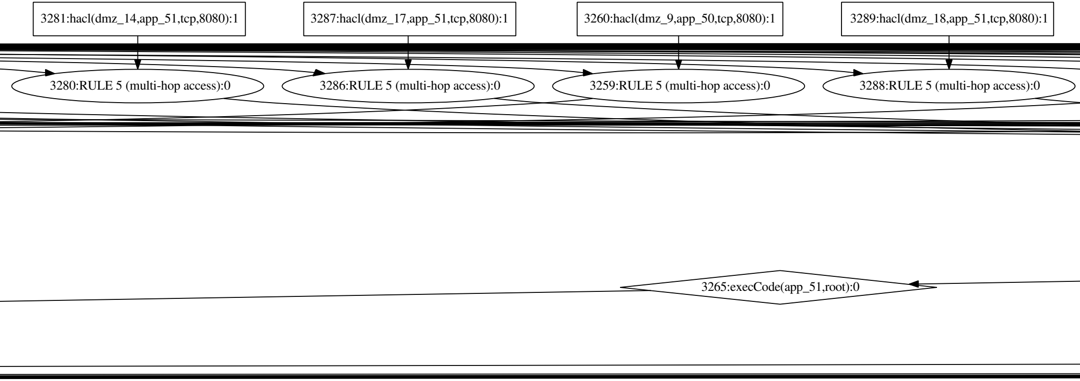

3232:hacl(dmz_21,app_50,tcp,8080):1
3126:hacl(dmz_7,app_49,tcp,8080):1
3204:hacl(dmz_0,app_50,tcp,8080):1
3234:hacl(dmz_22,app_50,tcp,8080):1
3231:RULE 5 (multi-hop access):0
3125:RULE 5 (multi-hop access):0
3203:RULE 5 (multi-hop access):0
3233:RULE 5 (multi-hop access):0

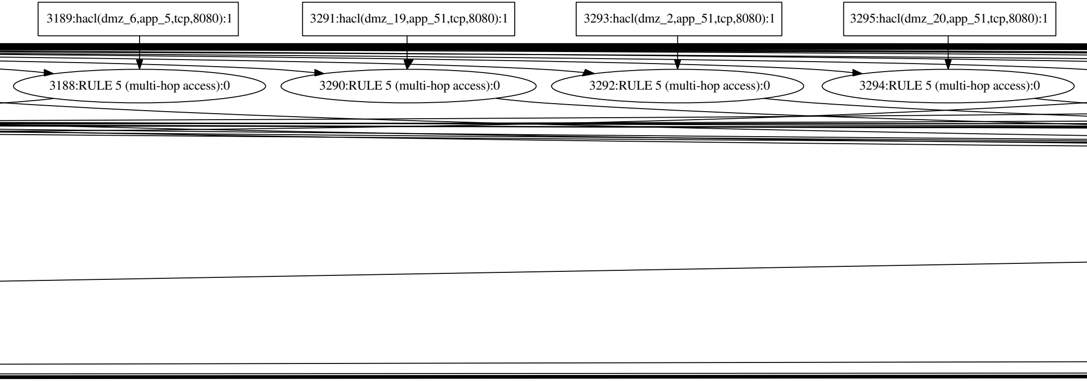

3206:hacl(dmz_1,app_50,tcp,8080):1
3128:hacl(dmz_8,app_49,tcp,8080):1
3208:hacl(dmz_10,app_50,tcp,8080):1
3210:hacl(dmz_11,app_50,tcp,8080):1
3205:RULE 5 (multi-hop access):0
3127:RULE 5 (multi-hop access):0
3207:RULE 5 (multi-hop access):0
3209:RULE 5 (multi-hop access):0
3202:netAccess(app_50,tcp,8080):0
3261:netwo
3201:RULE 2 (remote exploit of a server pr

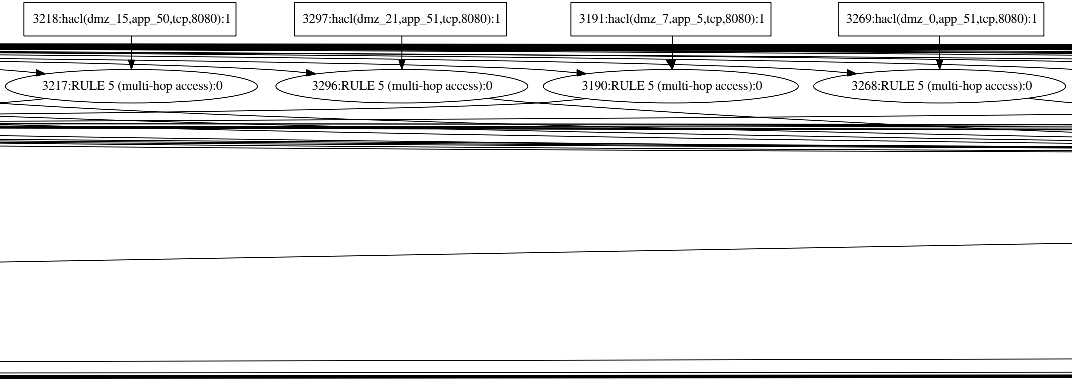

3185:hacl(dmz_4,app_5,tcp,8080):1
3149:hacl(dmz_13,app_5,tcp,8080):1
3212:hacl(dmz_12,app_50,tcp,8080):1
3177:hacl(dmz_26,app_5,tcp,8080):1
3184:RULE 5 (multi-hop access):0
3148:RULE 5 (multi-hop access):0
3211:RULE 5 (multi-hop access):0
3176:RULE 5 (multi-hop access):0
rkServiceInfo(app_50,tomcat,tcp,8080,root):1
3262:vulExists(app_50,vul_app,tomcat,remoteExploit,privEscalation):1
rogram):0
3264:hacl(app_51,db_59,tcp,5432):1
3263:RULE 5 (multi-hop access):0

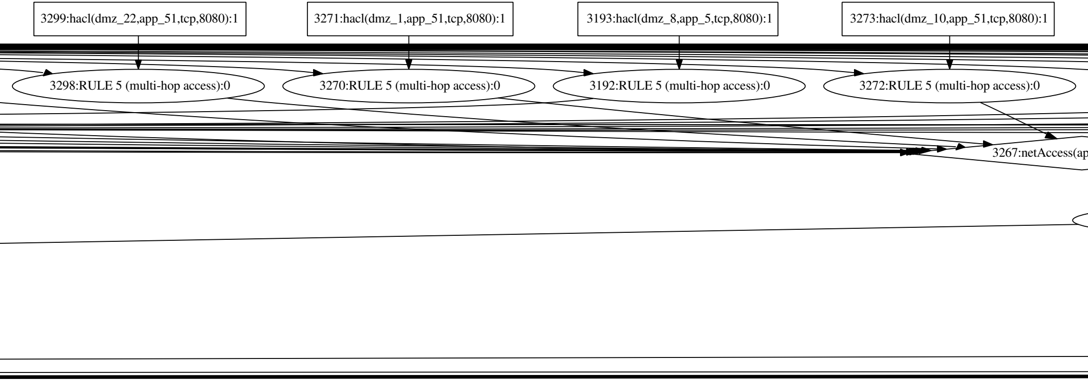

3236:hacl(dmz_23,app_50,tcp,8080):1
3155:hacl(dmz_16,app_5,tcp,8080):1
3238:hacl(dmz_24,app_50,tcp,8080):1
3187:hacl(dmz_5,app_5,tcp,8080):1
3235:RULE 5 (multi-hop access):0
3154:RULE 5 (multi-hop access):0
3237:RULE 5 (multi-hop access):0
3186:RULE 5 (multi-hop access):0

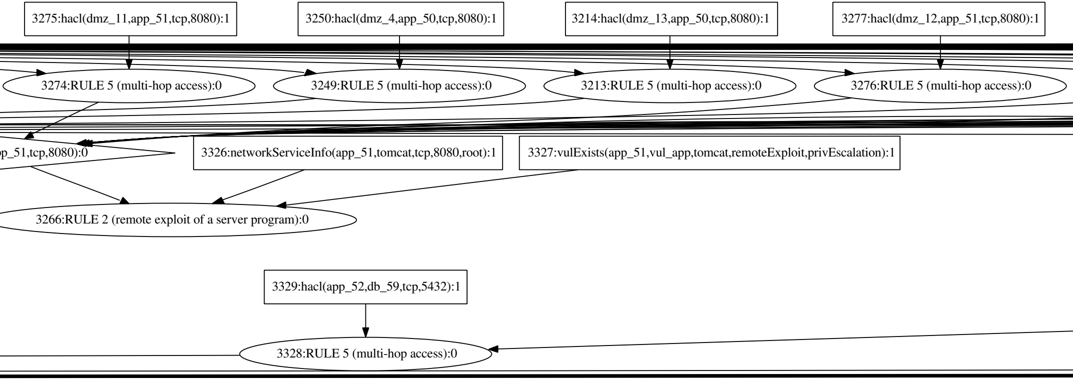

3309:hacl(dmz_27,app_51,tcp,8080):1
3311:hacl(dmz_28,app_51,tcp,8080):1
3313:hacl(dmz_3,app_51,tcp,8080):1
3240:hacl(dmz_25,app_50,tcp,8080):1
3308:RULE 5 (multi-hop access):0
3310:RULE 5 (multi-hop access):0
3312:RULE 5 (multi-hop access):0
3239:RULE 5 (multi-hop access):0

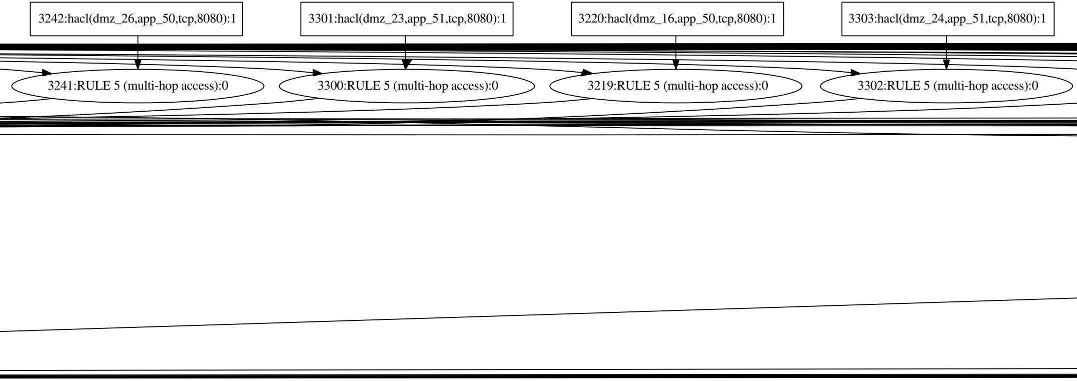

3281:hacl(dmz_14,app_51,tcp,8080):1
3287:hacl(dmz_17,app_51,tcp,8080):1
3260:hacl(dmz_9,app_50,tcp,8080):1
3289:hacl(dmz_18,app_51,tcp,8080):1
3280:RULE 5 (multi-hop access):0
3286:RULE 5 (multi-hop access):0
3259:RULE 5 (multi-hop access):0
3288:RULE 5 (multi-hop access):0
3265:execCode(app_51,root):0

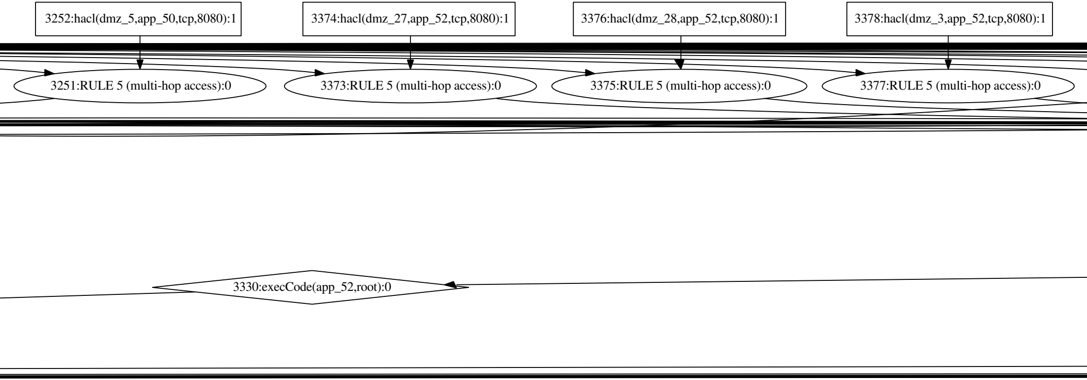

3189:hacl(dmz_6,app_5,tcp,8080):1
3291:hacl(dmz_19,app_51,tcp,8080):1
3293:hacl(dmz_2,app_51,tcp,8080):1
3295:hacl(dmz_20,app_51,tcp,8080):1
3188:RULE 5 (multi-hop access):0
3290:RULE 5 (multi-hop access):0
3292:RULE 5 (multi-hop access):0
3294:RULE 5 (multi-hop access):0

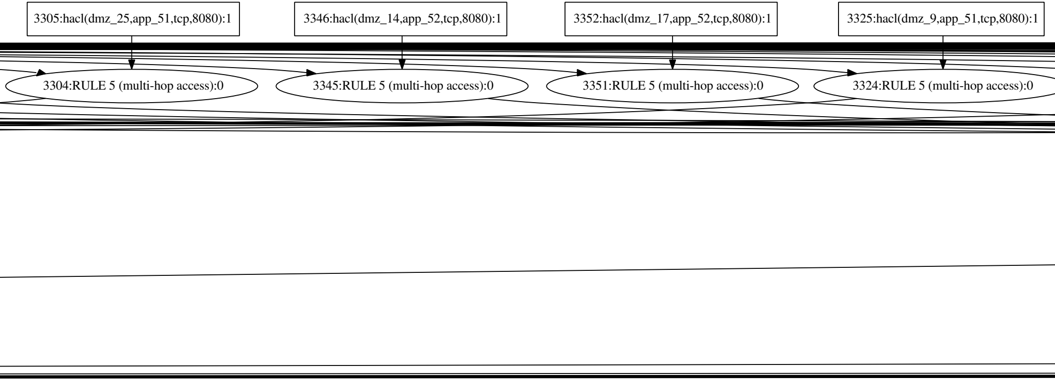

3218:hacl(dmz_15,app_50,tcp,8080):1
3297:hacl(dmz_21,app_51,tcp,8080):1
3191:hacl(dmz_7,app_5,tcp,8080):1
3269:hacl(dmz_0,app_51,tcp,8080):1
3217:RULE 5 (multi-hop access):0
3296:RULE 5 (multi-hop access):0
3190:RULE 5 (multi-hop access):0
3268:RULE 5 (multi-hop access):0

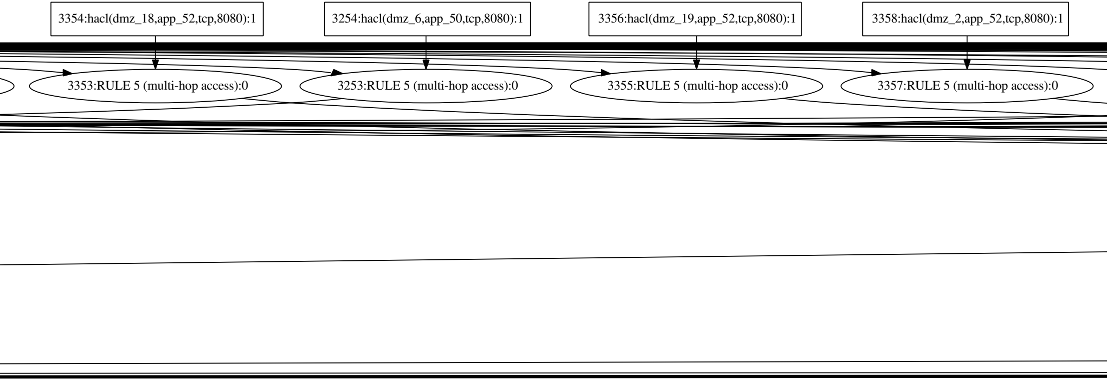

3299:hacl(dmz_22,app_51,tcp,8080):1
3271:hacl(dmz_1,app_51,tcp,8080):1
3193:hacl(dmz_8,app_5,tcp,8080):1
3273:hacl(dmz_10,app_51,tcp,8080):1
3298:RULE 5 (multi-hop access):0
3270:RULE 5 (multi-hop access):0
3192:RULE 5 (multi-hop access):0
3272:RULE 5 (multi-hop access):0
3267:netAccess(ap

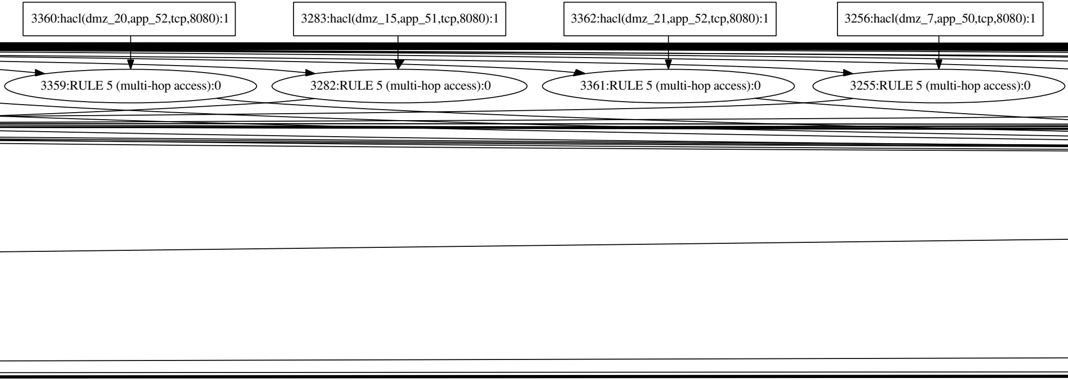

3275:hacl(dmz_11,app_51,tcp,8080):1
3250:hacl(dmz_4,app_50,tcp,8080):1
3214:hacl(dmz_13,app_50,tcp,8080):1
3277:hacl(dmz_12,app_51,tcp,8080):1
3274:RULE 5 (multi-hop access):0
3249:RULE 5 (multi-hop access):0
3213:RULE 5 (multi-hop access):0
3276:RULE 5 (multi-hop access):0
p_51,tcp,8080):0
3326:networkServiceInfo(app_51,tomcat,tcp,8080,root):1
3327:vulExists(app_51,vul_app,tomcat,remoteExploit,privEscalation):1
3266:RULE 2 (remote exploit of a server program):0
3329:hacl(app_52,db_59,tcp,5432):1
3328:RULE 5 (multi-hop access):0

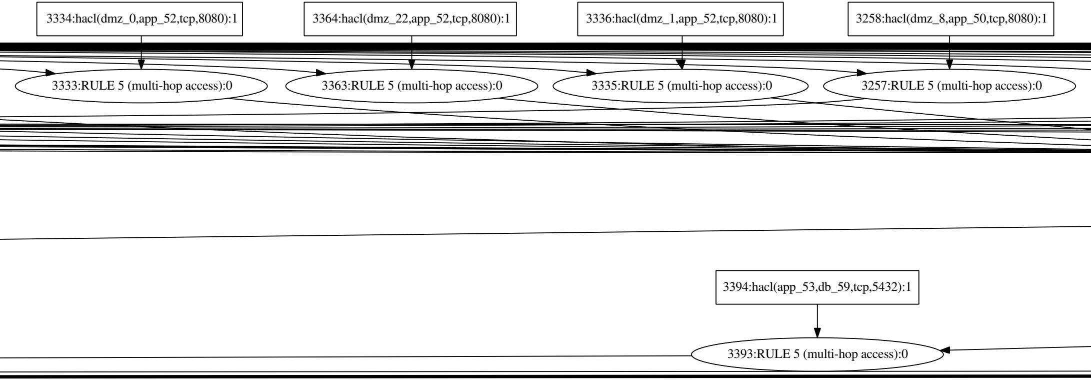

3242:hacl(dmz_26,app_50,tcp,8080):1
3301:hacl(dmz_23,app_51,tcp,8080):1
3220:hacl(dmz_16,app_50,tcp,8080):1
3303:hacl(dmz_24,app_51,tcp,8080):1
3241:RULE 5 (multi-hop access):0
3300:RULE 5 (multi-hop access):0
3219:RULE 5 (multi-hop access):0
3302:RULE 5 (multi-hop access):0

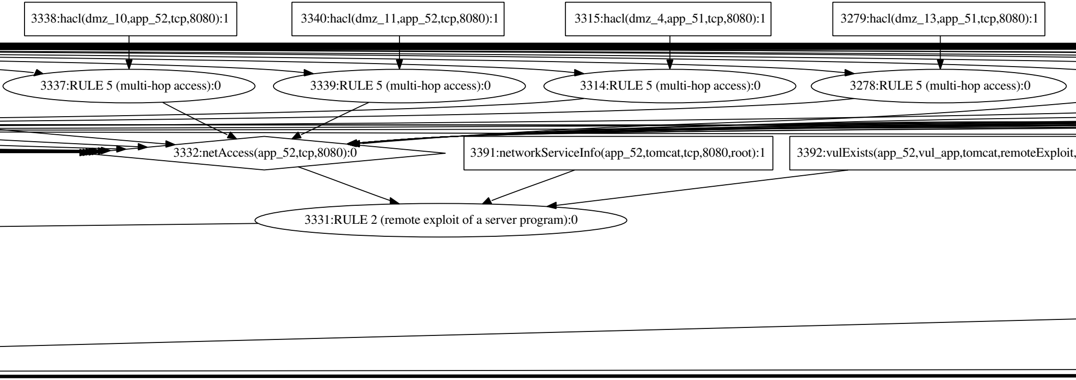

3252:hacl(dmz_5,app_50,tcp,8080):1
3374:hacl(dmz_27,app_52,tcp,8080):1
3376:hacl(dmz_28,app_52,tcp,8080):1
3378:hacl(dmz_3,app_52,tcp,8080):1
3251:RULE 5 (multi-hop access):0
3373:RULE 5 (multi-hop access):0
3375:RULE 5 (multi-hop access):0
3377:RULE 5 (multi-hop access):0
3330:execCode(app_52,root):0

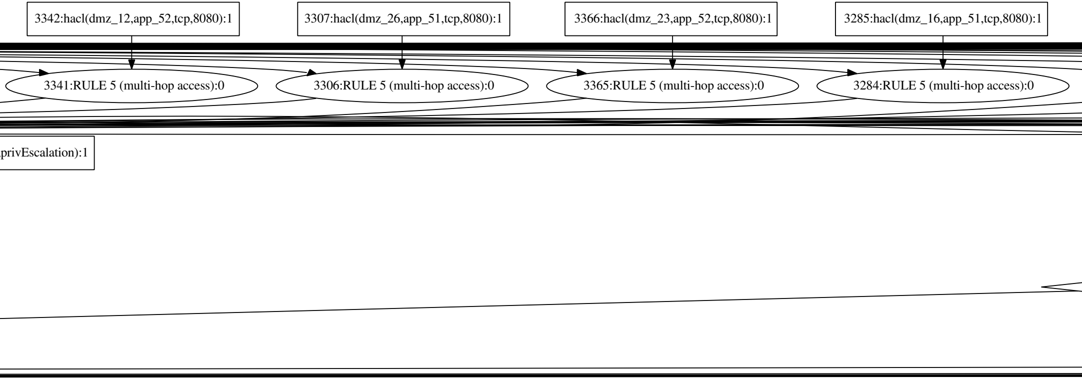

3305:hacl(dmz_25,app_51,tcp,8080):1
3346:hacl(dmz_14,app_52,tcp,8080):1
3352:hacl(dmz_17,app_52,tcp,8080):1
3325:hacl(dmz_9,app_51,tcp,8080):1
3304:RULE 5 (multi-hop access):0
3345:RULE 5 (multi-hop access):0
3351:RULE 5 (multi-hop access):0
3324:RULE 5 (multi-hop access):0

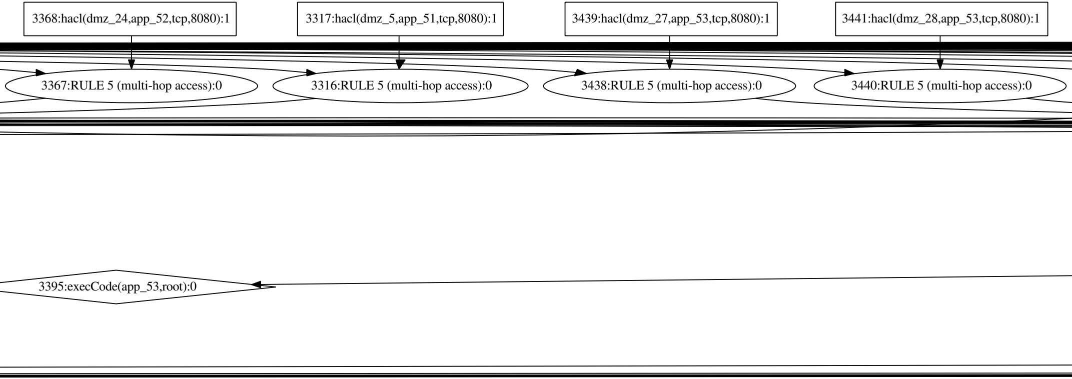

3354:hacl(dmz_18,app_52,tcp,8080):1
3254:hacl(dmz_6,app_50,tcp,8080):1
3356:hacl(dmz_19,app_52,tcp,8080):1
3358:hacl(dmz_2,app_52,tcp,8080):1
3353:RULE 5 (multi-hop access):0
3253:RULE 5 (multi-hop access):0
3355:RULE 5 (multi-hop access):0
3357:RULE 5 (multi-hop access):0

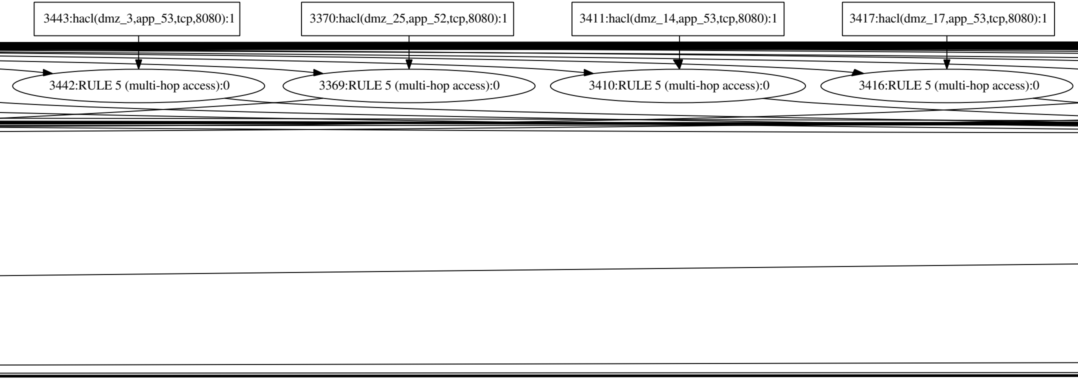

3360:hacl(dmz_20,app_52,tcp,8080):1
3283:hacl(dmz_15,app_51,tcp,8080):1
3362:hacl(dmz_21,app_52,tcp,8080):1
3256:hacl(dmz_7,app_50,tcp,8080):1
3359:RULE 5 (multi-hop access):0
3282:RULE 5 (multi-hop access):0
3361:RULE 5 (multi-hop access):0
3255:RULE 5 (multi-hop access):0

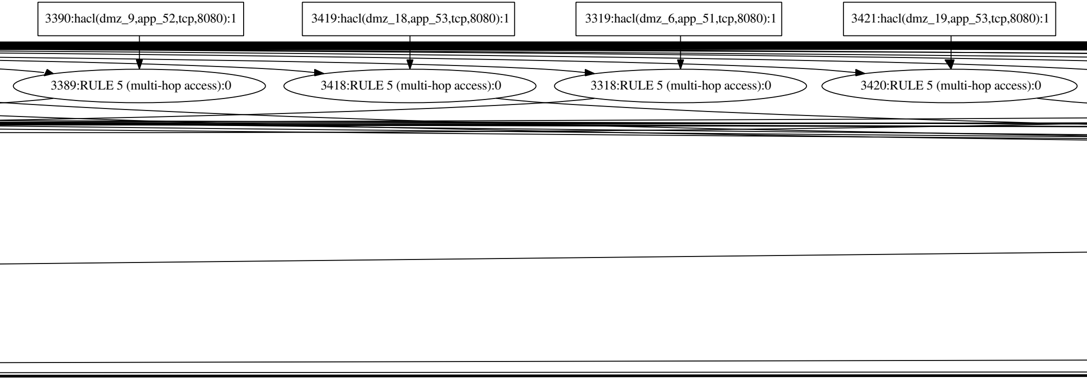

3334:hacl(dmz_0,app_52,tcp,8080):1
3364:hacl(dmz_22,app_52,tcp,8080):1
3336:hacl(dmz_1,app_52,tcp,8080):1
3258:hacl(dmz_8,app_50,tcp,8080):1
3333:RULE 5 (multi-hop access):0
3363:RULE 5 (multi-hop access):0
3335:RULE 5 (multi-hop access):0
3257:RULE 5 (multi-hop access):0
3394:hacl(app_53,db_59,tcp,5432):1
3393:RULE 5 (multi-hop access):0

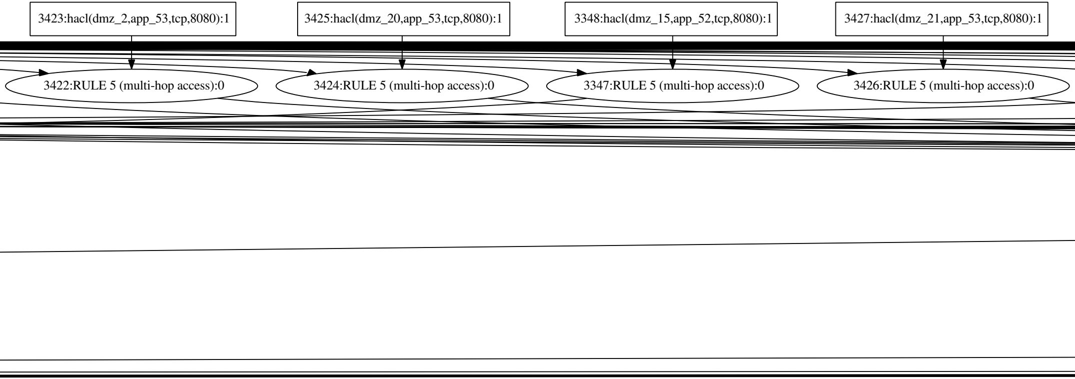

3338:hacl(dmz_10,app_52,tcp,8080):1
3340:hacl(dmz_11,app_52,tcp,8080):1
3315:hacl(dmz_4,app_51,tcp,8080):1
3279:hacl(dmz_13,app_51,tcp,8080):1
3337:RULE 5 (multi-hop access):0
3339:RULE 5 (multi-hop access):0
3314:RULE 5 (multi-hop access):0
3278:RULE 5 (multi-hop access):0
3332:netAccess(app_52,tcp,8080):0
3391:networkServiceInfo(app_52,tomcat,tcp,8080,root):1
3392:vulExists(app_52,vul_app,tomcat,remoteExploit,
3331:RULE 2 (remote exploit of a server program):0

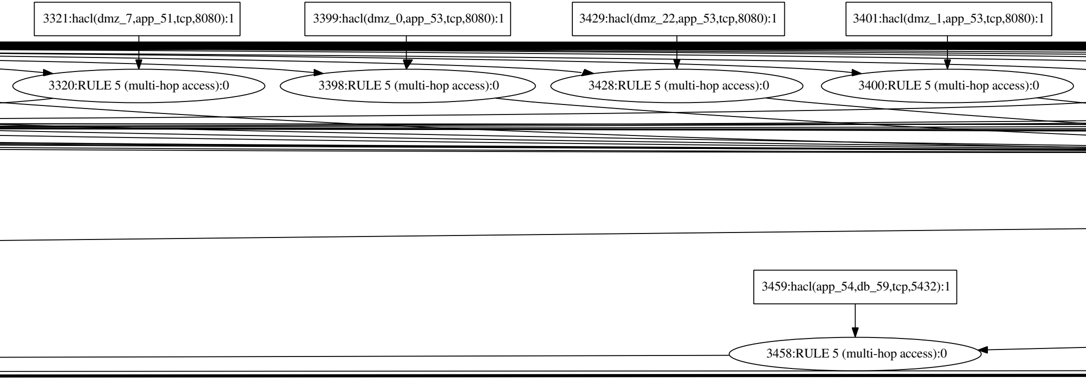

3342:hacl(dmz_12,app_52,tcp,8080):1
3307:hacl(dmz_26,app_51,tcp,8080):1
3366:hacl(dmz_23,app_52,tcp,8080):1
3285:hacl(dmz_16,app_51,tcp,8080):1
3341:RULE 5 (multi-hop access):0
3306:RULE 5 (multi-hop access):0
3365:RULE 5 (multi-hop access):0
3284:RULE 5 (multi-hop access):0
privEscalation):1

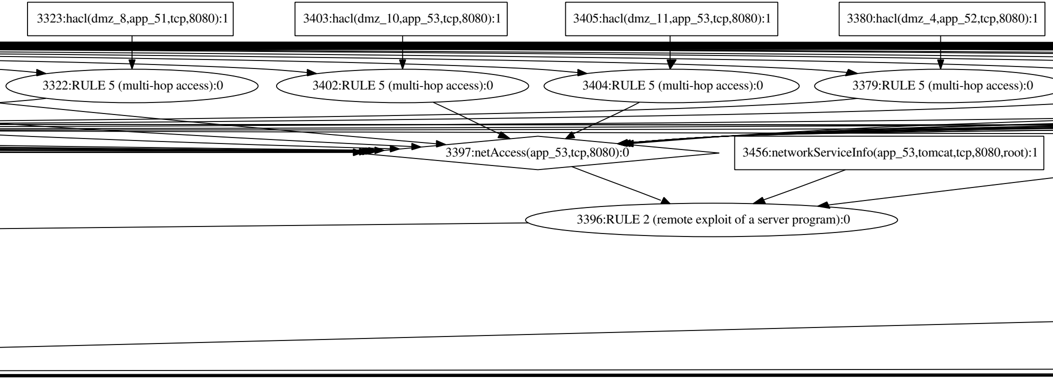

3368:hacl(dmz_24,app_52,tcp,8080):1
3317:hacl(dmz_5,app_51,tcp,8080):1
3439:hacl(dmz_27,app_53,tcp,8080):1
3441:hacl(dmz_28,app_53,tcp,8080):1
3367:RULE 5 (multi-hop access):0
3316:RULE 5 (multi-hop access):0
3438:RULE 5 (multi-hop access):0
3440:RULE 5 (multi-hop access):0
3395:execCode(app_53,root):0

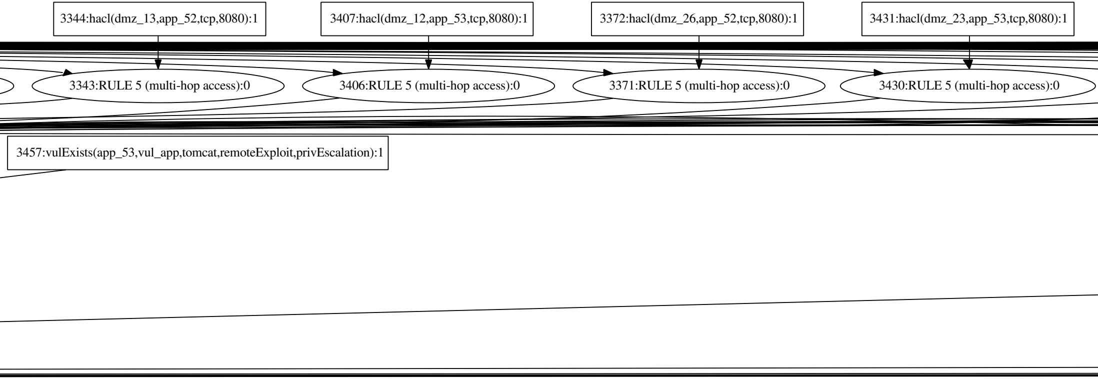

3443:hacl(dmz_3,app_53,tcp,8080):1
3370:hacl(dmz_25,app_52,tcp,8080):1
3411:hacl(dmz_14,app_53,tcp,8080):1
3417:hacl(dmz_17,app_53,tcp,8080):1
3442:RULE 5 (multi-hop access):0
3369:RULE 5 (multi-hop access):0
3410:RULE 5 (multi-hop access):0
3416:RULE 5 (multi-hop access):0

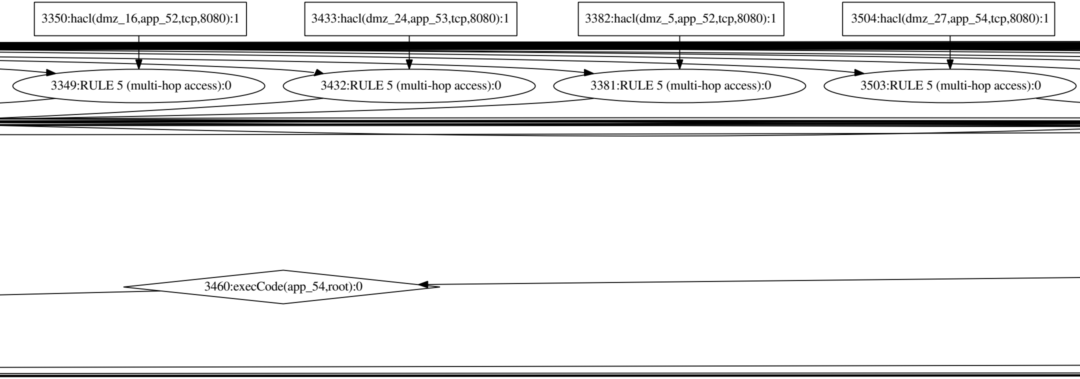

3390:hacl(dmz_9,app_52,tcp,8080):1
3419:hacl(dmz_18,app_53,tcp,8080):1
3319:hacl(dmz_6,app_51,tcp,8080):1
3421:hacl(dmz_19,app_53,tcp,8080):1
3389:RULE 5 (multi-hop access):0
3418:RULE 5 (multi-hop access):0
3318:RULE 5 (multi-hop access):0
3420:RULE 5 (multi-hop access):0

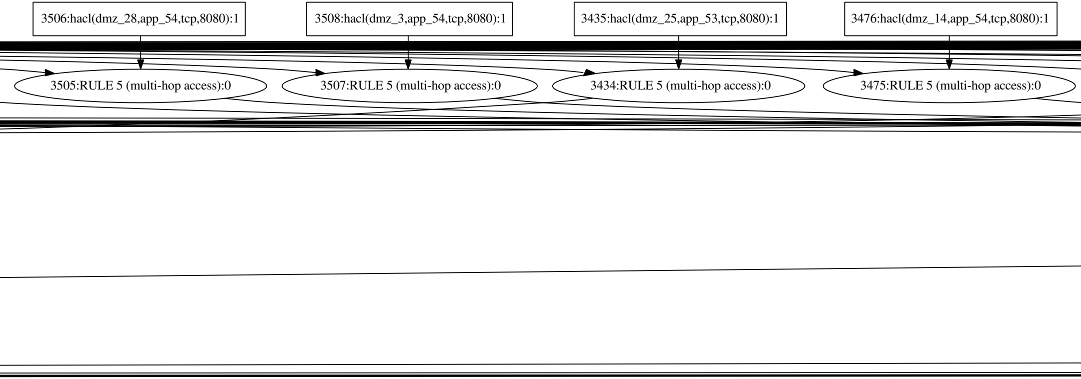

3423:hacl(dmz_2,app_53,tcp,8080):1
3425:hacl(dmz_20,app_53,tcp,8080):1
3348:hacl(dmz_15,app_52,tcp,8080):1
3427:hacl(dmz_21,app_53,tcp,8080):1
3422:RULE 5 (multi-hop access):0
3424:RULE 5 (multi-hop access):0
3347:RULE 5 (multi-hop access):0
3426:RULE 5 (multi-hop access):0

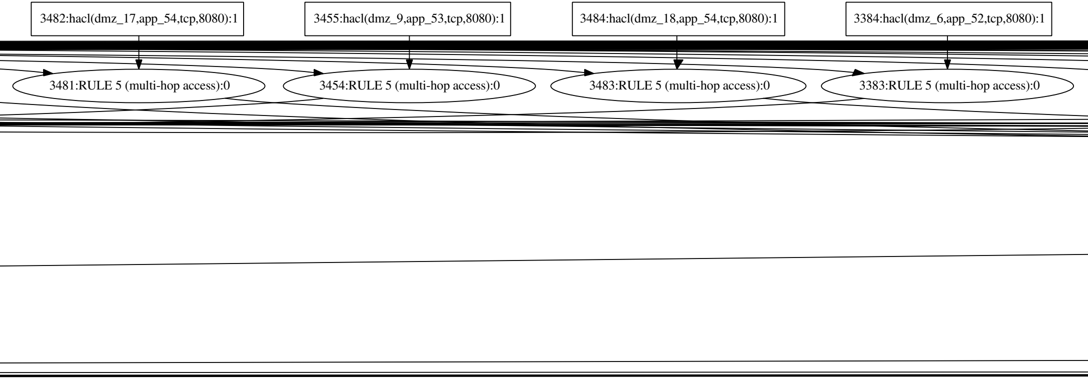

3321:hacl(dmz_7,app_51,tcp,8080):1
3399:hacl(dmz_0,app_53,tcp,8080):1
3429:hacl(dmz_22,app_53,tcp,8080):1
3401:hacl(dmz_1,app_53,tcp,8080):1
3320:RULE 5 (multi-hop access):0
3398:RULE 5 (multi-hop access):0
3428:RULE 5 (multi-hop access):0
3400:RULE 5 (multi-hop access):0
3459:hacl(app_54,db_59,tcp,5432):1
3458:RULE 5 (multi-hop access):0

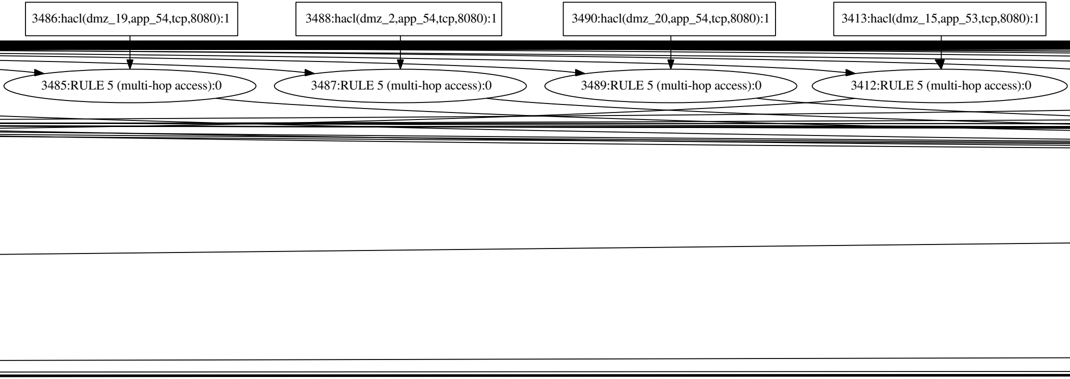

3323:hacl(dmz_8,app_51,tcp,8080):1
3403:hacl(dmz_10,app_53,tcp,8080):1
3405:hacl(dmz_11,app_53,tcp,8080):1
3380:hacl(dmz_4,app_52,tcp,8080):1
3322:RULE 5 (multi-hop access):0
3402:RULE 5 (multi-hop access):0
3404:RULE 5 (multi-hop access):0
3379:RULE 5 (multi-hop access):0
3397:netAccess(app_53,tcp,8080):0
3456:networkServiceInfo(app_53,tomcat,tcp,8080,root):1
3396:RULE 2 (remote exploit of a server program):0

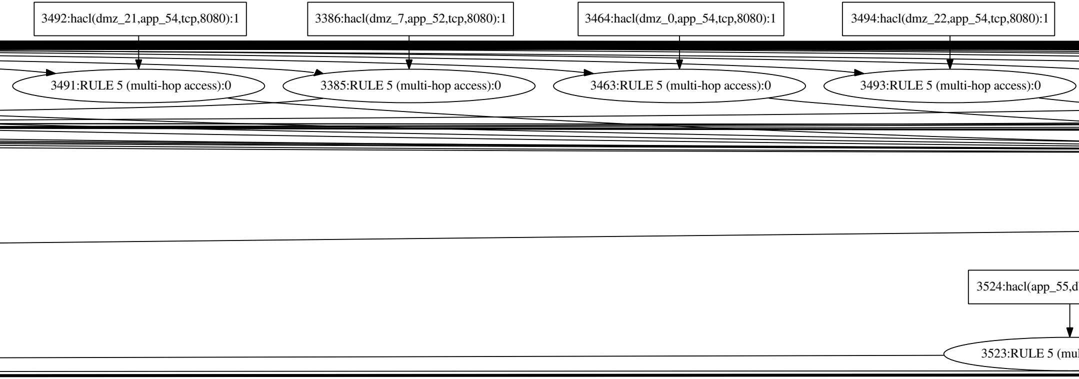

3344:hacl(dmz_13,app_52,tcp,8080):1
3407:hacl(dmz_12,app_53,tcp,8080):1
3372:hacl(dmz_26,app_52,tcp,8080):1
3431:hacl(dmz_23,app_53,tcp,8080):1
3343:RULE 5 (multi-hop access):0
3406:RULE 5 (multi-hop access):0
3371:RULE 5 (multi-hop access):0
3430:RULE 5 (multi-hop access):0
3457:vulExists(app_53,vul_app,tomcat,remoteExploit,privEscalation):1

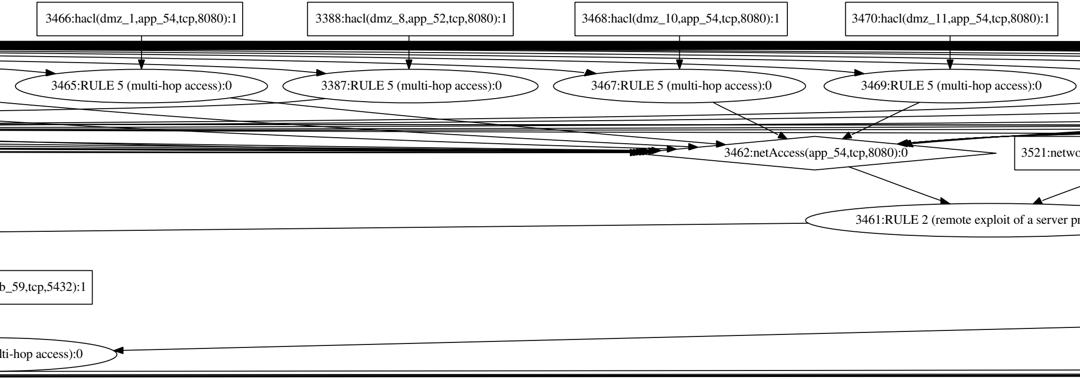

3350:hacl(dmz_16,app_52,tcp,8080):1
3433:hacl(dmz_24,app_53,tcp,8080):1
3382:hacl(dmz_5,app_52,tcp,8080):1
3504:hacl(dmz_27,app_54,tcp,8080):1
3349:RULE 5 (multi-hop access):0
3432:RULE 5 (multi-hop access):0
3381:RULE 5 (multi-hop access):0
3503:RULE 5 (multi-hop access):0
3460:execCode(app_54,root):0

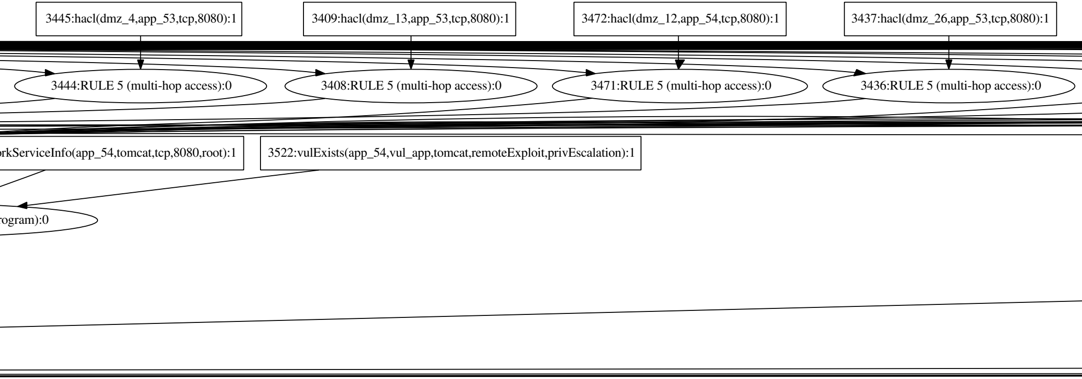

3506:hacl(dmz_28,app_54,tcp,8080):1
3508:hacl(dmz_3,app_54,tcp,8080):1
3435:hacl(dmz_25,app_53,tcp,8080):1
3476:hacl(dmz_14,app_54,tcp,8080):1
3505:RULE 5 (multi-hop access):0
3507:RULE 5 (multi-hop access):0
3434:RULE 5 (multi-hop access):0
3475:RULE 5 (multi-hop access):0

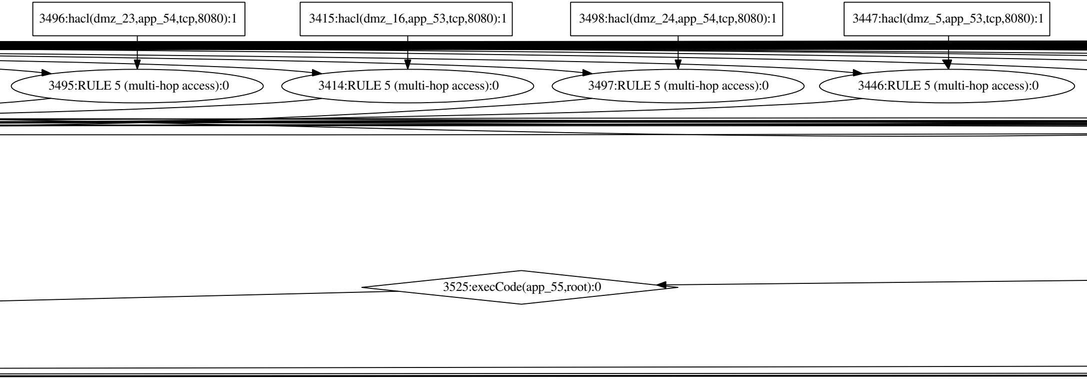

3482:hacl(dmz_17,app_54,tcp,8080):1
3455:hacl(dmz_9,app_53,tcp,8080):1
3484:hacl(dmz_18,app_54,tcp,8080):1
3384:hacl(dmz_6,app_52,tcp,8080):1
3481:RULE 5 (multi-hop access):0
3454:RULE 5 (multi-hop access):0
3483:RULE 5 (multi-hop access):0
3383:RULE 5 (multi-hop access):0

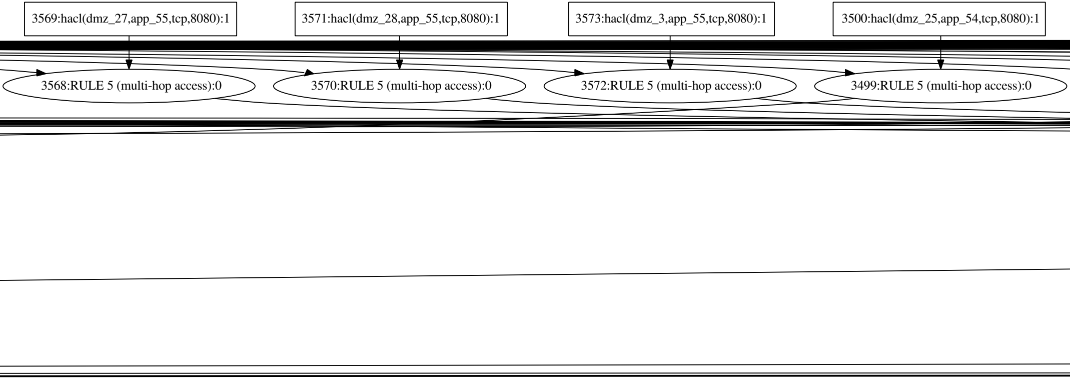

3486:hacl(dmz_19,app_54,tcp,8080):1
3488:hacl(dmz_2,app_54,tcp,8080):1
3490:hacl(dmz_20,app_54,tcp,8080):1
3413:hacl(dmz_15,app_53,tcp,8080):1
3485:RULE 5 (multi-hop access):0
3487:RULE 5 (multi-hop access):0
3489:RULE 5 (multi-hop access):0
3412:RULE 5 (multi-hop access):0

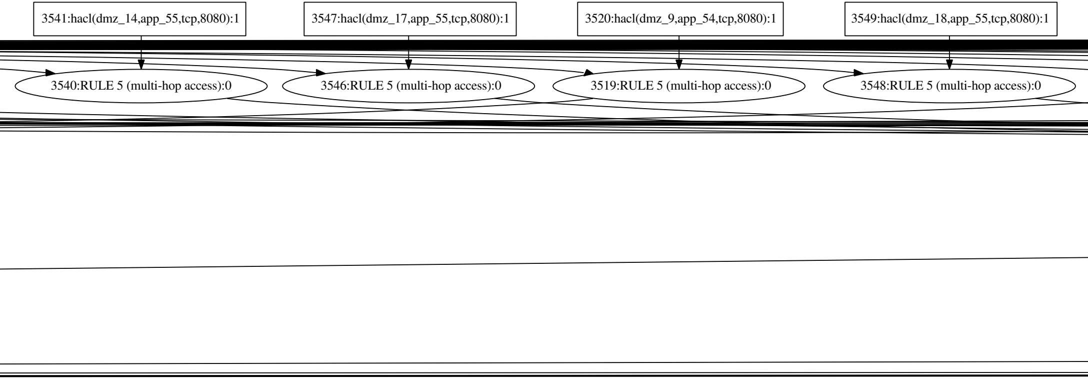

3492:hacl(dmz_21,app_54,tcp,8080):1
3386:hacl(dmz_7,app_52,tcp,8080):1
3464:hacl(dmz_0,app_54,tcp,8080):1
3494:hacl(dmz_22,app_54,tcp,8080):1
3491:RULE 5 (multi-hop access):0
3385:RULE 5 (multi-hop access):0
3463:RULE 5 (multi-hop access):0
3493:RULE 5 (multi-hop access):0
3524:hacl(app_55,d
3523:RULE 5 (mul

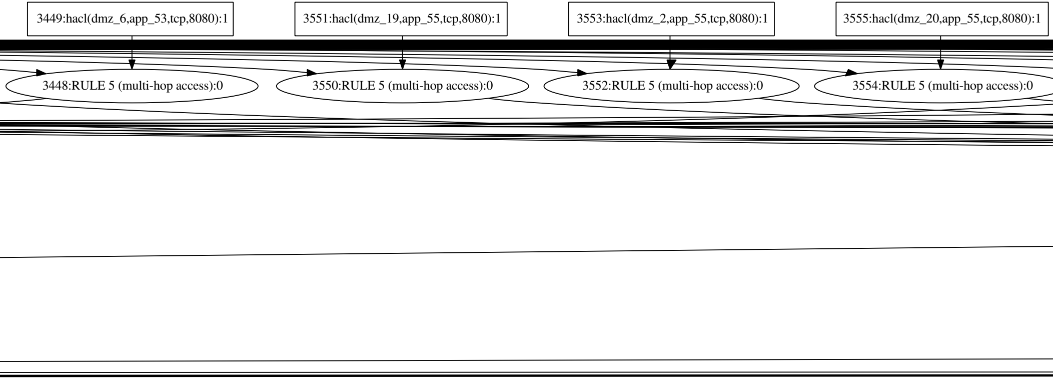

3466:hacl(dmz_1,app_54,tcp,8080):1
3388:hacl(dmz_8,app_52,tcp,8080):1
3468:hacl(dmz_10,app_54,tcp,8080):1
3470:hacl(dmz_11,app_54,tcp,8080):1
3465:RULE 5 (multi-hop access):0
3387:RULE 5 (multi-hop access):0
3467:RULE 5 (multi-hop access):0
3469:RULE 5 (multi-hop access):0
3462:netAccess(app_54,tcp,8080):0
3521:netwo
3461:RULE 2 (remote exploit of a server pr
b_59,tcp,5432):1
lti-hop access):0

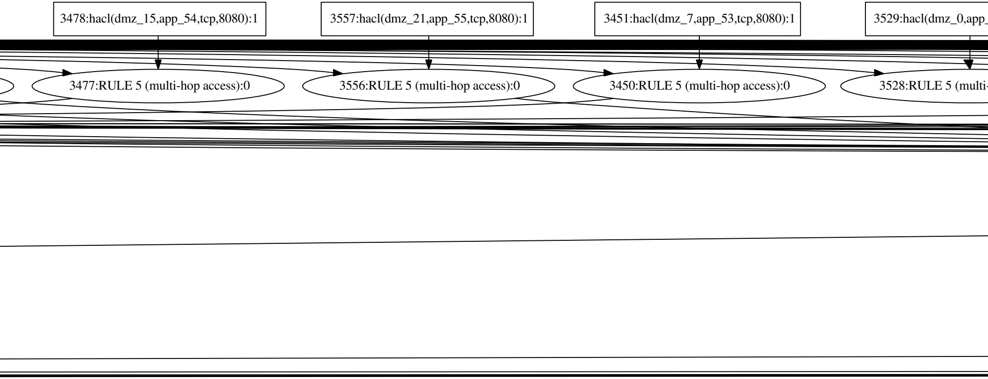

3445:hacl(dmz_4,app_53,tcp,8080):1
3409:hacl(dmz_13,app_53,tcp,8080):1
3472:hacl(dmz_12,app_54,tcp,8080):1
3437:hacl(dmz_26,app_53,tcp,8080):1
3444:RULE 5 (multi-hop access):0
3408:RULE 5 (multi-hop access):0
3471:RULE 5 (multi-hop access):0
3436:RULE 5 (multi-hop access):0
rkServiceInfo(app_54,tomcat,tcp,8080,root):1
3522:vulExists(app_54,vul_app,tomcat,remoteExploit,privEscalation):1
rogram):0

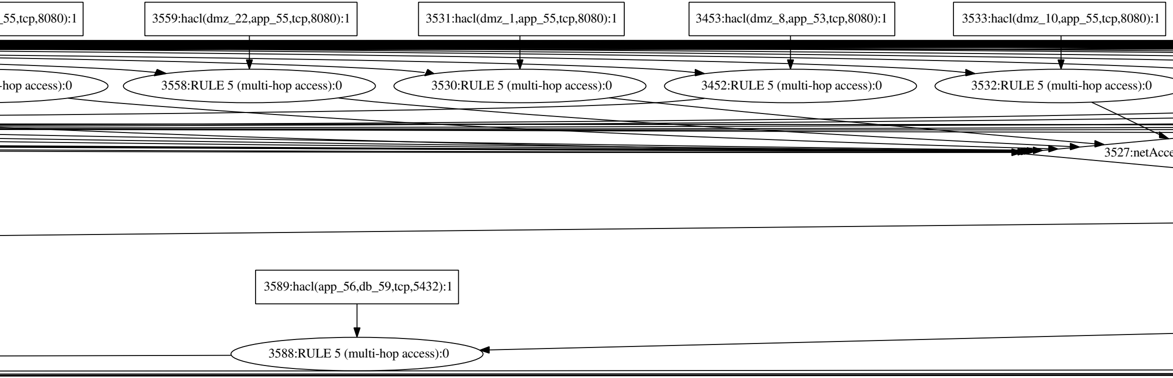

3496:hacl(dmz_23,app_54,tcp,8080):1
3415:hacl(dmz_16,app_53,tcp,8080):1
3498:hacl(dmz_24,app_54,tcp,8080):1
3447:hacl(dmz_5,app_53,tcp,8080):1
3495:RULE 5 (multi-hop access):0
3414:RULE 5 (multi-hop access):0
3497:RULE 5 (multi-hop access):0
3446:RULE 5 (multi-hop access):0
3525:execCode(app_55,root):0

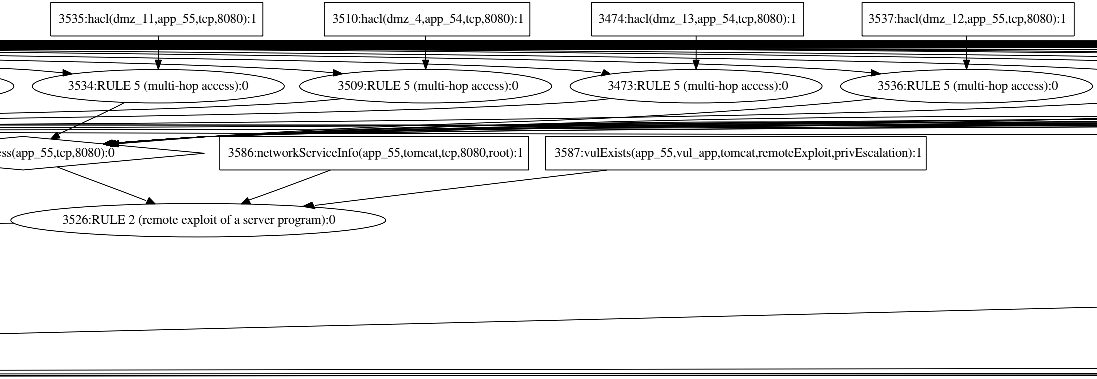

3569:hacl(dmz_27,app_55,tcp,8080):1
3571:hacl(dmz_28,app_55,tcp,8080):1
3573:hacl(dmz_3,app_55,tcp,8080):1
3500:hacl(dmz_25,app_54,tcp,8080):1
3568:RULE 5 (multi-hop access):0
3570:RULE 5 (multi-hop access):0
3572:RULE 5 (multi-hop access):0
3499:RULE 5 (multi-hop access):0

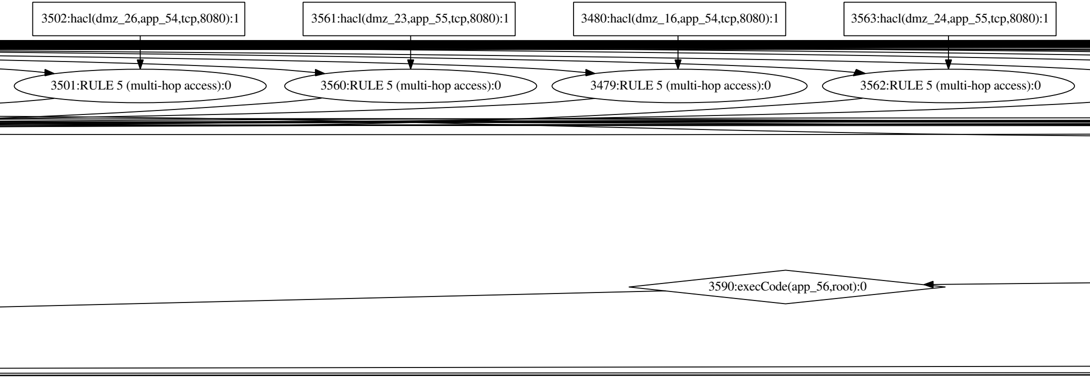

3541:hacl(dmz_14,app_55,tcp,8080):1
3547:hacl(dmz_17,app_55,tcp,8080):1
3520:hacl(dmz_9,app_54,tcp,8080):1
3549:hacl(dmz_18,app_55,tcp,8080):1
3540:RULE 5 (multi-hop access):0
3546:RULE 5 (multi-hop access):0
3519:RULE 5 (multi-hop access):0
3548:RULE 5 (multi-hop access):0

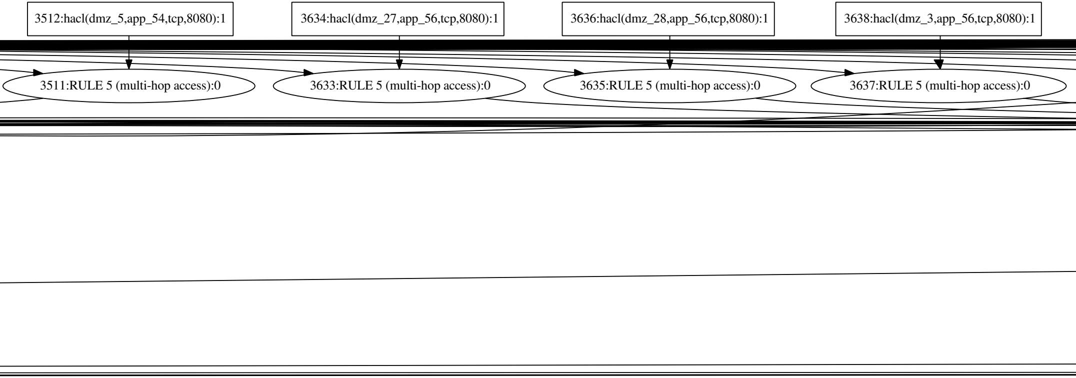

3449:hacl(dmz_6,app_53,tcp,8080):1
3551:hacl(dmz_19,app_55,tcp,8080):1
3553:hacl(dmz_2,app_55,tcp,8080):1
3555:hacl(dmz_20,app_55,tcp,8080):1
3448:RULE 5 (multi-hop access):0
3550:RULE 5 (multi-hop access):0
3552:RULE 5 (multi-hop access):0
3554:RULE 5 (multi-hop access):0

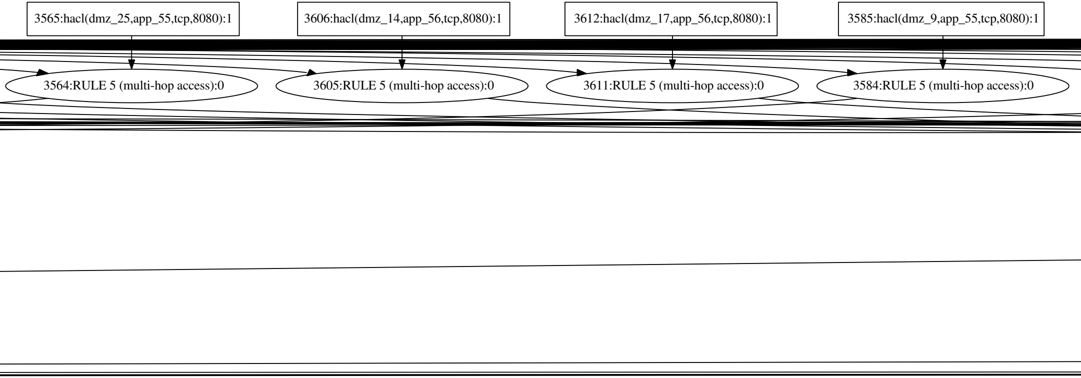

3478:hacl(dmz_15,app_54,tcp,8080):1
3557:hacl(dmz_21,app_55,tcp,8080):1
3451:hacl(dmz_7,app_53,tcp,8080):1
3529:hacl(dmz_0,app_
3477:RULE 5 (multi-hop access):0
3556:RULE 5 (multi-hop access):0
3450:RULE 5 (multi-hop access):0
3528:RULE 5 (multi-

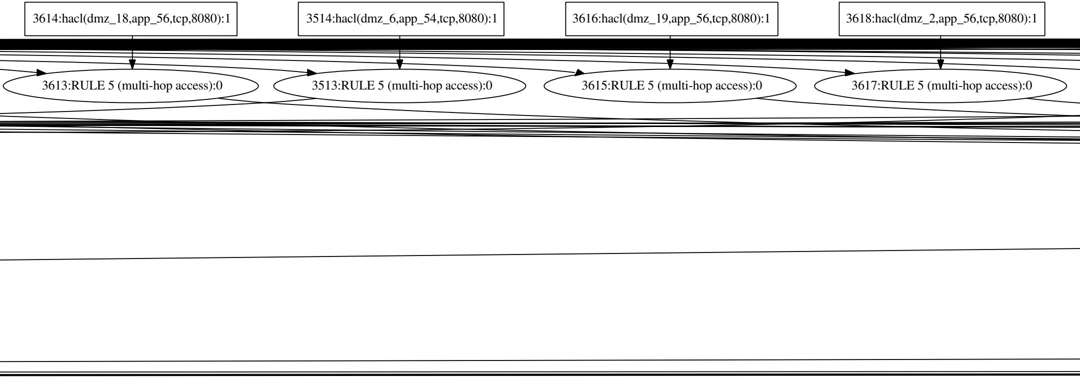

_55,tcp,8080):1
3559:hacl(dmz_22,app_55,tcp,8080):1
3531:hacl(dmz_1,app_55,tcp,8080):1
3453:hacl(dmz_8,app_53,tcp,8080):1
3533:hacl(dmz_10,app_55,tcp,8080):1
-hop access):0
3558:RULE 5 (multi-hop access):0
3530:RULE 5 (multi-hop access):0
3452:RULE 5 (multi-hop access):0
3532:RULE 5 (multi-hop access):0
3527:netAcce
3589:hacl(app_56,db_59,tcp,5432):1
3588:RULE 5 (multi-hop access):0

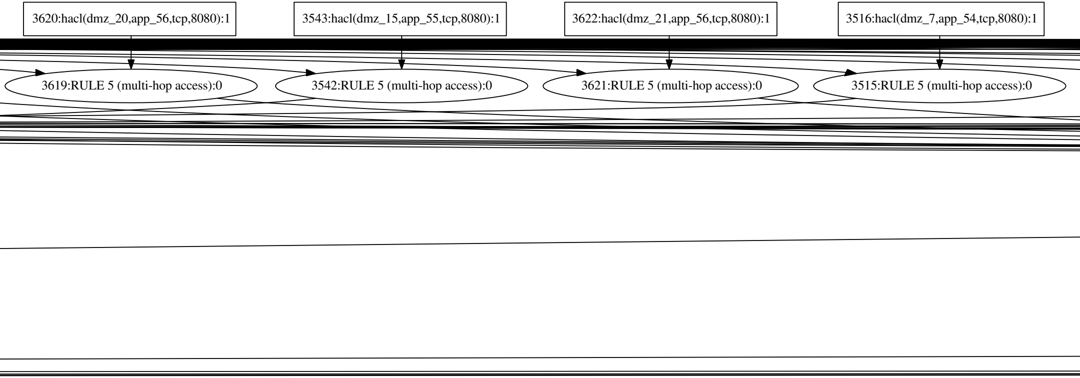

3535:hacl(dmz_11,app_55,tcp,8080):1
3510:hacl(dmz_4,app_54,tcp,8080):1
3474:hacl(dmz_13,app_54,tcp,8080):1
3537:hacl(dmz_12,app_55,tcp,8080):1
3534:RULE 5 (multi-hop access):0
3509:RULE 5 (multi-hop access):0
3473:RULE 5 (multi-hop access):0
3536:RULE 5 (multi-hop access):0
ess(app_55,tcp,8080):0
3586:networkServiceInfo(app_55,tomcat,tcp,8080,root):1
3587:vulExists(app_55,vul_app,tomcat,remoteExploit,privEscalation):1
3526:RULE 2 (remote exploit of a server program):0

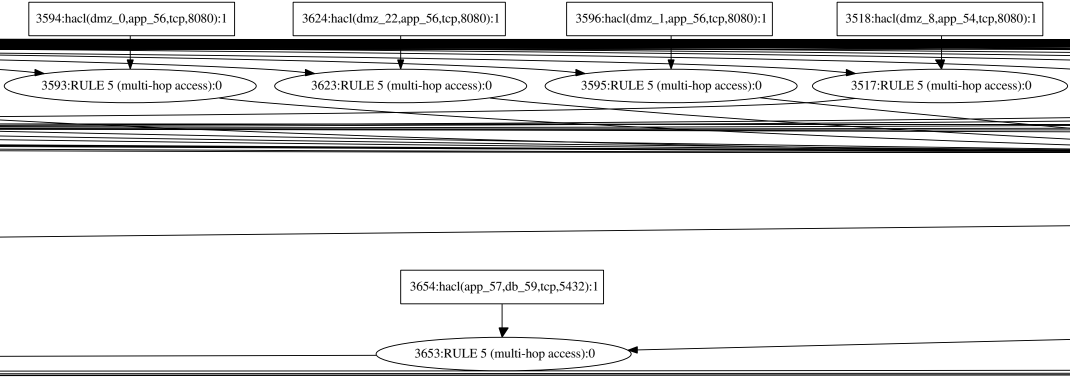

3502:hacl(dmz_26,app_54,tcp,8080):1
3561:hacl(dmz_23,app_55,tcp,8080):1
3480:hacl(dmz_16,app_54,tcp,8080):1
3563:hacl(dmz_24,app_55,tcp,8080):1
3501:RULE 5 (multi-hop access):0
3560:RULE 5 (multi-hop access):0
3479:RULE 5 (multi-hop access):0
3562:RULE 5 (multi-hop access):0
3590:execCode(app_56,root):0

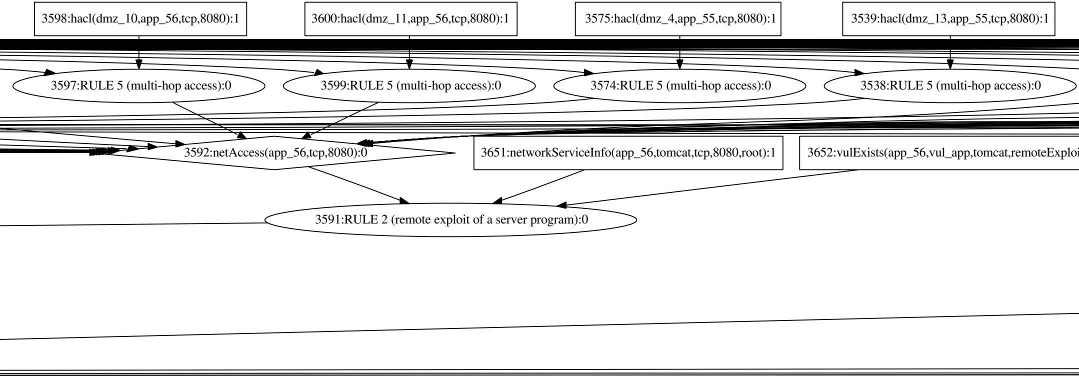

3512:hacl(dmz_5,app_54,tcp,8080):1
3634:hacl(dmz_27,app_56,tcp,8080):1
3636:hacl(dmz_28,app_56,tcp,8080):1
3638:hacl(dmz_3,app_56,tcp,8080):1
3511:RULE 5 (multi-hop access):0
3633:RULE 5 (multi-hop access):0
3635:RULE 5 (multi-hop access):0
3637:RULE 5 (multi-hop access):0

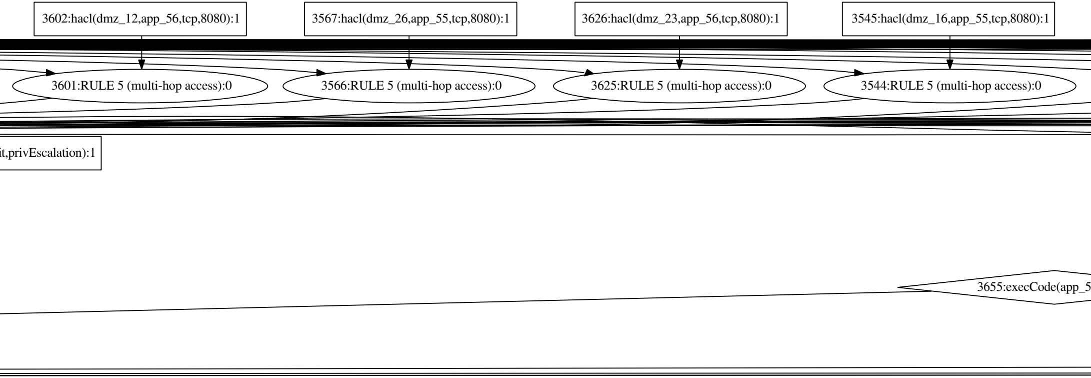

3565:hacl(dmz_25,app_55,tcp,8080):1
3606:hacl(dmz_14,app_56,tcp,8080):1
3612:hacl(dmz_17,app_56,tcp,8080):1
3585:hacl(dmz_9,app_55,tcp,8080):1
3564:RULE 5 (multi-hop access):0
3605:RULE 5 (multi-hop access):0
3611:RULE 5 (multi-hop access):0
3584:RULE 5 (multi-hop access):0

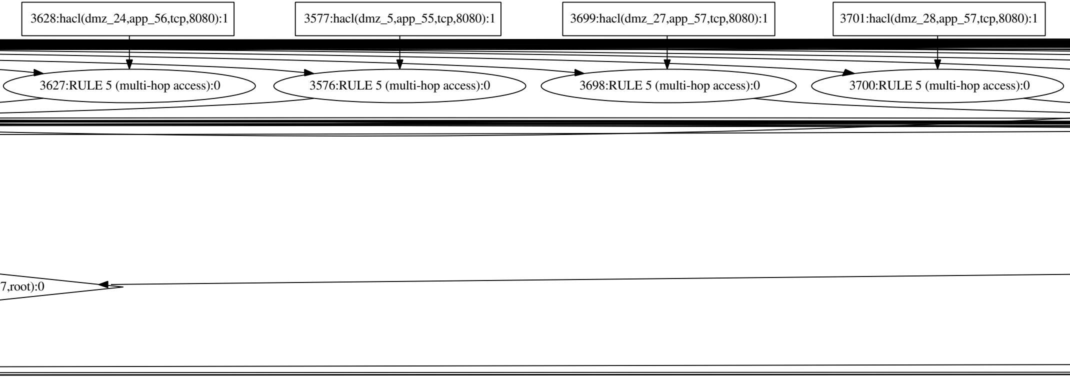

3614:hacl(dmz_18,app_56,tcp,8080):1
3514:hacl(dmz_6,app_54,tcp,8080):1
3616:hacl(dmz_19,app_56,tcp,8080):1
3618:hacl(dmz_2,app_56,tcp,8080):1
3613:RULE 5 (multi-hop access):0
3513:RULE 5 (multi-hop access):0
3615:RULE 5 (multi-hop access):0
3617:RULE 5 (multi-hop access):0

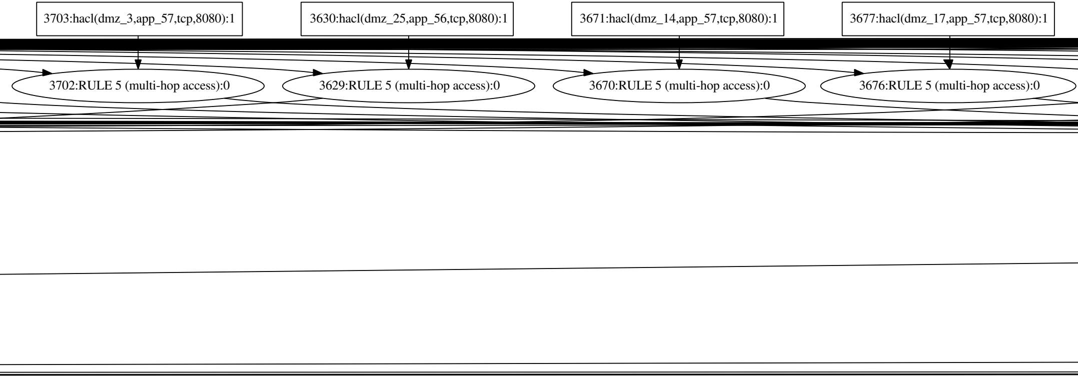

3620:hacl(dmz_20,app_56,tcp,8080):1
3543:hacl(dmz_15,app_55,tcp,8080):1
3622:hacl(dmz_21,app_56,tcp,8080):1
3516:hacl(dmz_7,app_54,tcp,8080):1
3619:RULE 5 (multi-hop access):0
3542:RULE 5 (multi-hop access):0
3621:RULE 5 (multi-hop access):0
3515:RULE 5 (multi-hop access):0

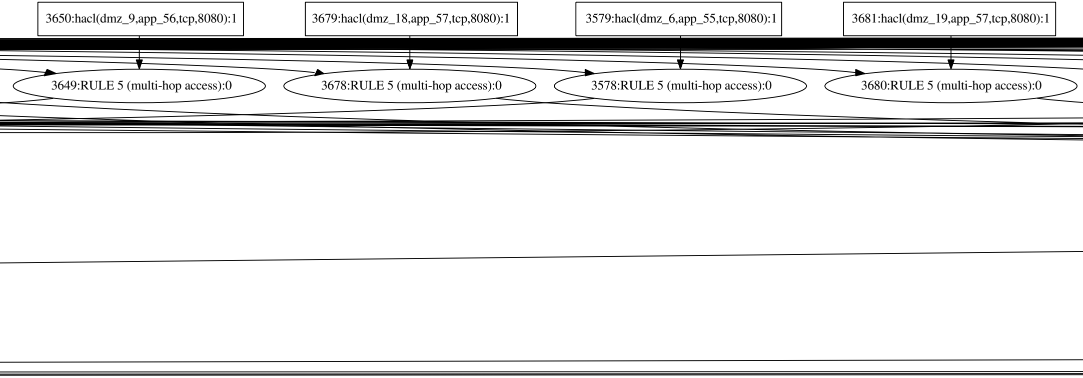

3594:hacl(dmz_0,app_56,tcp,8080):1
3624:hacl(dmz_22,app_56,tcp,8080):1
3596:hacl(dmz_1,app_56,tcp,8080):1
3518:hacl(dmz_8,app_54,tcp,8080):1
3593:RULE 5 (multi-hop access):0
3623:RULE 5 (multi-hop access):0
3595:RULE 5 (multi-hop access):0
3517:RULE 5 (multi-hop access):0
3654:hacl(app_57,db_59,tcp,5432):1
3653:RULE 5 (multi-hop access):0

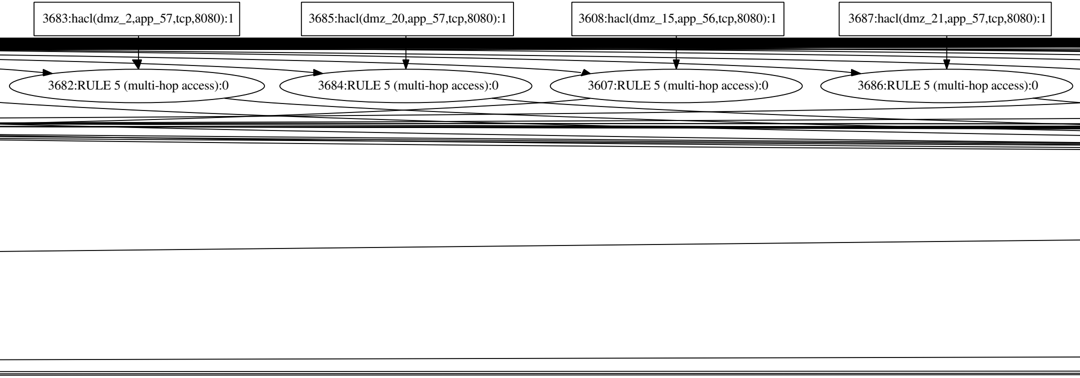

3598:hacl(dmz_10,app_56,tcp,8080):1
3600:hacl(dmz_11,app_56,tcp,8080):1
3575:hacl(dmz_4,app_55,tcp,8080):1
3539:hacl(dmz_13,app_55,tcp,8080):1
3597:RULE 5 (multi-hop access):0
3599:RULE 5 (multi-hop access):0
3574:RULE 5 (multi-hop access):0
3538:RULE 5 (multi-hop access):0
3592:netAccess(app_56,tcp,8080):0
3651:networkServiceInfo(app_56,tomcat,tcp,8080,root):1
3652:vulExists(app_56,vul_app,tomcat,remoteExploi
3591:RULE 2 (remote exploit of a server program):0

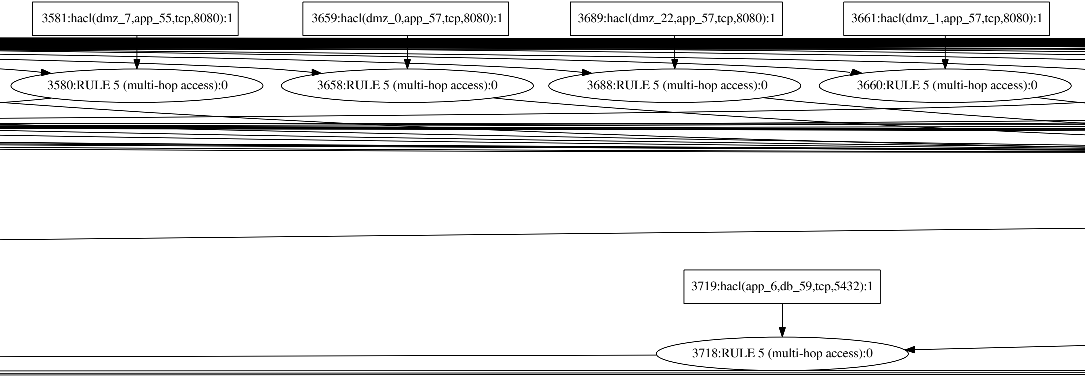

3602:hacl(dmz_12,app_56,tcp,8080):1
3567:hacl(dmz_26,app_55,tcp,8080):1
3626:hacl(dmz_23,app_56,tcp,8080):1
3545:hacl(dmz_16,app_55,tcp,8080):1
3601:RULE 5 (multi-hop access):0
3566:RULE 5 (multi-hop access):0
3625:RULE 5 (multi-hop access):0
3544:RULE 5 (multi-hop access):0
it,privEscalation):1
3655:execCode(app_5

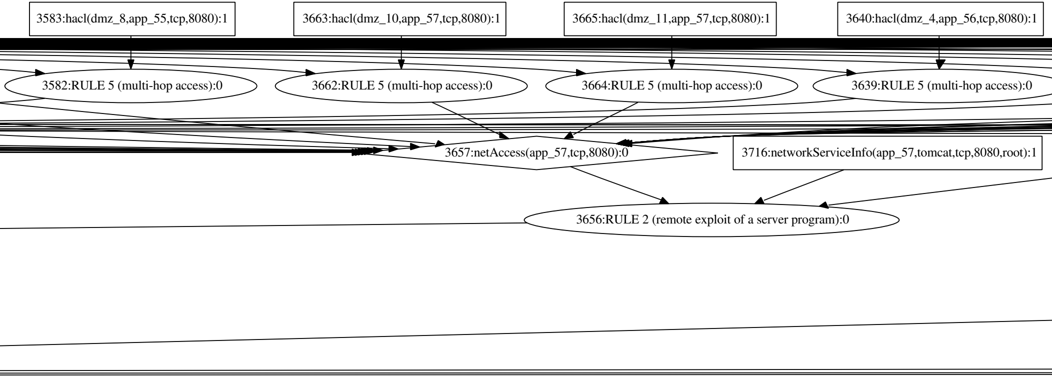

3628:hacl(dmz_24,app_56,tcp,8080):1
3577:hacl(dmz_5,app_55,tcp,8080):1
3699:hacl(dmz_27,app_57,tcp,8080):1
3701:hacl(dmz_28,app_57,tcp,8080):1
3627:RULE 5 (multi-hop access):0
3576:RULE 5 (multi-hop access):0
3698:RULE 5 (multi-hop access):0
3700:RULE 5 (multi-hop access):0
7,root):0

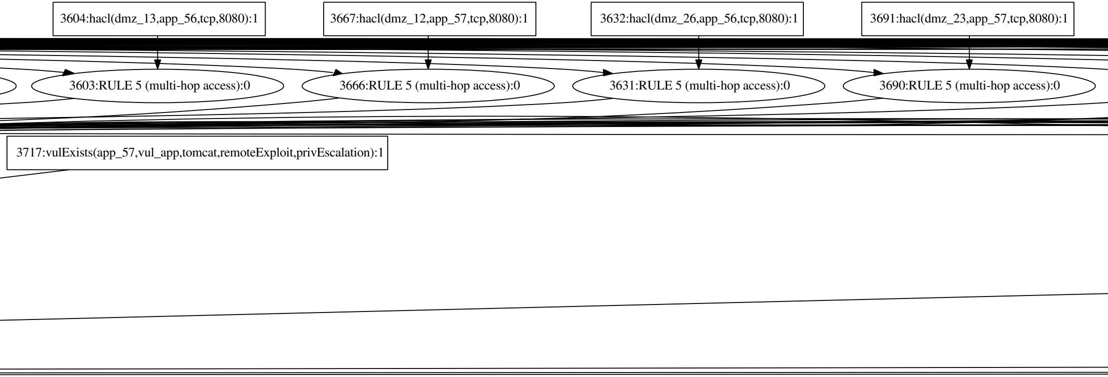

3703:hacl(dmz_3,app_57,tcp,8080):1
3630:hacl(dmz_25,app_56,tcp,8080):1
3671:hacl(dmz_14,app_57,tcp,8080):1
3677:hacl(dmz_17,app_57,tcp,8080):1
3702:RULE 5 (multi-hop access):0
3629:RULE 5 (multi-hop access):0
3670:RULE 5 (multi-hop access):0
3676:RULE 5 (multi-hop access):0

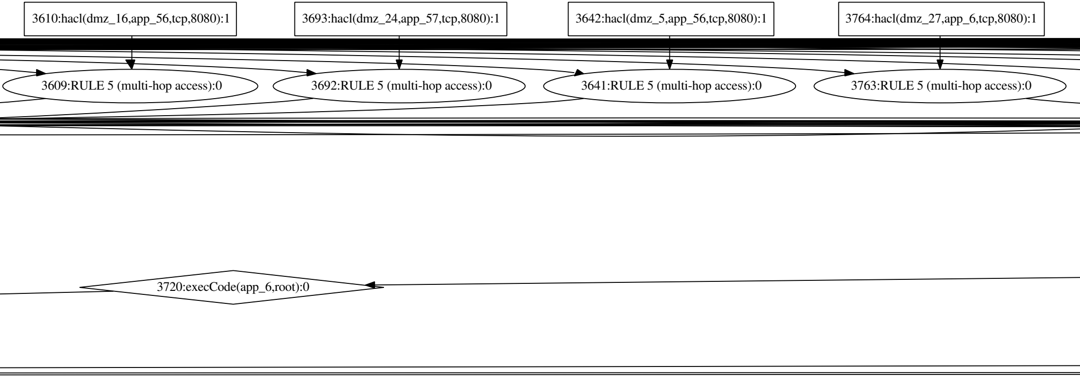

3650:hacl(dmz_9,app_56,tcp,8080):1
3679:hacl(dmz_18,app_57,tcp,8080):1
3579:hacl(dmz_6,app_55,tcp,8080):1
3681:hacl(dmz_19,app_57,tcp,8080):1
3649:RULE 5 (multi-hop access):0
3678:RULE 5 (multi-hop access):0
3578:RULE 5 (multi-hop access):0
3680:RULE 5 (multi-hop access):0

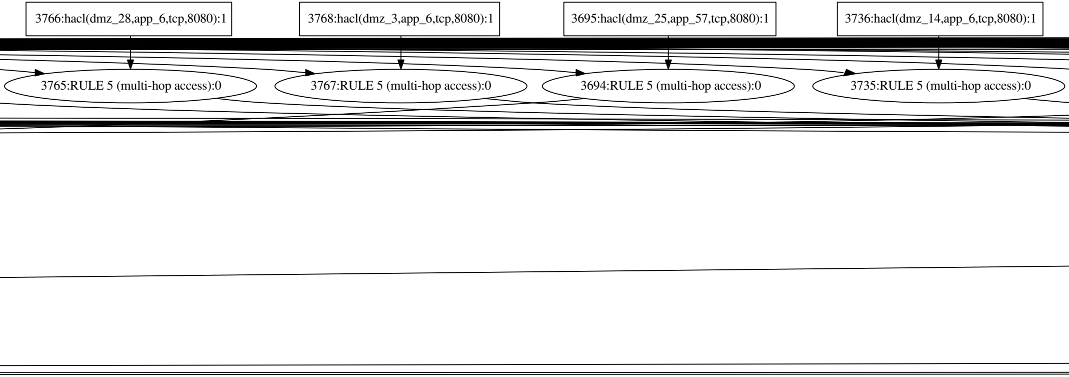

3683:hacl(dmz_2,app_57,tcp,8080):1
3685:hacl(dmz_20,app_57,tcp,8080):1
3608:hacl(dmz_15,app_56,tcp,8080):1
3687:hacl(dmz_21,app_57,tcp,8080):1
3682:RULE 5 (multi-hop access):0
3684:RULE 5 (multi-hop access):0
3607:RULE 5 (multi-hop access):0
3686:RULE 5 (multi-hop access):0

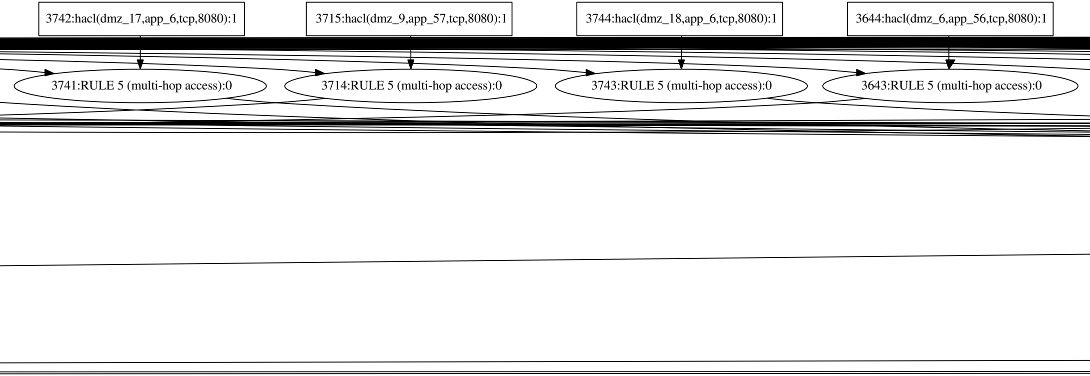

3581:hacl(dmz_7,app_55,tcp,8080):1
3659:hacl(dmz_0,app_57,tcp,8080):1
3689:hacl(dmz_22,app_57,tcp,8080):1
3661:hacl(dmz_1,app_57,tcp,8080):1
3580:RULE 5 (multi-hop access):0
3658:RULE 5 (multi-hop access):0
3688:RULE 5 (multi-hop access):0
3660:RULE 5 (multi-hop access):0
3719:hacl(app_6,db_59,tcp,5432):1
3718:RULE 5 (multi-hop access):0

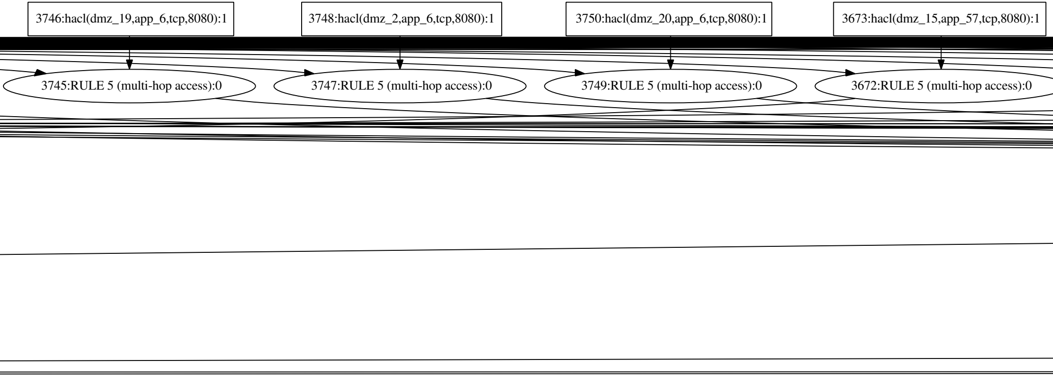

3583:hacl(dmz_8,app_55,tcp,8080):1
3663:hacl(dmz_10,app_57,tcp,8080):1
3665:hacl(dmz_11,app_57,tcp,8080):1
3640:hacl(dmz_4,app_56,tcp,8080):1
3582:RULE 5 (multi-hop access):0
3662:RULE 5 (multi-hop access):0
3664:RULE 5 (multi-hop access):0
3639:RULE 5 (multi-hop access):0
3657:netAccess(app_57,tcp,8080):0
3716:networkServiceInfo(app_57,tomcat,tcp,8080,root):1
3656:RULE 2 (remote exploit of a server program):0

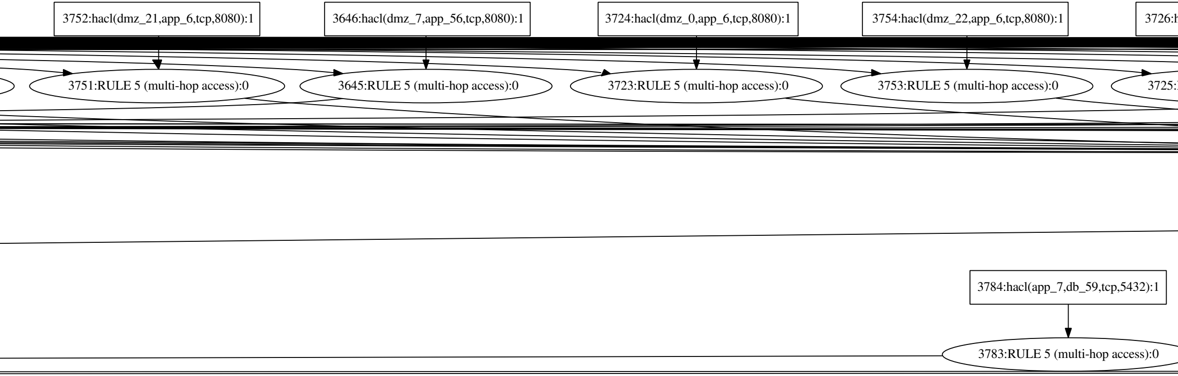

3604:hacl(dmz_13,app_56,tcp,8080):1
3667:hacl(dmz_12,app_57,tcp,8080):1
3632:hacl(dmz_26,app_56,tcp,8080):1
3691:hacl(dmz_23,app_57,tcp,8080):1
3603:RULE 5 (multi-hop access):0
3666:RULE 5 (multi-hop access):0
3631:RULE 5 (multi-hop access):0
3690:RULE 5 (multi-hop access):0
3717:vulExists(app_57,vul_app,tomcat,remoteExploit,privEscalation):1

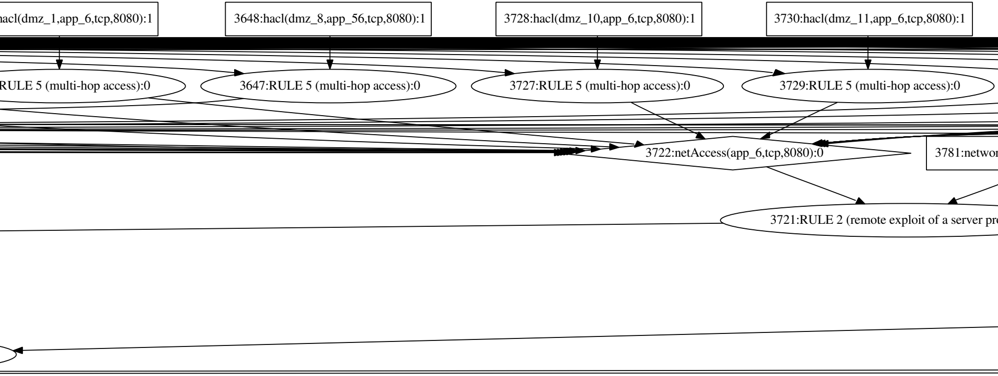

3610:hacl(dmz_16,app_56,tcp,8080):1
3693:hacl(dmz_24,app_57,tcp,8080):1
3642:hacl(dmz_5,app_56,tcp,8080):1
3764:hacl(dmz_27,app_6,tcp,8080):1
3609:RULE 5 (multi-hop access):0
3692:RULE 5 (multi-hop access):0
3641:RULE 5 (multi-hop access):0
3763:RULE 5 (multi-hop access):0
3720:execCode(app_6,root):0

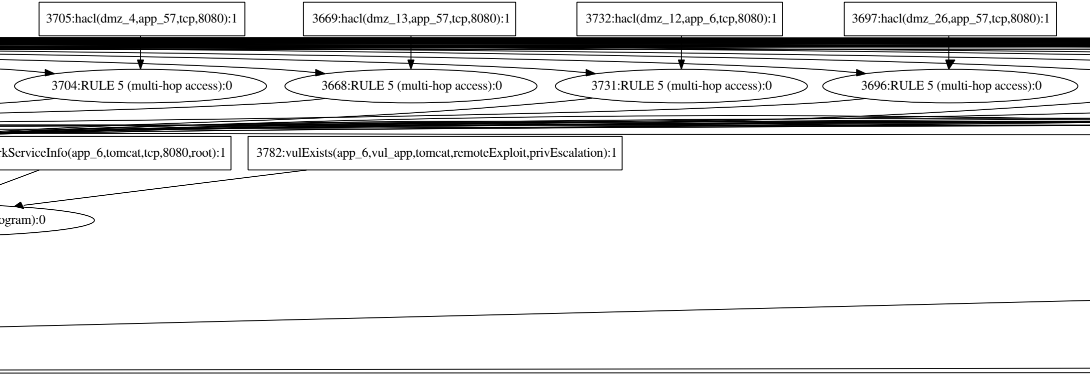

3766:hacl(dmz_28,app_6,tcp,8080):1
3768:hacl(dmz_3,app_6,tcp,8080):1
3695:hacl(dmz_25,app_57,tcp,8080):1
3736:hacl(dmz_14,app_6,tcp,8080):1
3765:RULE 5 (multi-hop access):0
3767:RULE 5 (multi-hop access):0
3694:RULE 5 (multi-hop access):0
3735:RULE 5 (multi-hop access):0

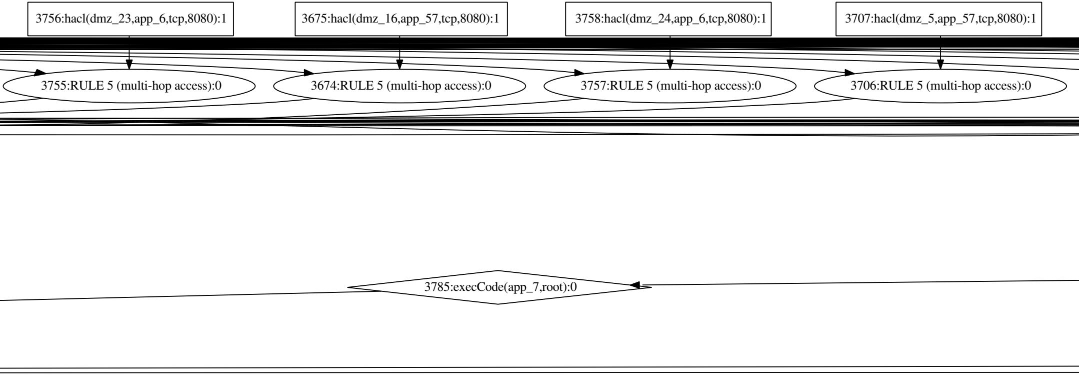

3742:hacl(dmz_17,app_6,tcp,8080):1
3715:hacl(dmz_9,app_57,tcp,8080):1
3744:hacl(dmz_18,app_6,tcp,8080):1
3644:hacl(dmz_6,app_56,tcp,8080):1
3741:RULE 5 (multi-hop access):0
3714:RULE 5 (multi-hop access):0
3743:RULE 5 (multi-hop access):0
3643:RULE 5 (multi-hop access):0

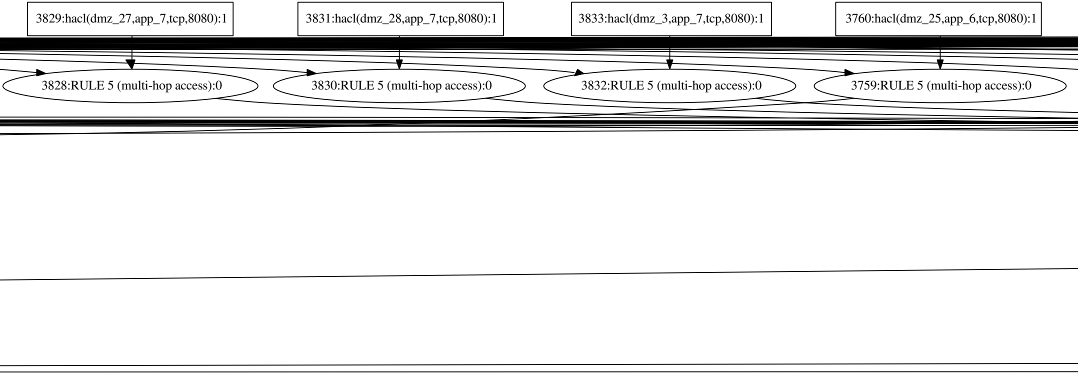

3746:hacl(dmz_19,app_6,tcp,8080):1
3748:hacl(dmz_2,app_6,tcp,8080):1
3750:hacl(dmz_20,app_6,tcp,8080):1
3673:hacl(dmz_15,app_57,tcp,8080):1
3745:RULE 5 (multi-hop access):0
3747:RULE 5 (multi-hop access):0
3749:RULE 5 (multi-hop access):0
3672:RULE 5 (multi-hop access):0

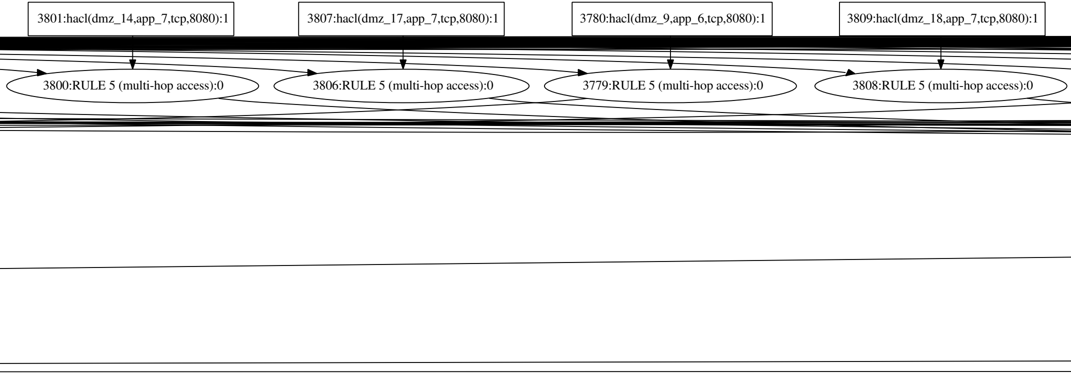

3752:hacl(dmz_21,app_6,tcp,8080):1
3646:hacl(dmz_7,app_56,tcp,8080):1
3724:hacl(dmz_0,app_6,tcp,8080):1
3754:hacl(dmz_22,app_6,tcp,8080):1
3726:h
3751:RULE 5 (multi-hop access):0
3645:RULE 5 (multi-hop access):0
3723:RULE 5 (multi-hop access):0
3753:RULE 5 (multi-hop access):0
3725:
3784:hacl(app_7,db_59,tcp,5432):1
3783:RULE 5 (multi-hop access):0

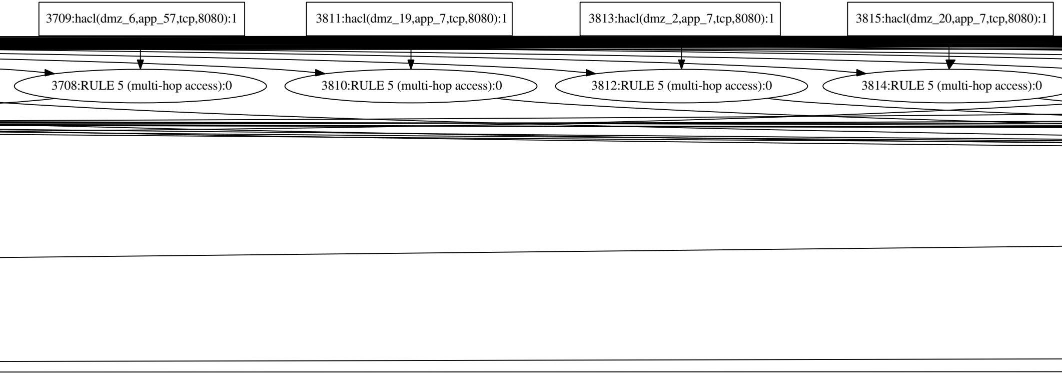

hacl(dmz_1,app_6,tcp,8080):1
3648:hacl(dmz_8,app_56,tcp,8080):1
3728:hacl(dmz_10,app_6,tcp,8080):1
3730:hacl(dmz_11,app_6,tcp,8080):1
RULE 5 (multi-hop access):0
3647:RULE 5 (multi-hop access):0
3727:RULE 5 (multi-hop access):0
3729:RULE 5 (multi-hop access):0
3722:netAccess(app_6,tcp,8080):0
3781:networ
3721:RULE 2 (remote exploit of a server pro

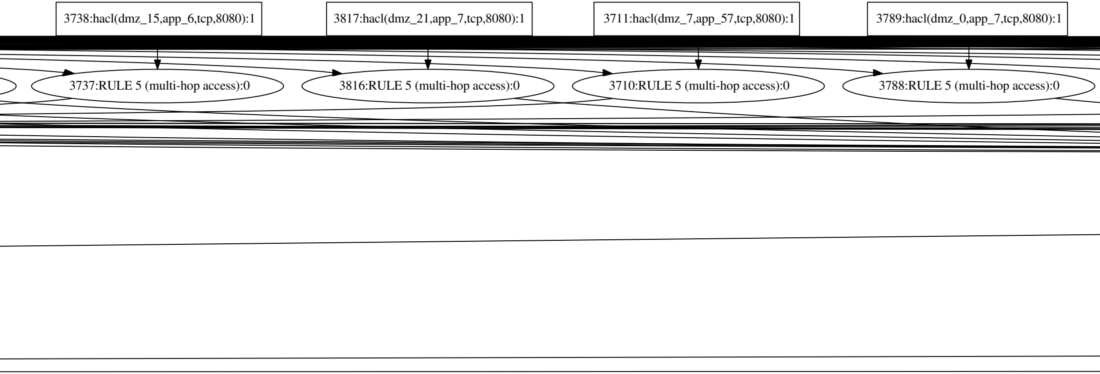

3705:hacl(dmz_4,app_57,tcp,8080):1
3669:hacl(dmz_13,app_57,tcp,8080):1
3732:hacl(dmz_12,app_6,tcp,8080):1
3697:hacl(dmz_26,app_57,tcp,8080):1
3704:RULE 5 (multi-hop access):0
3668:RULE 5 (multi-hop access):0
3731:RULE 5 (multi-hop access):0
3696:RULE 5 (multi-hop access):0
rkServiceInfo(app_6,tomcat,tcp,8080,root):1
3782:vulExists(app_6,vul_app,tomcat,remoteExploit,privEscalation):1
ogram):0

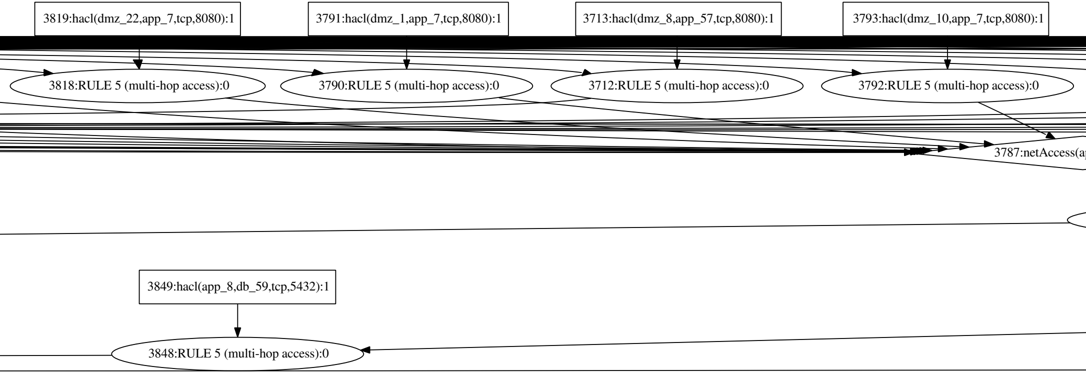

3756:hacl(dmz_23,app_6,tcp,8080):1
3675:hacl(dmz_16,app_57,tcp,8080):1
3758:hacl(dmz_24,app_6,tcp,8080):1
3707:hacl(dmz_5,app_57,tcp,8080):1
3755:RULE 5 (multi-hop access):0
3674:RULE 5 (multi-hop access):0
3757:RULE 5 (multi-hop access):0
3706:RULE 5 (multi-hop access):0
3785:execCode(app_7,root):0

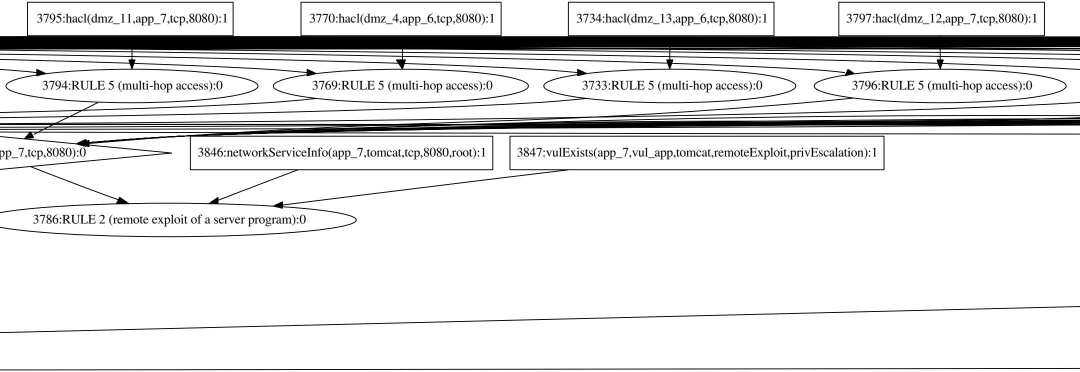

3829:hacl(dmz_27,app_7,tcp,8080):1
3831:hacl(dmz_28,app_7,tcp,8080):1
3833:hacl(dmz_3,app_7,tcp,8080):1
3760:hacl(dmz_25,app_6,tcp,8080):1
3828:RULE 5 (multi-hop access):0
3830:RULE 5 (multi-hop access):0
3832:RULE 5 (multi-hop access):0
3759:RULE 5 (multi-hop access):0

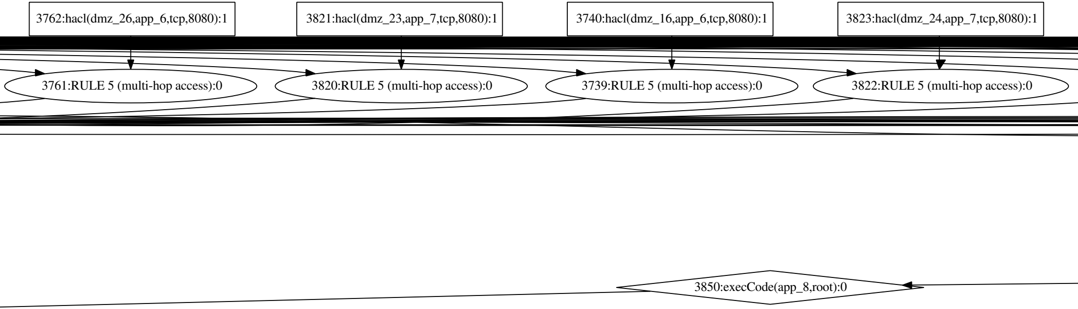

3801:hacl(dmz_14,app_7,tcp,8080):1
3807:hacl(dmz_17,app_7,tcp,8080):1
3780:hacl(dmz_9,app_6,tcp,8080):1
3809:hacl(dmz_18,app_7,tcp,8080):1
3800:RULE 5 (multi-hop access):0
3806:RULE 5 (multi-hop access):0
3779:RULE 5 (multi-hop access):0
3808:RULE 5 (multi-hop access):0

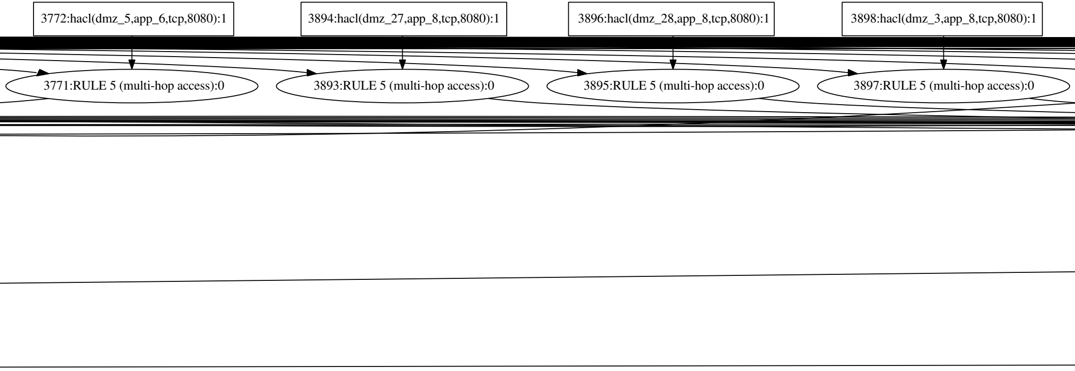

3709:hacl(dmz_6,app_57,tcp,8080):1
3811:hacl(dmz_19,app_7,tcp,8080):1
3813:hacl(dmz_2,app_7,tcp,8080):1
3815:hacl(dmz_20,app_7,tcp,8080):1
3708:RULE 5 (multi-hop access):0
3810:RULE 5 (multi-hop access):0
3812:RULE 5 (multi-hop access):0
3814:RULE 5 (multi-hop access):0

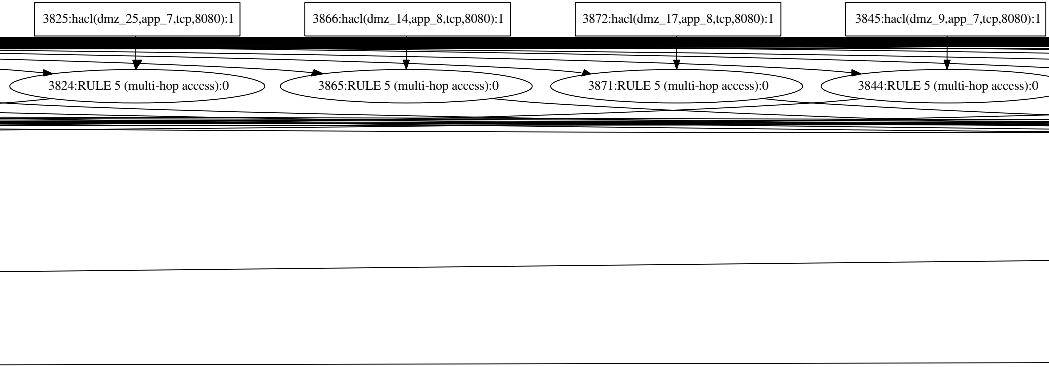

3738:hacl(dmz_15,app_6,tcp,8080):1
3817:hacl(dmz_21,app_7,tcp,8080):1
3711:hacl(dmz_7,app_57,tcp,8080):1
3789:hacl(dmz_0,app_7,tcp,8080):1
3737:RULE 5 (multi-hop access):0
3816:RULE 5 (multi-hop access):0
3710:RULE 5 (multi-hop access):0
3788:RULE 5 (multi-hop access):0

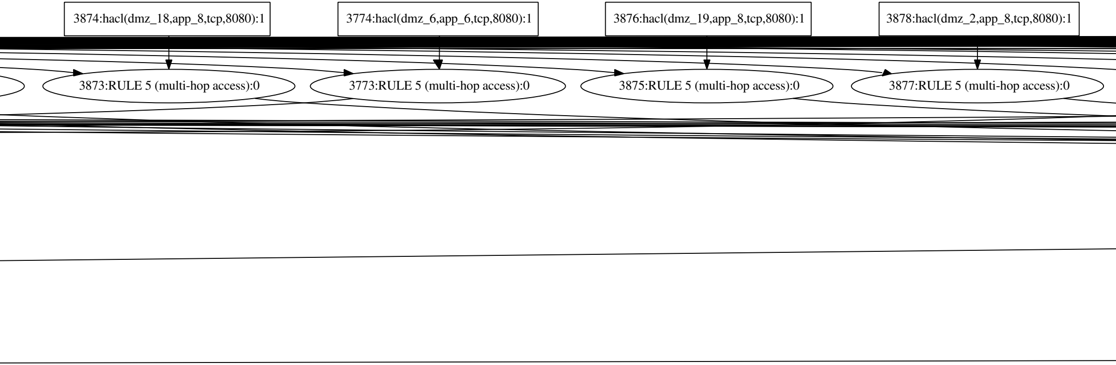

3819:hacl(dmz_22,app_7,tcp,8080):1
3791:hacl(dmz_1,app_7,tcp,8080):1
3713:hacl(dmz_8,app_57,tcp,8080):1
3793:hacl(dmz_10,app_7,tcp,8080):1
3818:RULE 5 (multi-hop access):0
3790:RULE 5 (multi-hop access):0
3712:RULE 5 (multi-hop access):0
3792:RULE 5 (multi-hop access):0
3787:netAccess(a
3849:hacl(app_8,db_59,tcp,5432):1
3848:RULE 5 (multi-hop access):0

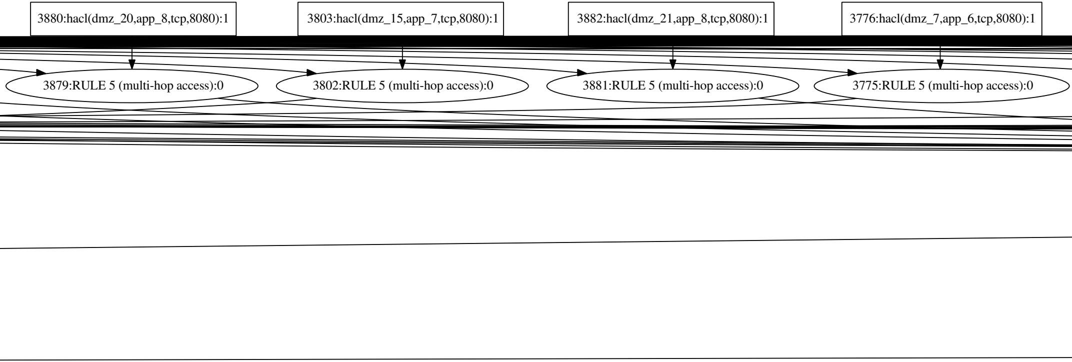

3795:hacl(dmz_11,app_7,tcp,8080):1
3770:hacl(dmz_4,app_6,tcp,8080):1
3734:hacl(dmz_13,app_6,tcp,8080):1
3797:hacl(dmz_12,app_7,tcp,8080):1
3794:RULE 5 (multi-hop access):0
3769:RULE 5 (multi-hop access):0
3733:RULE 5 (multi-hop access):0
3796:RULE 5 (multi-hop access):0
pp_7,tcp,8080):0
3846:networkServiceInfo(app_7,tomcat,tcp,8080,root):1
3847:vulExists(app_7,vul_app,tomcat,remoteExploit,privEscalation):1
3786:RULE 2 (remote exploit of a server program):0

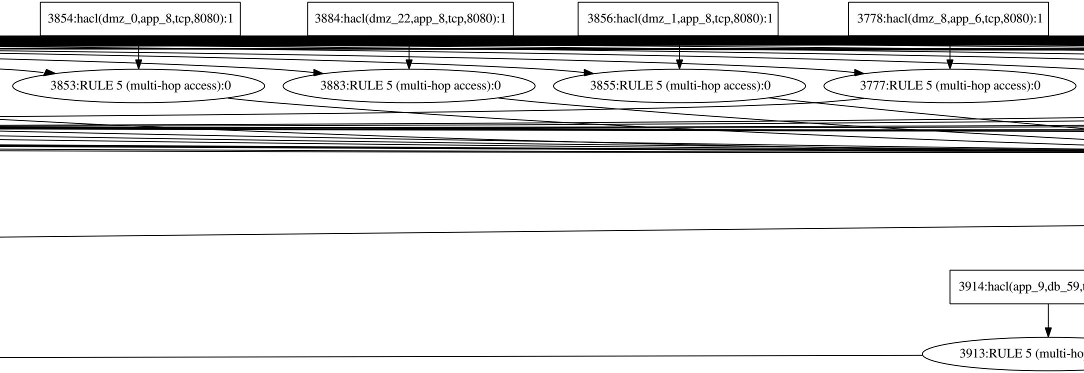

3762:hacl(dmz_26,app_6,tcp,8080):1
3821:hacl(dmz_23,app_7,tcp,8080):1
3740:hacl(dmz_16,app_6,tcp,8080):1
3823:hacl(dmz_24,app_7,tcp,8080):1
3761:RULE 5 (multi-hop access):0
3820:RULE 5 (multi-hop access):0
3739:RULE 5 (multi-hop access):0
3822:RULE 5 (multi-hop access):0
3850:execCode(app_8,root):0

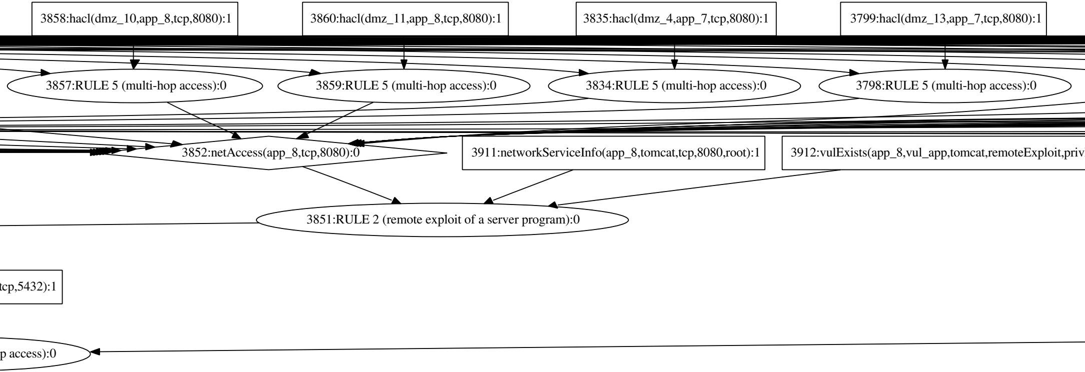

3772:hacl(dmz_5,app_6,tcp,8080):1
3894:hacl(dmz_27,app_8,tcp,8080):1
3896:hacl(dmz_28,app_8,tcp,8080):1
3898:hacl(dmz_3,app_8,tcp,8080):1
3771:RULE 5 (multi-hop access):0
3893:RULE 5 (multi-hop access):0
3895:RULE 5 (multi-hop access):0
3897:RULE 5 (multi-hop access):0

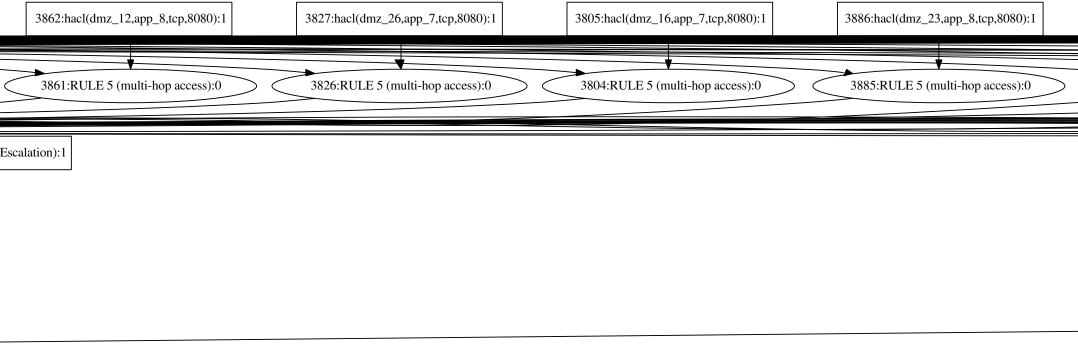

3825:hacl(dmz_25,app_7,tcp,8080):1
3866:hacl(dmz_14,app_8,tcp,8080):1
3872:hacl(dmz_17,app_8,tcp,8080):1
3845:hacl(dmz_9,app_7,tcp,8080):1
3824:RULE 5 (multi-hop access):0
3865:RULE 5 (multi-hop access):0
3871:RULE 5 (multi-hop access):0
3844:RULE 5 (multi-hop access):0

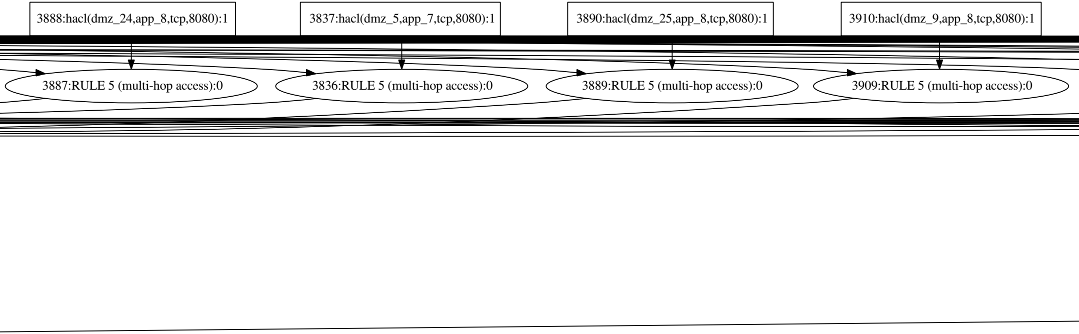

3874:hacl(dmz_18,app_8,tcp,8080):1
3774:hacl(dmz_6,app_6,tcp,8080):1
3876:hacl(dmz_19,app_8,tcp,8080):1
3878:hacl(dmz_2,app_8,tcp,8080):1
3873:RULE 5 (multi-hop access):0
3773:RULE 5 (multi-hop access):0
3875:RULE 5 (multi-hop access):0
3877:RULE 5 (multi-hop access):0

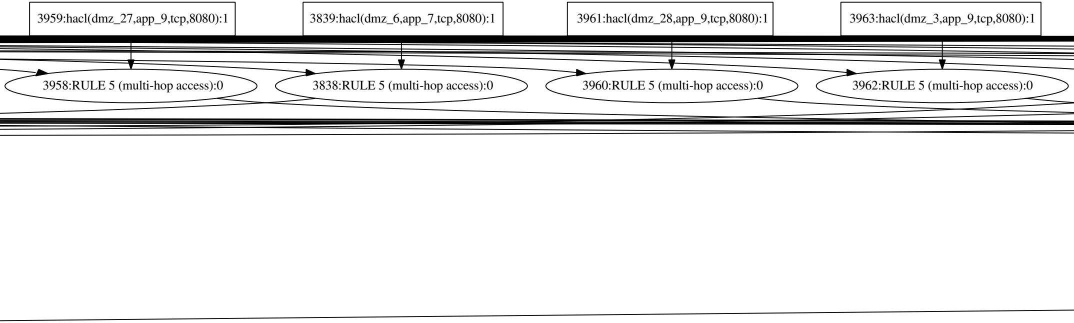

3880:hacl(dmz_20,app_8,tcp,8080):1
3803:hacl(dmz_15,app_7,tcp,8080):1
3882:hacl(dmz_21,app_8,tcp,8080):1
3776:hacl(dmz_7,app_6,tcp,8080):1
3879:RULE 5 (multi-hop access):0
3802:RULE 5 (multi-hop access):0
3881:RULE 5 (multi-hop access):0
3775:RULE 5 (multi-hop access):0

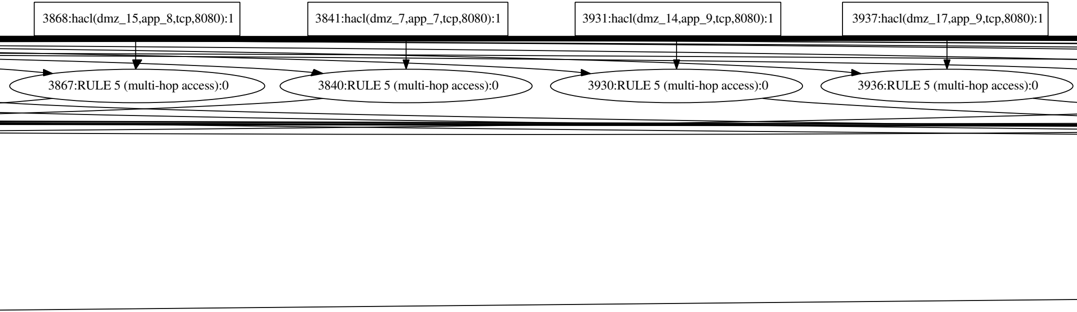

3854:hacl(dmz_0,app_8,tcp,8080):1
3884:hacl(dmz_22,app_8,tcp,8080):1
3856:hacl(dmz_1,app_8,tcp,8080):1
3778:hacl(dmz_8,app_6,tcp,8080):1
3853:RULE 5 (multi-hop access):0
3883:RULE 5 (multi-hop access):0
3855:RULE 5 (multi-hop access):0
3777:RULE 5 (multi-hop access):0
3914:hacl(app_9,db_59,t
3913:RULE 5 (multi-ho

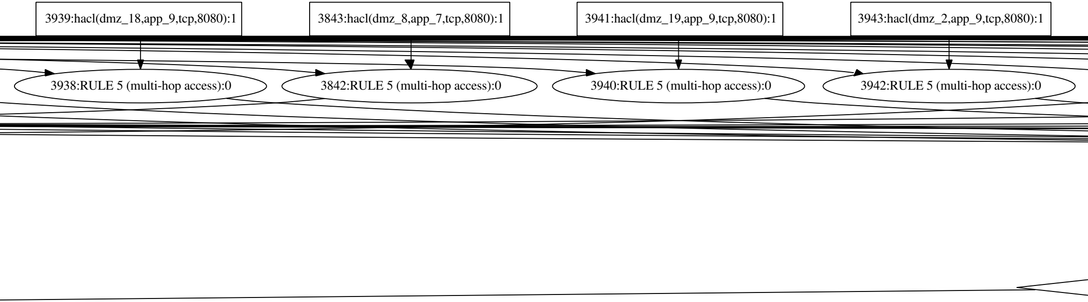

3858:hacl(dmz_10,app_8,tcp,8080):1
3860:hacl(dmz_11,app_8,tcp,8080):1
3835:hacl(dmz_4,app_7,tcp,8080):1
3799:hacl(dmz_13,app_7,tcp,8080):1
3857:RULE 5 (multi-hop access):0
3859:RULE 5 (multi-hop access):0
3834:RULE 5 (multi-hop access):0
3798:RULE 5 (multi-hop access):0
3852:netAccess(app_8,tcp,8080):0
3911:networkServiceInfo(app_8,tomcat,tcp,8080,root):1
3912:vulExists(app_8,vul_app,tomcat,remoteExploit,priv
3851:RULE 2 (remote exploit of a server program):0
tcp,5432):1
p access):0

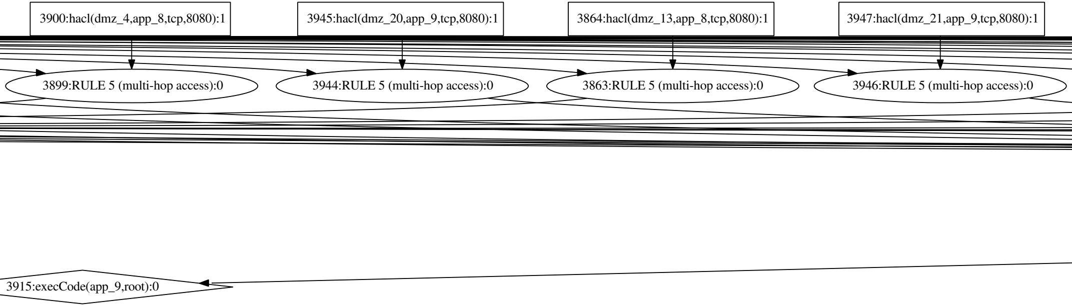

3862:hacl(dmz_12,app_8,tcp,8080):1
3827:hacl(dmz_26,app_7,tcp,8080):1
3805:hacl(dmz_16,app_7,tcp,8080):1
3886:hacl(dmz_23,app_8,tcp,8080):1
3861:RULE 5 (multi-hop access):0
3826:RULE 5 (multi-hop access):0
3804:RULE 5 (multi-hop access):0
3885:RULE 5 (multi-hop access):0
Escalation):1

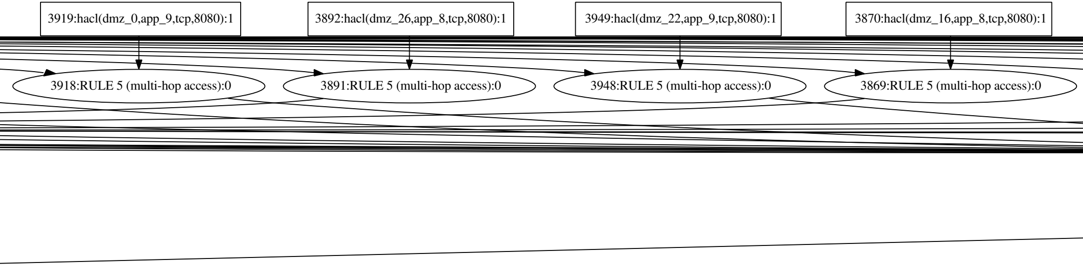

3888:hacl(dmz_24,app_8,tcp,8080):1
3837:hacl(dmz_5,app_7,tcp,8080):1
3890:hacl(dmz_25,app_8,tcp,8080):1
3910:hacl(dmz_9,app_8,tcp,8080):1
3887:RULE 5 (multi-hop access):0
3836:RULE 5 (multi-hop access):0
3889:RULE 5 (multi-hop access):0
3909:RULE 5 (multi-hop access):0

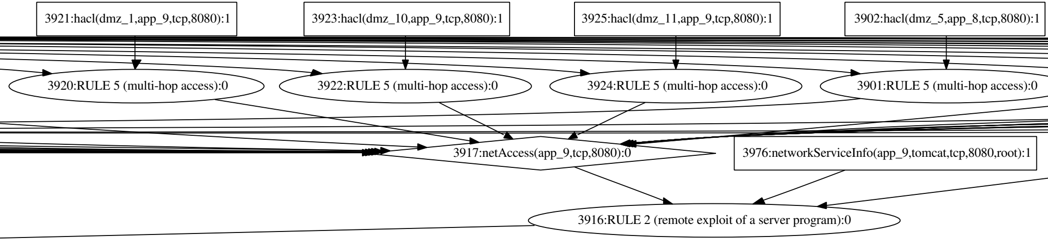

3959:hacl(dmz_27,app_9,tcp,8080):1
3839:hacl(dmz_6,app_7,tcp,8080):1
3961:hacl(dmz_28,app_9,tcp,8080):1
3963:hacl(dmz_3,app_9,tcp,8080):1
3958:RULE 5 (multi-hop access):0
3838:RULE 5 (multi-hop access):0
3960:RULE 5 (multi-hop access):0
3962:RULE 5 (multi-hop access):0

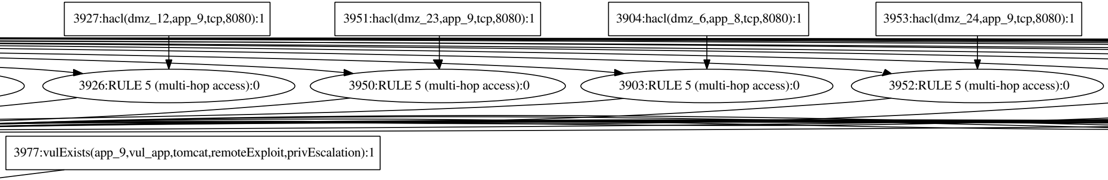

3868:hacl(dmz_15,app_8,tcp,8080):1
3841:hacl(dmz_7,app_7,tcp,8080):1
3931:hacl(dmz_14,app_9,tcp,8080):1
3937:hacl(dmz_17,app_9,tcp,8080):1
3867:RULE 5 (multi-hop access):0
3840:RULE 5 (multi-hop access):0
3930:RULE 5 (multi-hop access):0
3936:RULE 5 (multi-hop access):0

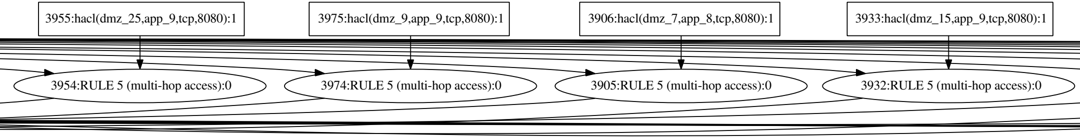

3939:hacl(dmz_18,app_9,tcp,8080):1
3843:hacl(dmz_8,app_7,tcp,8080):1
3941:hacl(dmz_19,app_9,tcp,8080):1
3943:hacl(dmz_2,app_9,tcp,8080):1
3938:RULE 5 (multi-hop access):0
3842:RULE 5 (multi-hop access):0
3940:RULE 5 (multi-hop access):0
3942:RULE 5 (multi-hop access):0

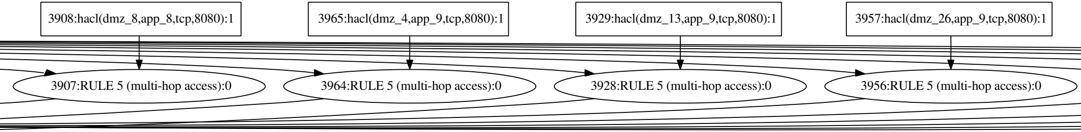

3900:hacl(dmz_4,app_8,tcp,8080):1
3945:hacl(dmz_20,app_9,tcp,8080):1
3864:hacl(dmz_13,app_8,tcp,8080):1
3947:hacl(dmz_21,app_9,tcp,8080):1
3899:RULE 5 (multi-hop access):0
3944:RULE 5 (multi-hop access):0
3863:RULE 5 (multi-hop access):0
3946:RULE 5 (multi-hop access):0
3915:execCode(app_9,root):0

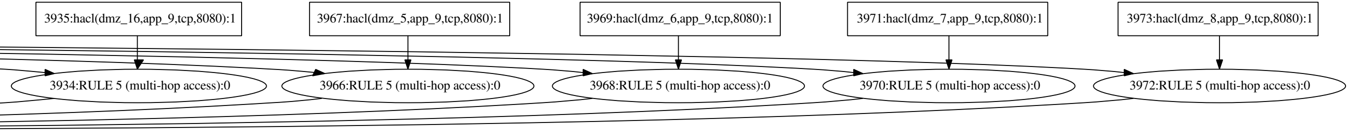

3919:hacl(dmz_0,app_9,tcp,8080):1
3892:hacl(dmz_26,app_8,tcp,8080):1
3949:hacl(dmz_22,app_9,tcp,8080):1
3870:hacl(dmz_16,app_8,tcp,8080):1
3918:RULE 5 (multi-hop access):0
3891:RULE 5 (multi-hop access):0
3948:RULE 5 (multi-hop access):0
3869:RULE 5 (multi-hop access):0


3921:hacl(dmz_1,app_9,tcp,8080):1
3923:hacl(dmz_10,app_9,tcp,8080):1
3925:hacl(dmz_11,app_9,tcp,8080):1
3902:hacl(dmz_5,app_8,tcp,8080):1
3920:RULE 5 (multi-hop access):0
3922:RULE 5 (multi-hop access):0
3924:RULE 5 (multi-hop access):0
3901:RULE 5 (multi-hop access):0
3917:netAccess(app_9,tcp,8080):0
3976:networkServiceInfo(app_9,tomcat,tcp,8080,root):1
3916:RULE 2 (remote exploit of a server program):0


3927:hacl(dmz_12,app_9,tcp,8080):1
3951:hacl(dmz_23,app_9,tcp,8080):1
3904:hacl(dmz_6,app_8,tcp,8080):1
3953:hacl(dmz_24,app_9,tcp,8080):1
3926:RULE 5 (multi-hop access):0
3950:RULE 5 (multi-hop access):0
3903:RULE 5 (multi-hop access):0
3952:RULE 5 (multi-hop access):0
3977:vulExists(app_9,vul_app,tomcat,remoteExploit,privEscalation):1


3955:hacl(dmz_25,app_9,tcp,8080):1
3975:hacl(dmz_9,app_9,tcp,8080):1
3906:hacl(dmz_7,app_8,tcp,8080):1
3933:hacl(dmz_15,app_9,tcp,8080):1
3954:RULE 5 (multi-hop access):0
3974:RULE 5 (multi-hop access):0
3905:RULE 5 (multi-hop access):0
3932:RULE 5 (multi-hop access):0


3908:hacl(dmz_8,app_8,tcp,8080):1
3965:hacl(dmz_4,app_9,tcp,8080):1
3929:hacl(dmz_13,app_9,tcp,8080):1
3957:hacl(dmz_26,app_9,tcp,8080):1
3907:RULE 5 (multi-hop access):0
3964:RULE 5 (multi-hop access):0
3928:RULE 5 (multi-hop access):0
3956:RULE 5 (multi-hop access):0


3935:hacl(dmz_16,app_9,tcp,8080):1
3967:hacl(dmz_5,app_9,tcp,8080):1
3969:hacl(dmz_6,app_9,tcp,8080):1
3971:hacl(dmz_7,app_9,tcp,8080):1
3973:hacl(dmz_8,app_9,tcp,8080):1
3934:RULE 5 (multi-hop access):0
3966:RULE 5 (multi-hop access):0
3968:RULE 5 (multi-hop access):0
3970:RULE 5 (multi-hop access):0
3972:RULE 5 (multi-hop access):0